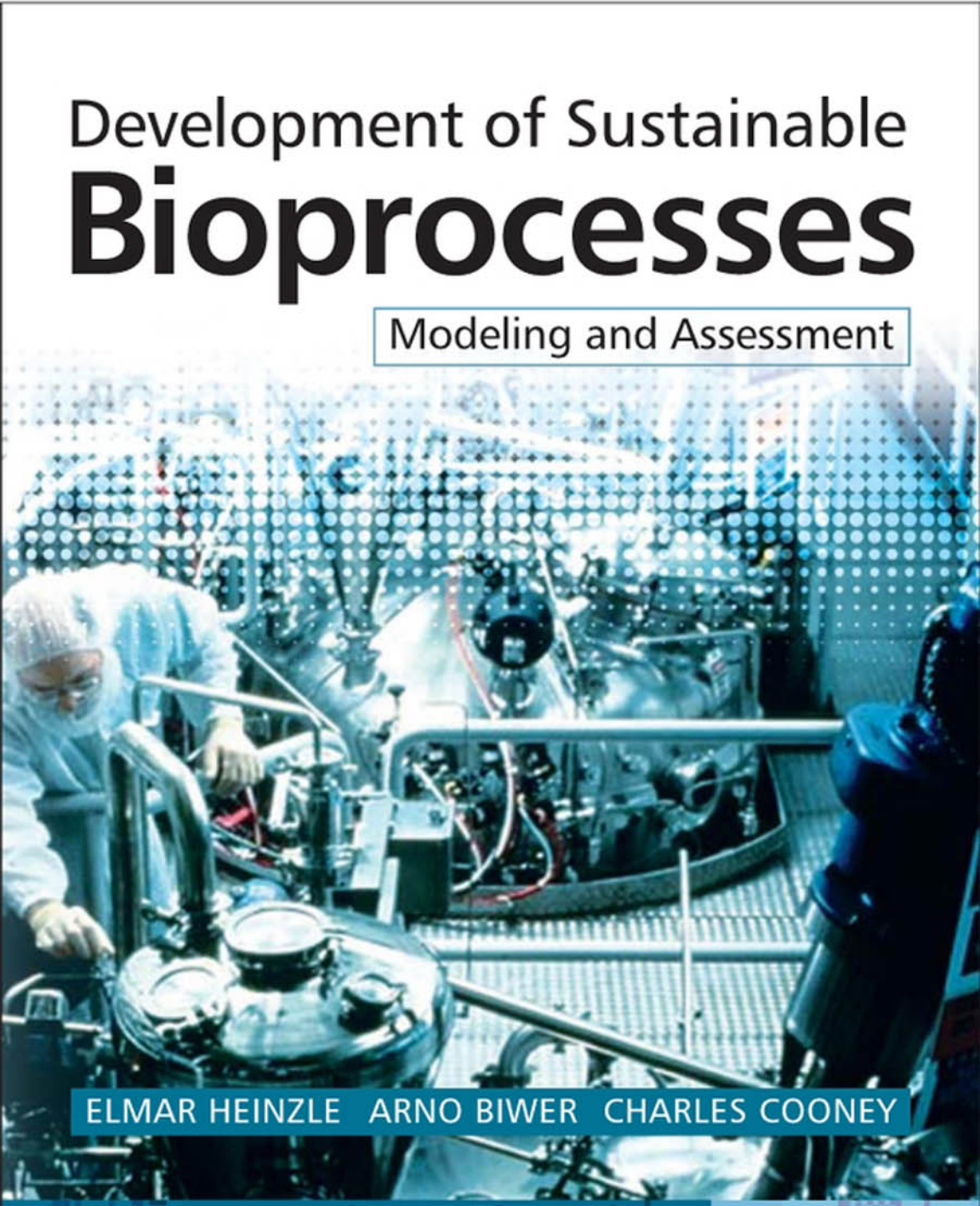
Development of Sustainable Bioprocesses

Modeling and Assessment



ELMAR HEINZLE ARNO BIWER CHARLES COONEY

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ELMAR HEINZLE

University Saarland, Saarbrücken, Germany

ARNO P. BIWER

University Saarland, Saarbrücken, Germany

CHARLES L. COONEY

Massachusetts Institute of Technology, Cambridge, MA, USA



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ELMAR HEINZLE

University Saarland, Saarbrücken, Germany

ARNO P. BIWER

University Saarland, Saarbrücken, Germany

CHARLES L. COONEY

Massachusetts Institute of Technology, Cambridge, MA, USA



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John Wiley & Sons Ltd, The Atrium, Southern Gate, Chichester,
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Telephone (+44) 1243 779777

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Preface

This book is intended to provide a framework for the development of sustainable bioprocesses. It includes methods for assessing both the economic and environmental aspects of biotechnological processes and illustrates their application in a series of case studies covering a broad range of products. Bioprocesses have accompanied human development from very early times. Currently, bioprocesses are gaining increased attention because of their enormous potential for the production of high-value products, especially in human health care and because of their inherent attribute as sustainable processes. New bio-industries have potential as efficient processes based on renewable resources characterized by minimal pollution. Modern methods of enzyme optimization and metabolic engineering are powerful tools for the development of novel efficient biocatalysts. The development of new bioprocesses is enhanced by the application of modern process modeling and simulation techniques, combined with assessment methods that are applied systematically in the very early phases of process development. Future sustainability essentially depends on the ability of industry to develop new processes which are (i) short- and long-term commercially successful, which (ii) at the same time are environmentally friendly using minimal resources that are preferably renewable and constitute a minimal environmental burden, and which (iii) generally satisfy the needs of society.

This book attempts to provide integrating frameworks in a manner useful to both the student in chemical and biochemical engineering, and the scientist and engineer engaged in process development. As time-to-market is a criterion of ever increasing importance, methods are needed which can deliver superior results in a short time. This is of central importance for professionals working in industries applying bioprocesses. Such professionals may be biochemical, chemical, and process engineers, but also biologists, chemists, environmental managers, and business economists. This book may also assist graduate and postgraduate students of economics, as well as environmental sciences. The intent is to

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assist both students and professionals by providing a condensed introduction into the basic theory of bioprocess modeling and sustainability assessment methods, combined with typical case studies. The book is intended to supplement more comprehensive texts on process economics, biochemical reaction engineering, and bioseparation processes. The case studies are supplemented with fully operational models, which are all supplied on the accompanying CD. The models are built using the software SuperPro Designer,™ which is kindly supplied by Intelligen, Inc. (Scotch Plains, NJ, USA) in a version that allows running all examples. These case studies make the book particularly attractive to practitioners who would like to start modeling from an already well developed similar case to shorten development time. The only prerequisites required to be able to follow the book immediately is a basic understanding of bioprocesses and basic economic principles. The reader lacking this background is guided to literature filling these knowledge gaps.

We believe the book is unique in providing (i) an introduction to bioprocess modeling in combination with economic and environmental assessment methods, which both are important in a world with limited resources and increasing environmental pollution; (ii) the book cuts across multiple process industries, including pharmaceutical, biochemicals, chemicals, and food production. The methods presented are broadly applicable in all these fields; (iii) the book also addresses risk and uncertainty analysis, which are particularly important in early process and product development. These methods will help to efficiently direct research and development efforts, to reduce the risk of later stage failures, and to put decision-making on a fundamental basis; (iv) the unique set of case examples from various parts of biotechnology improves the understanding of this technology and provides a starting point for developing one's own specific model.

Organization of the Book

The book consists of two parts. The first part presents the essential, necessary theory, and part two consists of 11 case studies covering a broad range of bio-industries.

Chapter 1 starts with a short introduction to bioprocesses, outlining the expected future potential of biotechnological processing. This chapter also highlights the importance of modeling and simulation for developing sustainable bioprocesses.

Chapter 2, characterizing the development of bioprocesses, describes types of bioprocesses, raw materials, and bioproducts. Then, essentials of bioreaction stoichiometry, thermodynamics, and kinetics are introduced. The elements of bioprocesses described comprise those of upstream processing, bioreaction, downstream processing, utilities, and also waste treatment and recycling. This chapter is concluded by the description of the development process including managerial issues.

Chapter 3 provides a hands-on approach on setting up a process model and simulating it. This starts with problem structuring, process analysis, and setting up a process scheme. Then the implementation into a computer model is illustrated. This chapter concludes with methods of uncertainty analysis comprising scenario analysis, sensitivity analysis, and Monte Carlo simulations.

An integral part of the book is sustainability assessment, and a problem-oriented approach to process development is described in Chapter 4. The economic assessment follows standard procedures, as already included in SuperPro Designer™. The environmental

assessment, which is primarily based on mass and energy balances of the process, uses an ABC method developed for such types of problems. Social assessment and safety are briefly addressed but not incorporated in the case studies.

The second part describes 11 case studies which originate from our own work and from various persons around the world who used modeling tools for bioprocesses and who kindly accepted our invitation to contribute to this book. All process model examples are implemented into SuperPro Designer™. An attached CD-ROM contains the process models described in the book. The models are selected such that characteristic examples of each application area covered are comprized. These major areas of bioprocess industries covered include bulk biochemicals, fine chemicals, enzymes, and low- and high-molecular-weight pharmaceuticals. These elaborate examples are of inestimable value in providing a quick hands-on approach, which will be highly welcomed both by students and professionals already working in bioprocess industries.

The authors' different backgrounds help to cover the broad field. Prof. Charles L. Cooney from the Chemical Engineering Department at MIT in Cambridge, Massachusetts, USA has extended experience in chemical and biochemical engineering. He initiated the creation of SuperPro Designer™ during the PhD work of Demetri Petrides, who is now chief executive of Intelligen, Inc. Throughout his career he closely cooperated with firms actively engaged in biochemical process development. Prof. Elmar Heinzle from the Biochemical Engineering Institute of the Saarland University, Germany studied Applied Chemistry at the Technical University of Graz, Austria and specialized in Biochemical Engineering. During his time at the Swiss Federal Institute of Technology (ETH), Zurich, Switzerland and at the Saarland University he also closely cooperated with various chemical and biochemical industries and was involved in process modeling and assessment. He was also engaged with modeling biochemical kinetics and reactors throughout his carrier and published two books with Drs I.J. Dunn, J. Ingham and J.E. Prenosil [Ingham, J., Dunn, I.J., Heinzle, E., Prenosil, J.E. (2000): Chemical Engineering Dynamics. An Introduction to Modelling and Computer Simulation, 2nd Edition, Wiley-VCH; Weinheim; Dunn, I.J., Heinzle, E., Ingham, J., Prenosil, J.E. (2003): Biological Reaction Engineering. Dynamic Modelling Fundamentals with Simulation Exercises. Wiley-VCH; Weinheim]. These books stimulated the organization of this book combining 50% basic theory with 50% case studies supplied as executable computer programs on an attached CD. Dr Arno Biwer studied biogeography at the Saarland University, where he made his PhD in the field of modeling and assessment of biotechnological processes. After a postdoctoral stay at MIT with Prof. C.L. Cooney, he moved back to the Saarland University to put together the book presented here.

The authors hope that they can contribute to the establishment of sustainable bioprocesses, which have a great potential to serve human needs and at the same time help to efficiently use renewable resources and to prevent pollution of our limited natural environment. The authors would be very grateful for any comments on the book. Please, use the corresponding web site http://www.uni-saarland.de/dsbp.

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List of Contributors

Jochen Buechs
Biochemical Engineering
RWTH Aachen University
Worringer Weg 1
52056 Aachen, Germany

Sindélia S. Freitas
Centre for Biological and Chemical Engineering
Instituto Superior Técnico
Av. Rovisco Pais
1049-001 Lisbon, Portugal

Justus von Geibler
Wuppertal Institute for Climate, Environment, Energy
Research Group Sustainable Production and Consumption
Döppersberg 19
42103 Wuppertal, Germany

M. Abdul Kholiq
Biochemical Engineering
Saarland University
P.O. Box 15 11 50
66041 Saarbrücken, Germany

Development of Sustainable Bioprocesses E. Heinzle, A. Biwer and C. Cooney
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Arnd Knoll
Biochemical Engineering
RWTH Aachen University
Worringer Weg 1
52056 Aachen, Germany

Christa Liedtke
Wuppertal Institute for Climate, Environment, Energy
Research Group Sustainable Production and Consumption
Döppersberg 19
42103 Wuppertal, Germany

Karen McDonald
Department of Chemical Engineering and Materials Science
One Shields Ave
University of California
Davis, CA 95616, USA

Rolf Metz
An der Bahn 11
76351 Likenheim, Germany

Demetri Petrides
Intelligen, Inc.
2326 Morse Avenue
Scotch Plains, NJ 07076, USA

Duarte M.F. Prazeres
Centre for Biological and Chemical Engineering
Instituto Superior Técnico
Av. Rovisco Pais
1049-001 Lisbon, Portugal

José A.L. Santos
Centre for Biological and Chemical Engineering
Instituto Superior Técnico
Av. Rovisco Pais
1049-001 Lisbon, Portugal

Winfried Storhas
Biochemical Engineering
Mannheim University of Applied Sciences MUAS
Windeckstraße 110
D-68163 Mannheim, Germany

Holger Wallbaum
Wuppertal Institute for Climate, Environment, Energy
Research Group Sustainable Production and Consumption
Döppersberg 19
42103 Wuppertal, Germany

Elizabeth Zapalac
Department of Chemical Engineering and Materials Science
One Shields Ave
University of California
Davis, CA 95616

Part I
Theoretical Introduction

1
Introduction

1.1 Bioprocesses

1.1.1 History of Biotechnology and Today's Situation

Biotechnological processes have been essential for human survival and for satisfying various needs throughout human culture. Table 1.1 gives a short overview of the history of biotechnology. Early biotechnological processes that use microorganisms to produce a certain product have been used for several thousand years. The Egyptians brewed beer and baked bread in the 4th millennium BC. A basic purification step, the distillation of ethanol, was applied in the 2nd millennium BC in China. Modern biotechnology was started in the 19th century when general knowledge about biological systems, their components, and interactions between them grew [1.1]. In the first half of the 20th century the first large-scale fermentation processes, namely citric acid and penicillin, were realized. The progress of recombinant gene technology then led to a substantial increase in the number of bioprocesses and their production volume starting with insulin, the first product manufactured with recombinant technology, in the early 1980s.

While the first bioprocesses exclusively used fungi, bacteria and yeasts, the industrial production was later extended with the application of enzymes and mammalian cells. Other biocatalysts like plant and insect cells, and transgenic plants and animals were added to the available platform of technologies but are much less used in production so far. In parallel, fermentation and downstream technologies were further developed and the engineering knowledge about designing bioprocesses grew significantly.

Today, the bioindustries have reached a critical size and are additionally based on a broad understanding of genomics, proteomics, bioinformatics, genetic transformation, and molecular breeding. Table 1.2 shows the industries where bioprocesses are applied today. These different industries are reflected in the case studies in the second part of the book. The present worldwide sales of bioprocess products are reported to range between 13 and 60 billion dollars, depending on the source [1.2–1.4]. The share of the different product

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Table 1.1 *Milestones in the history of biotechnology (data taken largely from [1.2] and [1.5])*

Time	Event
4th/3rd mill. BC	Baking, brewing (Egypt)
2nd mill. BC	Ethanol distillation (China)
17th century	Invention of microscope (A. von Leeuwenhoek, Netherlands)
18th century	First vaccination in Europe (cowpox) (E. Jenner, UK). Heat sterilization of food and organic material (Spallanzani, Italy)
1860–1890	Most amino acids isolated, first tyrosine (J. von Liebig, Germany)
1890s	*In vivo* synthesis and extraction of hormones from animal tissue
1921	Insulin isolated from pig pancreas (Toronto, Canada)
1920s	Mutation of microorganisms by X-rays and chemicals (e.g. H.J. Mueller, USA)
1923	Commercial production of citric acid (Pfizer, USA)
1940s	Production of penicillin by fermentation (USA)
1950s	Design and scale-up of large aerated fermenters. Elucidation of principles of sterile air filtration
1953	Discovery of the double helix of DNA (J. Watson and F. Crick, USA)
1972	Restriction enzymes (W. Arber, Switzerland)
1973	First recombinant DNA organism (S. Cohen and H. Boyer, USA)
1975	Monoclonal antibodies (G.J.F. Köhler and C. Milstein, UK/Germany)
1976	Genentech first specialist biotech company
1980s	Polymerase chain reaction (PCR). Large-scale protein purification from recombinant microorganisms
1982/1983	First genetically engineered product: human insulin (Eli Lilly/Genentech)
1982	First rDNA vaccine approved in Europe
1986	Release of genetically engineered plant
1995	First bacterial genome sequenced (*Haemophilus influenzae*)
1998	Isolation of human embryonic stem cells
2000/2001	Human genome sequenced

groups on these sales is shown in Table 1.3, where antibiotics and therapeutic proteins dominate due to their relatively high prices.

In 2000, there were 1270 bioscience companies in the U.S. and 1180 in the EU [1.5]. The six largest of them had revenues of $8 billion and invested 20–37% of their revenues in research and development (R&D). The average investment spending for the pharmaceutical industries is 9–18%. The overall R&D spending in biotechnology was $37 billion in 2000, with an expected growth rate of 30% per year [1.5].

The share of bioproducts differs from industry to industry. Some products are provided almost exclusively by bioprocesses, e.g. amino acids like lysine and glutamate, carboxylic acids, e.g. citric and lactic acid, and vitamins, e.g. vitamin B2 and vitamin C. One focus of bioprocesses is the pharmaceutical industry. Since the introduction of the centralized European drug-approval system in 1995, recombinant proteins count for 36% of all new drug approvals [1.6]. More than 100 new drugs and vaccines produced by bioprocesses have been brought to market since the mid 1970s and more than 400 are in clinical trials-the highest number ever [1.2, 1.5]. The average process development from laboratory to final

Table 1.2 *Process industries versus process types. MO = microorganisms (bacteria, yeasts, fungi)*

Industry	Scale	Downstream complexity	Biocatalyst	Products	Biotech market share
Basic chemicals	very large	low	MO/enzymes	organic small molecules	very low
Fine chemicals	medium	medium	MO/enzymes	organic small molecules	low
Detergents	large	low	MO	enzymes	medium
Health care/cosmetics	small–medium	medium–high	MO/enzymes/ mammalian cells	proteins and small molecules	medium
Pharma					
conventional	medium	medium–high	MO	organic small molecules	low–medium
biopharma	small	high	mammalian cells, MO	proteins	high
Food/feed	very large	medium	MO/enzymes	proteins and others	medium
Metal mining	very large	low	MO	metals/metal compounds	very low
Waste treatment	very large	low	MO	Purified water, air, and soil	high

Table 1.3 *Market volume of bioproduct groups. Estimated overall sales were $60 billion in 2000 (= 100%) (Data from [1.4])*

Bioproduct group	Share of bioproduct sales (%)	Typical products
Antibiotics	42	penicillins, cephalosporins
Therapeutic proteins	25	interferon, insulin, antibodies
Other pharma- and animal health products	17	steroids, alkaloids
Amino acids	8	lysine, glutamate
Enzymes	3	proteases, cellulases, amylases
Organic acids	3	lactic acid, citric acid
Vitamins	1	B2, B12, biotin
Polysaccharides	1	xanthan, dextran

approval takes 10–15 years and costs $300–800 million [1.5]. A short but comprehensive overview of present biotechnological production is provided in the book of R. Schmid [1.7].

1.1.2 Future Perspectives

The last decade brought an enormous stimulation from biological sciences combined with informatics, e.g. the genome sequences of man, plants, and microorganisms or the isolation of human stem cells. However, this knowledge waits to be transformed to technology and market products. The knowledge of molecular breeding, stem cell technology and pharmagenomics might lead to strongly personalized therapies and therapeutics.

It can be expected that biocatalysts such as insect and plant cells and transgenic plants and animals sooner or later will reach a much broader applicability, although this might not happen in the next decade. The increased use of extremophiles and their enzymes and biocatalysis in non-aqueous solution will broaden the technology platform for bioprocesses. Apart from the recombinant technology, the naturally occurring organisms also provide a huge reservoir of new products, e.g. the almost endless variety of plants, insects, and microorganisms in the tropical rain forests.

The share of bioprocesses in the different industries will rise substantially during the next decades. Additionally, bioprocesses will be used in industries where they are not used today or where only lab-scale processes are developed, e.g. the production of new materials with new properties that mimic natural materials. It is expected that the combination of biotechnology, nanotechnology, and information technology will lead to a substantial rate of progress and expansion [1.2]. The use of information technology has already led to improvements in the screening and development of new drugs and in the understanding of biological systems (bioinformatics). It might also lead to bio-chips for computers that replace silicon-based chips.

In the chemical industry it is expected that the sales from bioprocesses will rise to $310 billion in 2010 and will than account for more than 20% of the overall sales of that industry [1.3]. Here, an increase is mainly expected for fine chemicals, especially chiral products. Compared with the chemical industry the bioindustries are still immature and production costs are relatively high. Therefore, not only do the strains and fermentations

have to be optimized and production scales increased, but also a substantial progress in downstream technologies is necessary. Modeling, simulation, and accompanying sustainability assessment will play a crucial role in achieving a full exploitation of the potential of bioprocessing.

However, in some areas the expected positive development will reach its full potential only if the public acceptance of biotechnology can be improved considerably (see Section 4.4 and 4.5). The expending development of biofuel is an important example. Here, an open and constructive dialogue based on a sound sustainability assessment (see Chapter 4) is crucial, and scientists can make a valuable contribution to this discussion (see e.g. [1.8–1.10]). Furthermore, well-trained bioengineers are essential for the existing potential of biotechnology to be realized. A more detailed discussion of the future perspectives is given in the literature [1.2, 1.3, 1.5].

1.2 Modeling and Assessment in Process Development

In process development we want to gain an understanding of the actual future production process as early and as detailed as possible. The modeling of the process under development and a thorough assessment helps to improve this knowledge. Here an iterative assessment is essential in order to realize competitive industrial processes. Decisions have to be made based on sound estimates of costs and potentials of a process and the 'hot spots' in the process schedule have to be identified. The assessment should include economic and environmental evaluation; this is known as integrated development. Figure 1.1 illustrates the importance of an early evaluation. The more advanced the process design, the more the final production process with its cost structure and environmental burdens is already determined. The additional cost for redesign to solve a problem that was previously overlooked rises with the development stage. For environmental problems often only end-of-pipe technologies that cause additional cost are possible in a later stage of the development.

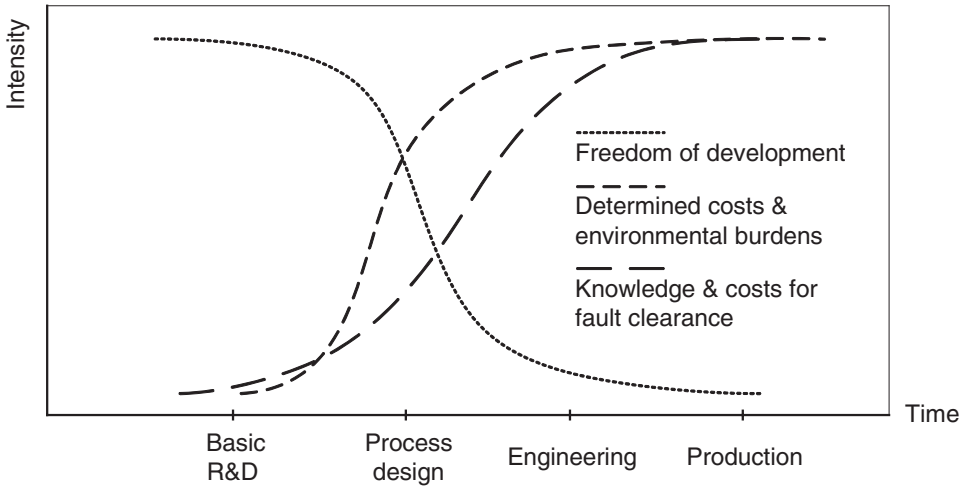


Figure 1.1 *Process knowledge and freedom of decision in the process development [1.11]*

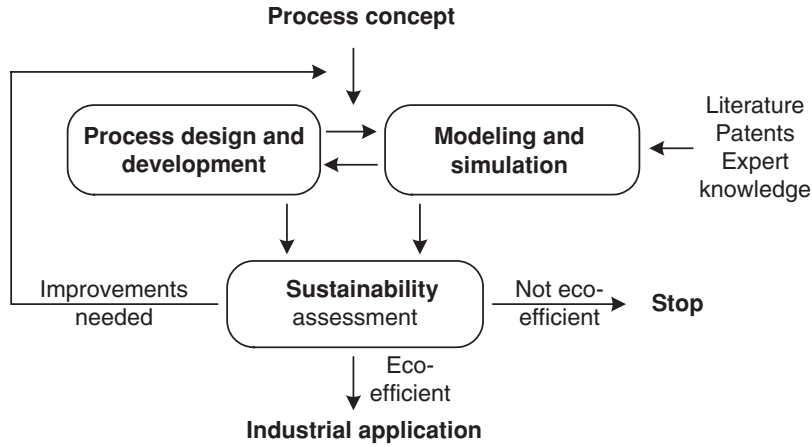


Figure 1.2 *Integrated development of bioprocesses*

In development gaps and uncertainty in data cause an incomplete picture of the expected production-scale process. The use of process modeling can fill this gap and provide a sound evaluation basis [1.11]. Figure 1.2 shows the iterative approach of modeling and assessment. The models should be developed in close collaboration with the process design, and additional information is taken from patents, literature, and other external sources. The simulation results are used to evaluate the process and to guide the R&D effort to the most promising directions and the most urgent problems. Thereby, it is important to look at the whole process and not only to optimize single parts, such as the fermentation step isolated from the whole process. The most competitive and sustainable process is the overall aim. The modeling and assessment process is repeated iteratively and demands an interdisciplinary effort. Using this approach, crucial problems that might impede a successful transformation to an industrial application can be identified earlier, thus avoiding the waste of R&D spending. Naturally, the created models and the assessment based on these models include a certain inherent uncertainty. This uncertainty has to be considered and quantified.

We live in a world of limited resources, with a fast growing population and a limited carrying capacity of our planet. Therefore, besides the economic structure of a process, environmental and social aspects should be considered (see e.g. [1.12–1.15]). The concept of sustainability connects these three aspects that interact in many ways with each other. As we will discuss in Chapter 4, the development of a more sustainable process improves the long-term success and leaves it usually well prepared for future regulatory demands.

In this book, we look at one specific product that might be produced in one or several processes. This product provides a certain human benefit or service. We do not discuss the general question whether it is sustainable to supply this service or not. We also do not discuss other ways that might meet this benefit and whether they are more sustainable. These aspects can be very relevant. However, the required product is usually determined before the process development starts and the discussion of these aspects goes far beyond the scope of this book. Looking only at one specific product, different processes that provide the same product are compared. However, if the product is the same, it can be assumed that

its behaviour during use and disposal is identical. Therefore, once the product is defined, one can concentrate on the production process itself, the supply chain of the raw materials, and the environmental impact of the wastes produced during manufacturing, and one does not have to look at the use and disposal of the product itself. This substantially reduces the necessary effort for modeling and assessment.

It is widely expected that the use of bioprocesses can contribute considerably to a more sustainable development. Biotechnology is seen as a 'powerful enabling technology for achieving clean industrial products and processes that can provide a basis for industrial sustainability' [1.16]. Bioprocesses are economically competitive in a growing number of industries and have advantages concerning several local and global environmental challenges. Bioprocesses are usually based on renewable resources and thus reduce the depletion of limited fossil raw materials. The mild reaction conditions with regard to temperature, pressure, and pH reduce the risk of accidents. Since bioprocesses work with biological systems, the by-products and other wastes have normally a low pollution potential. Nevertheless, the environmental performance has to be optimized and aligned with the economic performance during the development. Here, relatively low product concentrations and productivities are generally the major limitations. The use of agricultural raw materials puts bioprocesses in competition with food production. Furthermore, the aspects of bio-risks and related public acceptance have to be discussed.

The Rio conference and, more recently, the Kyoto Protocol [1.17], identified global warming as one of the most urgent environmental problems. The greenhouse effect is essentially determined by the carbon balance between the different carbon reservoirs. By using renewable carbon sources, bioprocesses usually have an equalized carbon balance. This is an important environmental asset and, with the starting trade of carbon dioxide emission allowances, also an economic advantage. However, in this context the energy requirements of a bioprocess have to be assessed critically.

References

[1.1] Fiechter, A. (2000): History of modern biotechnology I. Springer, Berlin.

[1.2] Sager, B. (2001): Scenarios on the future of biotechnology. *Technol. Forecasting Social Change*, **68**, 109–129.

[1.3] Festel, G., Knoell, J., Goetz, H., Zinke, H. (2004): Der Einfluss der Biotechnologie auf Produktionsverfahren in der Chemieindustrie. *Chem.-Ing.-Tech.*, **76**, 307–312.

[1.4] Storhas, W. (2003): Bioverfahrensentwicklung. Wiley-VCH, Weinheim.

[1.5] Hulse, J. (2004): Biotechnologies: past history, present state and future prospects. *Trends Food Sci. Technol.*, **15**, 3–18.

[1.6] Walsh G. (2003): Pharmaceutical biotechnology products approved within the European Union. *Eur. J. Pharm. Biopharm.*, **55**, 3–10.

[1.7] Schmid, R. (2003): Pocket guide to biotechnology and genetic engineering, Wiley-VCH, Weinheim.

[1.8] Young, A. (2004): The future of biotechnology in support of bio-based industries. *Environ. Sci. Pollut. Res.*, **11**, 71–72.

[1.9] Gaugitsch, H. (2004): The future of biotechnology in support of bio-based industries – a differentiated assessment of the future of biotechnology. *Environ. Sci. Pollut., Res.*, **11**, 141–142.

[1.10] Braun R., Moses V. (2004): A public policy on biotechnology education: What might be relevant and effective? *Curr. Opin. Biotechnol.*, **15**, 246–249.
[1.11] Heinzle, A., Hungerbühler, K. (1997). Integrated process development: The key to future production of chemicals. *Chimia*, **51**, 176–183.
[1.12] El-Halwagi, M. (1997): Pollution prevention through process integration – systematic design tools, Academic Press, London.
[1.13] Verfaillie, H., Bidwell, R. (2000): Measuring Eco-efficiency: A Guide to Reporting Company Performance, World Business Council for Sustainable Development, Geneva.
[1.14] OECD (1995): The life cycle approach: An overview of product/process analysis OECD, Paris.
[1.15] OECD (2001): OECD Environmental indicators: Towards sustainable development OECD, Paris.
[1.16] OECD (1998): Biotechnology for clean industrial products and processes – Towards industrial sustainability OECD, Paris.
[1.17] UNFCCC (1997): The Kyoto Protocol; United Nations Framework Convention on Climate Change, Bonn.

2

Development of Bioprocesses

2.1 Types of Bioprocess and Bioproduct

2.1.1 Biocatalysts and Process Types

The fundamental operational element in a bioprocess is the enzyme, while the scope of bioprocesses ranges from reactions with single purified enzymes to complex cellular and even animal and plant systems. To classify the different biocatalysts, one can distinguish between those that are enzymatic biotransformations versus metabolic bioconversions. In enzymatic biotransformations, only one or few specific reactions take place. Metabolic bioconversions, in contrast, need the metabolic system of the living and growing biocatalyst, either of single cultivated cells or the entire plant or animal. Table 2.1 provides an overview of the different biocatalysts. To select the appropriate biocatalyst to produce a desired product, multiple criteria are applied:

- What yield, product concentration, and productivity can be reached?
- What substrate can be utilized, what additional media components are required, and how does it all affect downstream processing?
- What by-products are formed and how do they affect yield and downstream processing?
- What are the challenges in biocatalyst preparation, storage, propagation, security, and safety?
- What are the optimal reaction conditions e.g. temperature, oxygen supply, shear sensitivity, foam formation, etc.?
- How well do we understand the reaction mechanisms, are they robust and genetically stable?
- If the product is expressed intracellularly, how is it extracted?
- How do we purify the desired product from the many impurities in the process?

Enzymatic Biotransformation. Enzymes are proteins with a unique three-dimensional structure able to bind a substrate, usually but not always a small molecule, and catalyse a

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Table 2.1 *Characteristics of biocatalysts. HSA = Human serum albumin, PHB = Poly (3-hydroxybutyrate)*

Biocatalyst	Production device	Raw material	Time-scale	Purification	Complex protien structure	Viral/prion risk	Process examples
Enzymes	bioreactor	pure substrates	short	simple	no	no	cyclodextrin, acrylamide, L-dopa
Bacteria and yeast	bioreactor	simple media	short	medium	no	small	lysine, vitamin B_2, insulin
Fungi	bioreactor	simple media	medium	medium	no	small	citric acid, antibiotics
Mammalian cells	bioreactor	complex media	medium	medium	yes	medium	monoclonal antibodies, interferons
Plant cells	bioreactor	simple media	medium	medium	possible	small	taxol, shikonin, methyldigoxin
Transgenic plants	whole plant	fertilizer, CO_2, various others	long	complex	possible	small	antibodies, antibody fragments, HSA, PHB
Transgenic animals	whole animal	various plant & animal materials	long	complex	yes	high	α_1-antitrypsin, HSA, lactoferrin
Extractive technology		certain parts of plants, animals and humans	long	complex	possible	high	plasma components, taxol

specific reaction, similar to chemical catalysis but under mild conditions of temperature and pressure. The enzyme forms a complex at its active site with the substrate, which is converted via an enzyme–product complex to yield product and free enzyme. Enzymes are classified in six groups according to the chemical reaction they catalyse: (i) oxido-reductases, (ii) transferases, (iii) hydrolases, (iv) lyases, (v) isomerases, (vi) ligases.

Enzymes are both highly specific and selective in the reaction they catalyse and the substrate they utilize. They are usually regio-, stereo- and enatioselective. Their ability to produce enantiopure chiral molecules makes them superior to chemical synthesis that usually produces racemic mixtures. The nature and specificity of their catalytic activity evolves from the three-dimensional structure of the folded protein.

The enzymatic biotransformation can be done by using one or a few enzymes that are purified from their natural source or by using whole cells. Whole cells are used when the product formation requires multiple reaction steps that are catalysed by different enzymes that are all present in the cell, or when the separation of the specific enzyme from the cell is either too complex or expensive, providing the other cell enzymes do not disturb the desired reaction.

The enzyme can be in solution and immobilized, i.e. attached to a solid support or entrapped within a macroscopic support matrix. When immobilized, it can be reused and easily separated from the product solution. Whole cells can be immobilized as well, often without losing much of their desired enzymatic activity.

Enzymatic biotransformations are widely used in the production of fine chemicals and pharmaceuticals, e.g. for vitamin C, amino acids, antibiotics, and steroids (see e.g. [2.1]). An overview of enzymatic processes in industry is given by Liese et al. [2.2]. There are five major categories of reactions where enzymes are used industrially: (i) hydrolysis of proteins, polysaccharides, esters, amides, nitriles, and epoxides; (ii) synthesis of esters, amides, and glycosides; (iii) carbon–carbon bond formation; (iv) reduction reactions; and (v) oxidation reactions [2.3]. The industrially dominating enzymes are hydrolases and oxido-reductases [2.4]. Examples of industrial processes employing enzymes include: high-fructose corn syrup (HFCS) via an immobilized isomerase (over 1 000 000 tons/a), acrylamide (nitrilase, over 10 000 tons/a), nicotinamide (3-stage batch reaction using a nitrilase), L-dopa (β-tyrosinase), L-aspartate (fixed-bed reaction using an immobilized aspartase), L-carnitine (whole cells; dehydratase and hydroxylase), and 7-aminocephalosporanic acid (glutaryl amidase). In this book, we use the enzymatic hydrolysis of penicillin G using immobilized penicillin amidase as an example for the environmental assessment (Section 4.3.6). The production of cyclodextrin using cyclodextrin glycosyl transferase (CGTase) is described as a full scale case study in Chapter 9.

Enzymes are not only used as biocatalysts in production processes but also as products in their own right for clothes-washing detergent additives, mainly proteases, lipases, amylases, and cellulases that are produced by fermentations in large amounts.

Metabolic Bioconversion using Cell Cultivation. Figure 2.1 shows a classification of living organisms. Theoretically, species from any class or parts thereof can be used as biocatalysts. Traditionally, prokaryotic bacteria and eukaryotic fungi have been used. Together with the algae and the protozoa, the fungi constitute the protists. Today, algae are only used to produce food stuff and food additives, mainly in Japan [2.5, 2.6], while protozoa have not been used industrially at all. Plants and animals have been sources of biocatalysts for a long time, and today bioprocesses are evolving with transgenic plants and animals to make

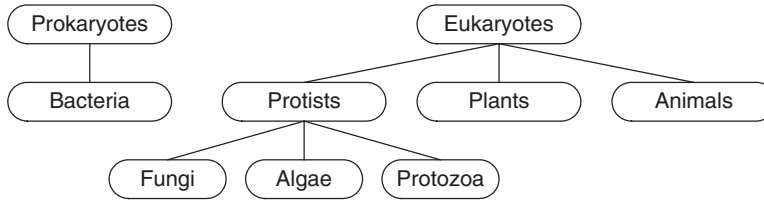


Figure 2.1 *A classification of the living organisms with particular attention to the groups that regularly provide biocatalysts*

recombinant proteins. In this book, single cells separated from plants or animals, as well as whole animals or plants, are considered.

Mostly pure cultures are applied in bioprocesses, i.e. only one species is cultivated. Defined mixed cultures of more than one species are relatively rarely used. However, the largest scale bioprocesses are undefined, mixed cultures used in environmental biotechnology for diverse applications such as waste water treatment and mineral leaching.

(i) *Bacteria*

Bacteria are unicellular prokaryotes with a rigid cell wall. Media composition, temperature, gaseous environment, and pH are key determinents for their growth. Bacteria show a range of responses to oxygen. Aerobic bacteria require oxygen for their growth, anaerobic ones grow only at the absence of oxygen, while facultatively-anaerobic bacteria are able to grow under both conditions. Depending on the temperature optimal for growth, one can distinguish between psychrophiles (20–30 °C), mesophiles (30–40 °C), thermophiles (45–60 °C), and extreme thermophiles (extremophiles) (80–105 °C). The optimum pH for most of the bacteria lies between pH 6.5 and pH 7.5, although there are extremophiles that live at higher or lower pH. Another important group of prokaryotes are the *Actinomycetes*. These organisms propagate as mycelia (similar to the molds) forming highly viscous fermentation broths that present a challenge for oxygen transfer. This group of prokaryotes are especially important in the production of antibiotics.

Only a relatively small number of bacteria that have been studied very well are used commercially as biocatalysts. Often-used genera are *Escherichia*, *Bacillus*, *Corynebacterium*, *Clostridium*, *Acetobacter*, *Pseudomonas*, *Lactobacillus*, and *Zymomonas*. In a bioprocess, either the wild type that is found in Nature or, increasingly often, a genetically modified strain of the species is used. The modifications are done either by classical random mutagenesis or more commonly today by genetic engineering.

Bacteria can be cultivated in large volume with inexpensive media, and high productivity is regularly realized. A wide variety of products is produced with bacteria, ranging from organic acids, amino acids, and vitamins to biopolymers and pharmaceutical proteins. Their use for enzyme production is common. They are constrained for biotherapeutic protein synthesis by an inability to implement post-translational modifications that are required for many therapeutic proteins. Furthermore, the proteins are usually expressed and accumulated intracellularly and tend to form insoluble inclusion bodies which complicates their purification as active molecules. Nevertheless, a number of proteins are produced industrially using bacteria, mainly in *Escherichia coli*. Examples include: insulin, interferons, interleukins, and human growth hormone. The

production of DNA for vaccination and gene therapy is of ever-increasing importance and is discussed in the case study described in Chapter 15.

The second part of the book contains three case studies using bacteria as biocatalyst: *E. coli* used to produce pyruvic acid (Chapter 6), *C. glutamicum* to produce lysine (Chapter 7), and again *E. coli* to produce human recombinant insulin (Chapter 12)

(ii) *Fungi*

It is convenient to divide the fungi into two subgroups: yeasts and molds. Yeasts are small, single cells that can grow as individual cells or clumps. The yeast most often used is *Saccharomyces cerevisiae*. It is well characterized and at industrial scale it can be grown quickly in inexpensive media. Yeasts are traditionally used to produce alcohol in anaerobic fermentations, baker's yeast, and yeast extract as a food additive. Yeast can also be used for recombinant protein production. Recently, yeast also has been engineered to produce hydrocortisone [2.7].

Molds develop a multicellular, vegetative structure called mycelium, a usually highly-branched system of tubules. They are mostly grown under aerobic conditions and the formation of a dense filamentous mycelium in the form of cell aggregates and pellets often causes oxygen-transfer problems. The two commercially dominant genera are *Aspergillus*, e.g. used for citric acid production (see Chapter 5) and *Penicillium*, used to produce antibiotics (see Chapter 10). For commercial production of riboflavin, three types of organisms are currently used: the bacterium *Bacillus subtilis*, the yeast *Candida famata*, and the filamentous fungi *Ashbya gossypii*. Chapter 8 describes a process using a close relative of *A. gossypii*.

Filamentous fungi are used at very large scale to produce enzymes like amylases, cellulases, and glucoamylases. The production of cellulase using *Trichoderma reesei* is described in a case in Chapters 3 and 4. Yeasts are applied for the expression of human proteins such as insulin, growth factors, and vaccines. The production of human serum albumin using *Pichia pastoris* is described in Chapter 11.

(iii) *Mammalian cells*

Starting in the 1980s, recombinant human therapeutics production represents now the core of human medical biotechnology industry, worth over $32 billion in 2003 [2.8]. Major therapy areas are haematology, diabetes and endocrinology, oncology, central nervous system, and infectious diseases. The majority of these drugs are produced by recombinant DNA mammalian cell cultivation [2.9]. Mammalian cells have been cultivated for about 100 years but only in the 1950s did the first production of poliomyelitis vaccine initiate industrial application of mammalian cells. [2.10, 2.11]. Monoclonal antibodies represent an increasing share of biopharmaceuticals. These are primarily derived from hybridoma cells following the pioneering work of Koehler and Milstein [2.12], who fused lymphocytes and myeloma cells to produce an immortal, reproducing cell line. In the initial virus-production processes baby hamster kidney cells (BHK) were of primary importance. Currently, recombinant Chinese hamster ovary (CHO) cells are probably the most frequently applied production cells.

Unlike most microorganisms, mammalian cells produce correctly folded proteins and secrete them to the culture environment. Additionally, they are unique in carrying out required post-translational modifications of proteins, e.g. glycosylations. Therefore, they are generally used to produce high-value proteins where a correct

(native) three-dimensional structure is crucial. Traditionally, production titers are very low but recent developments have yielded up to 5 g/L of product [2.11]. However, mammalian cell cultivation is generally much more delicate than microbial cultivation. The stability of recombinant mammalian cells is still an important problem. Mammalian cells have complex nutritional requirements often requiring serum, e.g. fetal calf serum. These media components bear a potential risk of contamination by adventitious agents such as viruses. Therefore, new media were more recently developed to allow cultivation in chemically defined media [2.11, 2.13]. Mammalian cells grow quite slowly, with typical doubling times of 12–20 h. Since mammalian cells do not have a cell wall, cells are more shear sensitive and fragile. Typically conditions are 37 °C and pH 7.3. Since they grow more slowly, the oxygen demand is usually lower than for microbial cells. Slow supply of nutrients in fed-batch culture or perfusion culture increases the efficiency of primary metabolism and allows a reduction in the formation of undesirable by-products such as lactate and ammonia [2.11, 2.14]. The rich media applied and the slow growth rate of mammalian cells make these cultures susceptible to infection. This requires specially manufactured equipment with cleaning-in-place (CIP) capability. The complexity of these processes leads to high manufacturing costs. Typical mammalian cell product examples are monoclonal antibodies (see Chapter 13), interferons, vaccines, and erythropoietin.

(iv) *Insect cells*

Besides mammalian cells, the cultivation of insect cells has been commercialized. They can produce recombinant proteins less expensively and more quickly than can mammalian cells and at high expression levels; e.g. 30–50% of the total intracellular protein [2.9] is possible. Insect cells typically grow at around 28 °C and pH 6.2. Two veterinary vaccines for the swine fever virus are produced commercially today [2.15]. However, the overall use of insect cells is limited; they are less well understood then mammalian cells and much more research is necessary before they may become a broadly applicable tool in the bioprocess industry.

(v) *Plant cells*

Plant cells are 10 to 100 times larger then microbial cells and more sensitive to shear; their metabolism is slower, with doubling times of 20–100 h resulting in low volumetric productivities even though high cell densities can be reached. As a consequence only higher-value products are reasonable targets for plant cell culture. Plant cells are cultivated as a callus or a lump of undifferentiated plant tissue growing on a solid nutrient medium or as aggregated plant cells in suspension. A comprehensive introduction to the field of plant cell culture is given by Chawla [2.16].

Plant cell culture shows a number of advantages compared with transgenic plants. The cultivation is independent of the geographical location and the season. Owing to the standardized conditions a more constant product quality is possible and at least for some products higher yields can be reached.

Plant cells are mainly used to produce secondary metabolites. An example is the dye shikonin that is produced commercially in Japan in a three-week batch cultivation [2.15]. The anticancer drug paclitaxel (taxol) that was originally extracted from plant materials (see Section 2.1.1.5.) is produced in plant cell culture in stirred tanks of about 30 m^3 volume [2.17]. Plant cells can be potentially used to produce recombinant

proteins of high value as discussed in the case in Chapter 14 for the production of α-1-antitrypsin.

Transgenic Plants. Genetically modified plants can be used to produce a wide variety of products. The expression can take place in the whole plant or only in a certain part as in the seeds. Commonly used plants for this purpose are tobacco, potatoe, rice, and wheat. The use of transgenic plants has a number of advantages compared with fermentation technology. The plant cultivation is inexpensive, easy to scale-up, and free of human pathogens. The harvest methodology is well established and inexpensive. Proteins expressed in seeds are often stable for a prolonged time. However, there a several significant constraints that have delayed industrial application: The expression levels realized today are low and unstable. The post-translational modification patterns differ from the native (human) protein. The plant cultivation depends on the season and the geographical location, and large amounts of genetically modified waste accumulate. A possible future application lies in the production of oral vaccines in plants or fruit such as tomatoes or bananas.

Transgenic Animals. The use of genetically modified animals reduces the dependency on the seasonal and geographical conditions for the case of protein production, and the post-translational modifications are more likely to mimic the native structure. However, there is a higher risk concerning viruses and prions. The genetic modification is usually done by injecting exogenous DNA into the egg cells to produce a vital embryo that is later able to express the desired product. Today, research concentrates on the expression of therapeutic proteins in the milk of transgenic goats or sheep or in the eggs of transgenic chickens. Although animal breeding is relatively inexpensive and well known, it has not yet reached commercial reality [2.18].

Extractive Technologies. Extractive technologies comprise all processes where a product is extracted from natural material. Two important areas are the extraction of pharmaceuticals from human or animal blood and from plant material. Several clotting factors and immunoglobins are extracted from plasma. Over 25% of the pharmaceuticals in the Western World [2.15] are extracted from plant material. In Asia this value is even higher. An example is the anticancer drug paclitaxel (taxol) that is extracted from the bark of the pacific yew tree (*Taxus brevifolia*). Besides pharmaceuticals, also dyes, food colors, flavors, fragrances, insecticides, and herbicides are extracted from plants. These products are usually chemically complex non-protein materials.

2.1.2 Raw Materials

One of the first and most crucial steps in bioprocess design is specification of the raw material requirements. Water is the dominant raw material although the one often receiving the least attention. The other components of the reaction medium can be described as macronutrients and micronutrients. Macronutrients are needed in concentrations larger than 10^{-4} M; they include the carbon-energy source, oxygen, nitrogen, phosphate, sulfur, and some minerals such as magnesium and potassium ions. In some processes there are specific nutrient requirements such as amino acids and vitamins.

The carbon-energy source is the dominant requirement as it provides the carbon for biosynthesis as well as energy derived by its oxidation. Heterotrophic organisms (all bacteria, fungi, animals) need organic compounds as a carbon source, while autotrophic plants and some bacteria can utilize carbon dioxide. Table 2.2 provides an overview of typically

Table 2.2 *Characteristics of commonly used substrates for fermentation*

Carbon source	Composition	Defined composition	Price range ($/kg)	Source	Remarks
Glucose	$C_6H_{12}O_6$	yes	0.10–0.35	starch	price depending on amount and necessary purity
Starch	$(C_6H_{10}O_5)_x$	yes	0.05–0.35	corn/maize/ grain, potato, rice	proteins, fats, fatty acids as impurities
Corn syrup	different sugars mainly glucose, dextrin	no	0.35–0.45	hydrolysed corn or potato starch	around 70–80% dry substance
High-fructose corn syrup	fructose, glucose, higher saccharides	yes	0.45–0.85	hydrolysed corn starch	around 50% fructose and 50% glucose, and higher saccharides
Molasses	mainly carbohydrates	no	0.08–0.12	sugar beet, sugar cane	around 50% fermentable sugars, 20% water, 10% organic acids, N-source, also vitamins, minerals
Cottonseed flour	mainly proteins, carbohydrates	no	0.12–0.55	cotton	ca. 40–50% proteins, 20–40% carbohydrates, also amino acids, fats, vitamins, minerals, also N- and P-source
Corn steep liquor	mainly proteins and peptides, lactic acids, sugar	no	0.05–0.15	by-product of corn wet milling process	around 50% dry substance; protein content varies depending on source (20–50%), also N- and P-source
Soybean oil	fat, fatty acids	yes	0.15–0.50	soybeans	almost 100% fats/fatty acids
Palm oil	fat, fatty acids	yes	0.15–0.50	oil palm tree	almost 100% fats/fatty acids
Glycerol	$C_3H_8O_3$	yes	0.2–0.3	natural oils & fats	often by-product of biodiesel production
Ethanol	C_2H_6O	yes	0.2–0.8	oil/gas or fermentative	
Methanol	CH_4O	yes	0.20–0.25	based on oil/gas	

used carbon sources. On average, 50% of the carbon source is incorporated in the biomass. The remaining 50% is used to derive energy for biosynthesis resulting in carbon dioxide production.

Nitrogen accounts for 10–14% of the dry cell mass. Most widely used nitrogen sources are ammonia and ammonium salts [NH_4Cl, $(NH_4)_2SO_4$, NH_3NO_3], but also proteins, amino acids, urea, and complex materials like yeast extract, soy meal, cotton seed extract, and corn steep liquor.

Oxygen amounts to 20% of the cell mass and hydrogen around 8%. Both are derived from the carbon source, and oxygen additionally from the aeration of the reactor. Phosphorus accounts for around 3% of cell dry weight and is provided by phosphate salts such as KH_2PO_4, organic glycerol phosphates, or complex media. Sulfur (0.5% of cell mass) is added as sulfate salts (e.g. ammonium sulfate) or with amino acids (methionine and cysteine) contained in complex media. Magnesium and potassium ions are provided as inorganic potassium and magnesium sulfate, respectively.

Micronutrients are required in low concentrations. Iron, zinc, and manganese are almost always needed. Other elements like copper, calcium, sodium, and boron are needed only under specific growth conditions. The trace elements are often added as inorganic salts. Additionally, depending on the biocatalyst, so called growth factors like vitamins, hormones, or amino acids are necessary to stimulate the growth and the synthesis of some metabolites. Chelating agents, e.g. citric acid or EDTA (ethylenediaminetetraacetic acid), can be used to prevent the precipitation of some ions like Mg^{2+} or Fe^{3+}. Buffers are often used to maintain a desired pH.

In general, one can distinguish between defined or synthetic media and complex or natural media. Defined media contain specific amounts of pure chemicals with a known composition. Complex media include one or more natural materials whose chemical composition is not exactly known and which may vary with source or time. Natural media are often cheaper (e.g. molasses); however, they often cause less reproducible fermentation and more complex downstream processing.

Bacteria and fungi usually only need a relatively simple medium that in the best case consists only of a carbon-energy source, a nitrogen source, and some mineral salts to provide both macro- and micronutrients. Thus, the medium's cost is relatively low. For the cultivation of mammalian cells a more complex medium is necessary. Typical components are glucose, glutamine and other amino acids, mineral salts, antibiotics, vitamins, growth factors, and buffer. Here, an important feature of the media is whether serum is a required ingredient (complex media) or not (synthetic media). Serum provides a number of often unknown organic supplements. However, the use of serum involves a number of disadvantages: Serum is expensive, and its composition is not precisely known and may be variable. Furthermore, it foams easily upon aeration, and the serum proteins can complicate the downstream processing. There is an increasing concern with the risk that viruses and prions can enter a process via serum. For all these reasons, serum-free media are increasingly used in industrial processes.

Plant cell cultures differ from the other cell cultures and usually require a carbohydrate source, typically sucrose, inorganic macronutrients (salts of N, K, Ca, P, Mg, and S) and micronutrients (e.g. Fe, Mn, Zn, Cu). Additionally organic supplements like amino acids, vitamins, and plant growth regulators are needed. The cultures are usually maintained in the dark.

2.1.3 Bioproducts

Product Classifications/Characteristics. There are several criteria that can be used to classify the wide range of bioprocesses by the products that are made. The scale of production affects process configuration, equipment selection, and economics. Usually, one distinguishes between bulk or commodity chemicals made at large scale, fine chemicals (and specialties), and pharmaceuticals made at smaller scale. Bulk chemicals are produced in very large amounts (e.g. more than 1 000 000 tons per year) with a usually simple downstream processing, sold at a relatively low price, and a medium purity. A biocatalyst that grows in inexpensive media and reaches a high productivity is necessary. In contrast, most pharmaceuticals are produced in small amounts, sometimes as low as a few kilograms per year. Since they have a high price, the use of expensive media and complex equipment with low productivities and complex product separation and purification is acceptable for economic commercial production. Downstream costs are strongly increased by the high purity required for human use. The fine chemicals are used as intermediates and have application in a variety of industries. Their annual production, price, and required purity lie between those of bulk chemicals and pharmaceuticals. Table 2.3 provides an overview of typical bioproducts and their market volume.

According to their size, bioproducts can be divided into small molecules, large molecules, and solid particles. Small molecules like sugars, amino acids, organic acids, or vitamins have a molecular weight of 30–600 Da and a radius that is smaller than 1 nm. Large molecules include proteins, nucleic acids, and polysaccharides. They have a molecular weight of 10^3–10^{10} Da and a radius typically larger than 1 nm. Whole cells like yeast or animal cells, ribosomes, or viruses have a radius of up to several μm.

Among the small molecules, one can distinguish between primary and secondary metabolites. Primary metabolites like sugars, organic alcohols, and acids are produced in the primary growth phase of the organism, while secondary metabolites are formed at or near the beginning of the stationary phase, e.g. antibiotics and steroids. This differentiation is, however, not always very clear.

The retention or secretion of the product molecule by the cell has important implications for downstream processing. To separate and purify a product that is retained by the cell requires disruption or extraction to access the intracellular product. Together with the product, a lot of different proteins, acids, and lipids are released into the solution. This causes

Table 2.3 *Typical bioprocesses and their market volume (data from [2.14]). Reproduced by permission from Wiley-VCH*

Product	Annual volume (metric tons)	Approximate value ($ billion)	Price ($/kg)
Ethanol	19 000 000	5	0.25
Citric acid	1 100 000	1.1	1
Glutamic acid	800 000	0.8	1
Detergent protease	100 000	0.3	3
Aspartame	10 000	0.05	5
Cephalosporins	5000	2.5	500
Tetracyclines	5000	0.3	60
Insulin	8	1	125 000
Erythropoietin	0.01	5	500 000 000

additional complexity for the product separation and purification. Additionally, product concentration is limited for most intracellular products. This leads to higher costs. Therefore, only high-price molecules can be produced economically via intracellular expression and retention. Examples are some biotherapeutic proteins. Most bioproducts are secreted into the media (extracellular) where product separation is usually much less complex.

Product Classes. A bioproduct is best described by its chemical composition or structure and its function or application. Proteins, organic acids, and lipids are typical structure classes, while the application can include food and feed additives, pharmaceuticals, detergents, chemical intermediates, or agriculturally used products, e.g. insecticides and herbicides [2.19]. The process designer faces a dilemma in the initial stage of the development because the particular structure of the molecule causes some constraints, and also the product's function causes other constraints. The process needs to be designed around the structure and the function of the product. For example, a therapeutic protein and an industrially used enzyme might have very similar structures and might be produced by the same organism but they have totally different functions. Resulting production processes will be very different. Therefore, when discussing bioproduct classes, one has to keep in mind both the chemical structure as well as the final application.

Organic alcohols and ketones are mainly produced in anaerobic fermentations, from inexpensive carbon-energy sources such as glucose, starchy materials, molasses or sucrose-containing materials. Examples are the production of ethanol using *Saccharomyces* or *Zymomonas*, and acetone and butanol or z-propanol using *Clostridium*.

Organic acids are used, for instance, as intermediates or as food additives. The three major organic acids produced via a bioprocess are citric, lactic, and gluconic acid. Citric acid is produced by fermentation using *Aspergillus niger* (see Chapter 5). The gluconic acid fermentation uses also *A. niger* or *Gluconobacter suboxidans*, while lactic acid is produced via different *Lactobacillus* species. Metabolic engineering creates a new opportunity to improve the production of other organic acids, such as pyruvic acid (see Chapter 6).

Amino acids are the building blocks of proteins and are connected via peptide bonds. The bioproduction of single amino acids started in the 1950s using *Corynebacterium glutamicum*; later also *E. coli* was applied. They are used as food additives (flavor enhancer, sweetener), feed additives, and in pharmaceuticals. The industrially most important amino acids are L-glutamic acid and L-lysine (see Chapter 7) that are produced from molasses and starch hydrolysates, and the chemically synthesized racemic DL- methionine [2.20, 2.21].

Nucleic acids are used as therapeutics, e.g. DNA vaccines, and in gene therapy. For a process example see Chapter 15. Short interference RNA molecules (sRNAi) also have a large future commercial potential as therapeutics and diagnostics. sRNAi molecules interfere with messenger RNA and can as such be applied for the silencing of specific genes. Additionally, these molecules can also interfere with genes and suppress a gene's expression [2.22]. Aptamers, another pharmaceutically interesting group of biochemicals, are small DNA, RNA, or peptide molecules that bind with high specificity and affinity to DNA, RNA, or proteins [2.23].

Antibiotics with a frequent use in human and animal health are produced in fungal fermentation. Penicillin G and V (*Penicillium chrysogenum*), cephalosporin (*Cephalosporium spp.*), and streptomycin (*Streptomyces griseus*) belong to the major antibiotics. Chapter 10 describes the production of Penicillin V.

A number of vitamins are produced in bioprocesses, e.g. vitamin A, C, E, and the B vitamins. *Propionibacterium* or *Pseudomonas* are fermented on glucose or molasses to obtain vitamin B_{12} while vitamin B_2 (riboflavin) is produced by *Ashbya gossypii*, *Candida spp*., and genetically engineered *Bacillus subtilis* [2.24]. *Eremothecium ashbyii* also can be used, as described in Chapter 8.

Biodegradable biopolymers are plastics derived from renewable material. A common form are the polyhydroxyalkanoates (PHA) accumulated as storage material in bacteria. The most common biopolymer is polyhydroxybutyrate (PHB) that is produced at large scale from glucose using recombinant *E. coli*.

Dextran and xanthan are industrially produced microbial polysaccharides. Xanthan is obtained from glucose or starch using the bacterium *Xanthomas campestris* as biocatalyst. Dextran is produced from sucrose by *Leuconostoc*, *Acetobacter*, and other genera. Polysaccharides can be used as thickening, gelatinizing, or suspending agents in food and pharmaceuticals [2.25]. Cyclodextrins are produced by enzymatic conversion of starch (see Chapter 9).

Carotenoids are natural pigments (yellow or red color). Different carotenoids are produced in different microorganisms. *Blakeslea trispora*, for example, is used to obtain β-carotene; xanthophylls are produced by bacteria and algae. Here, oils are often used as carbon source.

Pesticides, especially insecticides, are a relatively new group of bioproducts. The most prominent example is from *Bacillus thuringiensis* which produces an endotoxin selectively effective against a group of insects. The world production in 2003 was around 13 000 tons [2.26].

The group of lipids includes fats, oils, waxes, phospholipids, and steroids. Glycerol and fatty acids are important building blocks. Prostaglandins, leukotrienes, and thromboxane are commercially produced lipids.

Proteins are characterized by four levels of structure: the primary structure (linear amino acid sequence), the secondary, hydrogen-bonded structure (alpha helix and beta sheet), the tertiary (folding pattern of hydrogen-bonded and disulfide-bonded structures), and the quaternary structure (formation of homo- and hetero-multimeric complexes by individual protein molecules). Proteins are of interest predominantly because of their function that depends on a correctly formed structure. However, there are an increasing number of performance proteins of interest because of their physical properties. Proteins have two major applications, as industrial enzymes and as therapeutic and diagnostic proteins. Industrial enzymes are often produced from inexpensive carbon sources by filamentous fungi such as *Aspergillus*, *Fusarium*, *Pichia*, and *Saccharomyces*, and bacteria, mainly *E. coli*. Proteases, lipases, amylases, and cellulases (compare the training case in Chapters 3 and 4) are produced in large amounts at low prices and are applied as washing detergents and in the food, feed, leather, and textile industry. The emergence of the biofuels industry will have a major impact on the need for more of these enzymes at large scale.

Therapeutic and diagnostic proteins are of higher value but produced in very small amounts, using mainly mammalian cell culture but also bacteria and fungi. They require complex downstream processing. Typical groups of therapeutic proteins are vaccines, monoclonal antibodies, and hormones such as insulin, glucagon, and the human growth hormone (hGH). Cytokines are a diverse group of regulatory proteins. From this group, interferons are used to treat autoimmune diseases and cancer, interleukins for asthma, cancer, and HIV

treatment, and erythropoietin (EPO) is used as a growth factor. Chapters 11 to 14 deal with the production of therapeutic proteins.

A large but special field of bioprocessing is the bioleaching of metals, mainly copper, gold, and uranium from low-grade ores and mining wastes using acidophilic, chemolithotrophic iron- and sulfur-oxidizing microbes. The bacteria used for biomining, such as *Thiobacillus* and *Acidothiobacillus*, extract the metals from large heaps of sulfidic ore, e.g. several hundred thousand tons of copper per year [2.27–2.30].

2.2 Bioreaction Stoichiometry, Thermodynamics, and Kinetics

Central to the understanding and design of bioprocesses are the reaction kinetics of the biochemical conversions that are catalysed either by single enzymes or by whole cells. These reactions are described by their stoichiometry, thermodynamics, and kinetics. Together with mass and energy balances on the reactors these fundamental relationships provide a quantitative description for design of the process. The usual performance parameters are conversion yields, productivities or space–time yields, reaction time, and selectivity. From these parameters, one can calculate requirements of raw materials, utilities, and determine reactor size and associated investment and operation costs. These results are also the basis for design and dimensioning of downstream operations.

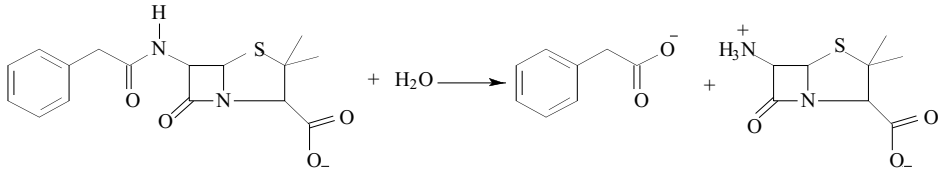
2.2.1 Stoichiometry

Stoichiometry is the basis for quantitative analysis of chemical and biochemical reactions. The stoichiometry of chemical reactions is used to relate the relative quantities of the reactants with products that are formed. Most chemical and biochemical reactions are relatively simple in terms of their molar relationship or stoichiometry. For single reactions, stoichiometric coefficients are well defined. The reaction shown below for components A and B reacting to form product C is an example:

$$\nu_A A + \nu_B B \rightleftarrows \nu_C C \tag{2.1}$$

Here ν_i is the stoichiometric coefficient for species i in the reaction. By convention, the value of ν is positive for the products and negative for the reactants. The stoichiometric coefficients relate the simplest ratio of the number of moles of reactant and product species involved in the reaction.

An example of a single biochemical reaction carried out in a large-scale commercial process is the hydrolysis of penicillin G to 6-aminopenicillanic acid using penicillin acylase (see also Section 4.3.6) and its reaction stoichiometry represented as:

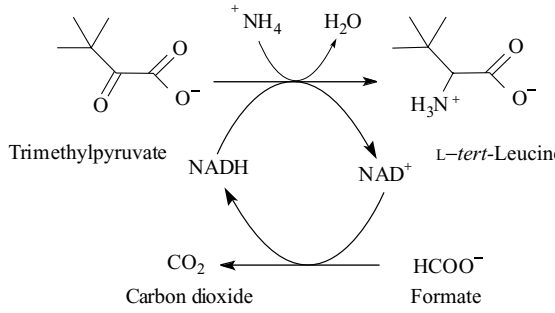


$C_{16}H_{17}N_2O_4S^-$ + H_2O ⟶ $C_8H_7O_2^-$ + $C_8H_{12}N_2O_3S$

Penicillin G salt + H_2O ⟶ Phenylacetate + 6-Aminopenicillanic acid

The stoichiometric coefficients of this reaction are all 1. A proof of the formal correctness of this equation is received by checking elemental and charge balances, which is fulfilled for this reaction.

An important case in biochemical catalysis is coupled reactions as seen in the application of oxido-reductases, which require the regeneration of co-factors [2.2]. An elegant solution is the application of formate dehydrogenase to regenerate NADH. In such cases two reactions are coupled, and a stoichiometric amount of formate has to be fed to the reactor. Here the oxidized product carbon dioxide is eventually released into the gas phase, which has to be considered in a process model.



The overall stoichiometry of this reaction is:

Trimethylpyruvate + *Ammonium* + *Formate* → L-tert-*Leucine* + *Water* + *Carbon dioxide*

(2.2)

In process modeling the net reaction can be treated as a single reaction. The amount of NADH required is not determined by the stoichiometry because it is only needed in catalytic amounts. Corresponding values have to be taken from practical experience or experiment.

A complete, well defined stoichiometric equation can be set up for a whole set of biochemical reactions, e.g. ethanol fermentation by yeast starting from glucose. This represents the net result of many coupled biochemical reactions which utilize multi-co-factors.

$$\begin{aligned} C_6H_{12}O_6 &\rightarrow 2\,CO_2 + 2\,C_2H_6O \\ \text{Glucose} &\rightarrow 2\,\text{Carbon dioxide} + 2\,\text{Ethanol} \end{aligned} \tag{2.3}$$

In the case of fermentation as presented above, the associated production of yeast biomass is neglected. Yeast biomass is the catalyst for the formation of ethanol from glucose, and it is produced from glucose and other nutrients during the fermentation. Thus, the rate and overall yield of ethanol will be influenced by the amount of yeast made but the stoichiometry for ethanol from glucose entering this reaction pathway is not affected.

Biomass synthesis is a complex process requiring the elements carbon, nitrogen, hydrogen, oxygen, sulfur, phosphorus, calcium, iron, magnesium, and many other trace elements in suitable chemical form. For many complex biological reactions, e.g. biomass formation and product synthesis by whole-cell biocatalysis, not all elementary reactions and their

contributions to the overall observed reaction stoichiometry are known [2.31–2.35]. Thus, the general case for fermentation is usually approximated by an overall reaction equation:

$$\begin{aligned}&\text{Substrates} + \mathrm{O_2} \rightarrow \text{Products} + \mathrm{CO_2} + \mathrm{H_2O}\\&\sum_{j=1}^{NS} \nu_{\mathrm{S}_j}\mathrm{C}_{\mathrm{S}_j\mathrm{C}}\mathrm{H}_{\mathrm{S}_j\mathrm{H}}\mathrm{O}_{\mathrm{S}_j\mathrm{O}}\mathrm{N}_{\mathrm{S}_j\mathrm{N}} + \nu_{\mathrm{O2}}\,\mathrm{O_2} \rightarrow \qquad (2.4)\\&\rightarrow \sum_{j=1}^{Np} \nu_{\mathrm{P}_j}\mathrm{C}_{\mathrm{P}_j\mathrm{C}}\mathrm{H}_{\mathrm{P}_j\mathrm{H}}\mathrm{O}_{\mathrm{P}_j\mathrm{O}}\mathrm{N}_{\mathrm{P}_j\mathrm{N}} + \nu_{\mathrm{CO2}}\,\mathrm{CO_2} + \nu_{\mathrm{H2O}}\,\mathrm{H_2O}\end{aligned}$$

where the jth substrate or product, such as metabolites or biomass, is given by a general formula. $\nu_{\mathrm{S}j}$ and $\nu_{\mathrm{P}j}$ are the stoichiometric coefficients. N_S and N_P are the numbers of substrates and products, respectively. It is generally recommended to formulate all equations in terms of C-moles, i.e. such that every organic molecular formula contains one atom of carbon, and then all $\mathrm{S}_j\mathrm{C} = 1$ and $\mathrm{P}_j\mathrm{C} = 1$. Examples are $\mathrm{CH_2O}$ for glucose, and lactic and acetic acids, or $\mathrm{CH_2O_{0.5}}$ for ethanol. The general formula for biomass grown under carbon-limited conditions is $\mathrm{CH_{1.8}O_{0.5}N_{0.2}}$ or $\mathrm{CH_{1.8}O_{0.5}N_{0.2}S_{0.002}P_{0.02}}$, if sulfur and phosphorus are also considered. This allows one to represent a 'mole of cells' with a molecular weight of 25.3 g/C-mol. While this mole of cells does not have a physical basis it does allow one to write the general fermentation balance on a molar basis. Average compositions of cellular polymeric materials are listed in Table 2.4. The ratio of stoichiometric coefficients directly provides C-molar yield values.

Some indication as to the relative magnitudes of the stoichiometric coefficients can be obtained from elemental balancing. Elemental balances of the above general reaction are:

$$\begin{aligned}&\mathrm{C}: \sum_{j=1}^{NS} \nu_{\mathrm{S}_j} S_j\mathrm{C} - \sum_{j=1}^{NP} \nu_{\mathrm{P}_j} P_j\mathrm{C} - \nu_{\mathrm{CO2}} = 0\\&\mathrm{H}: \sum_{j=1}^{NS} \nu_{\mathrm{S}_j} S_j\mathrm{H} - \sum_{j=1}^{NP} \nu_{\mathrm{P}_j} P_j\mathrm{C} - 2\nu_{\mathrm{H2O}} = 0\\&\mathrm{O}: \sum_{j=1}^{NS} \nu_{\mathrm{S}_j} S_j\mathrm{O} + 2\nu_{\mathrm{O2}} - \sum_{j=1}^{NP} \nu_{\mathrm{P}_j} P_j\mathrm{O} - 2\nu_{\mathrm{CO2}} - \nu_{\mathrm{H2O}} = 0\\&\mathrm{N}: \sum_{j=1}^{NS} \nu_{\mathrm{S}_j} S_j N \;-\; \sum_{j=1}^{NP} \nu_{\mathrm{P}_j} P_j N = 0\end{aligned} \qquad (2.5)$$

Table 2.4 *Average composition of S. cerevisiae excluding ash (4–8%) [2.35]. Data taken from Kluwer Academic Publishers*

Macromolecule	Elemental composition	Percent by weight	(g/C-mol)
Proteins	$\mathrm{CH_{1.58}O_{0.31}N_{0.27}S_{0.004}}$	57	22.5
RNA	$\mathrm{CH_{1.25}O_{0.25}N_{0.38}P_{0.11}}$	16	34.0
DNA	$\mathrm{CH_{1.15}O_{0.62}N_{0.39}P_{0.10}}$	3	31.6
Carbohydrates	$\mathrm{CH_{1.67}O_{0.83}}$	10	27.0
Phospholipids	$\mathrm{CH_{1.91}O_{0.23}N_{0.02}P_{0.02}}$	10.8	18.5
Neutral fats	$\mathrm{CH_{1.84}O_{0.12}}$	2.5	15.8
Pool of cellular metabolites	$\mathrm{CH_{1.8}O_{0.8}N_{0.2}S_{0.01}}$	0.7	29.7

In this general problem, there are too many unknowns for the solution method to be taken further, since the elemental balances provide only four equations and hence can be solved for only four unknowns. Assuming that the elemental formulae for substrates, biomass and products are known and hence all S_j and P_j values are defined, there still remain $N_S + N_P + 2$ unknown stoichiometric coefficients and only four elemental balance equations. Only in the case where both N_S and N_P are equal to 1, i.e. where only one substrate produces one product, e.g. biomass, can the system be solved. Further stoichiometric coefficients have to be determined by experiment. Thus, the elemental balances need supplementation by $N_S + N_P - 2$ additional parameters such as substrate, oxygen, and ammonia consumption rates (assuming controlled pH conditions), and carbon dioxide or biomass production rates, such that the condition is satisfied that the number of unknowns is equal to the number of defining equations. Alternatively, specific conversion yield values can be used as supplementary results. In principle, the problem then becomes solvable. In many industrial fermentations, where complex media like soy flour, oils, yeast hydrolysates, corn steep liquor, etc. are used, or where unknown products are formed, elemental balancing allows the completion of the mass balance, provided there are enough experimental data. An example is the pyruvate production described by Biwer et al. [2.36], which is the basis for the case study in Chapter 6. Another example is the citric acid production illustrated in Chapter 5.

Such analysis can be supported by degree of reductance balances [2.35]. For organic compounds the degree of reduction is defined as the number of equivalent available electrons per gram atom C that would be transferred to CO_2, H_2O, and NH_3 upon oxidation. Taking charge numbers: C = 4, H = 1, O = −2, N = −3, S = 6 and P = 5, reductance degrees γ_i, can be defined for a C-mole of:

$$\begin{array}{ll} \text{substrate (S)} & \gamma_S = 4 + m - 2e \\ \text{biomass (X)} & \gamma_X = 4 + p - 2n - 3q \\ \text{product (P)} & \gamma_P = 4 + r - 2s - 3t \end{array} \tag{2.6}$$

where m, p, and r are the number of hydrogen atoms; l, n, and s are the number of oxygen atoms; and q and t are the number of nitrogen atoms per C-mole of substrate, biomass, or product. The reductances for NH_3, H_2O, CO_2, H_2SO_4, and H_3PO_4 are zero by definition. If the carbon and nitrogen balances are not completely closed, it is often possible to determine the average degree of reductance of the missing compounds. If the number of missing carbon and nitrogen atoms is known, a hypothetical molecular formula can be identified for the missing substance. This can be further used in the downstream modeling. This hypothetical compound finally ends up in corresponding waste streams.

If the stoichiometry of the biochemical conversion and the degree of conversion are fixed, it is possible to calculate several important variables. Knowing the feed concentrations an estimate of final concentrations of all components of the equation is directly obtained for the complete conversion case. This is an important basis for the design and calculation of the downstream processing train. From the amount of oxygen consumed, the total amount of heat produced can be directly estimated using the relationship described by Cooney et al. [2.37], $Y_{Q/O2} = 460$ kJ/mol O_2. If ammonia is used as nitrogen source, or if specified organic acids are produced or consumed, a first estimate of the alkali or acid requirement for pH control can be made.

In some cases, particularly when complex media are used, it is very difficult to set up a reaction equation as specified above. In such cases one can directly use yield coefficients

derived from experimental data. Yields are variables, and are used to relate the ratio between various consumption and production rates of mass and energy. They are typically assumed to be time-independent and are calculated on an overall basis. Care is needed in making this assumption. The yield coefficients are usually determined as a result of a large number of elementary biochemical reactions, and it can easily be understood that their values might vary depending on environmental and operating conditions.

The biomass yield coefficient on substrate ($Y_{X/S}$) is defined as:

$$Y_{X/S} = \frac{\text{amount of biomass produced}}{\text{total amount of substrate consumed}} = \frac{\Delta X}{\Delta S} \tag{2.7}$$

Yield coefficients for biomass with respect to nutrients are listed in various publications [2.31, 2.33]. In many cases, these are useful values because the biomass composition is uniform, and often product selectivity does not change very much during an experiment involving exponential growth and associated production. Again, care and judgment are needed in making these simplifying but useful assumptions. Some useful typical values are given in Table 2.5.

Energy yield coefficients may be defined similarly to mass yield coefficients. In terms of oxygen uptake,

$$Y_{Q/O2} = \frac{\text{amount of heat released}}{\text{amount of oxygen consumed}} \tag{2.8}$$

In terms of carbon substrate consumed,

$$Y_{Q/S} = \frac{\text{amount of heat released}}{\text{amount of substrate consumed}} \tag{2.9}$$

A detailed description of some of these dependencies is given in the literature. Despite their limited accuracy, measured yield coefficients are often very useful for practical purposes of process description and modeling. A useful note in the design process is to document these assumptions for subsequent verification with data and results.

Table 2.5 *Typical mass and energy yield values [2.24, 2.42]. Note: The molecular weight of biomass, X, is taken here as 24.6 g/C-mol. Q indicates heat, S substrate. Data taken from Wiley-VCH*

Type of yield coefficient	Dimension	Value
$Y_{X/S,}$ aerobic	C-mol/C-mol	0.4–0.7
$Y_{X/S,}$ anaerobic	C-mol/C-mol	0.1–0.2
Y_{X/O_2} (Glucose)	C-mol/mol	1–2
$Y_{X/ATP}$	C-mol/mol	0.35
Y_{Q/CO_2}	kJ/mol	380–490
Y_{Q/CO_2}	kJ/mol	460
$Y_{Q/X,}$ aerobic (Glucose)	kJ/C-mol	325–500
$Y_{Q/X,}$ anaerobic	kJ/C-mol	120–190

2.2.2 Thermodynamics

Two major thermodynamic characteristics are important for the description of biochemical reactors in process modeling, i.e. heats of reaction and thermodynamic equilibrium. The heat of reaction determines the amount of heat to be removed by appropriate cooling since most biological reactions are run isothermally. Heat changes are determined by reaction enthalpies, ΔH. The heat of reaction, ΔH, can be calculated from the heats of formation or heats of combustion:

$$\Delta H = \sum_{i=1}^{n} \nu_i \, \Delta H_{\mathrm{F}i} = \sum_{i=1}^{n} \nu_i \, \Delta H_{\mathrm{C}i} \tag{2.10}$$

where $\Delta H_{\mathrm{F}i}$ is the heat of formation of component i, and $\Delta H_{\mathrm{C}i}$ is the heat of combustion of component i having stoichiometric coefficients ν_i. If heats of formation are not available, heats of combustion can be determined experimentally from calorimetric measurement. The resulting heat of reaction, ΔH, is negative for exothermic reactions and positive for endothermic reactions by convention.

Whole-cell growth and product formation is a more complex process, and we have available only empirical data, ideally from relevant experiments or by empirical correlation, e.g. typical energy yield coefficients, to calculate the total heat production as described earlier.

Chemical equilibrium is defined by the equilibrium constant, e.g. for the reaction specified in Equation (2.1):

$$K = \frac{C^{\nu_C}}{A^{\nu_A} \; B^{\nu_B}} \tag{2.11}$$

Gibbs Free Energy of a reaction, ΔG, is related to reaction enthalpy, ΔH, and reaction entropy, ΔS. At standard conditions indicated by superscript 0:

$$\Delta G^0 = \Delta H^0 - T \Delta S^0 \tag{2.12}$$

where T is the absolute temperature. The equilibrium constant is related to Gibbs Free Energy of a reaction by:

$$\Delta G^0 = -RT \ln K \tag{2.13}$$

where R is the universal gas constant. An example of an enzymatic equilibrium reaction is the isomerization of glucose to fructose used to produce fructose corn syrup. This is an endothermic reaction with $\Delta H = 2670\,\mathrm{J\,mol^{-1}}$, $\Delta G = 349\,\mathrm{J\,mol^{-1}}$, and $C_{\mathrm{P}} = 76\,\mathrm{J\,mol^{-1}\,K^{-1}}$ at 25 °C [2.38]. From this the calculated equilibrium constants at 30 and 60 °C are 0.886 and 1.034, respectively, as calculated by the van't Hoff equation. For this reaction the equilibrium conversion, x_{e}, is defined as:

$$x_e = \frac{1}{1 + K} \tag{2.14}$$

A temperature increase from 30 to 60 °C therefore allows an increase in the equilibrium conversion of about 8%. High temperature is thus desirable but may be limited by decreased enzyme stability at elevated temperature. While variation of temperature is feasible for simple enzyme-catalysed reactions, it is less often used for cell-based processes since the temperature-range optimal performance is quite narrow, not usually more then a few degrees Celsius. Furthermore, growth and fermentation processes are irreversible processes.

Therefore, thermodynamically possible conversion is not influenced by the usual temperature changes allowed.

2.2.3 Kinetics

The third major characteristic of biochemical reactions is kinetics. These determine the time needed for a desired conversion and therefore reactor size and associated investment costs. Kinetics also determine reaction selectivity and therefore the requirements in downstream processing and waste treatment. There aren't any general rules for kinetics. Neither their types of dependencies on environmental factors nor their magnitudes can generally be predicted using first principles. In this book, we provide a short introduction to enzyme and growth kinetics as needed for process design. More detailed descriptions can be found in various textbooks (e.g. [2.15, 2.32–2.35, 2.39]).

Enzyme Kinetics. Enzymatic bioconversions usually employ single enzymes. Most enzymes applied industrially have relatively simple kinetics, and they are typically applied in a well-defined medium reducing the probability of complex behavior. Major differences to their natural environment are caused by (i) much higher reactant concentrations, leading to substrate and product inhibition, (ii) the application of non-natural solvents, leading to alterations in reaction and deactivation rates and (iii) by immobilization on solid supports, leading to mass-transfer constraints and therefore alteration of the observed kinetics.

Enzymes often follow Michaelis–Menten-type kinetics with first-order dependency on reactant or substrate concentration in the lower, and zero-order dependency in the higher, concentration range.

$$v_S = v_{max} \frac{S}{K_M + S} \tag{2.15}$$

Here $v_{\max}$ is the maximum reaction rate, S is substrate concentration, and K_M is the saturation constant describing the affinity of the enzyme.

A typical example of substrate-inhibition kinetics caused by allosteric effects is:

$$v_S = v_{max} \frac{S}{K_M + S + \frac{S^2}{K_I}} \tag{2.16}$$

where K_I is the inhibition constant. Here, at high substrate concentration, i.e. $S > K_\mathrm{I}$, the rate is decreasing proportionally with $1/S$. In such cases, fed-batch operation or operation in a continuous well-mixed reactor will be beneficial. Another typical phenomenon is product inhibition which can be described with:

$$v_S = v_{max} \frac{S}{K_M + S + \frac{P}{K_I}} \tag{2.17}$$

In such cases it is advisable to use a batch reactor or a continuous reactor with plug flow characteristics.

Biocatalysts in reactors usually undergo irreversible conformational changes, generally known as denaturation or deactivation. This often causes an exponential decrease of activity with time and can be described by a first-order reaction rate process:

$$r_d = -k_d E \tag{2.18}$$

where k_d is the deactivation constant and E is the enzyme concentration. In immobilized systems the kinetic constants may be different due to mass transfer and other molecular reasons. Enzyme kinetics may be more complex and their impact on conversion and reactor choice and size are discussed in various textbooks [2.32–2.34, 2.39–2.41]. The parameters most important for process design and modeling are the type of reactor applied, its dimensions, and requirements for auxiliary equipment, e.g. for control, and flow rates and composition of streams entering and leaving the system. The latter determines the dimensions of surrounding unit operations, e.g. storage of substrate or acid or base for pH control. It is particularly recommendable to use computer simulation to determine optimal reactor design and operation [2.34]. This is also illustrated in the lysine production case study of this book (Chapter 7).

Whole-Cell Kinetics. Whole-cell kinetics is usually much more complex than the individual enzymatic reaction. A typical growth curve is depicted in Figure 2.2, where substrate concentration and the logarithm of biomass concentration are plotted against time.

Cellular growth is autocatalytic in nature and is often observed to be exponential, which is described in a batch reactor by:

$$\frac{dX}{dt} = \mu_{max} X \tag{2.19}$$

Integration yields:

$$X = X_0 e^{\mu_{max} t} \tag{2.20}$$

where X_0 is the initial biomass concentration. Exponential growth is characterized by the maximum specific growth rate μ_{max} which is in turn dependent on the environmental conditions of the process. Observed lag phases may introduce considerable uncertainty into process design. Lag phases can often be avoided or controlled by careful, reproducible

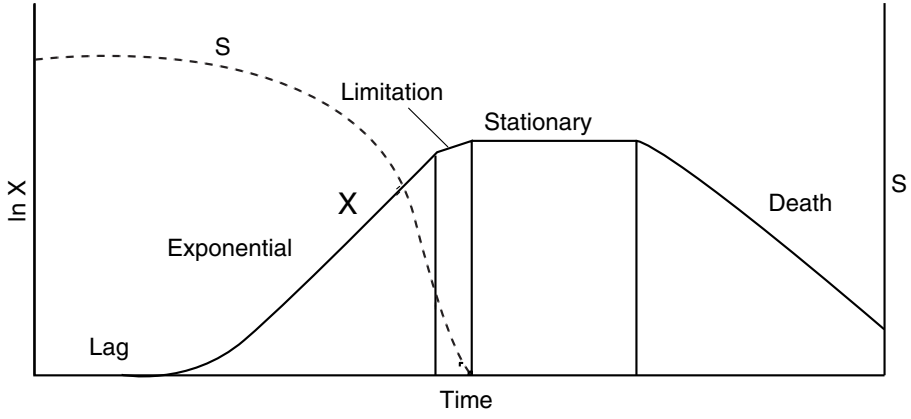


Figure 2.2 *Typical cellular growth phases. X = biomass concentration, S = limiting substrate concentration*

pre-cultivation. Substrate limitation is often described by Monod-type kinetics:

$$\mu = \mu_{\max} \frac{S}{K_M + S} \tag{2.21}$$

where S is the concentration of the limiting substrate, $\mu_{\max}$ is the maximum specific growth rate, and K_M the substrate concentration at half maximum rate.

Important for the design of bioreactors are maximum heat production and oxygen-transfer rates required. These can be calculated knowing maximum growth rates and heat and stoichiometric relationships (see Table 2.5). Anaerobic processes are much less costly in terms of heat removal as can be seen from typical heat yield data provided in Table 2.5, and aeration is not required at all. They are, however, only useful for the production of fermentation products such as ethanol, lactic acid, or butanol.

Three typical kinetic patterns of growth and product formation are frequently observed, as depicted in Figure 2.3. Most production processes operate only until the stationary growth phase, but in the case of secondary metabolites, and typically also heterologous proteins, production occurs only in late phases of cultivation in which growth has slowed to a low rate. The product is completely associated with cellular growth in case A. In case B the production starts already during growth but is prolonged into the stationary phase. Case C describes typical secondary metabolite production where production occurs predominantly during the stationary or even death phase.

Kinetics of product formation for all three cases can be described by the Luedeking–Piret equation:

$$q_P = a\,\mu + b \tag{2.22}$$

showing that the specific product formation rate, q_P, is linked to growth by parameter a. Non-growth-associated production is characterized by parameter b. More complex models are described throughout the literature but in the absence of justification from experimental data this simpler relationship is very useful, especially in the early stages of design. Growth and product-formation kinetics determine required reaction time and final product and by-product concentrations. These are essential parameters for the design of the downstream unit operation train. The lysine case study (Chapter 7) describes how growth and production formation kinetics influence the process performance.

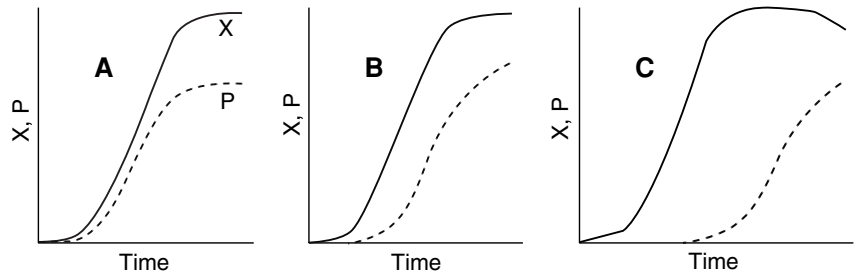


Figure 2.3 *Kinetic patterns of growth and product formation in batch culture. A = growth-associated formation, B = mixed-growth-associated formation, C = non-growth-associated product formation*

Cell deactivation or death is particularly important for sterilization processes used to pre-treat fermentation media. This is simply described by a first-order decay, where the rate constant is a function of temperature as originally introduced by Arrhenius.

$$\frac{\mathrm{d}X}{\mathrm{d}t} = -k_{\mathrm{d},0}\ \mathrm{e}^{\left(-\frac{E_\mathrm{a}}{RT}\right)} X \tag{2.23}$$

Here, $k_{\mathrm{d},0}$ is the pre-exponential rate constant, E_a is the activation energy, R is the universal gas constant, and T is the absolute temperature. Similarly, medium components are decomposed during heat sterilization following the same type of kinetics but typically lower activation energies, E_a. Integration of equations permits optimal design of heat sterilization. Heat sterilization is often carried out continuously in a counter-current way, which allows significant reduction of heat consumption and usually also a more gentle treatment of medium components with short-term high-temperature exposure (see also Chapter 2.3). A more detailed description of sterilization procedure can be found in [2.42].

2.3 Elements of Bioprocesses (Unit Operations and Unit Procedures)

A bioprocess can be divided into the bioreaction section, the upstream processing containing all operations running before the bioreactor step, and the downstream processing with the separation and purification of the product. Figure 2.4 depicts a schematic overview of

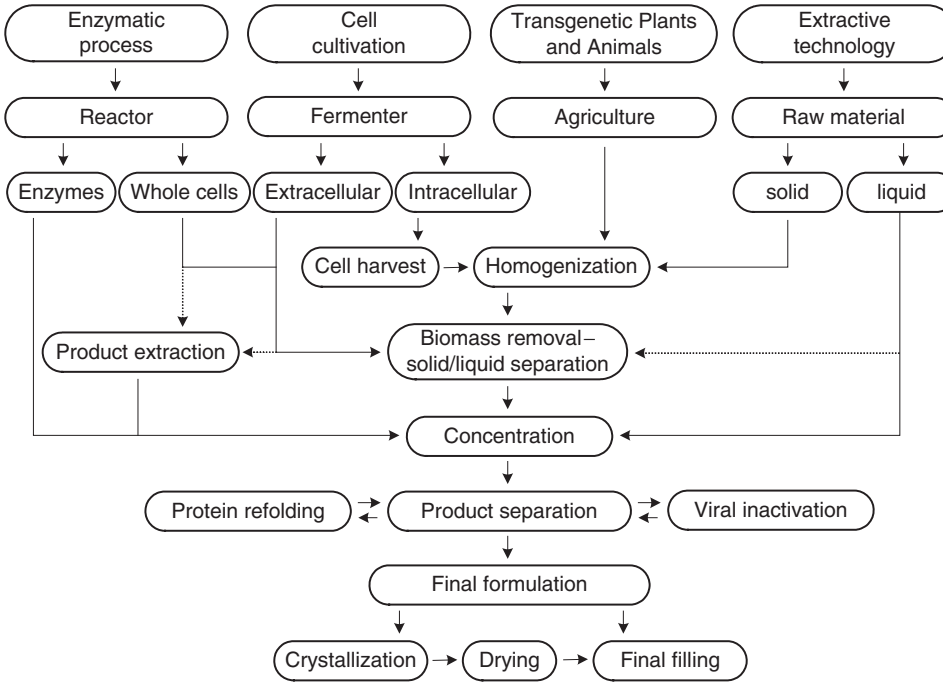


Figure 2.4 *General applicable process tree for the different classes of bioprocesses*

a general process tree for bioprocesses. As commonly done in process engineering, we consider unit operations as basic steps in a production process. Typical unit operations in bioprocesses are for example: sterilization, fermentation, enzymatic reaction, extraction, and filtration or crystallization. A unit procedure we define, analogously to SuperPro Designer,™ as a set of operations that take place sequentially in a piece of equipment, e.g. charging of substrate to a fermenter, addition of acid to adjust pH, reaction, transfer of fermentation broth to another vessel.

2.3.1 Upstream Processing

Upstream processing includes all unit operations that are necessarily performed before the bioreactor step. Typical upstream steps are the preparation of the medium, the sterilization of the raw materials, and the inoculum preparation.

Preparation and Storage of Solutions. Mixing and storage operations are used to provide and store solutions that are needed at some point in the process. Examples are the preparation of the medium for the bioreactor or the buffers needed in chromatography. Liquid and solid components are filled in a tank where they are mixed by agitation. After a homogeneous mixture is reached, the solution can be stored in the tank or transferred to a separate storage tank until it is needed in the process. Usually, the material is either sterilized in the tank or in a continuous sterilizer before its use. A decision needs to be made on which materials to store and how much for how long. This decision has a significant impact on the size of the capital investment for storage and the variable cost of materials inventory. It is also an important decision in risk management as it can allow one to absorb process variation in individual unit procedures.

If possible, raw material solutions are prepared with high concentrations to keep the volume of the preparation tanks small. The solution then is diluted in the bioreactor by adding sterilized water which might be made continuously and thus is not stored. Usually, carbon and nitrogen sources are prepared in separate tanks to avoid the formation of Maillard or non-enzymatic browning reactions during heat sterilization.

The desired volume of the solution has to be defined, e.g. 5 m^3 sugar solution, and the composition and the concentration of the components, e.g. 400 g/L glucose. The mixing conditions (temperature, agitation, etc.) and the order in which the components are added have to be carefully defined to avoid precipitations. One also identifies the need for automation and process control.

The storage conditions might be different from the mixing conditions, particularly with regard to temperature. Especially when using mammalian cell culture, it is necessary to define and validate a maximum storage time for a solution to minimize the risk of contamination or degradation of ingredients.

Sterilization of Input Materials. Input materials are pretreated or sterilized to preclude contamination of the bioreactor. Bacteria and viruses that might be included in the input materials as contaminants are largely destroyed or inactivated. It is important to recognize that inactivation is a probabilistic phenomenon and that one assumes sterile conditions when the possibility of survival of an adventitious agent is less than 10^{-3}. Usually the design is based on the death kinetics of heat-resistant bacterial spores. Sterilization by filtration or by heat are the dominant methods used in bioprocesses.

(i) *Filtration*

Gaseous streams are almost exclusively sterilized by filtration. Mostly membrane filters with pore sizes of 0.2–0.3 μm are used. A compressor usually creates the necessary pressure to assure air flow through the membrane filters that retain contaminants. Prefilters are used for dust and other particles. Air filters are also used to remove bioburden from the exhaust gas stream especially to prevent the release of recombinant or pathogenic microorganisms. Product solutions that contain heat-sensitive substances are also filter-sterilized. With the on going improvement of membrane filters, the general use of filtration for the sterilization of liquids has increased. In some cases, several consecutive membranes with decreasing pore size are used if there is a high particle load to minimize fouling.

(ii) *Heat sterilization*

Sterilization temperature and exposure time are the key parameters for heat sterilization. The higher the temperature, the lower the sterilization time required to reach the same level of sterilization.

Heat sterilization can be done batch-wise or continuously. In batch sterilization, the solution in a tank or the bioreactor is heated most often with steam (in a jacket or sparged directly into the vessel), held at the sterilization temperature for a period of time and then cooling water is used to bring the temperature back to normal operating conditions. Here, often a temperature of 121 °C (corresponding to one atmosphere of overpressure) and a holding time of 10 to 20 minutes are applied.

Continuous heat sterilization requires the necessary heat-exchanger network for heating and cooling. However, the time required to sterilize a given volume is much shorter and the energy consumption is up to 80% lower. Although the applied sterilization temperature is higher, usually around 140–145 °C, heat-sensitive materials are less damaged due to the short exposure time of 120–240 s; this is a consequence of a lower activation energy for thermal degradation than thermal death of bacterial spores. A case, where such sterilization is essential, is the production of riboflavin discussed in Chapter 8.

In both cases, the heat can be transferred either by direct injection of hot steam into the solution or by indirect heat transfer between the steam and the solution via a heat exchanger (e.g. the reactor wall or a tube). When the steam is injected directly, the sterilization temperature is reached more quickly. However, this method leads to dilution of the solution resulting from steam condensation. Therefore, the sterilization via a heat exchanger (tubular or plate-and-frame) is more often used, especially in continuous sterilizers. In a bioreactor, steam injection can be useful, if the solution has to be diluted anyway before the inoculation. For injection the steam has to be appropriately clean.

A continuous, counter-current heat sterilizer typically consists of three heat exchangers. The first heat exchanger heats the cold media using the hot, sterilized media that has been cooled down. The second heat exchanger brings the solution to the sterilization temperature by using steam. The solution then moves through a holding tube. The length of the holding tube is determined by the velocity of the solution and the exposure time necessary for sterilization. Thus, axial dispersion reduces the actual sterilization effect compared to that predicted for plug flow. This axial dispersion has to be considered in the sizing of the heat exchangers. In the following heat exchanger,

the hot, already sterilized solution transfers most of its heat to the cold, not yet sterilized, input stream. This step enables the high energy savings compared with batch sterilization to be obtained. The last heat exchanger cools the solution down to the desired exit temperature using cooling water or another cooling agent.

Inoculum Preparation. The inoculum preparation has to provide a sufficient amount of active cells to inoculate the production fermenter. A so-called cell banking system preserves the strain, e.g. in liquid nitrogen, of the cell line that is used in a bioprocess. Each biocatalyst is stored in a large number of vials or ampoules. One vial provides the inoculum for the starter culture of the seed train for each batch. The cells are grown under conditions that enable high cell densities of actively growing cells within a short time. When the cell concentration reaches a certain level, the entire volume is transferred to the second step where it is diluted with fresh medium. This is repeated, sometimes 2–4 more times, until the necessary amount of biomass is available to inoculate the production reactor.

The volume factor describes the increase of the volume from one inoculum preparation step to the next. For example, a volume factor of 10 means that the volume of one seed reactor is ten times larger than that of the preceding seed reactor. Mammalian cell cultures require relatively low volume factors of around 5 to 10, while bacteria and yeast can be prepared with higher volume factors. The volume factor defines the necessary number of inoculum preparation steps. A typical sequence of an animal cell seed train is: (i) T flask, (ii) roller bottle, (iii) disposable bag bioreactor, (iv) first seed reactor, (v) second seed reactor, and finally the production fermenter. The selection of the volume factor will have a significant impact on the size and cost of the seed preparation portion of the plant.

The medium's composition and the reaction conditions in the seed train can be different from that of the production stage in order to minimize product formation and to maximize cell growth. For example, mammalian cells can be first grown with serum-containing medium to reach high growth rates. In the last seed reactor, the cells are adapted to serum-free medium that is necessary to minimize the risk of contamination of the final product and to simplify the downstream processing.

The modeling of a seed reactor is quite similar to the modeling of the production bioreactor (see the following chapter). The carefully planned seed train is important for an optimized scheduling of a process. Especially for processes using mammalian cell culture, the seed train also occupies a considerable amount of the investment and labor costs (see Chapter 13).

Cleaning-in-Place (CIP). After the use of a piece of equipment, cleaning-in-place (CIP) is done to prepare it for the next batch or cycle. The cleaning may be done without removing the equipment or disconnecting it from the process system (in-place). Almost all bioprocess equipment requires CIP operations, often after every batch or cycle. For some consumables such as membranes or chromatographic resins, the harsh cleaning conditions are the main factor that limits their useful life.

The empty unit, e.g. a reactor, a tank, or a centrifuge, is rinsed with a cleaning agent. The type of cleaning agent, the necessary amount, and the required incubation time have to be defined. A typical CIP sequence is: water – H_3PO_4 (20% w/v) – water – NaOH (5 M) – water, as is applied in the simulation model for the production of insulin (Chapter 12). Other examples with only alkali cleaning are provided in Chapters 13 and 15. The consumed amount of cleaning agent is either expressed as overall demand, e.g. in L or L/m^3, or as a rate such as L/min. The necessary time can be important for the scheduling of the process.

The CIP of a unit normally consists of several steps that often run at different temperatures and the whole process can take between a few minutes and a few hours. A typical sequence could be: (i) washing with process water, (ii) rinsing with a acidic solution, (iii) washing with purified water, (iv) rinsing with a caustic solution, and (v) washing with purified water.

2.3.2 Bioreactor

Bioreactor Types

(i) *Stirred tank bioreactor*
The stirred tank bioreactor is the most commonly used reactor type in bioprocesses. Depending on the complexity of the bioreaction, they range from simple stirred tanks for enzymatic reactions to more sophisticated, aerated fermenters for metabolic bioconversions. The air, usually supplied by a compressor, enters the vessel at the bottom under pressure. The mixing and bubble dispersion are accomplished by mechanical agitation. This requires a relatively high energy input per unit volume. A jacket and/or internal coils allow heating and cooling.

The height/diameter quotient varies. The simplest vessels with the smallest surface area per unit volume have a ratio around 1 but in some large-scale fermenters this can exceed 3. For aerated bioreactors, higher ratios are chosen to prolong the contact time between the rising bubbles and the liquid phase.

(ii) *Airlift bioreactor*
In an airlift bioreactor, mixing is achieved without mechanical agitation by the convection caused by the sparged air. Thus, the energy consumption is lower than in a stirred tank reactor. Owing to the low shear levels, airlift bioreactors are used for plant and animal cell culture and for immobilized biocatalysts.

The gas is sparged only in one part of the vessel, the so called riser. The gas hold-up and the decreased density of the fluid let the medium move upwards in the riser. At the top of the reactor, the bubbles disengage and the now heavier medium moves downward through the non-sparged part of the vessel, the downcomer. The achievable transfer of oxygen is generally lower compared with stirred tank bioreactors.

(iii) *Packed-bed and fluidized bed bioreactor*
In a packed-bed bioreactor, the immobilized or particulate biocatalyst is filled in a tube-shaped vessel. The medium flows through the column (upwards or downwards). High velocity of the liquid phase promotes good mass transfer. Compared with a stirred tank reactor, possible particle attrition is small. Often, the medium is recycled and led several times through the column to improve conversion. In this case an intermediate vessel is needed for storage.

The medium flows upwards in expanded- or fluidized-bed bioreactors and causes an expansion of the bed at high flow rates. The biocatalyst particles have to have an appropriate size and density. Since the particles are in constant motion, channeling and clogging are avoided.

Unit Procedures. The bioreactor is the core of the flowsheet where the conversion of raw materials to desired product takes place. To run the bioreactor, a number of unit procedures are routinely carried out.

(i) *Filling and transfer of materials in vessels*

These operations are used to bring materials (liquids, solids) into the bioreactor and to transfer parts or the whole reactor volume to the next unit operation at the end of the bioreaction. The parameters that have to be defined for the filling are mass or volume of the input and its composition, or alternatively the concentration of a newly fed substance in the partially filled reactor. For filling and transfer, the duration of the operation should be specified, either by setting the overall filling time or by defining a filling rate, e.g. kg/min, to a vessel of known volume.

A bioreactor is usually filled up to only 70 or 90% of its overall volume to keep some headspace for foam build-up and the volume increase caused by aeration and subsequent substrate feeding. Additionally, the disengagement of droplets from the exhaust air in the headspace is attempted. The volume that is actually used is called the working volume of a reactor.

(ii) *Agitation*

A bioreactor is agitated to achieve and maintain homogeneity, to enable efficient heat transfer and, in the case of an aerated fermentation, for the uniform distribution of the gas phase and gas–liquid mass transfer. An agitator rotates by consuming electrical energy and keeps the fermenter content in motion. Key parameters are the energy demand, expressed either as overall consumption (kW) or as specific consumption (kW/m^3), the agitation or mixing time, and sometimes the impeller speed in revolutions per minutes (rpm). Usually, the agitator runs during most of the reaction time of the bioreactor. The energy consumption depends on the rotational speed and the geometry of the agitator, the working volume of the bioreactor, fluid density and viscosity, and baffling of the reactor. Additional equipment inside the reactor, such as heating coils or thermometer pipes, have a baffling effect and can therefore increase the demand.

The specific energy consumption of a bioreactor lies typically between 0.2 and 3.0 kW/m^3. At the same stirring rate, aerated fermenters have a lower consumption than do unaerated bioreactors. A good average value is 0.8 kW/m^3 (see Table 2.6). The plain mixing of liquids, for example in the medium's preparation, requires usually around 0.2–0.5 kW/m^3.

Table 2.6 *Average values of typical energy consumption steps, referred to 1 m^3 aqueous solution. For all, an efficiency factor of $\eta = 0.9$ is assumed. Unit energy prices are taken from Table 4.5. Evaporate and condensate consider the energy demand to vaporize water at 100 °C to steam at 100 °C, and vice versa, respectively. Assumption for cooling water: $\Delta T = 15$ °C; assumption for input power agitator: 0.8 kW/m^3*

Consumption step	Energy demand (MJ)	Energy-transfer agent	Average cost ($)
Heat by 10 °C	46.4	steam (22 kg)	0.10
Cool by 10 °C	−46.4	cooling water (740 kg)	0.06
Agitate for 10 h	32	electricity (9 kWh)	0.40–0.70
Evaporate	2510	steam (1185 kg)	5.20
Condense	−2510	cooling water (40 m^3)	3.20
Centrifuge	72	electricity (20 kWh)	0.90–1.50

(iii) *Aeration*

The aeration provides oxygen to meet the aerobic demand of the cells during the fermentation and removes gaseous by-product, mainly carbon dioxide. The aeration is specified by the gas used and the aeration rate. Owing to its low cost, air is used in industrial bioprocesses. However, also pure oxygen, pure nitrogen, or air enriched with oxygen or carbon dioxide can be used.

The aeration rate typically lies between 0.1 and 2 volume of gas (under atmospheric pressure) per volume of solution per minute (vvm). In large bioreactors the air utilization is more efficient. Here, a good average aeration rate is 0.5 vvm while in smaller reactors the average rate is around 1 vvm. The aeration rate also can vary during the fermentation, e.g. when the biomass concentration increases. For example, Kristiansen et al. [2.43] mention for the citric acid fermentation a starting rate of 0.1 vvm that is stepwise increased to 0.5–1.0 vvm.

(iv) *Heat transfer*

Heat-transfer operations are necessary to change and control the temperature of the bioreactor, or to keep the temperature constant while exothermic reactions take place in the fermenter. In the case of heating, the heat is transferred from a heat-transfer fluid via a heat-transfer surface to the reactor content or in the case of cooling from the fermentor content to the cooling fluid.

Steam is usually used for heating. The heating rate depends on the bioreactor volume, typically at 1.5–3.0 °C/min for a 10 m^3 reactor and at 1–2 °C/min for a 50 m^3 reactor. Commonly, used cooling agents are cooling water (around 20 °C), chilled water (5 °C), or for lower temperatures Freon, glycol, sodium chloride brine or calcium chloride brine. The final temperature of the cooling agent should be at least 5–40 °C below the final temperature of the cooled liquid.

The heat Q (J) necessary to heat up or cool down a substance i with mass m_i (kg) and specific heat capacity $c_{p,i}$ (J/kg K) from a starting temperature T_0 to an end temperature T_1 [temperature change ΔT (K)] is:

$$Q = m_i \cdot c_{p,i} \cdot (T_0 - T_1) = m_i \cdot c_{p,i} \cdot \Delta T \tag{2.24}$$

For a mixture of substances, a good approximation is:

$$Q = \sum m_i \cdot c_{p,i} \cdot \Delta T \tag{2.25}$$

In cases where specific heat capacities are not available for all compounds the heat capacity of water is used as an approximation. In heating operations, steam is the heat-transfer agent. It condenses on the heat-transfer surface without changing its temperature. The heat of condensation is:

$$Q = m_S \cdot h_C \tag{2.26}$$

where m_S = amount of steam (kg), h_C = condensation enthalpy (J/kg). The condensation enthalpy of steam at 150 °C is 2115 kJ/kg. The necessary amount can be calculated by combining Equations (2.25) and (2.26).

$$m_S = \frac{\sum m_i \cdot c_{p,i} \cdot \Delta T}{\eta \cdot h_C} \tag{2.27}$$

Thereby the efficiency number η is introduced to the equation to consider heat losses, with $\eta = 0.9$ as a good average.

In cooling operations, the heat transported by the cooling agent is:

$$Q = m_c \cdot c_{p,c} \cdot (T_{c,1,av} - T_{c,0}) = m_c \cdot c_{p,c} \cdot \Delta T_{c,av} \tag{2.28}$$

with $C_{p,c}$ = heat capacity of the cooling agent (J/kg K), $T_{c,0}$ = starting temperature of the cooling agent (K), $T_{c,1,av}$ = the average final temperature (K) and $\Delta T_{c,av}$ = the average temperature change of the cooling agent (K). By combining Equations (2.27) and (2.28), the necessary amount of cooling agent can be calculated by:

$$m_C = \frac{\sum m_i \cdot c_{p,i} \cdot \Delta T}{\eta \cdot c_{p,c} \cdot \Delta T_{c,av}} \tag{2.29}$$

Batch cooling, e.g. in a jacketed vessel, involves an unsteady heat transfer. That means the temperature difference between the cooling agent and the vessel content varies along the heat-transfer surface and at every point of the surface over time. However, the heat-transfer rate is proportional to this temperature difference and the heat removed by the cooling agent decreases with a decreasing difference during the cooling operation. Assuming a constant flow rate the final temperature of the cooling agent decreases during the operation. For a first estimation, it is sufficient to define an average temperature change of the cooling agent. Table 2.6 gives examples for the consumption of heating and cooling steps.

(v) *Foam control*

The combination of agitation and aeration with the presence of foam-producing and foam-stabilizing substances such as proteins, polysaccharides, and fatty acids can lead to substantial foam formation in the bioreactor. Particularly, aerobic fermentations with complex media tend to have significant foam formation. An overflow of foam can cause blocking of outlet gas lines and filters, a loss of fermenter content, and provide a route for contamination. The foam build-up can be controlled chemically or mechanically. The addition of antifoam agents, usually surface-tension-lowering substances, can deal with even highly foaming cultures. However, they also reduce the oxygen transfer to the cells. Mechanical foam breakers destroy the foam bubbles, e.g. by using a disk rotating at high speed at the top of the vessel. Mechanical devices are only efficient for moderately foaming fermentations, and for large bioreactors they can cause prohibitively high energy consumption. Therefore, the use of chemical antifoam agents often cannot be avoided. The foam problem increases with the fermenter size and cannot be easily predicted. Antifoam agents often have negative impact on oxygen transfer rates and on downstream processes by fouling of membranes.

(vi) *pH control*

Many bioreactions and biocatalysts require a constant pH. In industrial processes the medium is buffered and pH is adjusted and maintained by adding acids or bases to the bioreactor. If the necessary amounts are not known from experimental data, they can be estimated from the ion-charge balance for the reactor.

The sum of the positive charges of the cations is always equal to the sum of the negative charges of the anions. The equation is solved for the ion that is used for pH regulation. For example, if HCl is used, the equation is solved for the chloride

concentration. The following equation shows the ion-charge balance of a fermentation producing pyruvic acid (see case study Pyruvic Acid in Chapter 6) where ammonia is used (Ac = acetate, Pyr = pyruvate).

$$[NH_3^+] = [OH^-] + [Ac^-] + [Pyr^-] + [Cl^-] + 2[SO_4^{2-}] + [HSO_4^{2-}] + 3[PO_4^{3-}] + 2[HPO_4^{3-}] + [H_2PO_4^{3-}] - [H^+] - [Na^+] - [K^+] - 2[Mg^{2+}] \quad (2.30)$$

The concentrations of the added salts, acids, and bases are usually known. The H^+- and OH^-- concentration at the desired pH are also known. The dissociated and non-dissociated parts of an acid, especially weak acids and bases, and the degree of dissociation can be calculated using the following equation:

$$[H_{n-L}A^{L-}] = \frac{[H^+]^{n-L} \cdot [A]_{tot} \cdot \prod_{q=0}^{L} K_{Aq}}{\sum_{m=0}^{n} \left\{ [H^+]^{n-m} \cdot \prod_{q=0}^{m} K_{Aq} \right\}} \quad \text{with} \quad K_{A0} = 1 \quad (2.31)$$

n = number of acidic protons; L = number of dissociated protons; K_S = acidity constant of each species; $(A)_{tot}$ = Total concentration of the acid. At pH 7, 99.4% of the acetic acid is dissociated ($pK_a = 4.75$)

$$[Ac^-] = \frac{K_{Ac} \cdot [Ac^-]_{[tot]}}{[H^+] + K_{Ac}} \rightarrow [Ac^-] = \frac{10^{-4.75} \cdot [Ac^-]_{[tot]}}{10^{-7} + 10^{-4.75}} \rightarrow [Ac^-] = 0.99441 \cdot [HAc]_{[tot]} \quad (2.32)$$

After the concentrations of all ions are calculated, the necessary amount of acid or base to reach the desired pH can be estimated from the ion-charge balance [2.44].

(vii) *Cleaning-in-place (CIP)*
A bioreactor has to be cleaned after every batch. A typical CIP procedure is discussed in a subsection of Section 2.3.1, above.

2.3.3 Downstream Processing

In this section, we provide an overview of the downstream unit operations regularly used in bioprocesses. The reader should understand the basic principles and purpose of each unit. This is important for design of the process flow scheme, specification of operating parameters, and subsequent modeling. However, for a deeper understanding of these units and their key parameters, we highly recommend consultation with appropriate biochemical and chemical engineering books (e.g. [2.45–2.51]).

All unit operations in downstream processing use one or several differences in the chemical and physical properties of the desired product from other materials in the often complex mixture. Table 2.7 provides an overview of the separation principles of the most regularly used unit operations and the yields that are typically observed.

Production methods for bulk chemicals, fine chemicals, and pharmaceuticals differ in the complexity of their downstream processing. This causes differences in overall yield of separation and purification (see Table 2.8). In general, downstream processing is always a tradeoff between yield and purity. High purity is usually paid for with low yield and

Table 2.7 *Separation principles of the separation methods regularly used in bioprocesses*

Method	Separation principle	Typical yield (%)	Separated product
Centrifugation	specific density	90–99	cells, particles
Sedimentation	specific density	80–99	cells, particles
Microfiltration	size/phase	80–99	cells, particles
Ultrafiltration	size		cell debris, proteins & polymers
Chromatography		60–99	
gel filtration	size/shape		large molecules
ion exchange	ionic charge		ions
hydrophobic interaction	hydrophobicity		hydrophilic or hydrophobic molecules
reversed phase	hydrophobicity/diffusivity		hydrophilic or hydrophobic molecules
affinity	specific binding molecular recognition		molecules with specific epitopes
Electrodialysis	ionic charge/diffusivity	70–99	ions
Extraction	solubility/phase affinity	70–99	hydrophilic or hydrophobic molecules
Distillation	volatility	80–99	volatiles
Drying/evaporation	volatility	97–99	high-boiling molecules
Crystallization	phase change	60–95	crystallized solids

Table 2.8 *Typical downstream yields for different product classes*

Product class	Typical downstream yield (%)
Bulk chemicals, industrial enzymes	>90
Fine chemicals (organic acids, amino acids, antibiotics)	70–90
Therapeutic proteins	45–65

vice versa. Therefore, one should define early in downstream process design how pure the product needs to be.

It is important to realize that downstream processing methods are highly dependent on the bioreaction and upstream steps. High concentrations of the product and low concentrations of by-products and residual substances are always beneficial. The first step of downstream processing is the deliberate selection of the raw materials used in the bioreactor. Here, a lower product concentration from the bioreaction may be economically favorable if it allows a simplified downstream process. Every additional separation and purification step means additional capital and operating costs and an additional product loss. Therefore, as a general principle the number of downstream steps should be kept to a minimum to meet target purity as well as process robustness.

Often, different unit operations can be used to achieve a separation. To select the most appropriate alternative, many characteristics of the unit operations have to be considered such as purity/selectivity, yield, operating cost, necessary investment cost, possible denaturation of product, process robustness, separation conditions, and product concentration after the step.

Biomass removal. In most bioprocesses using cells, the first downstream step is the separation of the biomass from the fermentation broth. There are several unit operations available for this purpose. Widely used are centrifugation, microfiltration, rotary vacuum filtration, and decanting/sedimentation. These unit operations are described in the following Subsections. The choice of method for a given process depends on a number of parameters. The concentration, particle size, and density of the biomass and the density and viscosity of the broth determine design, scale of operation, and operating conditions. For small particles such as bacteria or yeast cells, centrifuges or membrane filtration are often the most efficient. The necessary time for the separation, the required yield of removal, the possible degradation or denaturation of the product, and the investment and operating costs of the unit have to be considered as well. In many cases, prior experience with or ownership of a piece of equipment influences the decision.

Homogenization/Cell Disruption . If the product is intracellular, it is necessary to break open the cells to release the product into the solution before further purification. The available techniques include mechanical and nonmechanical methods such as enzymatic digestion of the cell wall, treatment with solvents and detergents, freezing and thawing, and osmotic shock. Most often used are high-pressure homogenization and mechanical bead milling.

In the high-pressure homogenization (for an example see Chapter 12), the slurry is pumped through a narrow valve at a very high pressure (up to 1200 bar). The large pressure

drop behind the valve causes strong shear forces that lead to a disruption of the cells. Often several passes through a homogenizer are necessary to recover the product. The shear forces can lead to denaturation of intracellular proteins.

In the mechanical bead mill homogenization, the slurry is fed to a chamber with a rapidly rotating stirrer filled with steel or glass bead, or other abrasives. High shear forces and impact during the grinding cause cell disruption.

Concentration. After the bioreaction, the product concentration is usually relatively low. It may be reasonable to have first a concentration step to reduce the volume of the product stream that has to be processed through the subsequent units and thus reducing equipment size and energy consumption of these units. There are three methods available for this purpose:

- Partial evaporation of the solvent: The solution is heated up to vaporize some of the solvent, usually water. This method requires a heat-stable product with a low vapor pressure to keep the product loss small and causes high energy costs. At reduced pressure, evaporation is possible at lower temperature but vacuum equipment is required.
- Filtration: A semi-permeable membrane retains the product in the retentate but transfers most of the solvent through the membrane. This step can also remove some impurities with a lower molecule size. This is most useful for harvesting large molecules such as proteins. Energy for maintaining the pressure for the mass transfer is necessary.
- Precipitation: The product is precipitated by adding a precipitation agent or by changing chemical or physical conditions (temperature, pH, etc.) and is subsequently separated by filtration or centrifugation. Costs incur for the precipitation agent and the separation of the solid product. This method requires a product that can be easily and selectively precipitated without degradation and is especially useful when several impurities can be separated that do not precipitate.

Phase Separation. As a rule, the simplest separation should be applied first. Therefore, many downstream processes start with the separation of the different phases that leave the bioreactor. Furthermore, phase separations are often used later in the process as well. They include centrifugation, filtration, sedimentation, and condensation steps.

(i) *Centrifugation*
Centrifugation is based on density differences between solid particles and a solution or between two immiscible liquids. The sedimentation force is amplified by the particle or drop size in a centrifugal field in the centrifuge. In many bioprocesses, centrifugation is used for biomass removal and solid separation. Disk-stack centrifuges are applied most often, but also basket and tubular bowl centrifuges are used. Sometimes a pretreatment is necessary, e.g. heating, pH change, or addition of filter aids (see also Table 2.7) to increase particle size. The maximum throughput of a centrifuge is defined by the sigma factor and the settling velocity. The sigma factor describes the centrifuge in terms of an equivalent area referenced to a settling tank and is the basis for scaling the centrifuge. It is expressed in m^2 and equals the area of a sedimentation tank that would be necessary to realize the same separation work. The settling velocity is specific for the feed that has to be separated. It is determined by the size and density of the particles (e.g. the average cell size lies between 0.5 and 5 μm) and the density and viscosity of the solution. The best separation is realized at low viscosity, for large particles, and large

density differences. For most biological materials the density difference with water is usually small.

(ii) *Filtration*

Filtration is used to separate particles or large molecules from a suspension or solution. A semi-permeable membrane splits the components according to their size. The permeate includes most of the solvent and small molecules that pass through the membrane. The retentate is a concentrate of the particles and large molecules that are retained by the membrane. Pressure is the driving force for flow through the membrane.

Filtration is used for biomass and cell debris removal, concentration of product solutions, and sterile filtration of final product solutions. The different filter types vary in their pore sizes. Microfilters have a pore size of 0.1–10 μm. They are used to retain particles. Ultrafiltration uses pore sizes of 0.001–0.1 μm and keeps back large molecules like proteins, peptides, and other large, dissolved molecules. The molecular weight cutoff of a membrane is the molecular weight of a globular protein that is 90% retained. It determines the retention (or rejection) of a molecule that lies between 0 and 100%. Further unit parameters are the concentration factor (quotient feed/retentate) and the filtrate flux through the membrane. Depending on the particle concentration and viscosity of the feed, the flux typically lies between 20 and 250 L/m^2 h for microfiltration and between 20 and 100 L/m^2 h for ultrafiltration. According to their flow pattern, one distinguishes dead-end and cross-flow filtrations. In dead-end filtration the particles are retained as a cake through which solvent must pass. Thus the pressure drop increases with solids' accumulation. In cross-flow filtration, the feed is moved tangentially along the membrane to reduce concentration polarization or filter-cake thickness and associated pressure drop. The particles are obtained as concentrated slurry. Rotary vacuum filtration is used only for large-scale filtration with large particles. Here, the mass transfer through the membrane is caused by the pressure difference between outside ambient pressure and vacuum inside the drum at the permeate side of the membrane. A horizontal drum, covered with the membrane, is partly submerged in a tank that is filled with the feed slurry. During the filtration the particles accumulate on the surface of the membrane outside the drum. The drum slowly rotates and the cake is mechanically removed when the membrane is outside the feed solution. This approach is taken for biomass removal in large-volume fermentation processes with filamentous fungi.

Diafiltration is used to change the buffer solution. The solvent and the components of the old buffer are transported through the membrane while the desired (larger) product is retained. At the same time, a new buffer is added continuously or stepwise to the feed, resulting in a complete buffer change after a certain time period.

(iii) *Sedimentation and decanting*

Sedimentation and decanting, like centrifugation, utilize the density differences of substances. In contrast to a centrifuge, only gravity is the driving force. Therefore, sedimentation needs a longer settling time and larger density difference and particle size of the substances than does centrifugation. Sedimentation is applied for large-scale biomass removal mostly in wastewater treatment. Flocculating agents can be added to enhance the sedimentation rate by increasing particle size.

Decanting is used for the separation of liquid phases, e.g. water and organic solvent. Three layers are usually formed: The solid or heavy liquid phase at the bottom and the light liquid phase on top and a dispersion phase in between. The key parameters are density and viscosity of the two phases. They determine the settling velocity of the heavy phase and thus the necessary settling time and consequently the required tank size. The residence time lies typically between 5 and 10 minutes.

(iv) *Condensation*

In condensation, vapor is condensed into liquid by cooling. Condensation is used to liquefy the distillate in distillation (e.g. in product separation or solvent recycling) and to turn vaporized steam to liquid water after a crystallization or concentration step. A typical condenser is a shell-and-tube surface condenser. Here, the coolant flows in the tube while the condensation of the vapor occurs at the shell side. Heat is transferred from the vapor through the tube wall to the cooling agent, typically cooling water (see also Table 2.6).

Heat of vaporization, boiling point, and partition coefficient of the vapor components are the key parameters. The partition coefficient of a condensation describes the mole fraction of a component in the gaseous and the liquid phase. The initial temperature and the temperature change of the cooling agent are also important and can be economically optimized (for an example see [2.52]). All these parameters, together with the heat-transfer coefficient of the system, determine the necessary heat-transfer area and thus the equipment size. For the system steam and cooling water, a heat-transfer coefficient of 2000 kcal/h m^2 °C (2325 J/s m^2 K) is a typical value.

Product Separation and Purification. Following solids removal, the target product is further separated form impurities and purified to meet predetermined specifications. The most often applied unit operations include: extraction, adsorption, chromatography, electrodialysis, and distillation.

(i) *Extraction*

In an extraction step a molecule is separated from a solution by transferring it to another liquid phase. The separation is based on the different solubilities of the product and the impurities in the feed phase, e.g. an aqueous solution and an organic extract solvent phase, and thus the selective partitioning of the product and impurities in the two liquid phases. Extraction is applied in the purification of antibiotics and organic acids and even occasionally proteins. It is regularly used when the product concentration is comparably low or when distillation cannot be applied.

The simplest extraction equipment is the so called mixer/settler. Here, the two liquid phases are mixed in a tank to enable the transfer across the phase boundaries of the product and then a sufficient time is allowed until the phases are separated. However, more often used are differential extraction columns that work continuously with countercurrent liquid flows and consist of several stages (e.g. see Chapter 6) or a centrifugal extractor. Here, the heavy phase, usually the aqueous solution, is added at the top of the column and the light phase, normally an organic solvent, is added at the bottom and moves upwards. Special equipment is used to disperse the solvent into small droplets that flow through the continuous phase to enable a maximum mass transfer. The density differences of the phases determine upward and downward velocities.

A centrifugal extractor often used in antibiotic purification works in principle like a centrifuge (e.g. see Chapter 10). The density differences are amplified by the centrifugal force.

The key parameter of an extraction is the partition coefficient. It is defined as the equilibrium concentration of a substance in the extract phase divided by its concentration in the feed phase. The partition coefficient finally determines the product loss of the step. It is usually strongly influenced by temperature, ionic strength, and pH. The maximum solubility of the product in the extract phase and the solubility of the solvents in each other are also important parameters. Since the volume of the extract phase is usually smaller, the extraction also leads to an increase of the product concentration.

(ii) *Distillation*

Differences in the volatilities of substances are prerequisites for distillation. Typically, the feed is preheated and enters a continuous distillation column that consists of several (theoretical) stages. The volatile compounds evaporate and the vapor moves upwards and leaves the column at the top as distillate. The high-boiling compounds remain in the liquid phase, move downwards, and leave the column at the bottom. The distillate is liquefied in a condenser. Parts of the distillate can be recycled to the column to improve separation. A sequence of columns that work at different temperatures can be used when more than one volatile fraction has to be separated.

Distillation is an alternative to extraction and adsorption. It is extensively used in the chemical, especially the petrochemical, industries. In bioprocesses, it is employed for the purification of large-volume, low-boiling products such as ethanol and other alcohols. Distillation requires heat stability of the product. The boiling point of the substances and the linear velocity of the vapor are the key parameters. At a smaller scale also batch distillation is applied. For a crude separation a so-called flash distillation can be used that consists of only one stage. Distillation is frequently applied for the recovery of organic solvents used in downstream processing.

(iii) *Electrodialysis*

In electrodialysis, an electromotive force is used to transport ions through a semipermeable, ion-selective membrane by ion diffusion and thus separate them from an aqueous solution. From the feed, the cations move through a cation membrane into the supplied acid stream. Additionally, or alternatively, the anions move through an anion membrane into the supplied base stream. The remaining stream is the diluate.

Electrodialysis is applied for the purification of organic acids, e.g. lactic acid (see also Chapter 6). Key parameters are the membrane flux and the transport number. The membrane flux is typically between 100 and 300 g/m^2 h. The transport number is the ratio of the flux of the desired ion and the flux of all ions through the membrane. The product concentration in the acid or base stream can be up to 5 molar.

(iv) *Adsorption*

Adsorption is used to retain either the product or impurities on a solid matrix. The solution is led through a column where the target molecules bind to the resin. If impurities are retained, they are immediately eluted from the column with a buffer. If the product is retained, usually a washing step is added in between.

The column can be operated as a packed bed or an expanded bed. Several columns are often used to enable a quasi-continuous processing. Key parameters are the binding capacity and selectivity of the resin, the binding yield of the target and non-target molecules, and the volume of the eluent. The performance is usually influenced by parameters such as pH and temperature. High recovery yield can be realized with adsorption columns (e.g. 70–90%), even at quite low product concentrations. Adsorption columns are used e.g. in the purification of vitamins and cyclodextrins (see Chapter 9). A special application is the use of activated carbon for decolorization of liquids (e.g. Chapters 5 and 9).

(v) *Chromatography*

Chromatography is used to resolve and fractionate a mixture of compounds based on differential migration, i.e. the selective retardation of solutes during the passage through a chromatography column. The basic principles are identical to purification by adsorption. The solvent (mobile phase) flows through a bed of resin particles (stationary phase), and the solutes travel at different speeds depending on their relative affinity for the resin. Thus, they appear at different times at the column outflow, either directly after the load of the column or the product initially remains retained by the resin and is later eluted with an eluent. Before the elution step, a buffer is used to displace the void fraction of the column. After the elution, a buffer is applied for regeneration and equilibration of the column.

The elution is carried out either isocratically or by gradient elution. In an isocratic elution, the composition of the elution buffer is kept constant. In a gradient elution, the composition of the eluent, e.g. the salt concentration, is changed continuously or stepwise to improve the fractionation of the attached molecules. The portion of the output stream that contains the desired product is separated from the residual that ideally contains most of the impurities.

Several forms of chromatography can specified. They differ in the mechanism by which the desired substances are retarded or retained in the column; thus the chemical or physical property differences that are exploited to fractionate a mixture. In bioprocesses, five types commonly used are:

- Gel or exclusion chromatography with molecular sieving that separates molecules according to their size. The column is packed with gel particles of a defined porosity. Large molecules cannot enter these pores and are eluted first, while smaller molecules enter the pores at a rate that is inversely proportional to their size, which increases their elution time. Gel filtration is often used as a polishing step at the end of protein purification. Its capacity is typically low but its resolving power is high.
- In affinity chromatography, the separation is based on the stereoselective binding of the solute to immobilized molecules, the so-called ligand. The target molecules are retained in the column and then eluted by a change of pH, ionic strength, or buffer composition. Affinity chromatography is highly selective. Examples are the purification of monoclonal antibodies using a protein A ligand or the purification of a recombinant therapeutic protein using a monoclonal antibody as ligand.
- Ion-exchange chromatography uses the electrostatic attraction between the target molecule that is charged at the given pH and the charged resin. The product is first

retained and then eluted by changing the pH or the ionic strength, often using a gradient elution.

- Hydrophobic interaction chromatography (HIC) is mainly used for the separation of proteins. Differences in their hydrophobicity are caused by the amino acids exposed at the surface of the molecule. HIC uses hydrophobic interactions between the solute and the resin to separate the substances. The product is eluted by a reduction of the (hydrophilic) salt concentration of the mobile phase.
- Separation in reversed-phase chromatography is based on the uneven distribution of the solutes between two immiscible liquid phases. The less polar of the two solvents is fixed on the column and provides the stationary phase. Such stationary phases are hydrophobic alkyl chains, typically C_4, C_8, and C_{18}. The column is loaded by applying an aqueous solution. The elution is based on an increase in the concentration of hydrophobic, organic solvents in the mobile phase and occurs in the order of hydrophobicity of the substances, with the most hydrophobic substance at the end. Here, methanol and acetonitrile are often used.

Chromatography can be operated in a packed-bed or in an expanded bed column (e.g. see Chapter 11). Key parameters are the binding capacity of the resin, the flow rate of the mobile phase through the column, the specific binding of components to the resin, the necessary volume of eluent, and the volume of the product fraction.

Chromatography is used for example in the purification of pharmaceuticals, mainly proteins (see Chapters 11–15). Since it is usually more expensive than extraction, distillation, or filtration methods, it is mainly used for high-price products.

Viral Inactivation. In the production of pharmaceuticals, inactivation of pathogenic bacteria, viruses, and prions that might occur as contaminants or impurities in the product is necessary. Particular attention is paid to viral inactivation when the product is derived from mammalian cell culture, blood plasma, or transgenic animals. An efficient inactivation step must reduce the concentration of active viruses by greater the 10^6 orders of magnitude. To meet the regulatory requirements, usually a combination of methods is necessary because none of the known methods inactivates all possible contaminants. Standard purification steps like extraction, filtration, and chromatography already lead to marked virus reduction. Additional steps, explicitly designed for virus reduction and applied at different points in the flowsheet, include:

- Micro- and ultrafiltration (not sufficient for small viruses)
- Heat: either continuous (high temperature, short time) or batch (lower temperature, longer time)
- UV radiation
- Chemical substances, e.g. with a high acid or base concentration

The methods are very similar to the methods used for the sterilization of raw materials (see Section 2.3.1). However, therapeutic proteins are very sensitive to such treatments. The optimal choice for the process is a combination of methods that guarantee a sufficient viral reduction and keep the denaturation of the protein product, and thus the activity loss, at a minimum (e.g. see Chapter 13).

Protein Solubilization and Refolding. Heterologous proteins produced in bacteria and fungi often form inclusion bodies or water-insoluble pellets inside the cell. While their primary structure, the amino acid sequence, and often also secondary structures are correct, their three-dimensional structure is usually incorrect. Therefore, they are biologically inactive. They are precipitated in a relatively pure form as inclusion bodies. It is, however, possible to solubilize and refold the proteins to their active form [2.53].

At the end of a cultivation, the cells are inactivated and separated from the broth, e.g. by centrifugation. Then the cells are disrupted to release the intracellular material and inclusion bodies. In the next step inclusion bodies are isolated, usually by centrifugation. The inclusion bodies are recovered in the heavy phase while most of the cell debris remains in the light phase. The inclusion body sludge is washed often while applying mild detergent, e.g. Triton-X 100, to remove lipids, proteins, and other impurities. In the next step the pellets are dissolved by adding high concentration of chaotropic reagents such as urea or guanidine hydrochloride and detergents such as SDS (sodium dodecyl sulfate). Additionally, reducing agents like 2-mercaptoethanol or dithiothreitol are applied to reduce disulfide bridges. Chelating agents such as EDTA (ethylenediaminetetraacetic acid) are added to prevent metal-catalysed oxidation of cysteines and methionines. By disruption of disulfide and non-covalent bonds, the proteins are unfolded and dissolved in the buffer. Mild dissolution allows retention of secondary structures intact and thus improving subsequent refolding. In the next step the concentration of the denaturants is substantially reduced. Different methods to do this are possible, for example dilution, electrodialysis, or diafiltration. At low concentrations of the denaturant the proteins can refold to their native form and be further purified. Low concentration of proteins promotes the fidelity of the refolding whereas at high concentration the formation of aggregates is favored. A successful strategy is the slow addition of solubilized protein to the renaturation buffer. This keeps the concentration of unfolded protein low and the renatured protein does not form new aggregates. An example of protein refolding is contained in the insulin case study (Chapter 12).

Final Product Processing. After most of the impurities have been removed from the product solution, the product has to be prepared for final formulation. This can include crystallization, stabilization, drying, and final formulation with materials to assure stability.

(i) *Crystallization*
In a crystallization step the desired product is converted from its soluble form into its crystallized (solid) form. After crystallization the crystals are separated from the liquid solution, for example by filtration. The mother liquor is often recycled to the crystallization tank to increase the yield. Crystallization is usually done at the very end of the downstream processing when only a very few impurities remain. However, crystallization also can be used as a first purification step right after the bioreaction if other components of the broth do not precipitate and are not incorporated into the crystals.

Crystallization is initiated either by a volume reduction of the solution or by reducing the solubility of the target molecules by addition of a crystallizing agent, or by changing the physical or chemical conditions (pH, temperature, etc.). Often, crystallization is a combination of both approaches. Key parameters are the crystallization yield, the crystallization heat, and the necessary residence time. The purity and shape of crystals are dependent on many parameters including rate of crystallization. Crystallization is

difficult to predict and to scale up. Therefore, well designed experiments to map the experimental space are very important.

(ii) *Product stabilization*
For products such as therapeutic proteins, it is necessary to stabilize the product to avoid premature degradation or denaturation. The shelf life of the product is usually extended by addition of stabilizing agents or a complete buffer exchange before final filling into vials.

(iii) *Drying*
In a drying operation water or another solvent is removed from a solid product. It is commonly used if the product is to be sold as powder. Two classes of dryers are used: contact dryers and convection dryers. For instance, in a drum dryer, an example of a contact dryer, the heat necessary to vaporize the water is provided via the drum wall from hot water, air, or steam that flows at the outer side of the wall. The drying agent and the product do not come into direct contact. Convection dryers are used more often. Here, the preheated drying gas is mixed with the solid and the solvent evaporates into the drying gas. Fluidized-bed and spray dryers are regularly used in bioprocesses. Both are characterized by a short residence time. In a spray dryer, the feed is sprayed as small droplets into a stream of hot gas. In a fluidized-bed dryer the wet solid is transported through the dryer and is fluidized by the drying gas that is led in cross flow through the powder. The discharged air is usually saturated with solvent vapor. The specific air consumption depends on the exit temperature of the drying gas. At 50 °C, typically 13 kg of air are required per kg of evaporated water, at 70 °C around 5 kg/kg.

A gentle way to dry heat-sensitive products, like proteins and vitamins, is freeze drying, also known as lyophilization. In a first step the wet product is frozen. The frozen material is introduced into a vacuum chamber and water starts to sublime. Owing to the heat required for sublimation, sublimation is usually accelerated by controlled heating.

(iv) *Filling, labeling, and packing*
The final step of a process is to get the product ready for the customer or patient. This part can be readily considered in a process model. It should be included if enough information is available as to how the product is formulated and packed, and if the product is traded as discrete entities. Then the price of a pharmaceutical is quoted as \$/100 vials or similar.

In the filling step, the product is filled in containers of a defined volume. Labels are attached in the labeling steps, and they are put into boxes or on a pallet in the final packaging step.

2.3.4 Waste Treatment, Reduction and Recycling

Waste treatment is an important operation in today's industrial processes and a comprehensive literature is available [2.54–2.63]. In this section, we look briefly at methods for waste reduction.

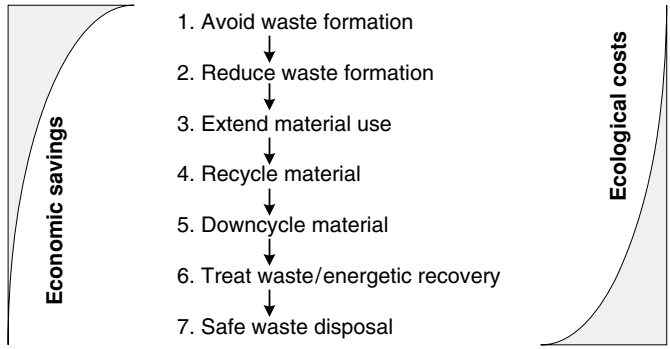


Figure 2.5 *Steps of waste avoidance and treatment*

Figure 2.5 shows the different steps for waste prevention and treatment in an integrated process development. The first step is always to avoid the formation of waste. If this is feasible and cost-effective, subsequent treatment is unnecessary. If waste formation cannot be prevented completely, one should try to reduce it as much as useful. The reuse of material is one approach; for example, if a chromatography resin can be used for multiple cycles, the annual amount of waste is significantly reduced.

The recycling of an organic solvent used in an extraction step is a good example of cost-effective recycling (see e.g. Chapters 6, 9, and 10). To decide if the recycling is really environmentally and economically favorable, the amount recycled and the amount of materials and energy necessary for the recycling should be compared. If the material cannot be recycled because the purification becomes too expensive, it might be used for another purpose that requires less purity (downcycling).

The materials that remain after waste reduction and recycling steps have to be treated or disposed of safely. Thereby, treatment should be preferred to disposal. Ideally, some energy is produced during the treatment (e.g. incineration). There are a number of books recommended to further study pollution prevention and integrated waste reduction (e.g. [2.62–2.65]).

The waste created in bioprocesses is often less a problem than in chemical processes. However, the amount can be quite large. The waste leaves the process boundaries as solid, liquid, and gaseous streams. The exhaust air from a bioreactor is the most common gaseous waste stream in bioprocesses. It usually contains air, carbon dioxide, and water. A filtration of the stream prevents the release of aerosols that might contain spores or other forms of the biocatalyst. This is especially relevant if pathogenic or recombinant organisms are used, even if considered as harmless. Gaseous waste streams are also formed in distillation and evaporation steps, e.g. associated with crystallization. Most of the vapor is liquefied in a condensation step and then further processed. The exhaust air from a drying operation does not require treatment as long as water is the solvent that is removed. However, if organic solvents are removed, they have to be separated from the air stream to avoid volatile organic emissions.

Solid waste is categorized as hazardous and non-hazardous waste. Hazardous waste, e.g. containing heavy metals or highly toxic substances, needs special treatment or disposal with high-safety measures. Both cause higher costs. Compared with chemical processes,

hazardous wastes are generated much less in bioprocesses. Wet biomass is the most common solid waste in bioprocesses. If a recombinant organism is used, sterilization of the material is necessary, usually by heat. The biomass can be used as animal feed or organic fertilizer or disposed as landfill. Owing to its high water content, it often can be added to a wastewater treatment plant. This is especially useful if the plant lacks organic carbon, nitrogen, or phosphorus, e.g. when processing mainly chemical wastewater.

Most bioreactions take place in an aqueous system and the product is dissolved in a liquid throughout most of the downstream processing. Thus, it is not a surprise that most waste streams in bioprocesses are liquid. They are treated in a biological sewage treatment plant at the production site of the bioprocesses, or they are released to the municipal sewer system. Under certain conditions a pretreatment is necessary. At a high or low pH, the liquid waste has to be neutralized by adding base or acid. Besides sterilization, as discussed above, pretreatment is necessary if the stream contains specific contaminants such as pharmaceutically active substances that cannot be handled in a standard sewage plant.

The raw materials used in the bioreaction and downstream processing influence the composition and complexity of the waste, which can cause higher costs and thus have to be considered when one compares different raw material alternatives. For example, molasses contains a wide range of impurities. If it is used as a carbon source in fermentation the waste streams are much more complex when compared with the use of pure glucose or starch hydrolysates.

Recycling of materials is regularly applied in bioprocesses. Biocatalysts are often immobilized to reuse them several times. Similar to other industrial processes, organic solvents are recycled to a high degree because they are relatively expensive and often environmentally critical. They usually have to be purified, e.g. by distillation, before reentering the process. Water can also often be partly recycled. However, it is usually more economic to discharge an aqueous waste stream. Whenever a material stream is recycled, one has to validate whether there is a possible enrichment of undesired substances in the recycling loop or whether hygiene problems may arise.

2.4 The Development Process

2.4.1 Introduction

The development of a process may take several years, require many steps, and involve many different participants. The cost of development will depend on the specifications for the product, the complexity of the process, and the demands of the application. The development of new biopharmaceuticals is the most expensive, with an average cost of $300–800 million and the longest with 10–15 years from the product idea to the final approval of the drug. The development of products for the chemical, food, or feed application industries is less costly and quicker but still requires a substantial investment of time and money.

The basis of the development process should be an R&D agenda and associated roadmap that focuses the effort on the most relevant problems and the most promising opportunities. A clear agenda helps to reduce the time and improve the chances to create a competitive and environmentally sound process that can be realized at industrial scale. The agenda must be regularly adjusted to the newly gained knowledge obtained during the development.

Throughout product and process development, many decisions have to be made. The sum of these decisions and their timeliness dictates whether the process will be successful or not. A successful process requires the best possible basis for decision-making at every point of the development, from the creation of the product idea to the realization of the industrial production plant. Two critical aspects of this process are, first, for every important decision all relevant information about the process and its socio-economic environment must be collected or estimated. Therefore, it is crucial to involve the relevant stakeholders of the development process in a timely manner. Depending on the decision, this might include people from marketing, the legal and patent department, or the environmental experts, in addition to biologists, chemists, and biochemical engineers that work on the biocatalysis project. Early phases of process development determine most of the cost structure as well as the environmental impact of the final industrial process. Therefore, it is essential to find a sound design basis and engage the various R&D participants from the very beginning of the development process [2.66].

The goal here is to create an overall optimal process for the production of the desired product. This explicitly includes consideration that single steps of the process, such as the bioreaction or the different downstream units, might deviate from their optimal operation. For example, the use of serum in mammalian cell cultivation can improve growth and product yield. However, the serum components can complicate downstream processing such that it can be favorable to accept a lower yield in a serum-free fermentation to enable a simplified purification. Process modeling with tools such as SuperPro Designer™ used here are very effective in evaluating the tradeoffs and making informed decisions early in the process of development.

2.4.2 Development Steps and Participants

The process and product development includes several steps. As illustrated in Figure 2.6, they do not form a linear sequence of independent steps, but at every point several steps run in parallel and interact with each other.

At the goal of every process development project is a product. The product must have a market, or a potential market, of a sufficient size that economically justifies the required investment in the process development. The desired product must be clearly defined and specified (quality, purity, etc.); it is the product specifications that establish the goal of the process development project.

After product definition, an extensive literature and patent review is required. This review should clarify if there are similar products already on the market or in development. A series of questions have to be addressed. Is a competitor working on the same product? Are there patents that prevent the use of technologies that might be needed to produce the desired product? In general, is there freedom to operate on the one hand and can one exclude others from the market on the other? The review also includes the search for appropriate biocatalysts and unit operations for the product formation and purification as these are the alternative tools available for production.

The biocatalyst plays a central role in the process. An organism or an enzyme that catalyses the formation of the desired product is needed. Once such a biocatalyst has been found, it has to be optimized to reach an economically feasible product yield and concentration. In principle there are two possible ways to realize this optimization. Either

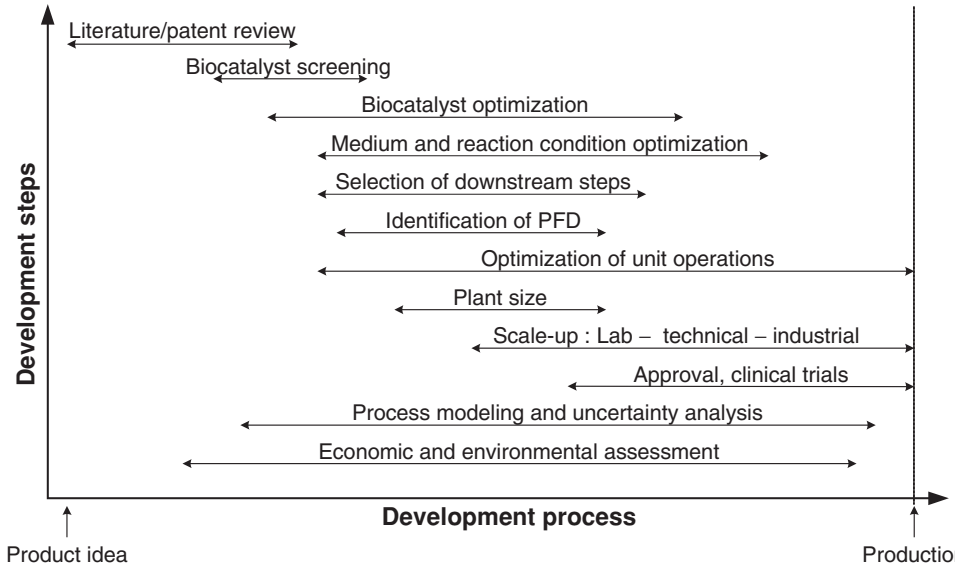


Figure 2.6 *Steps in the development of a bioprocess from the product idea to the production plant. PFD = Process flow diagram*

the native organism, where the product formation was originally found, is improved or the corresponding genes are transferred and over-expressed in a host organism that is well characterized and can be grown on inexpensive media, e.g. *Escherichia coli*. Both paths can include classical strain improvement as well as genetic engineering. Today, modern methods such as metabolic engineering are applied [2.67].

In parallel to the biocatalyst optimization, the medium and reaction conditions are adjusted to enable the best performance of the catalyst. The medium should be as simple and inexpensive as possible but still allow an optimal performance of the biocatalysts concerning growth and product formation. Different compositions and concentrations can be tested, such as the use of different carbon and nitrogen sources (several sugars, starch, molasses, yeast extract, corn steep liquor, etc.). In addition, the impact of the medium's components on the later product separation and purification should be considered. Also the supply chain should be taken into account, e.g. if the required raw material is available in sufficient amounts at the required quality and acceptable price.

The selected reaction conditions should provide the best environment for the biocatalyst (temperature, pH, pressure, oxygen supply, etc.) and maintain a homogeneous mixture in the bioreactor. This involves the reactor design as well as the operation of the reactor. In the reactor design different impeller shapes and height: diameter ratios might be tested. In most cases, however, the reactor and its geometry are already given and only operational conditions can be modified. The best aeration and agitation conditions have to be found for aeration rate, aeration with air, pure oxygen, carbon dioxide and/or oxygen-enriched air, etc. Feeding profiles have to be optimized for optimal performance. The medium and the reaction conditions of enzymatic processes are usually simpler than in fermentations.

Once the composition and concentrations at the end of the bioreaction are known, the appropriate separation and purification steps are selected (see Section 2.3.3). The process flow diagram (PFD) that includes the upstream operations such as medium preparation and sterilization, the bioreaction, and the downstream operations for product separation and purification is put together. For the downstream processing, the unit operations have to be chosen and connected in an efficient and robust manner. All the unit operations are based on differences in the chemical or physical properties of the product and the other components of the product stream. The most efficient way is to use the largest differences first, e.g. the phase difference in the separation of the solid biomass from the dissolved product. Usually, one compares different alternatives in the selection process and the process flow diagram might change during development and scale-up. Every unit is optimized towards the goal of maximizing the overall yield.

During the process scale-up, the plant size has to be determined. The market size and market share of the product estimated at the beginning of the development is validated. They determine the necessary annual production. The expected product concentration and the duration of the bioreaction and the expected downstream yield are used to estimate the necessary size and number of bioreactors.

The process is scaled up from laboratory experience, often via a pilot plant, to the industrial production plant. The lab scale includes several steps from laboratory flasks to lab bioreactor with usually less than 5 L volume.

Pilot plants have usually a volume of up to 500 L, or require a flow rate of up to 100 L/h. After the optimization, the pilot plant should be more or less identical to the production plant including recycling loops, scheduling, and the selection of the materials for the large-scale equipment. The production plant differs only in the capacity that is usually 10- to 1000-times larger. The pilot plant already provides the first samples for the market or, in case of a pharmaceutical product, the amounts required for the clinical trials.

Pilot plants are expensive to build and to run. They can cost 3–30% of the production plant cost [2.68]. Therefore, often so called mini-plants are used. They are like a pilot plant in the way they map the expected production plant. Material selection, scheduling, and recycling can be done in a mini-plant. However, the volume of the mini-plant is identical with the lab scale. A mini-plant is cheaper and more flexible than a pilot plant and the knowledge gained can reduce the necessary time and effort in the pilot plant. Under ideal conditions, the mini-plant can be directly scaled up to production size. However, a scale-up factor of 10 000 embraces higher risks. In parallel to the scale-up of the plant, the approval of the product for its intended use must be filed. For pharmaceuticals, clinical trials have to be planned and implemented. After clinical trials it is very difficult and costly to make any further changes in the process. Therefore, appropriate early process design is even more important in pharmaceutical production.

As soon as the first process data are available, process models can be built to estimate the material balance, energy consumption, labor requirement, and equipment needed in the production process. The models are improved stepwise through the development. The impact of possible variability in the process, assumptions, and estimates made in the modeling have to be validated in an uncertainty analysis. Modeling and uncertainty are discussed in detail in Chapter 3.

In batch production, the various pieces of equipment are occupied for different durations at different times during the process. To optimize the annual production and to minimize

the investment cost per product unit, the idle time of the different plant components should be minimized. Typically, the bioreaction step is the bottleneck of batch processes. The idle time of the downstream units can, for example, be reduced by using several small bioreactors rather than one big reactor to feed the downstream section. The procedure to optimize the occupation of equipment is called scheduling and is most efficiently carried out using appropriate process simulators. Scheduling aspects are, for example, addressed in Chapters 12, 14, and 15.

From early phases of development, the experimental and modeling results are assessed under economic and environmental aspects to realize a sustainable bioprocess. The assessment of sustainability is discussed in Chapter 4.

The development steps and the process have to be documented in detail. A clear documentation of the assumptions, estimates, problems, and alternatives in the development process helps in the decision-making. The process description is necessary in the build-up of the production plant, for process validation, and often also in the approval process of the product.

Successful process development involves many different participants as illustrated in Figure 2.7. The identification, engineering, and cultivation of the production strain or the enzyme used involves specialists in molecular biology, microbiology, biochemistry, genetic engineering, and cell-culture techniques interacting in a development team. Biochemical, chemical, and process engineers design and optimize the process. Environmental specialists have to make sure that the process is environmentally friendly and assure that the waste from the process should be treated. The marketing department assesses the possible market for the product. It also can give helpful advice about the required product quality and whether certain raw materials might create a negative image of the product on the market. For example, the use of animal serum in the fermentation of therapeutic proteins can reduce

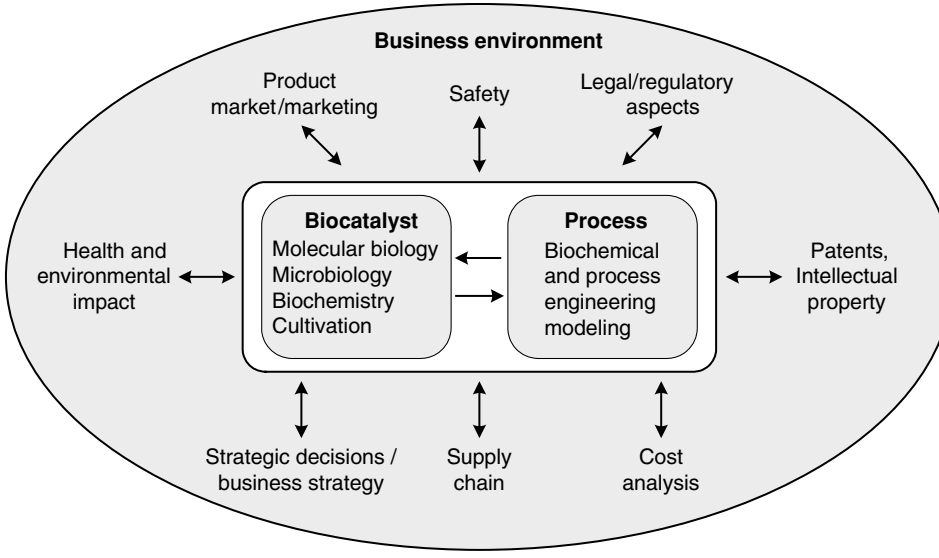


Figure 2.7 *Participants and interactions in the development of bioprocesses*

the sales potential in countries where many people do not eat meat or animal products for religious reasons.

The patent department helps to identify possible competitors and to see whether techniques that could be used in the process are protected by patents. It also prepares the patenting of its own process. Process modeling and cost analysis are important tools in the process development. Partly, they should be applied by the process developer, but usually also the collaboration of specialists is necessary. Every bioproduct, whether it is a drug, a food or feed additive, or a chemical intermediate, requires some form of approval. The legal department deals with this aspect. Finally, the management has to decide if the process fits into the business strategy of the company and if synergistic effects with other business units are possible.

References

[2.1] Roberts, S. (1999): Biocatalysts for fine chemicals synthesis. John Wiley & Sons, Ltd, Chichester.

[2.2] Liese, A., Seelbach, K., Wandrey, C. (2000): Industrial biotransformations. Wiley-VCH, Weinheim.

[2.3] Faber, K. (2004): Biotransformations in Organic Chemistry, 5th edn, Springer, Berlin.

[2.4] Bommarius, A., Riebel, B. (2003): Biocatalysis – Fundamentals and applications. Wiley-VCH, Weinheim.

[2.5] Pulz, O., Gross, W. (2004): Valuable product from biotechnology of microalgae. *Appl. Microbiol. Biotechnol*., **65**, 635–648.

[2.6] Torzillo G., Pushparaj B., Masojidek J., Vonshak A. (2003): Biological constraints in algal biotechnology. *Biotechnol. Bioprocess Eng*., **8**, 338–348.

[2.7] Szczebara, F., Chandelier, C., Villeret, C., Masurel, A., Bourot, S., Duport, C., Blanchard, S., Groisillier, A., Testet, E., Costaglioli, P., Cauet, G., Degryse, E., Balbuena, D., Winter, J., Achstetter, T., Spagnoli, R., Pompon, D., Dumas, B. (2003): Total biosynthesis of hydrocortisone from a simple carbon source in yeast. *Nature Biotechnol*., **21**, 143–149.

[2.8] Pavlou, A.K., Reichert, J.M. (2004): Recombinant protein therapeutics – success rates, market trends and values to 2010. *Nature Biotechnol.*, **22**, 1513–1519.

[2.9] Walsh, G. (2003): Biopharmaceuticals: Biochemistry and biotechnology. John Wiley & Sons, Inc., New York.

[2.10] Kretzmer, G. (2002): Industrial processes with animal cells. *Appl Microbiol. Biotechnol*., **59**, 135–142.

[2.11] Butler, M. (2005): Animal cell cultures: recent achievements and perspectives in the production of biopharmaceuticals. *Appl. Microbiol. Biotechnol*., **68**, 283–291.

[2.12] Koehler, G., Milstein, C (1975): Continuous culture of fused cells secreting antibody of predefined specificity. *Nature*, **256**, 495–497.

[2.13] Hesse, F., Wagner, R. (2000): Developments and improvements in the manufacturing of human therapeutics with mammalian cell cultures. *Trends Biotechnol*., **18**, 173–180.

[2.14] Eyer, K., Oeggerli, A., Heinzle, E. (1995): On-line gas analysis in animal cell cultivation: II. Methods of oxygen uptake rate estimation and its application to controlled feeding of glutamine. *Biotechnol. Bioeng*., **45**, 54–62.

[2.15] Shuler, M., Kargi, F. (2002): Bioprocess engineering – Basic concepts. Prentice Hall, New Jersey.

[2.16] Chawla, H. (2002): Introduction to plant biotechnology. Science Publisher, Enfield.

[2.17] Tabata H. (2004): Paclitaxel production by plant-cell-culture technology. *Adv. Biochem. Eng. Biotechnol*., **87**, 1–23.

[2.18] Faber D., Molina J., Ohlrichs C., Vander Zwaag D., Ferre L. (2003): Commercialization of animal biotechnology. *Theriogenology*, **59**, 125–138.
[2.19] Schmid, R. (2003): Pocket guide to biotechnology and genetic engeneering. Wiley-VCH, Weinheim.
[2.20] Faurie, R., Thommel, J. (2003): Microbial Production of L-Amino Acids. Springer, Berlin. Foerstner, U. (1998): Integrated Pollution Control. Springer, Berlin.
[2.21] Hermann T. (2003): Industrial production of amino acids by coryneform bacteria. *J. Biotechnol.*, **104**, 155–172.
[2.22] Morris, K.V., Rossi, J.J. (2006): Antiviral applications of RNAi. *Handb. Exp. Pharmacol.*, **173**, 105–116.
[2.23] Proske, D., Blank, M., Buhmann, R., Resch, A. (2005): Aptamers – basic research, drug development, and clinical applications. *Appl. Microbiol. Biotechnol.*, **69**, 367–374.
[2.24] Stahmann, K., Revuelta, J., Seulberger, H. (2000): Three biotechnical processes using *Ashbya gossypii*, *Candida famata*, or *Bacillus subtilis* compete with chemical riboflavin production. *Appl. Microbiol. Biotechnol.,* **53**, 509–516.
[2.25] Ramesh H., Tharanathan R. (2003): Carbohydrates – The renewable raw materials of high biotechnological value. *Crit. Rev. Biotechnol.*, **23**, 149–173.
[2.26] Rowe, G., Margaritis, A. (2004): Bioprocess design and economic analysis for the commercial production of environmentally friendly bioinsecticides from *Bacillus thuringiensis* HD-1 kurstaki. *Biotechnol. Bioeng.*, **86**, 377–388.
[2.27] Rawlings, D. (2002): Heavy metal mining using microbes. *Annu. Rev. Microbiol.*, **56**, 65–91.
[2.28] Acevedo, F., Gentina, J. (1999): Process engineering aspects of the bioleaching of copper ores. *Bioprocess Eng.*, **4**, 223–229.
[2.29] Bosecker, K. (1997): Bioleaching: Metal solubilization by microorganisms. *FEMS Microbiol. Rev.*, **20**, 591–604.
[2.30] Brierley, C. (1982): Microbiological mining. *Scientific American*, **247**, 42–50.
[2.31] Roels, J. (1983): Energetics and kinetics in biotechnology. Elsevier Biomedical Press, Amsterdam.
[2.32] Bailey, J., Ollis, D. (1986): Biochemical engineering fundamentals. McGraw-Hill, New York.
[2.33] Moser, A. (1988): Bioprocess technology. Springer, New York.
[2.34] Dunn, J., Heinzle, E., Ingham, J., Prenosil, J. (2003): Biological reaction engineering. Wiley-VCH, Weinheim.
[2.35] Nielsen, J., Villadsen, J., Lidén, G. (2003): Bioreaction engineering principles. Kluwer Academic/Plenum, Dordrecht.
[2.36] Biwer, A., Zuber, P., Zelic, B., Gerharz, T. Bellmann, K., Heinzle, E. (2005): Modeling and analysis of a new process for pyruvate production. *Ind. Eng. Chem. Res.*, **44**, 3124–3133.
[2.37] Cooney, C., Wang, D., Mateles, R. (1969): Measurement of heat evolution and correlation with oxygen consumption during microbial growth. *Biotechnol. Bioeng.*, **11**, 269–281.
[2.38] Tewari, Y., Goldberg, R. (1985): Thermodynamics of the conversion of aqueous glucose to fructose. *Appl. Biochem. Biotechnol.*, **11**, 17–24.
[2.39] Blanch, H., Clark, D. (1996): Biochemical engineering. Dekker, New York.
[2.40] Taylor, K. (2002): Enzyme kinetics and mechanisms. Kluwer Academic Publishers, Dordrecht.
[2.41] Leskovac, V. (2003): Comprehensive enzyme kinetics. Kluwer Academic / Plenum Publishers, New York.
[2.42] Raju, G.K., Cooney, C.L. Media and air sterilization. In Biotechnology (2nd Edn) – Vol. 3, edited by Stephanopoulos, G. VCH, Weinheim, 1993, pp. 157–184.
[2.43] Kristiansen, B., Mattey, M., Linden, J. (1999): Citric acid biotechnology. Taylor & Francis, London.
[2.44] John, G., Heinzle, E. (2001): Quantitative screening method for hydrolases in microplates using pH indicators: Determination of kinetic parameters by dynamic pH monitoring. *Biotechnol. Bioeng.*, **72**, 620–627.

[2.45] Ladisch, M. (2001): Bioseparation engineering: Principles, practice, and economics. Wiley Interscience, New York.
[2.46] Harrison, R., Todd, P., Rudge, S., Petrides, D. (2003): Bioseparations science and engineering. Oxford University. Press, New York.
[2.47] Perry, R., Green, D., Maloney, J. (1997): Perry's chemical engineers' handbook. McGraw-Hill, New York.
[2.48] McCabe, W., Smith, J., Harriott, P. (2001): Unit operations of chemical engineering. McGraw-Hill: New York.
[2.49] Doran, P. (1995): Bioprocess engineering principles. Academic Press, London.
[2.50] Atkinson, B., Mavituna, F. (1991): Biochemical engineering and biotechnology handbook. Stockton Press, New York.
[2.51] Ingham, J., Dunn, I.J., Heinzle, E., Prenosil, J.E. (2000): Chemical engineering dynamics. 2nd Edition. Wiley-VCH, Weinheim.
[2.52] Peters, M., Timmerhaus, K., West, R. (2003): Plant design and economics for chemical engineers. McGraw-Hill, Boston.
[2.53] Singh, S.M., Panda, A.K. (2005): Solubilization and refolding of bacterial inclusion body proteins. *J. Biosci. Bioeng*., **99**, 303–310.
[2.54] Thomé-Kozmiensky, K., Willnow, S., Fleischer, G. et al. (1995): Waste. In: Ullmann's encyclopedia of industrial chemistry, Vol. B8. Wiley-VCH, Weinheim, pp. 559–770.
[2.55] Brauer, H. (1996): Handbuch des Umweltschutzes und der Umwelttechnik. Springer Verlag, Berlin.
[2.56] Watts, R. (1998): Hazardous wastes: Sources, pathways, receptors. John Wiley & Sons, Ltd, Chichester.
[2.57] Lee, C., Lin, S. (2000): Handbook of environmental engineering calculations. McGraw-Hill, New York.
[2.58] Henze, M., Harremoes, P., Cour Jansen, J., Arvin, E. (2002): Wastewater treatment. Springer, Berlin.
[2.59] Tchobanoglous, G., Burton, F., Stensel, D. (2003): Wastewater engineering: treatment and reuse. McGraw-Hill, New York.
[2.60] Bagchi, A. (2004): Design of landfills and integrated solid waste management. John Wiley & Sons, Ltd, Chichester.
[2.61] Joerdening, H., Winter, J. (2004): Environmental biotechnology: Concepts and applications. John Wiley & Sons, Ltd, Chichester.
[2.62] Williams, P. (2005): Waste treatment and disposal. John Wiley & Sons, Ltd, Chichester.
[2.63] Bishop, P. (2000): Pollution prevention: Fundamentals and practice. McGraw-Hill, Boston. Bisswanger, H. (2002): Enzyme kinetics. Wiley-VCH, Weinheim.
[2.64] Foerstner, U. (1998): Integrated pollution control. Springer Verlag, Berlin.
[2.65] El-Halwagi, M. (1997): Pollution prevention through process Integration – systematic design tools. Academic Press, London.
[2.66] Heinzle, A., Hungerbühler, K. (1997). Integrated process development: The key to future production of chemicals. *Chimia*, **51**, 176–183.
[2.67] Stephanopoulos, G., Aristidou, A., Nielsen, J. (1998): Metabolic engineering: Principles and applications. Academic Press, London.
[2.68] Storhas, W. (2003): Bioverfahrensentwicklung. Wiley-VCH, Weinheim.

3

Modeling and Simulation of Bioprocesses

Process modeling and simulation enhances our insight and understanding of a process and helps to identify potential improvements as well as possible difficulties. In process development, simulation can supplement experiments to broaden the basis for sound decision-making, as illustrated in Figure 3.1.

There are a number of books on chemical engineering that deal with modeling of chemical processes [3.1–3.10]. While the general approach is similar, typical bioprocesses differ in their kinetics of product formation, process structure, and operating constraints when compared with chemical processes. In this chapter we provide a brief introduction to bioprocess modeling and simulation. First, we discuss the principles of process analysis and modeling, then model creation, and finally the consideration of uncertainties in the model.

To illustrate the different steps in bioprocess modeling we use the production of cellulase as a training case which is highlighted throughout this chapter.

Cellulases are a mixture of enzymes that can hydrolyse plant biomass, consisting mainly of cellulose and hemicellulose, to glucose. An overview of this process is given by Rabinovich et al. [3.11] and Zhang and Lynd [3.12]. Cellulases consist of two major groups, endoglucanases and cellobiohydrolases (for details see [3.13]). Cellulases are used today in the food, animal feed, textile, and pulp and paper industry and account, together with hemicellulases, for around 20% of the world enzyme market [3.14]. Cellulosic plant material is cheap and readily available in huge abundance. Economically feasible conversion into ethanol or other low-value, high-volume commodities would provide an important environmental and strategic benefit. Cellulases convert cellulosic material into glucose that is converted into ethanol by fermentation. This requires large amounts of inexpensive cellulases. Although such an ethanol-production process is not yet economically competitive, in part due to the high cellulase price, there is a high expectation for this process in the future [3.15–3.18].

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Most commercial cellulases are produced using the aerobic fungus *Trichoderma reesei* [3.19]. The fermentation uses insoluble cellulose as carbon source. For its use in ethanol production, biomass is removed after fermentation and the enzyme solution is concentrated. The model we use in this training case is based on data from Himmel et al. [3.17], Wooley et al. [3.16] and Saez et al. [3.20]. The fermentation model and the process flow diagram are kept simple to help the reader concentrate on the modeling process. Nevertheless, the model is a realistic representation of cellulase production.

3.1 Problem Structuring, Process Analysis, and Process Scheme

3.1.1 Model Boundaries and General Structure

Before moving into the detailed steps of modeling, we discuss the components of a process model. Figure 3.2 provides an overview of process components. Raw materials enter the process and are converted into a final product. In bioprocesses, typically complex raw materials are used as reactants or substrates for the bioreaction. Additionally, a range of additional materials like solvents and mineral salts are consumed in the fermentation as well

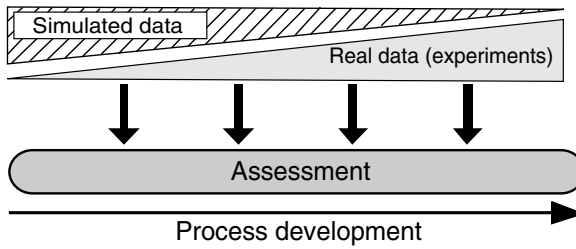


Figure 3.1 *Role of modeling and simulation to broaden the data basis for decision-making in the process development. Reproduced by permission of Wiley-VCH*

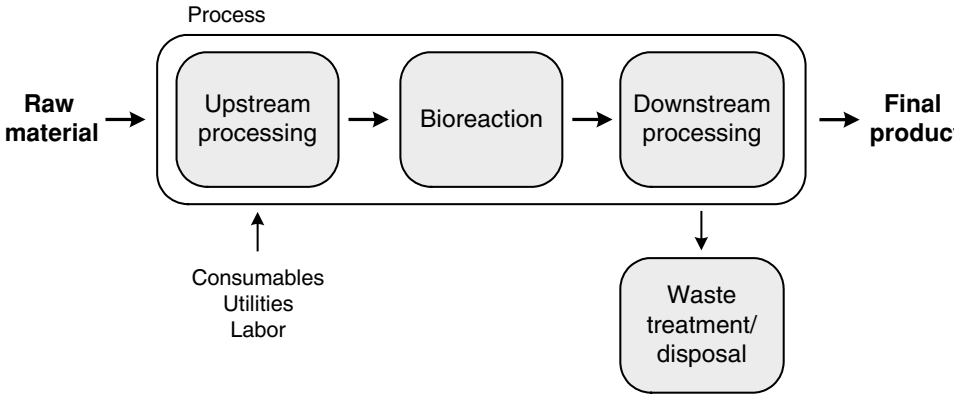


Figure 3.2 *Process boundaries and material balance regions of a process*

as product separation and purification. Apart from the raw materials, the process requires consumables like chromatography resins and membranes, utilities like electricity, steam, and cooling water, and finally human labor to run the process. The process can be divided conveniently into three sections: Upstream, bioreaction, and downstream. The upstream processing includes the seed train to provide the necessary amount of inoculum and the preparation of the media for the bioreaction. The bioreaction section includes a bioreactor and all related equipment, such as the compressor and air filter for sterilization of the air to a fermenter. The bioreaction is the central part of the process that converts the raw materials into the desired product. Usually, by-products are formed and raw materials are not completely consumed; thus waste is generated in the process. The following downstream processing section includes all steps necessary to separate and purify the product from the other materials to provide a sufficiently pure final product.

All materials not converted into the final product, nor sold as a by-product or recycled within the process, become waste that requires waste treatment or disposal. Usually the model boundaries enclose the three core parts of the process (upstream, bioreaction, downstream), but not the waste-treatment steps. Often the costs to treat or dispose waste are known and considered directly rather than including the necessary equipment in the model. However, certain pretreatment steps required to assure that the waste fulfills necessary quality standards are routinely covered in the model. For instance, a high-pH solution has to be neutralized before it can be discharged to a municipal sewage treatment plant. A process model should represent all relevant steps and streams within its boundaries.

3.1.2 Modeling Steps

Goal Definition and Model Boundaries. Figure 3.3 provides an overview of the steps in the modeling process. For successful modeling, it is crucial to define the modeling goal right at the beginning. This includes the final product specification, the plant size, usually also

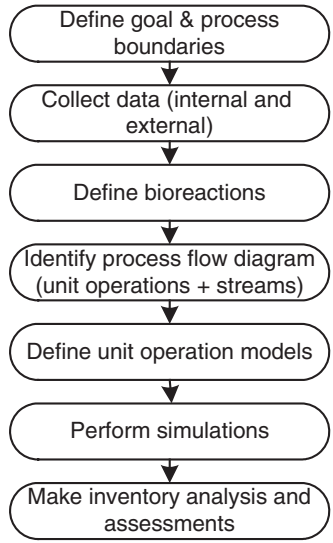


Figure 3.3 *Working steps in process modeling and assessment*

the biocatalyst, and the model boundaries. For the final product, it is important to define not only the molecule but also the necessary purity and other specifications.

In our training case, cellulase is the final product that is produced using the fungus *Trichoderma reesei*. Since it will be used as a catalyst to provide glucose for ethanol production, it is not necessary to separate fermentation by-products like glucose or non-consumed raw materials like cellulose or ammonia, because these materials are used in the ethanol fermentation. This is very important for specification of downstream processing.

The economy of scale has a strong impact on process cost. Therefore, it is important to choose a realistic plant size in the model. The plant size can be derived either from the volume and number of fermenters or from an expected annual production. The decision is determined by the current or the expected market volume, the technical feasibility of the process, the company's business plan, and the influence of competitors. In general, each model has to include all necessary process steps but keeping complexity at a minimum.

For our training case we assume an annual production of 300 tons of cellulase. The model will include the seed train, the fermentation, and the complete downstream processing. Before the cellulose can be fermented, it has to be pretreated with dilute acid. This pretreatment is not covered in the model for reasons of simplicity. Instead, we allocate a price to the pretreated cellulose and use it as the raw material in our model.

Data Mining. Once the goals and the model boundaries are defined, the necessary data have to be collected. In the best case, one can rely on data from one's own experiments. However, usually external data are needed to fill data gaps. Table 3.1 lists common data sources and possible difficulties involved in acquiring such data. Often, parameter values have to be estimated from different sources or extrapolated from conditions that differ from the expected process, e.g. in scale, process conditions, biocatalyst used, etc. Critical expert assessment of data reliability and applicability is necessary.

Bioreaction Model. One usually starts modeling with the bioreaction. From the collected data and the general bioprocess knowledge we discussed in Chapter 2, the reaction equations and conditions are derived. First, the raw materials needed for the applied biocatalyst

Table 3.1 *Possible data sources and problems usually connected with them*

Data source	Possible difficulties
Own experiments	scale, existence/availability
Previous project of a similar process	transferability, outdated information
Literature	accuracy, up-to-dateness, transferability
Patents	accuracy, (legal) usability
Expert opinions	actual availability, range of opinions
Own estimates	validation

are listed. In the next step, parameters like yields, fermentation time, final product concentration, by-product formation etc. are determined. Either reaction data are already known from experiment or a kinetic or a stoichiometric model can be applied to calculate these parameters' characteristics. Additionally, the reaction conditions for the process model have to be defined.

Table 3.2 provides an overview of the parameters chosen for the bioreaction model of our training case. The cellulase production with *T. reesei* in our example requires a medium with pretreated cellulose and corn steep liquor as carbon sources, ammonia as nitrogen source and for pH regulation, and some other nutrients and trace elements.

Table 3.2 *Key parameters of the fermentation model of the cellulase production as an example for the definition of model parameters. CSL = Corn steep liquor; dcw = dry cell weight*

Model parameter	Value	Source
Bioreaction		
Initial cellulose concentration (g/L)	45	[3.17,3.20]
Yield (g cellulase/g cellulose)	0.33	[3.17]
Productivity (g cellulase/L h)	0.125	[3.17]
Utilization cellulose (%)	90	own estimate
Initial CSL concentration (g/L)	7.5	[3.15,3.16]
Nutrients/trace elements (g/L) (sum)	4.1	[3.15,3.16]
Utilization CSL + nutrients (%)	75	own estimate
Ammonia added (g/L)	1.0	own estimate
CO_2 formation (g/L fermenter volume)	18	[3.20]
Final cellulase concentration (g/L)	13.4	calculated
Fermentation time (h)	107	calculated
Final biomass concentration (g dcw/L)	15	[3.20]
Bioreaction conditions		
Inoculum volume (% of working volume)	5.0	[3.15,3.16]
Working volume vessel (%)	80	[3.15,3.16]
Aeration rate (vvm)	0.58	[3.17]
Specific agitator power (W/m^3)	500	[3.17]
Fermentation temperature (°C)	28	[3.20]

Process Flow Diagram and Unit Operations. In the next step the process flow diagram (PFD) is identified. All unit procedures and the process streams of the model become defined. Every unit operation has to be described in a model and the model parameters have to be defined.

The model of our training process consists of three seed reactors and a production fermenter. A heat sterilizer for the raw materials and a compressor for aeration are connected to each reactor. After the bioreaction the biomass is removed in a rotary vacuum filter. The resulting enzyme solution is concentrated via an ultrafiltration step.

Documentation. Every model contains assumptions, estimates, and simplifications; their influence on individual steps and the overall performance can be addressed in an uncertainty analysis. However, it is essential to document all assumptions, estimates, and simplifications, in a written format, and to explain why certain values were chosen. Transparent documentation of a model serves as an anchor or reference point and enables others to comprehend and interpret the simulation results and identify uncertainties.

The model is created and finally transferred into suitable software where simulations are performed. This procedure is discussed in the next chapter. Apart from improving the general understanding of the process, simulation results are used for sustainability assessment and optimization as explained in Chapter 4.

3.2 Implementation and Simulation

3.2.1 Spreadsheet Model

Less complex models can be easily built in spreadsheet software like Microsoft Excel. In principle, it is possible to map a complete bioprocess in a spreadsheet model. All necessary calculations can be programmed in a spreadsheet environment.

A spreadsheet model of the seed train and the bioreaction step of the cellulase production is available on the CD. The model is based on the parameters shown in Table 3.2. A more detailed description is given on the CD in the file 'Fermentation model'. Such a model is convenient to calculate the mass balance of a batch or to estimate the annual production of a fermenter. Basic data that are used to calculate the model results are defined in Table 3.2.

However, such calculations become very complex when larger processes with multiple unit operations are implemented in spreadsheets. Process simulation software allows more efficient modeling. It supports clear structuring of the model and provides a large set of typical unit operations and procedures. Thus, the time to create and validate a model is significantly reduced and the analysis of the simulation results is greatly facilitated.

3.2.2 Modeling using a Process Simulator

In this book we use the process simulator SuperPro Designer™ from Intelligen, Inc. (New Jersey, USA). A demo version of the software is available on the CD and allows running of all models provided on that CD. In the following text, we provide a general introduction to the creation of a process model in a process simulator. A more specific introduction to the software is given in the SuperPro Manual, also available on the CD. Furthermore, the CD includes a SuperPro model of the cellulase production. Alternative process modeling tools are available, e.g. the products of Aspentech (Massachusetts, USA) (see e.g. [3.21]).

Figure 3.4 gives an overview of the consecutive modeling steps. The first step is to draw the process flow diagram on the flowsheet interface of the process simulator. The simulator provides models for most unit procedures and equipment typically used in bioprocesses. The equipment icons are placed in the flowsheet window in the order of their occurrence

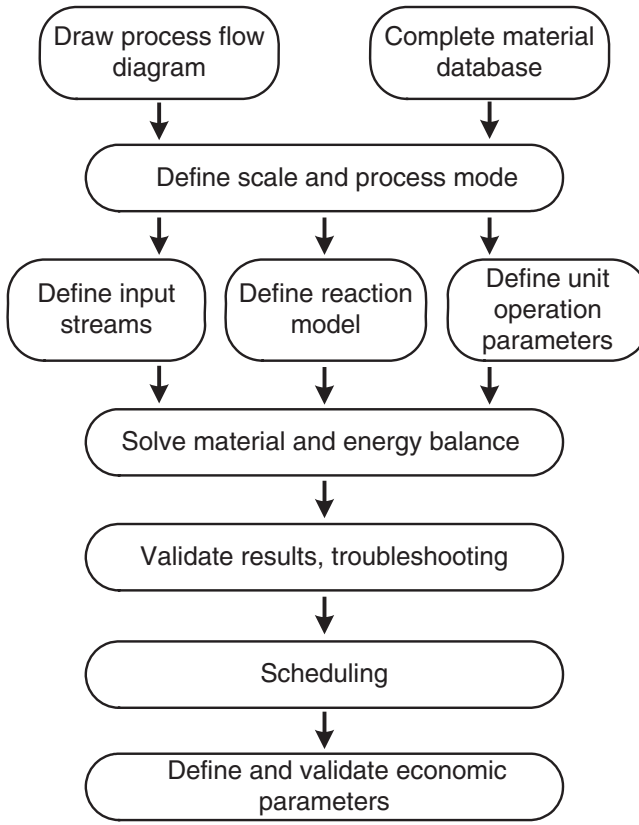


Figure 3.4 *Steps to build a model in process simulation software*

in the process. Then the PFD is completed by drawing the input streams, the connecting streams between the units, and the output streams that cross the model boundary.

It is recommended to define different process sections, e.g. upstream, bioreaction, and downstream. This facilitates the analysis of the simulation results and it enables the setting of different values for general model parameters in the various sections. For example, the level of detail may vary between the bioreaction section and the subsequent purification. This can be considered by using different values in the cost estimation of unlisted equipment (for details see Chapter 4).

Figure 3.5 shows the flowsheet for the cellulase model. The seed trains include the bioreactors P-1, P-2, and P-6 that are aerated with the compressors P-12, P-3, and P-7. The input materials (for P-1: S-101 to S-103) are sterilized in P-14, P-4, and P-9 and led to the bioreactor where ammonia is fed both as a nitrogen source and for pH regulation. In the seed reactors mainly biomass is produced. The seed reactor P-6 provides the inoculum for the production fermenter P-15. The input materials water, cellulose, corn steep liquor, and trace elements are sterilized in the continuous heat sterilizer P-10.

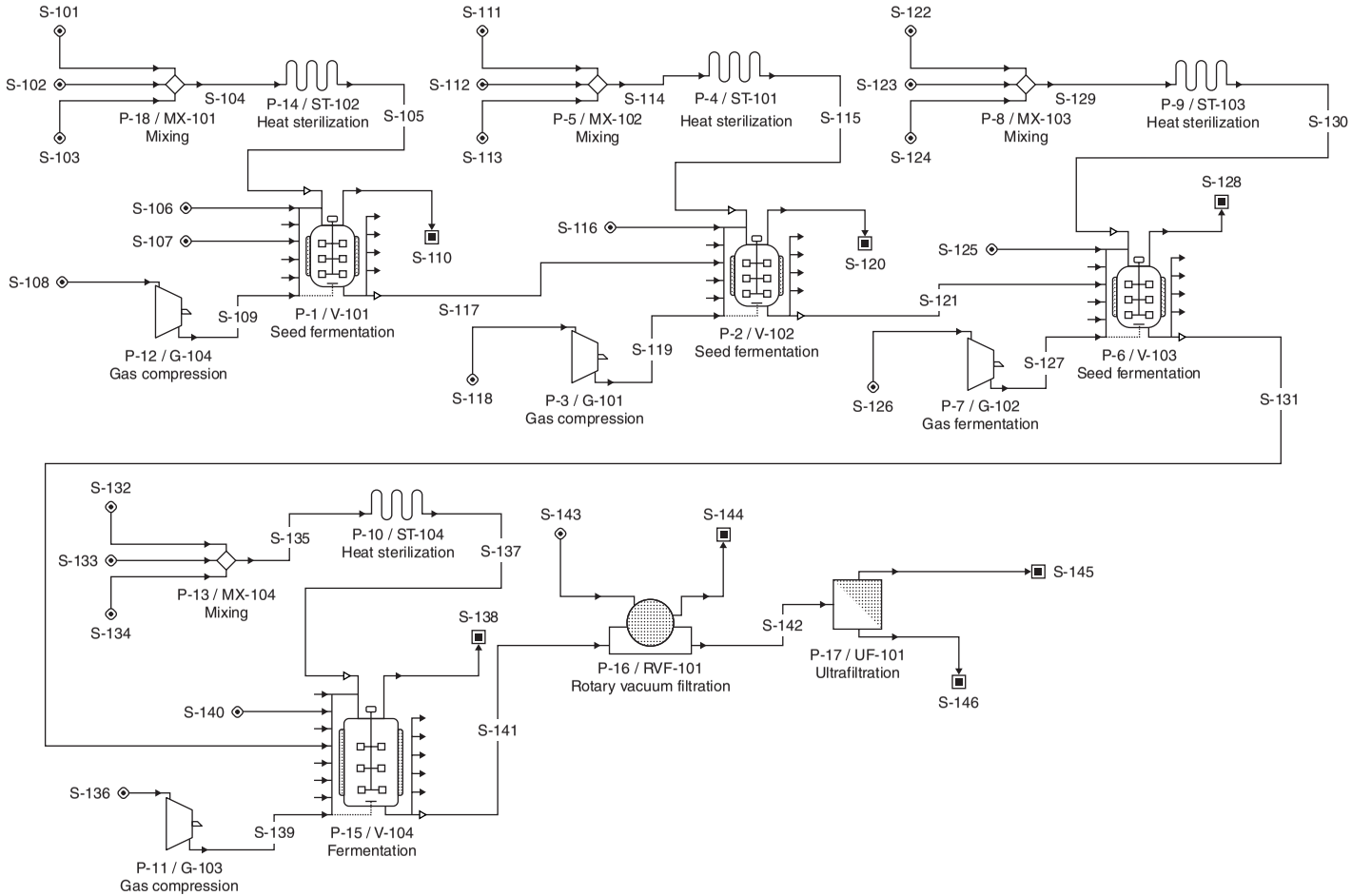


Figure 3.5 *Process flow diagram of cellulase production*

The fermenter is aerated by compressor P-11. Over a period of 107 h cellulase is produced. After the fermentation the biomass is separated with rotary vacuum filtration P-16 and the resulting enzyme solution is concentrated via the ultrafiltration P-17. The final solution can be directly used in the ethanol fermentation.

Before the detailed modeling of the individual units begins, a material database for the process should be compiled. This database includes all materials that enter the process or are formed during the process. Usually, the software provides a backup database that contains the most commonly used compounds. Other materials have to be defined. The database can contain a wide range of material properties. To reduce the necessary time, the entered properties can be restricted to those that are relevant for the process. Regularly used combinations of materials can be defined as stock mixtures, e.g. 5 molar hydrochloric acid.

In the next step the PFD and all the unit operations involved are specified. Before the specification of input streams and the unit models, the scale of the process has to be determined, e.g. by establishing the required fermenter volume. When specifying the unit models it is recommended to keep the PFD as a backup and start a new flowsheet with the same material database. Here, the first unit procedure and its related streams are drawn, specified, and this part of the model is calculated until all errors are fixed. Then the next process step is added and specified and so on. For each step the input streams that enter the process are defined. The material composition of a stream is set. Then the overall mass or volume of the stream is defined directly in the stream specifications. Afterwards the size of the stream determines the size of the subsequent equipment, e.g. a blending tank. Alternatively, the overall mass/volume is kept variable and the model parameters of the receiving equipment entail the input amount. For example, a blending tank of a defined size is filled to a defined volume.

Most unit procedures consist of several consecutive steps. For a blending tank a typical sequence would be: (i) feeding water and other materials, (ii) mixing of the tank content, and (iii) transfer out to a subsequent unit. This sequence has to be defined first. Then the model parameters of the single unit operation are specified.

A process simulator includes pre-defined models for most bioprocess unit operations. Thus, only the model parameters have to be specified. The unit model is usually explained in the help files. The parameter values are taken from the collected process data or the user's general engineering knowledge. Additionally, the software usually provides default values. They represent average values for the unit procedures and are thus of great help in very early development stages. For basic estimates, these values can be assumed. However, they may vary substantially from the situation in the process model. Therefore, in a detailed model, it is recommended to restrict the use of default values to a minimum.

The number of parameters needed depends on the complexity of the unit operation. For a simple step, like charging a raw material to a bioreactor, it is sufficient to specify the input stream, the start time, and the filling rate. In the bioreaction step, one or several reaction equations are defined on a mass or a molar basis to describe the product and by-product formation.

Often, a process includes several pieces of the same equipment, e.g. several fermenters that work in a staggered mode. To keep every modeling step simple, we recommend to model first a process with only one piece of equipment. When this single-batch model shows satisfying results it can then be expanded. Efficient scheduling is crucial for the

optimal production and, thus, for an economically feasible process. However, in the modeling process it is recommended not to pay too much attention to scheduling until one has a working batch model with all unit operations represented as single units. Only then should the model be expanded to the expected number of bioreactors or downstream units such as chromatography columns. Then scheduling and debottlenecking can be executed by working through the Gantt charts of the process model (see e.g. [3.22], and Chapters 12 and 15).

Besides raw materials, a process requires utilities, consumables, and labor. The various types of utilities and consumables are already defined in the software. After the basic model works properly, these definitions should be validated with respect to their suitability for the specific process under consideration. For example, the hourly labor cost might be different at the expected location of the modeled process, or the steam used may probably have a different temperature and thus provide a different amount of heat than assumed in the default settings. The annual consumption of a consumable, e.g. a membrane or a chromatography resin, is defined by the amount needed per batch, the maximum operating hours, and service life. Possible sources of information about such parameters can be experimental data, equipment supplier information, or literature. The amount of labor is defined for every unit operation and determines the number of people per shift and the number of shifts.

The modeling process is highly iterative. Usually many runs are necessary during the setup of a realistic model. The results are usually difficult to validate precisely. It is, therefore, indispensable to regularly check, at least, the plausibility of the results using order-of-magnitude calculations. This is done by checking the values of the magnitudes of streams and their compositions, as well as the values of model parameters going through the generated reports. It is necessary to check the assumptions concerning the utilities, consumables, and labor. When the basic model is built, again several rounds are needed to determine the number of fermenters and to optimize scheduling. The final technical model results in a material and energy balance of the process.

As an example, Table 3.3 shows the material balance of the cellulase production. The material and energy balance provide the basis for the environmental assessment (see Chapter 4).

Table 3.3 *Material balance of the cellulase production. (kg/kg P) = kg material per kg cellulase produced*

Component	Input (kg/kg P)	Output (kg/kg P)
Ammonia	0.08	
Biomass		1.17
Carbon dioxide		1.48
Cellulase		
in final product		1.0
product loss		0.04
Cellulose	3.62	0.35
Corn steep liquor	0.61	0.15
Nutrients	0.33	0.08
Water	77.4	77.8
Sum	432	432

After the technical model is validated, the economic parameters of the process and the model are edited. The software provides basic price information and a set of tools for an economic assessment. However, in parallel to the technical model, the economic model has to be validated. Before we turn to the detailed discussion of economics in Chapter 4, we address the different ways to assess the uncertainty in a process model.

3.3 Uncertainty Analysis

The understanding of the uncertainties in a process under development is crucial for a realistic assessment of a project. Thus, it is necessary to identify the risks and opportunities within a process and to quantify them. During process modeling there usually remain a number of open questions. These underline the need for uncertainty analysis. Furthermore, a solid documentation of the assumptions made while process modeling helps to identify the uncertain areas. Alternative process flow diagrams can be compared in a scenario analysis. The impact of single input variables, like medium cost or fermentation time, is studied with sensitivity analyses. However, for sound decision-making a quantification of the overall variability is critical. This can be assessed by Monte Carlo simulation where the probability distributions for a set of variables are specified and one can examine how these variabilities propagate through the model to effect economic and environmental performance parameters.

When discussing the term ‘uncertainty’, one can differentiate between variability and uncertainty. Variability is the effect of chance as seen in the actual variation. It is an intrinsic feature of the system. It cannot be reduced by further studies, although it may be reduced by changing the process settings. The variation of the product yield from batch to batch in an existing plant is a good example of variability. Uncertainty in the narrower sense is caused by a lack of knowledge about a parameter, e.g. the level of ignorance. The parameter itself does not show variability in reality but its exact value is not yet known. Further studies can reduce this type of uncertainty. An example might be the cost of a raw material that will be fixed by a long-term contract with a supplier but the price is not yet negotiated. Often, the variation of model parameters involves both variability and uncertainty. For example, the expected fermentation yield of the final process includes some uncertainty because it is not yet known what average yield can be realized. There is also a certain variability in the yield from batch to batch. In the following text we will use the term uncertainty to describe both types because the term is the most commonly used. However, it can be important to discuss whether an expected variation is due to variability or uncertainty. If uncertainty in the narrower sense is the main reason, additional studies should be done to reduce the uncertainty. If variability dominates, the process settings should be revised to reduce the overall uncertainty. A valuable discussion of these terms and more detailed introduction to risk analysis is provided by Vose [3.23].

Uncertainty that influences the process includes variation in the process itself, as well as in the supply chain and the market for the product (see Figure 3.6). In the supply chain, prices and quality of raw materials, consumables, labor, and utilities can show variability. Uncertainty in the market usually involves the selling price of the product and the market size. The uncertainty in the process itself concerns the structure of the process flow diagram that is studied in scenario analyses and the values of technical parameters of

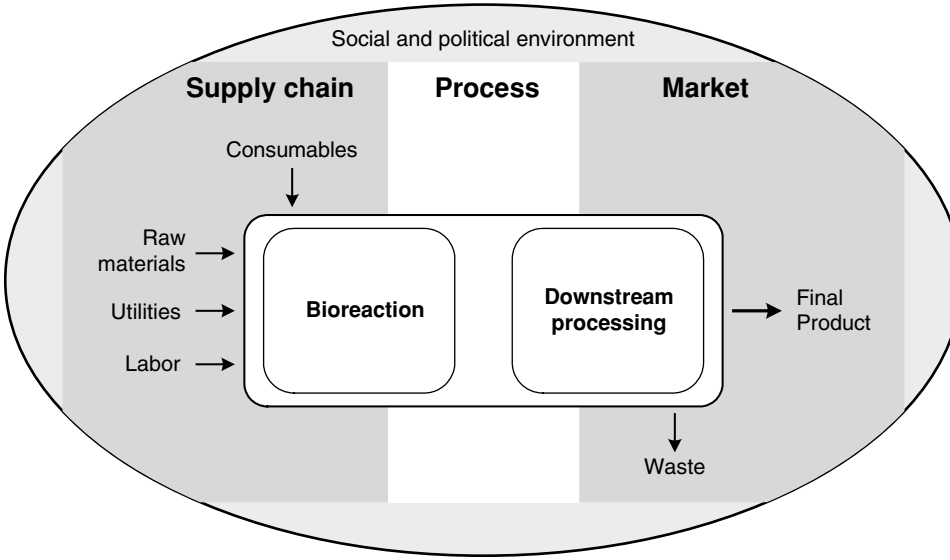


Figure 3.6 *Areas of uncertainty that affect a process*

the unit procedures. In addition to the differentiation in supply chain, technical, and market parameters, it can also be useful to differentiate between parameters that affect the different sections of a process (e.g. upstream, bioreaction, downstream).

Beside these uncertainties that directly affect the process there are also uncertainties in the social and political environment where the process is realized. For example, the social acceptance or new legal guidelines can strongly impact the success of a process as one can see for the use of genetically modified organisms in agriculture. However, it is very difficult to quantify, predict, and incorporate these variables in the model. Therefore, we do not include them in the following uncertainty analysis. They nevertheless should be considered and kept in mind (see Chapter 4.4).

Before starting the analysis, those parameters should be defined that are used as objective functions to describe the effect of uncertainty. Typically, a parameter that describes the technical performance of the process, e.g. the annual production, is chosen, and economic and environmental performance is characterized as discussed in Chapter 4.

3.3.1 Scenario Analysis

Variations of the process flow diagram and the process scale can be examined in scenario analyses, as exemplified in the Chapters 6, 9, 11, and 13. Especially in early process development, there might be a need to compare alternative process flowsheet topologies. An extraction step might replace a distillation column or the order of the downstream steps might vary. For such changes the economic and environmental impact can be derived in a scenario analysis. Furthermore, variation in size and number of pieces of key equipment, namely the fermenters, can be studied with scenarios. The base model can be used as a benchmark. For instance, if an extraction column is a theoretical alternative to the distillation used in the base model, one can identify the performance values like distribution

coefficients, possible yield, or number of theoretical stages that the extraction must reach to be economically viable.

Practically, one starts from the base model developed before and defines a number of new models (= new files). Scenarios normally refer to process flowsheet modifications but also scenarios concerning the supply chain can be made, e.g. if a key raw material is available in different qualities.

Table 3.4 shows the results of two scenario analyses for the cellulase model (corresponding model files are on the CD). In the base model, the inoculum volume is 5% of the fermenter volume. This defines the necessary volume of the seed reactors. If the inoculum volume is increased, the starting cell concentration is higher, and thus the time to reach the maximum biomass concentration and product formation might be shorter. In this scenario we assume the fermentation time to be 10 h shorter when the inoculum volume is increased to 10%. This enables a higher annual production. However, it requires an increase in the size of the seed reactors, which causes higher investment cost. This additional cost outweighs the higher annual production and causes higher unit production costs (see Table 3.4). The economic terms used are discussed in Chapter 4.

Table 3.4 *Scenario analyses of the cellulase production model. For a description of the scenarios see the text*

Scenario	Annual production (metric tons)	Capital investment ($ million)	Unit production cost ($/kg cellulase)
Base case	456	20.7	15.4
10% Inoculum	475	23.4	16.4
Additional chromatography	385	22.1	20.4

The second scenario describes the situation when an additional ion-exchange adsorption step is necessary to remove some interfering by-products. This additional step not only raises the investment cost but also reduces the annual production (product loss). Thus, it has two negative effects on the unit cost (see Table 3.4). The scenario analysis helps to quantify this impact.

3.3.2 Sensitivity Analysis

Sensitivity analyses study the impact of a single process parameter on the objective functions of the model. The analysis is usually done within the existing PFD. By comparing the sensitivity of different parameters, the most sensitive ones can be identified. Special attention must be paid to these parameters in the process development. Sensitivity analyses can be done for supply chain, and both technical and market parameters. Examples of such analysis are illustrated in the Chapters 7 and 13.

The first step in performing a sensitivity analysis is to select the right parameters to study and then define a reasonable value range for each parameter. A value range can be derived from the experimental results, from literature, or from one's own expectations and assumptions.

The easiest way to perform a sensitivity analysis is to change the model parameter and, if needed, corresponding but not automatically adjusted parameters stepwise in the model and plot resulting values of the objective functions versus the varied parameter values. This is appropriate if one wants to study only a few parameters. In a more advanced approach, one can predefine the settings in a spreadsheet and let the analysis perform automatically using the COM interface in SuperPro Designer™. Once the necessary Visual Basic script is written, the analysis can be performed and varied as often as necessary. The COM function is explained in more detail in the next chapter.

Before starting the calculations, it is important to check the model for parameters that are influenced by the varied parameter but are not automatically adjusted. Those parameters have to be adjusted manually. For example, if the product is an acid, and a base is used to adjust the pH at some point during the downstream processing. Then, in the model often the amount of base added to the product stream cannot be directly linked to the amount of acid that is contained in the stream. If one varies a parameter that changes the amount of product (acid) then the amount of base has to be adjusted manually.

As an example we study the impact of the reaction yield of cellulase formation on the unit production cost with a sensitivity analysis. All other parameters, such as start concentrations of cellulose and corn steep liquor, final biomass concentration, and CO_2 production, remain unchanged. Owing to the varying yield, the final product concentration varies as well. This is a certain simplification, because a proper C-balance is not possible under these settings. However, for our purpose, the possible error can be neglected. The base case yield is 33% and the yield is varied between 10% and 50%. As shown in Figure 3.7, the unit production cost (UPC) is highly sensitive at low yields and low corresponding final product concentrations. Then, the annual production is low and allocated fixed costs per unit product are high. At higher yields the impact of fixed costs becomes small and the sensitivity curve almost levels off. This behavior is often observed for fermentation parameters like yield and final product concentration (see e.g. [3.24]).

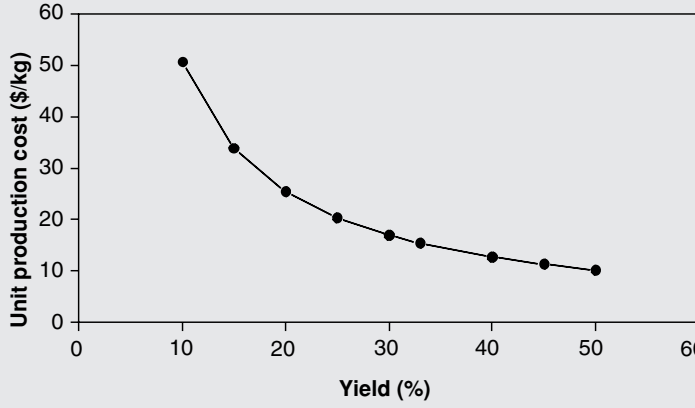


Figure 3.7 *Sensitivity of the unit production cost to the yield of cellulase production*

Sensitivity analysis quantifies the dependency of the objective functions on single parameters; it may not capture nonlinearities between multiple parameters that may vary simultaneously. However, it does not provide any information about the probability of certain values of the examined parameter.

3.3.3 Monte Carlo Simulation

Using the process model as the basis for a Monte Carlo simulation (MCS), we can explore how variance propagates through the entire process to impact both economic and environmental results (application examples in Chapters 10 and 13). The general procedure of an MCS is illustrated in Figure 3.8. The probability distributions of a set of uncertain parameters are defined. Values are selected randomly out of these distributions and the model is recalculated using this set of variables. This is repeated for a large number of iterations, resulting in probability distributions of the objective functions. MCS is widely recognized as a valid technique and appropriate software is commercially available. The level of mathematics is quite basic, and changes in the model can be done quickly. For a more detailed description see Vose [3.23] and Martinez and Martinez [3.25].

The implementation of a Monte Carlo Simulations consists of five steps. It is shown in the following for the use of SuperPro Designer™, MS Excel, and Crystal Ball 2000™:

(i) The selection of the objective functions: As discussed at the beginning of the chapter, it is important to define the relevant objective functions that are used to describe the impact of uncertainty.

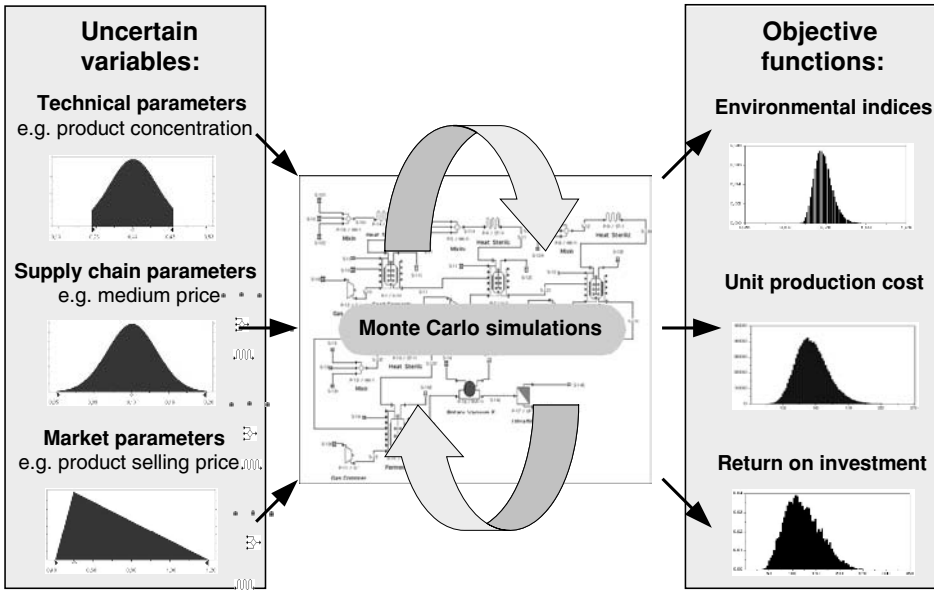


Figure 3.8 *General procedures for Monte Carlo simulations*

(ii) The selection of the uncertain input variables: From the model assumptions, the experimental results or one's own experience and expectations, those technical, supply chain and market parameters are identified that have relevance for the process and exhibit uncertainty. Here, a fast sensitivity analysis of the model parameters can help in the selection process.

(iii) Definition of the probability distribution: The realistic definition of the probability distributions of the input variables is essential to assure utility of the simulation results. Depending on the parameter, different data sources are available. For technical parameters, a distribution can be derived from a large number of experiments. For some supply chain parameters, like sugar or electricity prices, official statistics are available that can be used to derive a distribution. For other parameters, like the replacement frequency of a chromatography resin, suppliers might provide statistical information. However, there will always be parameters where direct data are not available and their distributions have to be estimated. Here, it is important to validate these estimates via literature and expert opinions.

(iv) Simulation: After all necessary data are defined, the simulations are performed. Figure 3.9 illustrates the calculation procedure for the Monte Carlo simulations using the COM function of SuperPro. A COM interface allows the model to interact with other software. The software Crystal Ball 2000™ (Decisioneering, Co., USA) and MS Excel are used here in connection with SuperPro Designer™. All parameters that will be varied are saved in an Excel spreadsheet. The probability distribution for every variable is defined in Crystal Ball and allocated to the corresponding cell in the spreadsheet. In each trial, Crystal Ball creates random values for the selected parameter set, according to the parameters' probability distributions. Via a Visual Basic (VBA) script these values are transferred to SuperPro Designer™, a simulation is initiated, and the simulation result for this set of parameters is transferred back to the spreadsheet, where the values of the different objective functions are saved by Crystal Ball. A high number of iterations is selected to reach a low standard error (<1%) for all objective functions. However, the time necessary to calculate the iterations can restrict their number.

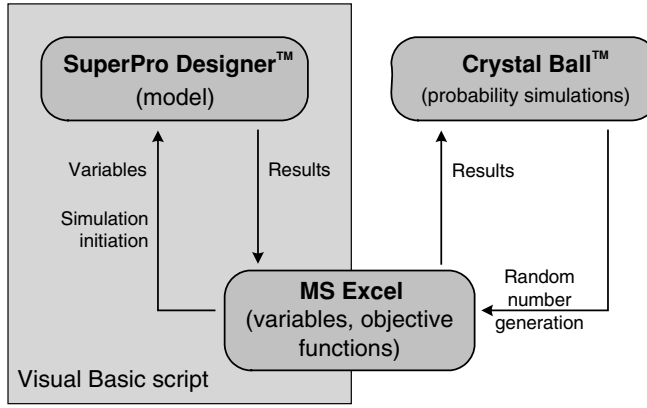


Figure 3.9 *Computational scheme to realize Monte Carlo simulations using MS Excel, Crystal Ball™ and SuperPro Designer™*

An alternative to Crystal Ball to perform MCS is the also-commonly-used software @RISK (Palisade, New York, USA).

(v) Analysis of the results: The MCS provides two important results. First, the probability distributions of the objective functions. The comparison of their mean values with the base model shows whether the 'best guess' in the base model was realistic. Their variation quantifies the uncertainty of the economic success and the environmental performance of the process. The second result is the identification of those input variables that contribute most to the existing uncertainty. Special attention must be paid to them in the process development, optimization and control.

Similar to the creation of the base model, an iterative refinement is usually necessary to reach the best possible results. In addition to simulations using all identified parameters, it can be useful to run the simulations with smaller sets, e.g. only the supply chain parameters or only those parameters that affect the bioreaction, and so on. If necessary, it is possible to define a correlation between input variables, e.g. if the aeration can be correlated to the biomass concentration.

To illustrate this approach we programmed an MCS with four parameters for the cellulase case. Table 3.5 shows the selected parameters and probability distributions. For simplicity, we only used technical parameters for the fermentation step. For the yield (see also the sensitivity analysis) and for the productivity we have chosen a normal distribution, for the aeration rate an even distribution, and for the specific power input a triangular distribution. These three distribution types are the most common. The variations of the parameters defined in Table 3.5 lie in a typical range that these parameters usually exhibit in fermentations.

Table 3.5 *Variables and their probability distributions used in the MCS of the cellulase process. V = Coefficient of variability*

Parameter	Value base model	Probability distribution	Variation
Yield (g/g)	0.33	normal	$V = 20\%$; range: 0.22–0.44
Productivity (g/L h)	0.125	normal	$V = 20\%$
Aeration rate (vvm)	0.58	even	0.3–0.8
Specific power (kW/m^3)	0.5	triangular	0.4–1.2, 0.5 as the most likely

The MCS is available in the Excel file on the CD. Figure 3.10 shows the probability distribution of the unit production cost (UPC). The mean is $ 16.20/g and the median $ 15.90/g. This is slightly higher than the base case. Whereas the yield, the productivity, and the aeration rate are evenly distributed around their base case value, the mean of the distribution chosen for the specific energy consumption is higher than its base case value. On average, this causes higher energy costs. This is a good example of how the definition of the parameter distribution affects the objective function. The standard deviation of the UPC is $ 2.30/g. That equals a coefficient of variability of 14%. The value range is $ 10.50/g to $ 35.20/g. With a 90% probability (90% percentile), the UPC are lower than $ 19.20/g, and higher than $ 13.40/g, respectively.

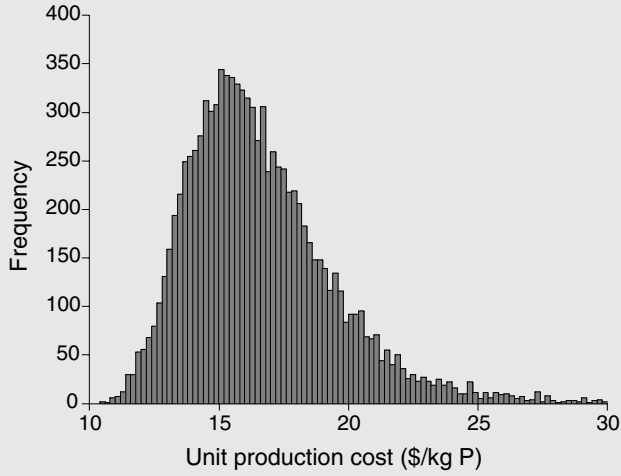


Figure 3.10 *Probability distribution of the unit production cost (UPC) of the cellulase model using 10 000 iterations and 100 classes in the graph*

The contribution of the three parameters that add more than 1% to the measured uncertainty is shown in Figure 3.11. The yield variation contributes most. It changes the reaction equation and, thus, the amount of product per batch. The productivity influences the fermentation time and hence the number of batches per year. Thus, both parameters influence the annual production and hence the UPC. The aeration rate and the specific power (not shown) have a far lower impact. They affect the energy consumption (utility cost) and the necessary equipment size of compressor and air filtration (capital investment).

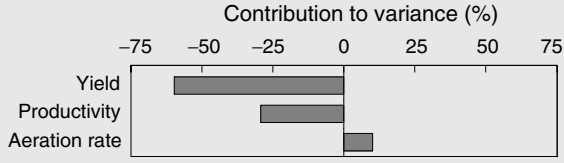


Figure 3.11 *Parameter contribution to the uncertainty of the unit production cost of the cellulase model*

References

[3.1] Biegler, L., Grossmann, I., Westerberg, A. (1997): Systematic methods of chemical process design. Prentice Hall, Upper Saddle River, USA.

[3.2] Oezilgen, M. (1998): Food process modeling and control: Chemical engineering applications. OPA, Amsterdam.

[3.3] Aris, R. (1999): Mathematical modeling: A chemical engineer's perspective. Academic Press, London.

[3.4] Keil, F., Mackens, W., Vob, H., Werther, J. (1999): Scientific computing in chemical engineering II: Simulation, image processing, optimization and control. Springer, Berlin.
[3.5] Anderson, N. (2000): Practical process research & development. Academic Press, London.
[3.6] Ingham, J., Dunn, J., Heinzle, E., Prenosil, J. (2000): Chemical engineering dynamics. Wiley-VCH, Weinheim.
[3.7] Luyben, W. (2002): Plantwide dynamic simulators in chemical processing and control. Dekker, New York.
[3.8] Dimian, A. (2003): Integrated design and simulation of chemical processes. Elsevier, Amsterdam.
[3.9] Elnashaie, S., Garhyan, P. (2003): Conservation equations and modeling of chemical and biochemical processes. Dekker, New York.
[3.10] Lewin, D. (2003): Using process simulators in chemical engineering: A multimedia guide for the core curriculum. John Wiley & Sons, Inc., New York.
[3.11] Rabinovich, M., Melnik, M., Bolobova, A. (2002): Microbial cellulases. *Appl. Biochem. Microbiol*., **38**, 305–321.
[3.12] Zhang, Y., Lynd, L. (2004): Toward an aggregated understanding of enzymatic hydrolysis of cellulose: Noncomplexed cellulase systems. *Biotechnol. Bioeng*., **88**, 797–824.
[3.13] Schuelein, M. (2000): Protein engineering of cellulases. *Biochim. Biophys. Acta*, **1543**, 239–252.
[3.14] Bhat, M. (2000): Cellulases and related enzymes in biotechnology. *Biotechnol. Adv*., **18**, 355–388.
[3.15] Wooley, R., Ruth, M., Glassner, D., Sheehan, J. (1999): Process design and costing of bioethanol technology: A tool for determining the status and direction of research and development. *Biotechnol. Prog*., **15**, 794–803.
[3.16] Wooley, R., Ruth, M., Sheehan, J., Ibsen, K., Majdeski, H., Galvez, A. (1999): Lignocellulosic biomass to ethanol process design and economics utilizing co-current dilute acid prehydrolysis and enzymatic hydrolysis: current and futuristic scenarios. Report NREL/TP-580-26157, National Renewable Energy Laboratory, Golden, Colorado.
[3.17] Himmel, M., Ruth, M., Wyman, C. (1999): Cellulase for commodity products from cellulosic biomass. *Curr. Opin. Biotechnol*., **10**, 358–364.
[3.18] Fujita, Y., Takahashi, S., Ueda, M., Tanaka, A., Okada, H., Morikawa, Y., Kawaguchi, T., Arai, M., Fukuda, H., Kondo, A. (2002): Direct and efficient production of ethanol from cellulosic material with a yeast strain displaying cellulolytic enzymes. *Appl. Environ. Microbiol.*, **68**, 5136–5141.
[3.19] Juhasz, T., Szengyel, Z., Szijarto, N., Reczey, K. (2004): Effect of pH on cellulase production of *Trichoderma reesei* RUT C30. *Appl. Biochem. Biotechnol*., **113**, 201–212.
[3.20] Saez, J., Schell, D., Tholudur, A., Farmer, J., Hamilton, J., Colucci, J., McMillan J. (2002): Carbon mass balance evaluation of cellulase production on soluble and insoluble substrates. *Biotechnol. Prog*., **18**, 1400–1407.
[3.21] Shanklin, T., Roper, K., Yegneswaran, P., Marten, M. (2001): Selection of bioprocess simulation software for industrial applications. *Biotechnol. Bioeng*., **72**, 483–489.
[3.22] Petrides, D., Koulouris, A., Siletti, C. (2002): Throughput analysis and debottlenecking of biomanufacturing facilities. *BioPharm,* (Aug) 2–7.
[3.23] Vose, D. (2000): Risk analysis. John Wiley, & Sons, Ltd, Chichester.
[3.24] Biwer, A., Griffith, S., Cooney, C. (2005): Uncertainty analysis of penicillin V production using Monte Carlo simulation. *Biotechnol. Bioeng*., **90**, 167–179.
[3.25] Martinez, W., Martinez, A. (2002): Computational statistics handbook with MATLAB. Chapman & Hall/CRC, Boca Raton.

4

Sustainability Assessment

4.1 Sustainability

In mid 1980s the Brundtland report started the contemporary discussion around the concept of sustainability [4.1]. However, the concept of sustainability management is much older and finds its origin in German forestry where, in 1713, the Saxonian Hans Carl von Carlowitz introduced the expression in his Sylvicultura Oeconomica. At that time it basically meant not to cut more timber in a certain year than was added to the stock by the natural growth.

The Club of Rome initiated the public discussion about the Earth having limited resources and capacity to absorb man-created pollution. In the Brundtland report, sustainability or sustainable development is defined as 'the development that meets the needs of the present without compromising the ability of the future generations to meet their own needs'. Others define it as the optimal growth path that maintains economic development while protecting the environment and optimizing the social conditions with the boundary of relying on limited, exhaustible natural resources [4.2]. All these definitions do explicitly see changes as an inherent characteristic of any living natural and social system. Therefore, sustainability clearly does not mean to preserve but to develop responsibly. Thinking in terms of sustainability also becomes more and more important in our modern economy, and in 1999 the Dow Jones Sustainability Indices were started. Corporate sustainability is considered a business approach that creates long-term shareholder value by embracing opportunities and managing risks deriving from **economic, environmental, and social development**.

Nowadays, these three dimensions constitute sustainability and might be considered the three pillars carrying this concept (Figure 4.1). All three parts are equally important in truly sustainable development. In the following three chapters we present methods to assess sustainability with respect to these three dimensions. However, they are not independent of each other but rather there are manifold interactions between them. In the last subchapter we discuss some of them to illustrate the complexity of these interactions.

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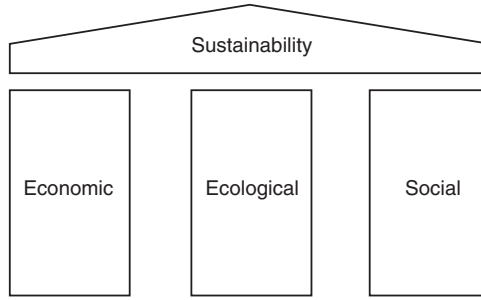


Figure 4.1 *The three pillars of sustainability*

4.2 Economic Assessment

We provide a basic description of economic assessment and several tools for cost and profitability analysis that are usually applied during process development. This background is essential to our understanding of economic assessment, as is illustrated in the case studies contained in the book. There are already a number of books, especially in the chemical engineering field, that cover cost and profitability assessment in detail. Here, we particularly recommend Peters et al. [4.3] as a standard reference book as well as several other texts [4.4–4.6].

The first step is the estimation of the capital investment that is usually based on the cost of the necessary equipment. After the capital investment is determined, the operating costs of the process can be derived from the different cost items like raw materials, energy, etc. These are the two parts of cost analysis. An overview of the estimation procedure is given in Figure 4.2. Complementary to this, profitability analysis examines the expected revenues

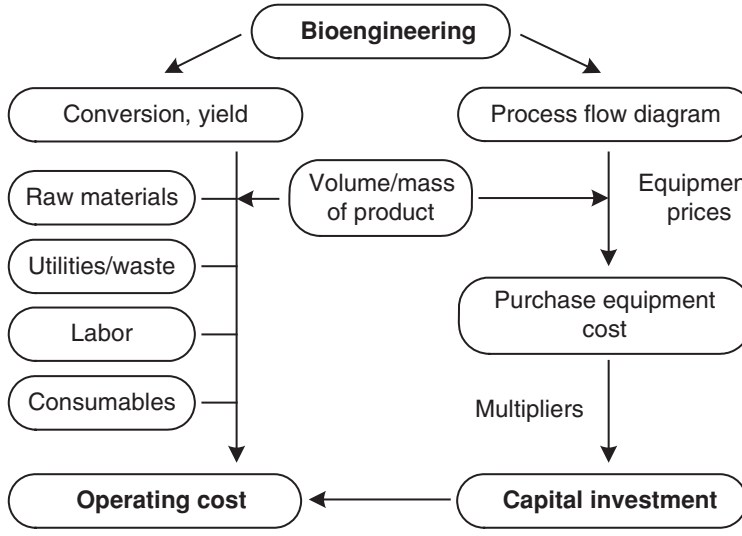


Figure 4.2 *Steps in the estimation of capital investment and operating costs*

and sets them in proportion to the costs and a number of other factors like the time-value of money.

4.2.1 Capital-Cost Estimation

Introduction. Capital cost, capital investment, or capital expenditure (often called CapEx) of a bioprocess facility is the total amount of money that has to be spent to supply the necessary plant (the fixed capital investment) plus the working capital that is needed for the operation of the facility.

Different methods exist to estimate the necessary capital investment of a planned facility. They are applied at different times during the life cycle of a project and vary in accuracy and time needed to establish the estimation. During process development preliminary estimates are usually made based on the purchased equipment cost. In the following section we give a short introduction to a method that uses multipliers to the purchased equipment cost to calculate the expected capital investment. For other methods we refer to Peters et al. [4.3], Perry, Green and Maloney [4.7], and Atkinson and Mavituna [4.8].

Equipment Purchase Cost. Since the equipment cost provides the basis for the capital cost estimation, the determination of a realistic value is crucial for the accuracy of the assessment. In the previous chapters we discussed how to identify and model the necessary unit operations and procedures. The resulting process flow diagram provides us with a list of the major equipment for the process. The starting point is often the fermenter size or the expected annual production; from this starting point and with the process model, we obtain the required size of the different pieces of equipment. Thus, we have the basic information to calculate the purchase cost of equipment.

The most accurate source of equipment prices is vendor quotations; these may require considerable effort to obtain. Another quite accurate source is the prices that were paid for the same or a similar piece of equipment in a previous project. However, the old prices must be updated to today's price level (see Section Price Indices).

If such data are not available, generalized values can be taken from literature. Here again, Peters et al. [4.3] is a good source and also Atkinson and Mavituna [4.8]. The equipment cost estimation in SuperPro Designer™ is based on a combination of vendor and literature data. While values in the literature are easier to obtain, they involve a higher uncertainty relative to vendor-supplied quotes.

In general, it is important to specify whether a given price of a piece of equipment includes delivery or is free on board (f.o.b.), i.e., the transportation cost must be added to the purchase price. For the estimation of the capital investment, the delivered purchased equipment cost is used. If the price is f.o.b. and exact freight costs are not available, an additional 10% of the purchase price can be assumed as an average value.

Auxiliary equipment that is necessary but not itemized in the major equipment list can be estimated by using a multiplier for unlisted equipment. The purchased equipment cost used in the following estimation of the capital investment becomes the sum of the cost for listed and unlisted equipment. In process modeling, different sections often have different levels of detail. For example, the bioreactor and all related equipment such as media tanks, sterilizer, air filters, etc. are completely covered in the model, while in the downstream section only the main separation steps are included. These differences can be compensated

for by choosing a small multiplier for unlisted equipment in the bioreaction section (e.g. 0.05) and a large value for the downstream section (e.g. 0.25).

Estimation of Total Capital Investment. The purchased equipment cost is the basis for the estimation of the total capital investment. Fixed capital cost items, such as process piping, insulation, electrical systems, etc. are estimated as percentage values of the purchased equipment cost. From the direct plant cost the indirect plant cost and subsequently the fixed and total capital investment are calculated. Table 4.1 shows the calculation of the capital investment for our training case of the production of cellulase. This method is commonly used for cost estimations in process development and the expected accuracy is ± 30% [4.3].

In the following section we discuss the different cost items contained in the capital investment. Thereby, we follow the structure of Table 4.1. The values used for the different multipliers are discussed in Section 'Multiplier Values'.

Table 4.1 *Calculation of the total capital investment based on purchased equipment cost and multipliers, shown for the training case of cellulase production*

Cost item	Multiplier	Base	Cost ($ thds.)
Delivered purchased equipment cost (PC)			3290
Installation	variable	PC	1060
Process piping	0.35		1150
Instrumentation/control	0.4		1320
Insulation	0.03		100
Electrical systems	0.10		330
Buildings	0.45		1480
Yard improvement	0.15		490
Auxiliary facilities	0.4		1320
Total plant direct cost (TPDC)			10 550
Engineering	0.25	TPDC	2640
Construction	0.35		3690
Total plant indirect cost (TPIC)			6330
Total plant cost (TPC) = TPDC + TPIC			16 880
Contractor's fee	0.05	TPC	840
Contingency	0.1		1690
Direct fixed capital cost (DFC)			19 410
Land	0.015	DFC	290
Start up and validation	0.05		970
Working capital	30 days	–	270
Total capital investment (TCI)			20 650

(i) *Direct cost*
The purchased equipment needs to be installed. The erection of the equipment involves labor costs and costs for foundation, platforms, support, construction, and other expenses that are represented in this multiplier; these can add up to an additional 100% of the purchased equipment cost.

The plant requires instrumentation and control facilities. The cost multiplier for these expenses includes, e.g., instrument and auxiliary equipment costs and the labor costs for the installation. The more complex a process, the higher this factor should be set.

The piping cost multiplier covers the construction material and labor to provide a complete piping of the process. This includes the pipes that are included in the PFD and that connect the different pieces of equipment as well as other items, such as plant piping for steam, cooling water, waste water, and others. Furthermore a plant needs an electrical system. In this multiplier, the cost for substations and transmission lines, motor switch gear and control centers, emergency power supplies, area lighting, and others are considered. Additionally, costs for insulation and painting have to be taken into account. Usually, these costs are relatively low. In low-temperature facilities, however, insulation cost can become unusually high.

The erection of all process-related buildings results in expenses for labor, materials, and other necessary supply. Additionally, the multiplier for yard improvement includes various costs, e.g. for excavation, site grading, roads, fences, railroad spur lines, fire hydrants, parking spaces. Satellite process-oriented service facilities vital to the proper operation of the process facility itself, e.g. a steam plant, are considered by the auxiliary facilities multiplier.

The sum of the purchased equipment cost and the other cost items derived from it gives the total plant direct cost. Additionally there may be indirect costs that cannot be allocated directly to a specific piece of equipment but which also contribute a substantial part of the capital investment.

(ii) *Indirect Cost*

The multiplier for engineering covers a number of planning costs, like the preparation of design books that document the process, the design of equipment, the specification sheets for equipment, instruments, auxiliaries, and the design of control logic and computer software. The construction multiplier accounts for costs associated with the organization of the total construction effort like temporary construction, construction tools and rentals, construction payroll, travel and living, taxes, and other construction overheads. The costs for engineering and construction are added to the plant direct costs to obtain the total plant cost. Besides the total plant cost, contractor's fees and an amount of money for contingencies contribute to the direct fixed capital (DFC). The inclusion of a contingency amount considers the fact that normally unexpected events during the project life cause additional costs, and it also takes into account the fact that a key element of the process might have been overlooked which can happen especially in early process development.

The cost for land cannot be depreciated. Therefore, it is normally not included in the calculation of the direct fixed capital but rather is added as a separate line item in the estimation of the capital investment. The costs for land vary dramatically depending on the location of a plant, but usually the cost lies in the range of a few percent of the DFC. The economic calculations in SuperPro Designer™ do not consider land cost.

Before a plant can come on stream, additional costs are incurred for the validation and start-up of the facility. A set of activities that include Installation Qualification, Operational Qualification, and Process Qualification are used to assure operability to meet product specifications and safety. Together with the cost for land and the working capital, these costs are part of the total capital investment. The working capital of a

plant consists of the costs for a 30-day supply of raw materials, consumables, energy, and the cost for labor and waste treatment in the same period. Sometimes, the value of a 30-day stock of finished products is also included in the working capital. If the information to calculate the costs for a 30-day supply is not available, the working capital can alternatively be charged as 10–20% of the fixed capital investment.

Depending on the product and R&D portfolio of a company, up-front R&D costs and up-front royalties can have an important economic impact. However, they are determined largely by the specific situation of a company. Therefore, these costs are not considered in the case studies presented in the second part of the book.

Multiplier Values. In the estimation method, the multiplier values for the different cost items are crucial for a realistic estimation of the capital investment. These multipliers are derived from empirical data and are different for different process types. Table 4.2 shows value ranges and average multiplier values for processes using mammalian cell cultures, microbial cell systems, and enzymatic or chemical processes. The complexity of a process using mammalian cell culture is usually higher than when using microbial cells. In the majority of cases enzymatic processes show the lowest complexity of biotechnological process, mainly because they do not have to deal with living cells. The degree of complexity is reflected in the multiplier values (see also [4.9–4.12]). These values have an inherent uncertainty and contribute substantially to the uncertainty in the estimated capital investment.

However, the multiplier values are not only influenced by the biocatalyst but also by the kind of product. For instance, the start-up and validation costs of a biotechnical plant are usually around 5% of DFC, but for a biopharmaceutical plant these costs lie at around 20% of DFC.

The multiplier for the installation cost is an average value. However, installation costs tend to be equipment specific. To get more realistic results, one can define this factor separately for every unit in the model. For example, bioreactors and centrifuges have usually a relatively high installation cost, while that for installation of chromatography columns on prefabricated skids may be low. The same is true for the maintenance cost of the different equipment types.

In SuperPro Designer™ there is an additional multiplier that accounts for the cost of unlisted equipment. (see Table 4.2). The value of this multiplier highly depends on the completeness of the model, e.g. whether all tanks are considered that are necessary for the preparation of the different solutions used in the process. The multiplier can be determined separately for the different process sections. For example, if the fermenter and all equipment related to it is modeled in great detail, but only the key downstream units are described, the multiplier for the fermentation section might be set to 0.05 that and for the downstream section to 0.4.

Price Indices. Equipment prices change over time due to inflation/deflation or market conditions. However, quite often the estimation of equipment cost has to be based on equipment prices that are already a few years old, e.g. from a previous project or from the literature. To align prices from different years and to update them to today's price level, price or cost indices are used. The present price is calculated by multiplying the original price by the ratio of today's index value divided by the index value of the time the original price was obtained (t_0):

$$\text{Present prices} = \text{Price at } t_0 \times \frac{\text{Index Value Today}}{\text{Index Value at } t_0} \tag{4.1}$$

Table 4.2 *Average values of the economic multipliers for different process types. The values for chemical and enzymatic processes are mainly taken from Peters et al. [4.3]. PC = Equipment purchase cost; TLC = Total Labor Cost; * The multipliers for these cost items often depend more on the type of product (pharmaceutical, etc.) than on the process type (mammalian cell culture, microbial system, chemical process). The multipliers for mammalian cell culture and microbial production are own estimates based on experience and educated guesses*

	Mammalian cell culture			Microbial systems			Chemical and enzymatic processes		
Cost item	Average	Min	Max	Average	Min	Max	Average	Min	Max
Total plant direct cost (TPDC)									
Installation	0.6 × PC*	0.2	1.5	0.5	0.2	1.2	0.47	0.25	0.55
Process piping	0.75× PC	0.3	0.8	0.7	0.3	0.8	0.68	0.30	0.8
Instrumentation/control	0.8 × PC	0.2	1.0	0.5	0.1	0.8	0.26	0.08	0.5
Insulation	0.05 × PC	0.01	0.08	0.05	0.01	0.08	0.085	0.08	0.09
Electrical systems	0.15 × PC	0.1	0.2	0.15	0.1	0.2	0.11	0.1	0.4
Buildings	2.5 × PC	1.0	4.0	0.5	0.8	2.0	0.18	0.1	0.7
Yard improvement	0.15 × PC	0.05	0.2	0.15	0.05	0.2	0.1	0.1	0.2
Auxiliary facilities	0.8 × PC	0.2	1.2	0.7	0.2	1.2	0.55	0.4	1.0
Overall factor	*5.8*	*2.1*	*9.0*	*3.3*	*1.8*	*6.5*	*2.4*	*1.4*	*4.2*
Total plant indirect cost (TPIC)									
Engineering	0.25 × TPDC	0.2	0.55	0.25	0.2	0.55	0.3	0.05	0.3
Construction	0.35 × TPDC	0.3	0.55	0.35	0.3	0.55	0.35	0.3	0.55
Total plant cost (TPC) = TPDC + TPIC									
Contractor's fee	0.06 × TPC	0.03	0.08	0.06	0.03	0.08	0.06	0.03	0.08
Contingency	0.1 × TPC	0.07	0.15	0.1	0.07	0.15	0.08 x DFC	0.05	0.15
Direct fixed capital cost (DFC)									
Start-up and validation	0.05 × DFC	0.03	0.08	0.05	0.03	0.08	0.05	0.03	0.08
for pharmaceuticals	*0.2 × DFC*	*0.2*	*0.3*						
Working capital	30 days			30 days			0.15 × DFC	0.1	0.2
Operating cost multipliers									
Insurance	0.01 × DFC	0.004	0.01	0.01	0.004	0.01	0.01	0.004	0.01
Local tax	0.02 × DFC	0.01	0.04	0.02	0.01	0.04	0.02	0.01	0.04
Maintenance	0.07 × DFC	0.02	0.1	0.07	0.02	0.1	0.07	0.02	0.1
Laboratory/QA/QC*	0.6 × TLC	0.5	1.0	0.6	0.5	1.0	0.15	0.1	0.2

Price indices usually used in chemical and biochemical engineering are, e.g., the *Marshall and Swift all-industry and process-industry equipment indexes* (M&S Index) and *Chemical Engineering plant cost index*. Values for both indexes are published monthly in *Chemical Engineering*. In SuperPro Designer™, an annual inflation rate can be specified to update equipment prices for years for which the Chemical Engineering cost index is not available.

Scale-up Factors. The cost of a single piece of equipment or a complete plant changes when its capacity is changed. However, a doubling of the capacity does not cause a doubling of the cost, but the cost increases slower than the capacity. This is known as the economy of scale. From empirical data an average capacity exponent of 0.6 was derived for vessels. Therefore, this cost estimation at increasing (or decreasing) capacity is also known as the *six-tenth factor rule*. When C_1 is the known cost of a plant with a certain capacity q_1, the cost C_2 of this plant at a capacity of q_2 can be calculated as:

$$C_2 = C_1 \times \left(\frac{q_2}{q_1}\right)^{0.6} \tag{4.2}$$

For example, if the investment cost of an antibody production plant is \$175 million at an annual capacity of 380 kg, then the cost of a similar plant with an annual capacity of 500 kg can be estimated from:

$$C_2 = \$175 \text{ million} \times \left(\frac{500}{380}\right)^{0.6} = \$206 \text{ million} \tag{4.3}$$

The exponent 0.6 is an average value. Specific exponents for different equipment types have also been derived that can be used for a more accurate estimate (see e.g. [4.7]).

4.2.2 Operating-Cost Estimation

The operating or manufacturing costs is the total of all costs of operating the plant and recovering the capital investment, i.e. the annual amount of money necessary to produce the product and pay back the investment cost. The operating cost can be divided into variable, fixed, and plant overhead costs. Variable costs largely depend on the amount of product that is produced. In contrast, the fixed costs are largely independent of the production volume. Variable and fixed costs are directly related to the production operations. However, there are additional expenses necessary to run a plant, e.g. storage facilities or safety measurements. These expenses are summarized under the plant overhead costs or factory expenses. In the following section we discuss the different items of the operating costs. Table 4.3 shows the operating cost estimation of the cellulase training case.

Variable Costs

(i) *Raw materials*
The list of raw materials and the amounts consumed are obtained from the material balance for the process. This requires a material balance that is as complete and accurate as possible. The raw material cost is derived by multiplying the amount by its prices. The best source for realistic raw material prices are quotations from suppliers or historical data if the material is already used within the company. If these data are not available, published prices can be taken. A good source for commodities

Table 4.3 *Annual total production cost of cellulase production (annual production: 456 metric tons)*

Cost item	Multiplier	Cost ($ thds./year)
Variable costs		
Raw materials		256
Consumables		112
Labor	25 840 h/year	
Basic labor cost (BLC)	$26/h	672
Fringe benefits	0.4 BLC	269
Supervision	0.2 BLC	134
Administration	0.5 BLC	336
Total labor cost (TLC)		1410
Operating supply	0.1 BLC	67
Laboratory/QC/QA	0.15 TLC	222
Utilities		1161
Waste treatment and disposal		64
Royalties		
Fixed costs		
Depreciation period	9.5 years	
Depreciation	0.095 DFC	1840
Insurance	0.01 DFC	194
Local tax	0.02 DFC	388
Maintenance and repair	equipment specific	1280
Plant overhead cost	0.05 DFC	970
General expenses		
Distribution and marketing		
Research and development		
Total product cost		6990

is the *Chemical Market Reporter*. Regular sales catalogues are only of limited use. Their prices are usually much higher than the prices that are paid for industrial quantities. For a very rough cost estimate, the catalogue price of a compound can be divided by 10 to estimate a large-volume contract price. However, other price sources should always be preferred. When selecting published values for materials prices it is also important to note if they are spot or contract prices and to investigate if the price is sensitive to time-dependent factors driven by seasonal supply and demand or competing uses.

(ii) *Consumables*

The category of consumables or auxiliary materials includes all material and equipment parts that have to be replaced from time to time. Typical consumables are filtration membranes, chromatography resins, and activated carbon. While the consumables in a chemical plant normally contribute little to the overall operating costs, they can be very important in bioprocesses, mainly because bioprocesses often use expensive membrane, adsorption units, and disposables in the downstream processing. Furthermore, the trend to use disposables is increasing.

The annual cost is defined by the price of the consumable (e.g. price per liter of resin), the amount per batch (e.g. 15 L of resin in a chromatography column), and the replacement frequency. The replacement frequency is usually expressed as a number of cycles (e.g. 100 chromatography cycles) or as operating hours.

The amount required per batch is derived from experiment and is defined in the process model. The replacement frequency and the price are usually obtained from the supplier of the consumable under consideration and from experiments. Alternative sources are average values from literature or the default values in SuperPro Designer™. Average costs for resins are, for example, \$7000/L for protein A resin, \$1000–2000/L for hydrophobic interaction resins, and \$700–1500/L for ion-exchange resins.

(iii) *Labor*

The labor cost is determined by the operator hours and the hourly wage. The necessary operating labor is calculated in the model for each unit. Here, usually average values are considered that are provided by the process simulator and can also be found in the literature. To run a fermenter for instance, one operator hour is necessary per operation hour of the unit. The sum for all units results in the number of people per shift and the number of shifts.

The hourly cost varies tremendously from location to location. Ideally, an internal company average value is used. Alternatively, literature values can be taken. Peters et al. [4.3] cite an average value of \$26/h for common labor and \$34/h for skilled labor (2001 prices in USA). These rates are used to calculate the basic labor cost. Fringe benefits are additional benefits paid by the company that are not part of the basic labor cost. They are estimated by multiplying the basic labor cost by a factor (e.g., 0.4). Labor expense will also change with time according to local inflationary effects.

Besides the work of the operators running the process units, some supervision by non-operational staff is necessary. This cost is also calculated from the basic labor cost and lies at 15–20% of the cost for operating labor.

(iv) *Operating supplies*

This category includes clothing, tools, and protective devices for the workers and also everyday items needed to run the plant. The cost of operating supply is estimated by multiplying the basic labor cost by a factor. The operating supply usually lies at ca. 10% of the basic labor cost. Alternatively, Peters et al. [4.3] multiply the total maintenance cost by the factor 0.15 to estimate the cost for operating supplies. This can be a notable expense when significant amounts of protective clothing are required in operation of hazardous or very clean processes.

(v) *Laboratory, quality control, and quality assurance*

Another cost item derived largely from the labor cost is the cost for laboratory (off-line analysis), quality control (QC), and assurance (QA). Chemical, biological, and physical analysis from the raw materials to the final product is an important part of a process. While this cost can be taken as 10–20% of the operating labor cost in a chemical plant [4.3], in bioprocesses, especially when producing pharmaceuticals,

Table 4.4 *Average utility and waste-treatment costs in the bioprocess industry*

Utility	Cost
Electricity	
US [4.13]	\$0.047/kWh
Europe [4.14]	\$0.077/kWh
Steam, saturated [4.3]	\$4.40/metric ton
Cooling water [4.3]	\$0.08/m^3
Wastewater	
US [4.3]	\$0.53/m^3
Germany [4.15]	\$2.14/m^3
Waste	
Hazardous [4.3]	\$145/metric ton
Nonhazardous [4.3]	\$36/metric ton

this factor goes up to 60% of the total labor cost. If the various assays and their detailed costs are already known, the laboratory/QC/QA cost can be calculated directly. One often observes that the ratio of headcount for quality operations to direct manufacturing operations range from 0.5:1 to 1.0:1.

(vi) *Utilities*
In bioprocesses, energy is typically consumed for heating, cooling, evaporation/distillation, aeration, agitation, and centrifugation. The energy is provided mainly by electricity, steam, and cooling water. The required types and amounts are determined in the process model and include the sum of the demand of all unit operations plus an additional amount for the general power load and unlisted equipment (multiplying factors). Table 4.4 shows average utility unit costs. However, they depend on the geographical location and the efficiency of the energy supply within the plant site. Power costs are very sensitive to local conditions and can vary significantly with time; this is a notable expense for the more commodity-type products as seen in Chapter 5.

(vii) *Waste treatment and disposal*
The treatment or disposal of wastewater, emissions, and solid wastes causes costs. The waste treatment is usually not part of the process model (apart from some preliminary steps like neutralization). Therefore, a treatment or disposal price is allocated to every output stream that is identified as waste. These costs depend on the phase, the composition of the waste, and the geographical location. Table 4.4 gives some average values.

(viii) *Royalty expenses*
Single unit operations or even the whole process can be covered by a patent owned by others. In order to have freedom to operate, it is necessary to pay licensing fees for the right to practice the method or to manufacture and sell a product. This cost can lie between zero and 10% or more of the unit production cost, depending on the specific patent situation for the process or product.

Fixed Cost (Facility-dependent Cost)

(i) *Depreciation*
A capital investment is necessary to build a plant and this investment has to be paid back; this is done by charging an annual amount of money (the depreciation) as a fixed operating cost. Usually, the recovery period over which a piece of equipment or a building can be depreciated is determined by the useful life of the equipment or plant and the tax laws of the country where the plant is built. The service life is the lifetime of a facility during which it deteriorates and declines in usefulness until its use is not economically feasible anymore. Theoretically, recovery period and service life should be similar; however, in reality they can be quite different.

The overall amount that can be depreciated is fixed by the direct fixed capital and maybe other depreciable spending. There are several methods to calculate this amount over the lifetime of a project. The simplest is the straight-line method which allocates the same amount of money to every year of the recovery period. Under US tax law the recovery period for chemical plants is 9.5 years [4.3]. For non-tax related assessments, other periods (e.g. service life) can be chosen. In the economic evaluation of a project this method results conveniently in a constant depreciation cost. However, it does not consider the time-value of money.

Therefore, companies usually depreciate their investment over a shorter time period with annually changing amounts. The declining-balance method, and the modified accelerated cost recovery system (MACRS) that is derived from it, depreciate most of the investment in the first part of the recovery period. For a chemical plant, which has a recovery period of 5 years when using MACRS (US tax law), over 70% of the investment is depreciated in the first 3 years. For details see Peters et al. [4.3].

(ii) *Maintenance and repair*
Every piece of equipment and the plant in general need to be maintained and repaired. Thereby, this category has a fixed part and a variable part that depends on the production rate of the plant. The cost can be derived from the direct fixed capital, or more accurately can be defined separately for every unit operation, typically as a percentage of the equipment price.

(iii) *Insurance and local taxes*
The cost for insurance and taxes is derived from the direct fixed capital (DFC). The cost to insure the plant lies around 1–1.5% of the DFC depending on the inherent risks of the process. The local property tax (not income tax!) is in the range of 2–4% of the DFC. Insurance and taxes will vary greatly by location.

(iv) *Rent and interests*
Some parts of the plant, the buildings, or the land may be rented and cause annual rental cost. However, in preliminary cost estimates, rent is not included. If the required capital needs to be borrowed completely or partly from an external source, annual interests have to be paid. There are different opinions whether the interests are part of the operating costs or should be listed under the general expenses of a company. In the case studies presented we did not consider interests in the operating costs because

they depend exclusively on the current situation of the company and are not influenced by the process.

Plant Overhead Costs. Factory expenses, or plant overhead costs, are part of the operating cost but they are caused by the operation of facilities that are not directly related to the process; for example, medical service, safety and protection, storage facilities, plant superintendence, packaging, cafeteria, and others. The annual amount can be estimated as 5% of the DFC [4.3].

General Expenses. The general expenses cover the cost to manage the company, to sell the product, and to develop new processes. They are not part of the operating cost but contribute substantially to the cost structure of a company. The sum of the operating cost and the general expenses result in the total product cost.

(i) *Administration*
The cost for administration includes the salaries for administrators, accounting, legal support and computer support, as well as office supply and equipment, administrative buildings, etc. The administration cost varies from company to company. As an estimate, 15–25% of operating labor cost can be assumed [4.3].

(ii) *Distribution and marketing*
The products of a company have to be advertised, sold, and shipped to the customer. All costs for the administration of these steps and the necessary equipment are summarized in this category. The cost varies from product to product. These costs are usually not included in a cost analysis during the process development although they would certainly be included in a business plan for the operation.

(iii) *Research and development*
To maintain or reach a competitive position, a company usually spends a high amount of money for research and development. The annual spending can be allocated to the whole product portfolio or to the up-front R&D costs of the process itself and can be considered in the cost analysis. Only a small percentage of R&D projects actually leads to an industrial production. Therefore, the revenues of one process must finance several R&D projects to create a pipeline of products and improve the process to maintain the competitiveness of a company. For estimate or preliminary studies, however, the R&D costs are usually not included.

Unit Production Cost. The unit production cost (UPC) is the total product cost allocated to the annual amount of product. For example, an annual production of 200 kg of a pharmaceutical and $40 million total product cost result in a unit production cost of $200/g final product.

The calculation and allocation methods for the operating cost presented in this chapter follow standard text book protocols, especially Peters et al. [4.3]. The calculations in SuperPro Designer™ differ in some details from this method. The cost for administration, for example, is part of the total labor cost in SuperPro Designer™ while it is usually a category of its own within the general expenses. The overall operating cost is identical in both cases. Only the allocation to the different cost categories varies slightly.

4.2.3 Profitability Assessment

Revenues. The revenue is the sum of all sales of the main and side products of a process within a certain time period, usually a year. For a single-product facility, the revenue r for year j is:

$$r_j = m_j \cdot p_j \tag{4.4}$$

where m_j is the amount of product sold in year j and p_j the (average) price realized in this year. It is not trivial to estimate the future price and sales of a product. The amount of product is limited by the capacity of the plant. Usually, 300–330 operating days are assumed. It can be reduced by technical problems within the production, as well as by a lower market demand that can be, for example, caused by high competition from other producers, substitution with another product, or a general decline in economic activity. The same market factors influence the price. Additionally, price and amount of product sold are related to each other as well.

Measurements of Profitability. There are a number of indices that are used to evaluate the profitability of a process. To obtain a comprehensive picture, several alternative methods might be considered in the assessment of a project.

The gross profit in year j (G_j) is the annual revenue r_j minus the annual total product cost c_j including depreciation:

$$G_j = r_j - c_j \tag{4.5}$$

The net profit in year j (N_j) is the gross profit minus the income tax. The income tax is determined by the tax rate Φ. For the tax rate an average value of 35% is usually assumed in the assessment. One can clearly see the impact of operating in a geographic region offering tax incentives.

$$N_j = \left(r_j - c_j\right) \cdot (1 - \Phi) = G_j \cdot (1 - \Phi) \tag{4.6}$$

The net cash flow in year j (A_j) is the sum of net profit and the depreciation d_j of that year. It is the amount of money that flows back to the corporate capital reservoir from which new investments, repayment of loans, dividends etc. are paid.

$$A_j = N_j + d_j \tag{4.7}$$

The gross margin is the ratio of gross profit to revenues, usually expressed as a percentage value. It is a measure of a company's efficiency in turning raw materials into income.

The return on investment (ROI) is the ratio of profit to investment and measures how effectively the company uses its invested capital to generate profit. It is usually calculated using the net profit and the total capital investment (TCI) and is shown as a percentage value:

$$\text{ROI} = \frac{N_j}{\text{TCI}} \cdot 100 \tag{4.8}$$

If the net profit is different for different years, an average ROI can be calculated.

The payback period (PBP) is the length of time necessary to pay out the capital investment by using the annual net cash flow that returns to the company's capital reservoir. In most

cases, the direct fixed capital (DFC) is used for this index.

$$\text{PBP} = \frac{\text{DFC}}{A_j} \tag{4.9}$$

Alternatively, the payback period can be calculated using the TCI and the net profit. Then, it is the reciprocal of the return on investment.

$$\text{PBP} = \frac{\text{TCI}}{N_j} = \frac{100}{\text{ROI}} \tag{4.10}$$

Time-value of Money. The profitability measures discussed so far do not consider the time-value of money. However, a dollar that is earned in five years time has a lower value than a dollar earned today. The net present value (NPV) considers this time-value of the earned money. To illustrate the basic principle of this index, assume that the dollar earned is put in a savings account where an annual interest rate of 7% is received. After five years the original dollar appreciates to \$1.40. To have exactly one dollar after five years it would have been enough to earn \$0.71, or from today's perspective one should earn \$1.40 in five years to have the same value as the dollar got today. The net present value now takes all expected annual earnings, i.e. the annual net cash flows, and discounts them to today's value:

$$\text{NPV} = \sum_{j=1}^{n} \frac{A_j}{(1+i)^j} \tag{4.11}$$

where j is the year of the net cash flow, i is the interest rate assumed, and n is the expected project lifetime. It shows the present value of all cash flows of the complete lifetime of the project. The estimate is very sensitive to the interest rate and the selection of the interest rate depends on the average interest rate in the capital market and on the expectation of the company and how it assesses the risk involved in the project. The internal rate of return (IRR), also known as the discounted cash rate of return, is the interest rate at which the net present value is zero (see e.g. Chapters 12 and 15).

4.3 Environmental Assessment

4.3.1 Introduction

The consideration of the environmental aspects of the process and the plant plays an ever-increasing role in the bioindustries. Many methods for environmental assessment have been published (e.g. [4.16–4.26]). With the method used in this book, we try to provide an approach that allows the scientist or engineer in the process development to make an environmental, health, and to a limited degree also a safety assessment within a reasonable time. Therefore, the method has a simple structure and is based on material data that can be accessed easily [4.27].

The purpose of this environmental assessment is to identify the environmental 'hot spots' of the process. That means it should draw attention to those materials or process steps that cause most of the potential environmental burden. Since the method can be applied from early phases of the process development onwards, these environmental burdens can be reduced from the beginning. Thus, a more sustainable process can be created, and costs

for waste treatment or possible regulatory penalties can be avoided or at least reduced [4.28]. If there is a substantial environmental problem that cannot be solved but prevents the successful realization of the process, it should be identified as soon as possible to avoid the loss of R&D spending.

By using this simplified assessment method, smaller differences between materials or also between different processes might not be identified. To overcome this limitation, much more complex methods like the life cycle assessment (LCA) must be applied and may help to a certain extent. However, such methods are time-consuming and require a much deeper knowledge of the considered process. They are important tools for the optimization of large production processes that already exist or are in the final stages of scale-up, and they require additional specialists in the development team. However, as a hands-on approach for the bioengineer in the process development it is too complex and time-consuming on the one hand, and not necessary on the other hand, because here the identification of the 'hot spots' is in the focus that can be reached by a simpler method.

Furthermore, the method concentrates on the process itself. Whether it is reasonable, or sustainable, to produce a specific product is not part of this assessment method and should be discussed separately.

4.3.2 Structure of the Method

The general structure of the environmental assessment method is shown in Figure 4.3. The method has two starting points. The first is the process and its characteristics that are represented by the SuperPro Designer™ model. A result of the simulation is the material balance of the process. From the material balance, the so called Mass Index (*MI*) defined by Heinzle et al. [4.29] is calculated for all input and output components (Table 4.5). For input materials, the Mass Index states how much of a component is consumed to produce a unit amount (e.g. 1 kg) of the final product. For output components, the *MI* defines how much of a component is formed per unit final product, e.g. how many kilograms biomass have to

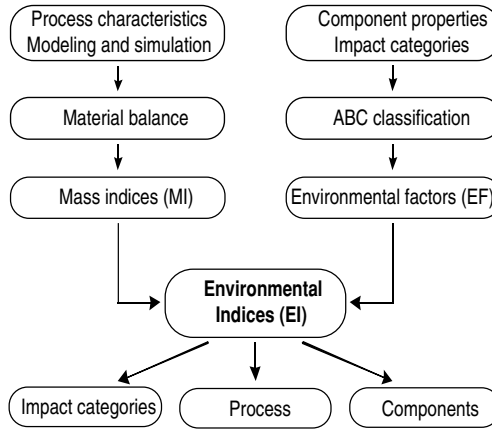


Figure 4.3 *Assessment structure of the method. Reproduced by permission of John Wiley & Sons, Ltd*

Table 4.5 *Calculation of weighting factors and indices. In = Input, Out = Output. Reproduced by permission of John Wiley & Sons, Ltd*

Weighting factors/indices	Calculation
Mass Index component i, MI_i (kg/kg P) m_i = amount of component i (kg); m_p = amount of final product (kg P)	$MI = \frac{m_i}{m_p}$
Mass Index process, MI_{process} (kg/kg P)	$MI_{\text{process, In}} = \sum_{1}^{i} \frac{m_i}{m_p}$ $MI_{\text{process, Out}} = 1 + \sum_{1}^{i} \frac{m_i}{m_p}$
Environmental Factor component i, EF_i (index points/kg) via arithmetic average; as $EF_{\text{Mv,i,In}}/EF_{\text{Mv,i,Out}}$ $IG_{j,i}$ = Value of component i in Impact Group j; j = Number of Impact Groups	$EF_{\text{Mv,i}} = \frac{IG_{1,i}+IG_{2,i}+IG_{3,i}+IG_{4,i}}{j}$
Environmental Factor component i, EF_i (index points/kg) via multiplication; as $EF_{\text{Mult,i,In}}$ and $EF_{\text{Mult,i,Out}}$	$EF_{\text{Mult,i}} = \prod_{1}^{j} IG_{j,i}$
Environmental Index component i, EI_i (index points/kg P) (as $EI_{\text{i,In}}$ or $EI_{\text{i,Out}}$)	$EI_i = \frac{EF_i \cdot m_i}{m_p} = EF_i \cdot MI_i$
Environmental Index process, EI_{process} (index points/kg P) (as EI_{In} or EI_{Out})	$EI_{\text{process}} = \sum_{1}^{i} EI_i$
General Effect Index process, GEI (nondimensional)	$GEI = \frac{EI_{\text{process}}}{MI_{\text{process}}}$
Impact Category Index impact category j, ICI_j (nondimensional) $IC_{j,i}$ = Value component i in impact category j	$ICI_j = \frac{\sum_{1}^{i} IC_{j,i} \cdot MI_i}{MI_{\text{process}}}$
Impact Group Index impact group j, IGI_j (nondimensional) $IG_{j,i}$ = Value component i in impact group j	$IGI_j = \frac{\sum_{1}^{i} IG_{j,i} . MI_i}{MI_{\text{process}}}$

be produced to get 1 kg of purified enzyme. The sum of all input *MIs* (or output *MIs*) gives the Mass Index of the process, which is a metric for the material intensity of the process.

The mass-based indices can be used for a first rough assessment. However, it is obvious that not all components have the same environmental relevance. Therefore, the possible environmental impact of the components has to be considered in the evaluation. Hence, the component properties are the second starting point of the method (Figure 4.3). There is a wide range of negative effects a compound can have on the human health and the environment [4.30]. We tried to represent these effects in 15 impact categories that we discuss in the following chapter. In each category, a component is allocated to the classes A, B, or C

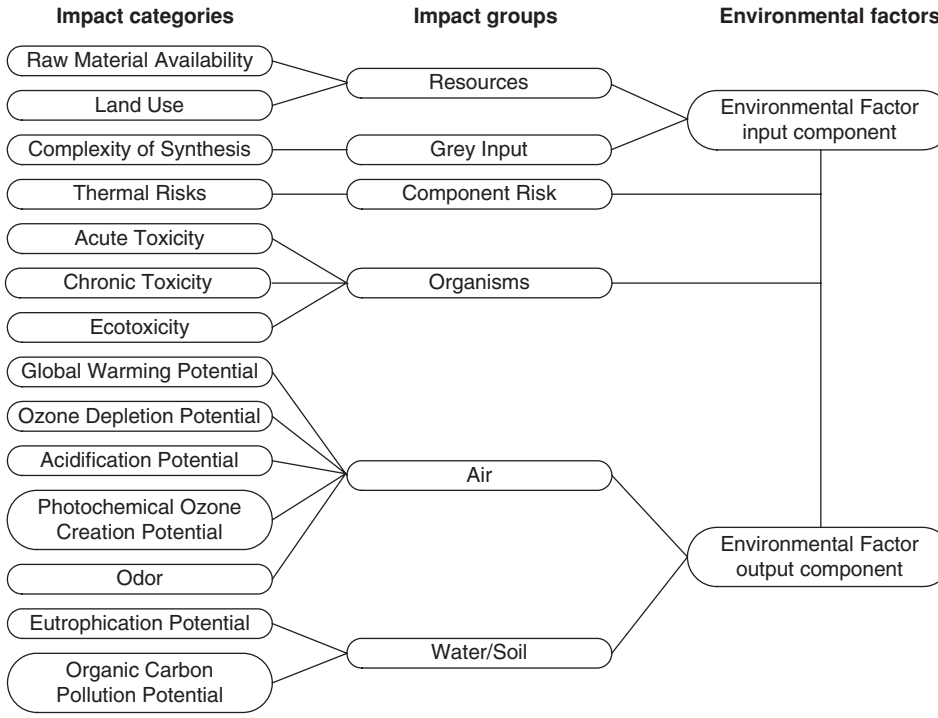


Figure 4.4 *Impact categories, allocation of the categories to the impact groups, and derivation of the Environmental Factors for input and output components. Reproduced by permission of John Wiley & Sons, Ltd*

that represent its relevance for this category (high, medium, or low relevance). For example, a highly toxic material will be allocated to class A in the impact category 'Acute Toxicity' while a non-toxic component will be put in class C. Class B then contains compounds with a medium toxicity. For every A classification it should be checked where this component occurs in the process and whether its negative property will be relevant under the process conditions.

These impact categories (IC) are allocated to six impact groups, each representing an important field concerning environmental, health, or safety aspects (Figure 4.4). According to their allocated impact categories they are also allocated to one of the three impact classes. In the next step, numerical values are defined for the classes A, B, and C and for every component a weighting factor (= Environmental Factor) is derived from its classifications in the impact groups. The exact method of calculation is explained in Section 4.3.4. For the present, it is only important to know that we derive an environmental weighting factor from the material properties. This factor represents the potential environmental relevance of a compound.

In the next step we link the amount of the components in the mass balance with their potential environmental impact by multiplying their Mass Indices with their Environmental Factors. The resulting Environmental Index (*EI*) helps to identify those components that are environmentally most relevant in the process. In addition, we derive a few other indices

that can be used to compare different processes and to identify the environmental impact categories that contribute most to the calculated indices.

4.3.3 Impact Categories and Groups

In the following we briefly discuss the scientific principles that lie behind each impact category. The definitions of the class limits for the classification of all ICs are given in Table 4.6. The definition of classes from a continuous distribution is always a point for discussion. We tried to look at the whole range of possible impacts in a particular category and set the class limits reasonably.

Table 4.6 *Parameters and class limits of the impact categories. In each category, literature cited indicates possible sources for relevant data. I = Category used to evaluate input components, O = Category used to evaluate output components. Reproduced by permission of John Wiley & Sons, Ltd*

Impact category	I/O	Class A	Class B	Class C
Raw Material Availability	I	only fossil, predicted exhaustion within 30 years	only fossil, predicted exhaustion in 30–100 years	exclusively renewable, or guaranteed long term supply (>100 years)
Land Use	I	$\geq$100m^2/kg	$\geq$10m^2/kg and <100 m^2/kg	<10 m^2/kg
Critical Material Used [4.31, 4.32]	I	critical materials like heavy metals, AOX, PCB used or produced in stoichiometric amounts	critical materials involved in sub-stoichiometric amounts	no critical compounds involved
Complexity of the Synthesis [4.31, 4.32]	I	>10 stages	3–10 stages	<3 stages
Thermal Risk [4.33, 4.34]	I/O	R 1–4, 9, 12, 15–17, 44; EU: F^+, E; NFPA F+R: 3, 4	R 5–8, 10, 11, 14, 18, 19, 30; EU: F, O; NFPA F+R: 2	NFPA F+R: 0, 1
Acute Toxicity [4.33]	I/O	EU: T^+; R 26–28, 32; CH-poison class: 1, 2; NFPA H: 4; WGK 3; ERPG: <100 mg/m^3; IDLH: <100 mg/m^3	EU: T, X_n, X_i, C; R 20–25, 29, 31, 34–39, 41–43, 65, 66, 67; CH-poison class: 3, 4; NFPA H: 2, 3; WGK 2; ERPG: 100–1000 mg/m^3; IDLH: 100–1000 mg/m^3	CH-poison class: 5; NFPA H: 0, 1; WGK 1; ERPG: >1000 mg/m^3; IDLH: >1000 mg/m^3

Table 4.6 *cont.*

Chronic Toxicity	I/O	MAK: <1 mg/m^3; IARC: 1, 2A; R 45–49, 60, 61, 64	MAK: 1–10 mg/m^3; IARC: 2B, 3; R 33, 40, 62, 63; EU: T, T$^+$, X$_n$; CH-poison class: 1, 2	MAK: >10 mg/m^3; IARC: 4; CH-poison class: 3, 4, 5
Ecotoxicity	I/O	EU: N; R 50; WGK 3	R 51–58; WGK 2	WGK1 or no water hazard
Global Warming Potential[4.35]	O	GWP > 20	GWP < 20	no global warming potential
Ozone Depletion Potential [4.36]	O	ODP > 0.5	ODP < 0.5	no ozone depletion potential
Acidification Potential [4.23]	O	AP > 0.5	AP < 0.5	no acidification potential
Photochemical Ozone Creation Potential [4.37, 4.38]	O	POCP > 30 or NO_x	30 > POCP > 2	POCP < 2 or no effect known
Odor [4.39]	O		odor threshold < 300 mg/m^3	odor threshold > 300 mg/m^3 or no odor
Eutrophication Potential	O	N-content > 0.2 or P-content > 0.05	N-content < 0.2 and P-content < 0.05	compound without N and P
Organic Carbon Pollution Potential	O		ThOD > 0.2 g O_2/g substrate	ThOD < 0.2 g O_2/g substrate or no organic compound

Some of the six Impact Groups (IG) are relevant for input and output components, some for both of them (Figure 4.4). For an input material, we ask:

- What basic resource is the compound based on and what is its availability (Impact Group Resources)?
- What environmental burden has the compound already caused on its way from the basic resource to the process (IG Grey Inputs)?
- Has the compound the potential to cause safety problems within the process, during transport, storage, handling, or reaction (IG Component Risk)?
- Has the compound the potential to harm human or other living organisms when they are exposed (IG Organisms)?

For an output component, the thermal risk (IG Component Risk) and the toxicity (IG Organisms) are relevant in the same way, while availability and grey inputs are not applicable.

However, it is important to consider their possible impact when they are emitted to the environment, either as emissions (IG Air) or as liquids or solids (IG Water/Soil).

In general, the 15 impact categories connect a large range of data that varies strongly regarding quality, availability, and usability. Here, ABC analysis is a common method in economics and other disciplines where numbers with high uncertainty have to be dealt with. Possible synergistic and additive interactions of the components in the environment are not considered due to the complexity and variability of such interactions and the limited knowledge about them.

Availability and Grey Inputs. The IC Raw Material Availability considers whether the input component is produced from a renewable or a nonrenewable resource. If it is a nonrenewable source, the period until the predicted exhaustion is taken into account. For this estimation, only the production processes predominantly used today are regarded. The cultivable land of the world is limited. By using renewable (agricultural) raw materials for biotechnical production, the area for food production is reduced. The IC Land Use considers how much land area (m^2) of agricultural soil is needed to produce one kilogram of a raw material.

Grey inputs are resource depletions and environmental burdens caused during the preparation of the input component, before it enters the process itself. A complete life cycle analysis would be needed to evaluate their impact in detail. However, such data are only available for very few compounds. Therefore, this impact has to be estimated using generally available information. Here, it is assumed that a component needing several synthesis steps causes more grey inputs than a component needing only one or two steps (IC Complexity of Synthesis). Although life cycle data are often not available, data about critical materials involved like heavy metals or adsorbable organic halides (AOX) can be found in the literature. Such materials are a crucial part of grey inputs and are therefore included in the IC Critical Materials Used.

In a typical chemical or biotechnical process, the energy consumption contributes significantly to the environmental impact of a process [4.40]. However, the energy consumption cannot be assessed with the ABC classification. Therefore, it is not included in the calculation of the Environmental Indices but it is discussed separately in the assessment process. This approach is similar to that of Glauser and Mueller [4.41].

Component Risk. An extensive risk assessment is an important part of process development. The IC Thermal Risks used here will explicitly not replace such an assessment. However, this IC provides an indication of potential risks on which a later risk assessment could concentrate. A similar approach comprising risk aspects in the environmental assessment is given by both Koller [4.17] and Elliott et al. [4.22].

The classification is based on international classifications like R-codes, the EU hazard symbols, and the flammability hazard classes and reactivity hazard classes of the US National Fire Protection Agency (NFPA) that consider flammability, thermal stability, reactivity, and incompatibility with air, water, and other compounds and are available for almost every compound. This IC considers input and output components. However, materials that are formed during the process and further react to form another compound are not included. In other words, this analysis is completely based on the input–output material balance.

In addition to the thermal risk there can be a biological risk when genetically modified organisms (microorganisms, plants, animals) are used. Biotechnological facilities are usually closed systems and normally only organisms with the risk classification S1 are used that are generally regarded as safe (GRAS). Here, the risk is limited. Therefore, the biological

risk is not considered in the calculation of the Environmental Indices. In public discussions the focus is on the use of genetically modified (micro) organisms (GMO) in open systems such as agriculture (see e.g. [4.42]). These discussions and the resulting regulatory laws influence especially the use of transgenic plants and animals for biotechnological production (see Section 4.5). In rare cases when harmful naturally occurring organisms are applied, biological risk would have to be taken into account.

Organisms (Toxicity). All impact categories somehow have an influence on organisms (humans, animals, plants). However, the categories summarized in the IG Organisms consider only direct toxic effects. The IG Organisms includes the impact on human health (Acute Toxicity, Chronic Toxicity), as well as on the plants and animals (Ecotoxicity). The toxicity is a measurement of the toxic potential of a compound. The toxic effect depends on the material properties, the concentration (dose), duration and frequency of the exposition, and the bioavailability and the type of exposition [4.43].

The toxicity is termed 'acute' when the toxic effect occurs after a single application or a short exposition within a short time frame that lasts, depending on the organism, between a few hours and a few days. Chronic toxicity needs a long term exposition or a large number of single applications over a long period of time. The reason for the final toxic effect is the accumulation of the compound in the organism or the combined impact of many small amounts of damage. Chronic toxicity can affect the organism in different respects,: physiology (growth, development), biochemistry (plasma, enzyme activity), cell structure (histology), and reproduction. This is then expressed as mutagenicity, carcinogenicity, immunotoxicity, or tissue damage.

The chronic toxicity of a compound cannot be derived from its acute toxicity. Globally, chronic impacts have played a bigger role than single, big events like a chemical incident and their acute toxic effects. The chronic toxicity of compounds has often not been recognized before they showed their toxic potential in the environment, e.g. DDT or PCBs.

There is no general consensus as to how to evaluate toxicity [4.44]. Therefore, different parameters have to be considered for the classification in these categories (Table 4.6). All of them are nationally or internationally recognized classifications and are usually easily accessible. In the IC Ecotoxicity only a few parameters are considered. Many of the parameters used in the (human) Acute and Chronic Toxicity classifications could also be considered in the IC Ecotoxicity. To avoid double counting, they are not listed again for the IC Ecotoxicity.

Environmental Compartment Air. The impairment of the environmental compartment air is covered by five ICs. The categories Global Warming Potential, Ozone Depletion Potential, and Photochemical Ozone Creation Potential use internationally well accepted data (Table 4.6). For these categories reference compounds are defined to which the impact of all other compounds is related.

The Global Warming Potential (GWP) considers the impact of a compound on climate change (Greenhouse effect). The combustion of fossil fuels, intensive agriculture, large waste landfills, and the ongoing destruction of the tropical forests are leading to an increased emission of greenhouse gases to the atmosphere where they increase the absorption of heat radiation. The International Panel on Climate Change (IPCC) has defined the Global Warming Potential (reference substance CO_2; $GWP_{CO2} = 1$) and regularly publishes updated lists (e.g. [4.35]). The GWP is used for the classification of this category.

The ozone layer of the atmosphere lies 30 to 50 km above ground level and protects the surface from dangerous UV-B radiation. However, different human-based gases, mainly chlorofluorocarbons (CFC) and halogenated hydrocarbons, lead to an increased degradation of ozone in the ozone layer [4.36]. In the Montreal Protocol, the Ozone Depletion Potential was defined [reference substance: trichlorofluoromethane (R-11), $ODP_{\text{R-11}} = 1$]. The UNEP – Ozone Secretariat regularly publishes a material list with *ODP* values (e.g. [4.45]) that are used for the classification of the IC Ozone Depletion Potential.

In the higher layers of the atmosphere the ozone fulfills an important function. However, at the earths surface it is an aggressive gas. In the presence of nitrogen oxides (NO_x) and sunlight, volatile organic compounds (VOC) form photochemical ozone that causes the so-called summer smog. The Photochemical Ozone Creation Potential (*POCP*) describes the photochemical potential of VOC to create ozone (reference substance: ethylene, $POCP_{\text{Ethylen}} = 100$). For the classification in the IC *POCP*, we use the *POCP* list published by Derwent et al. [4.37, 4.38].

Acidification describes the reduction of pH in the environment, mainly in the soil and in rivers and lakes. It is mainly caused by the combustion products sulfur dioxide and NO_x, and by ammonia from agriculture. They are emitted to the atmosphere where they react to form sulfuric or sulfurous acid, and nitric or nitrous acid, respectively, and are deposited in soils and water bodies. There, they cause leaching of nutrients and a combined toxic effect of protons and dissolved metal ions. The term Acidification Potential, while not defined in international treaties, is also widely used to evaluate acid-forming emissions. The class limits are defined in a way that the three most important acid-forming substances (sulfur dioxide, NO_x, ammonia) are allocated to class A.

In the IC Odor, odor thresholds are used to evaluate bad smells. Though malodors are locally unpleasant, they have neither long-term nor long-distance negative impacts on health and environment. Therefore class A (high potential environmental burden) is not defined for this IC.

Environmental Compartment Water/Soil. The impact on the environmental compartments water and soil (IG Water/Soil) is considered by two impact categories. The content of nitrogen and phosphorus is used to evaluate the Eutrophication Potential of a compound. Since phosphorus limits the biomass growth in inland waters and because the phosphorus content of phytoplankton is much lower than the nitrogen content, the class limits for phosphorus are set lower than for nitrogen (Table 4.6). The emission of organic compounds into lakes and rivers and their following decomposition leads to a strong oxygen consumption. The theoretical oxygen demand (ThOD) specifies how much oxygen is theoretically needed per amount of substance. If the chemical oxygen demand (COD) is not known, the theoretical oxygen demand (ThOD) calculated from the molecular composition can be used instead to characterize a compound with respect to its IC Organic Carbon Pollution Potential. During wastewater treatment, the COD is normally reduced. Therefore, a class A indicating high potential environmental burden is not defined in this category.

4.3.4 Calculation of Environmental Factors

All the information collected in the impact categories has to be summarized to reach a measurement of the overall environmental relevance of a component. These weighting factors, the Environmental Factors (*EF*), are calculated separately for input and output components.

The ABC classifications in the impact categories are the basis for the calculation of the *EF*. In the Impact Groups, a component is also allocated to one of the three classes (A, B, C). The highest classification in the referred ICs defines the class of the IG, for example if the five impact categories referred to the IG Air are three times class C, one class B and one class A, then the IG Air will be assigned to class A.

In the next step, the *EF*s are calculated from the impact groups. As discussed before, the *EF* of an input component considers the impact groups Resources, Grey Inputs, Organisms, and Component Risk, while the *EF* of an output component comprises the groups Air, Water/Soil, Organisms, and Component Risk (Figure 4.4). Consequently, a compound that is an input and an output component can have two totally different *EF*s.

To merge the four impact groups into one environmental factor, numerical values have to be defined for the three classes A, B, and C. The calculation of the *EF* is determined by two factors: The numerical values of the classes and the way they are aggregated to one value. In the method presented two options are offered. The EF_{Mult} uses the values $A = 4$, $B = 1.3$, and $C = 1$ and these values are aggregated by multiplication. Thus, possible values of EF_{Mult} are between 1 and 256. The alternative EF_{MV} uses the values $A = 1$, $B = 0.3$, and $C = 0$. There, the EF_{MV} is calculated by averaging (Table 4.5). Values lie between 0 and 1. The calculation of the Environmental Factors and the different indices is summarized in Table 4.5.

The EF_{Mult} highly emphasizes compounds with one or more groups allocated to class A. Since $C = 1$, every component has an EF_{Mult} bigger or equal to 1. This means that components allocated to class C in all four impact groups are nevertheless considered in the assessment. The EF_{MV} also emphasizes class A components, but it shows a more even value distribution of possible weighting factors. Thus, class B components are weighted relatively more strongly. Since $C = 0$, harmless components (class C in all four groups) are not considered in the assessment. Especially in biotechnological processes, there are usually several harmless compounds. Therefore, the evaluation results using EF_{Mult} and EF_{MV} can differ to a certain extent. In both cases, there is no additional weighting factor comparing the relevance of the four groups with each other. That means that each impact group is assumed to have the same importance.

Both *EF*s are weighting factors of the environmental relevance of a component. They represent two of several possible ways to summarize the different environmental impacts of a component. This necessary aggregation is not possible on an exclusively scientific, objective basis. Every aggregation method includes subjective evaluations of the relative importance of the different impacts. Therefore, it is important to show transparently the method of aggregation employed. Future users may use different weighting factors more appropriate for their particular case without significant modification of the method. Although the methods of weighting are somewhat arbitrary, such factors have to be derived in order to identify the most relevant compounds and to allow an eventual decision and enable a significant assessment of a process to be made. Fortunately, in most cases the details of weighting are not really of that high importance because the method concentrates on the identification of the most crucial environmental hot spots, and here results obtained lead to similar conclusions. In the case studies presented in the second part of the book, the compounds identified as the environmentally most relevant are usually the same, even though the relative importance of the compounds compared with each other varies. However, the results obtained after the application of *EF*s differ significantly from an evaluation based only on material balances. Therefore, a consideration of the environmental relevance of the compounds involved is crucial.

4.3.5 Calculation of Indices

To describe the environmental performance of a process, a number of indices are derived. The Mass Index that we have already discussed is derived from the mass balance and provides a rough measure of the impact of a component. It is the basis for the calculation of all other Indices (Table 4.5).

The Environmental Index (*EI*) of a component is derived from weighting its Mass Index with its Environmental Factor. Thus, the *EI* connects the mass consumed or formed to the environmental relevance of a compound. The *EI* is calculated for input und output components. The indices make it possible to identify the environmentally most crucial components of the mass balance. The sum of all *EI*s (input or output) is the *EI* of the process, and indicates the environmental relevance of the whole process. It can be used to compare alternative processes or process steps.

The General Effect Index (*GEI*) of the process specifies the ratio of *EI* to *MI*. It represents a weighted average of the Environmental Factors of all components involved. Therefore, the index does not show individual critical compounds. If the EF_{Mult} is used to calculate the *EI* (EI_{Mult}), the value of the *GEI* will vary between 1 and 256; if the EF_{MV} is used, GEI will be between 0 and 1. The *GEI* can also be used to compare alternative processes. However, the material intensity is not indicated by the *GEI*.

The indices shown so far indicate the general environmental performance of a component or the whole process. They do not show which impact categories or groups contribute to this environmental performance. The Impact Category Indices (*ICI*) and the Impact Group Indices (*IGI*) show the contribution of an impact category or an impact group to the overall environmental burden of the process. They provide additional information for the comparison of process alternatives.

Bioprocesses usually consume high amounts of water. When the EF_{Mult} ($C = 1$) is used for the calculation of the Environmental Indices, the high water amount dominates all other materials even if the latter have a high *EF*. Therefore, two separate presentations of the results, with and without water, are recommended.

4.3.6 Example Cleavage of Penicillin G

Penicillin G produced by fermentation is converted into 6-aminopenicillanic acid (6-APA) by splitting off the side chain of penicillin. 6-APA acid is the starting material for the production of semi-synthetic penicillins like ampicillin or amoxycillin. Two process alternatives for the splitting of penicillin are considered here: An older chemical process needs three intermediate stages; a more recent biocatalytic process using immobilized penicillin amidase needs only one synthesis step (Figure 4.5). The material balance was taken from Wiesner et al. [4.46].

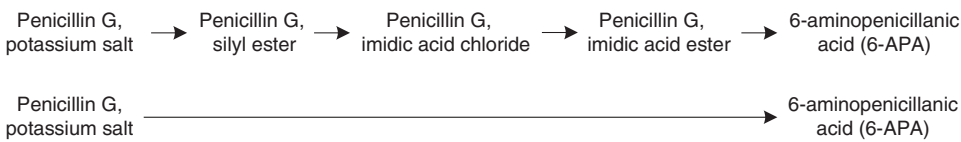


Figure 4.5 *Reaction schemes of chemical and enzymatic cleavage of penicillin G to form 6-aminopenicillanic acid (6-APA)*

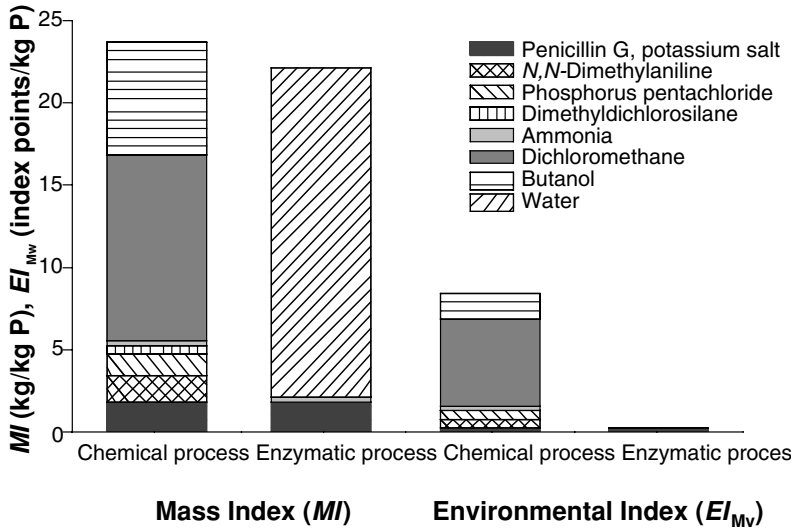


Figure 4.6 *Comparison of Mass Indices (*MI*) and Environmental Indices (*EI_{Mv}*) of the input. Reproduced by permission of John Wiley & Sons, Ltd*

The Mass Indices and the Environmental Indices (EI_{MV}) of the input materials are shown in Figure 4.6 and in Table 4.7. The Mass Indices of the alternative procedures are similar. However, if the environmental relevance of the input components is considered, big differences become obvious. The chemical process involves three substances with at least one class A rating. Phosphorus pentachloride has a high acute and chronic toxicity. Ammonia is also allocated to class A in the impact category Acute Toxicity. Dichloromethane used in the chemical process receives an A rating because during its production from methane by thermal chlorination, trichloroethylene and hexachloroethane are formed. These are highly toxic by-products (IC Critical Materials Used).

Although the Mass Indices of the processes are similar, the Environmental Index (EI_{MV}) of the chemical process is much bigger (Figure 4.6). The EI_{Mult} and the General Effect Index show very similar results. Thus, the environmental performance of the biocatalytic

Table 4.7 *Environmental assessment results for the chemical process and the enzymatic process of penicillin G cleavage*

Assessment metric	Chemical process	Enzymatic process	
		With water	Without water
Mass Index *MI* (kg/kg P)	23.7	22.1	2.1
Number of A-components	2	1	
Environmental Index EI_{Mv} (index points/kg P)	8.5	0.34	
Environmental Index EI_{Mult} (index points/kg P)	135	24	4.0
General Effect Index GEI_{Mv} (0–1)	0.36	0.015	0.16
General Effect Index GEI_{Mult} (1–256)	5.7	1.1	1.9

process is clearly superior to the chemical alternative. The most crucial substances in the chemical process are dichloromethane and butanol (both used as solvents), and to a lesser extent penicillin G (raw material), phosphorus, pentachloride, and dimethylaniline. In the biocatalytic process, penicillin G and a small amount of ammonia are the environmentally most relevant components.

The Impact Group Indices of the chemical process show that the IG Grey Inputs has the strongest impact caused by the critical materials used in the production of dichloromethane. Furthermore the IGs Resources and Organisms play a major role, while the IG Component Risk is less important. The bigger influence of the IG Organisms is determined by the toxic potential of butaniol, dichloromethane, and phosphorus pentachloride. Since most input materials are based on oil or natural gas, the IG Resources is also affected.

The composition of output components is not specified by Wiesner et al. [4.46]. However, concerning the input materials used and the reactions performed, the environmental performance of the biocatalytic processes can be assumed to be also superior at the output side. This case shows that involving the environmental relevance of components can help in identifying differences that cannot be seen by considering only the mass balance and the Mass Indices.

4.4 Assessing Social Aspects

Justus von Geibler*, Holger Wallbaum, Christa Liedtke
*Corresponding author: justus.geibler@wupperinst.org, ++49/202/2492-168

4.4.1 Introduction

As emphasized already in the Introduction, the assessment of the early product-design phase is of major importance since these early stages influence the cost spent for a product to a large extent (i.e. production costs, maintenance costs, and end-of-life costs). Similarly, the environmental and social effects are also determined in early stages of process development as illustrated in Figure 1.1 in Chapter 1.

Indicators also play a key role in the social assessment of effects of evolving technologies. They are accepted as management tools and used throughout business. Although the assessment of social sustainability has already entered scientific debate, it lacks a broad consensus on adequate indicators or a consistent method of their identification. Whereas in the ecological or economic area more or less widely accepted indicators have been developed, a consensus on indicators for the evaluation of the social side to sustainability is still to be developed, in particular for specific industrial sectors or specific technologies [4.47, 4.48].

Addressing these challenges, the Research Group 'Sustainable Production and Consumption' at the Wuppertal Institute has elaborated a social assessment model of processes/production in the biotechnology sector. Companies can use this model for assessing and steering potential sustainability risks and opportunities of biotechnological production. Furthermore, the data gathered and compiled with the indicator set enhance the ability to respond to growing information demands regarding sustainability performance of companies of all sizes [see e.g. the Global Reporting Initiative (GRI)]. The discussion here presents the criteria that are relevant for the social assessment of biotechnological production processes and how they have been identified.

4.4.2 Indicators for Social Assessment

In order to identify relevant social aspects and to compile a set of indicators, four basic perspectives on technology assessment have been taken into consideration, drawing on the methodology approach of concept specification developed in social science [4.49]. On a macroscopic scale the political relevance of the issue has been dealt with by regard of single political initiatives such as the sustainability strategies of the German government or the European Union [4.50, 4.51]. On a more systemic level the relevance of stakeholders in the biotech sector has been addressed through an international survey of both regional and global stakeholders. The entrepreneurial and product relevance has been considered through a survey of biotech companies and the consideration of the information demands from rating agencies of the financial market. In addition, international sustainability reporting demands from the GRI have been included.

In this context a stakeholder survey was used to address a wide array of different groups, such as suppliers, customers, unions, industry and employers' associations, national and international competitors, financial institutions and investors, regulatory and legislative bodies, international organizations, academia, and research, as well as NGOs. By doing so, the survey identified relevant social aspects in different phases of the process/product life cycle, covered the possible contribution of biotechnological products to the satisfaction of human needs, and addressed challenges and chances in the social field enhanced by the biotech industry. As the influence of stakeholders on a corporation's process of decision-making is growing, the integration of stakeholders' views on social aspects of bioproduction is of increasing importance.

Taking into account the results gained from the multi-perspective approach to technology assessment, including the implications of an international stakeholder survey, it has been possible to identify eight aspects that are significant for the social assessment of biotechnological operations: health and safety; quality of working conditions; impact on employment policy; education and advanced training; knowledge management; innovative potential, customer acceptance and societal product benefit; and societal dialogue. These aspects and their relevance are briefly explained below:

Health and Safety. The term 'health and safety' refers to all measures that improve the employees' safety and well-being at work – such as the prevention of working accidents, occupational diseases, or work-caused dangers to health. As health and safety is more than just an instrument to protect the employees' health and well-being, a consistent and conscious health and safety management grants companies a competitive advantage. In the context of biotechnological production, improved health and safety can lead to a higher motivation of the employees, reduced risk of damage to the public image of the enterprise, as well as cost reduction. Health and safety management is well advised to surpass compulsory legal measures [4.52, 4.53]. There are also benefits to using global standards for health and safety within individual firms and across industries.

Quality of Working Conditions. In the light of a current structural change in economy and society, the demographic development as well as socio-political demands on the working environment, the quality of working conditions is a competitive factor of growing importance. In detail this implies aspects such as work-related scopes of options, labor time

arrangements, operational regulations of remuneration, social benefits, or the elevation of the employee's psychological level. After all, positive working conditions result in better working satisfaction, motivation, and efficiency, and thereby evoke economic impacts for the enterprise [4.54].

Impact on Employment. As a consequence of the high level of innovation and technology, the biotechnology sector offers new opportunities for employment. This leads to an improved societal and political acceptance and positively influences the granting of public subsidies. Besides the sheer number of jobs created, it is relevant as to where and how long-ranging places of employment are secured and created [4.55].

Education and Training. In the biotechnology sector, the qualification of the employees is an important factor, since academic research and development form a key activity of the companies. The qualification includes, e.g., the consistency of advanced training, a frequent check-up of basic training needs, opportunities of apprenticeships, advanced training by the executive management level, or consideration of the employees' demands [4.56, 4.57].

Knowledge Management. Knowledge is an important factor of biotechnological production. Strategic knowledge management aims at the deliberate and systematic handling of knowledge, covering the creation, collection, distribution, advancement, and application of knowledge. Knowledge Management addresses the quality of experience and information exchange, analysis of this exchange's efficiency, the integration of electronic information systems, or the employees' participation in internal processes of company decision-making [4.58–4.61].

Innovative Potential. Biotechnology offers a wide array of new development and application opportunities. For biotech companies the innovative potential is especially relevant because it determines commercial exploitation and future income. This innovative potential is especially shaped by questions of national and international patenting (Figure 4.7). Innovative companies are able to adjust faster to societal change and thus securing places of employment in the long run. This can contribute to a progression of prosperity [4.62].

Customer Acceptance and Societal Product Benefit. The acceptance of products by customers is significantly influenced through product characteristics and information as well as production conditions. Regarding biotechnological production the utilization of methods of genetic engineering and the compliance with social standards play a key role. From a sustainability point of view products should also have a societal use and help securing and increasing everyone's quality of life. A higher value for society can be ascribed, e.g., to products to combat malaria or HIV/AIDS, rather than the development of a new artificial sweetener that does not bear an extensive societal use or financial advantage [4.63].

Societal Dialogue. The most recent development in the area of biosciences, particularly regarding work with genetically modified organisms (GMOs), has attracted public attention and initiated an intense debate. Sustainability demands a sincere dialogue, which includes all societal segments. This societal dialogue can also optimize a company's competitive ability, e.g., when it is applied in the field of marketing strategies. Correspondingly a sincere societal dialogue surpasses the 'mere' exchange of information with the public.

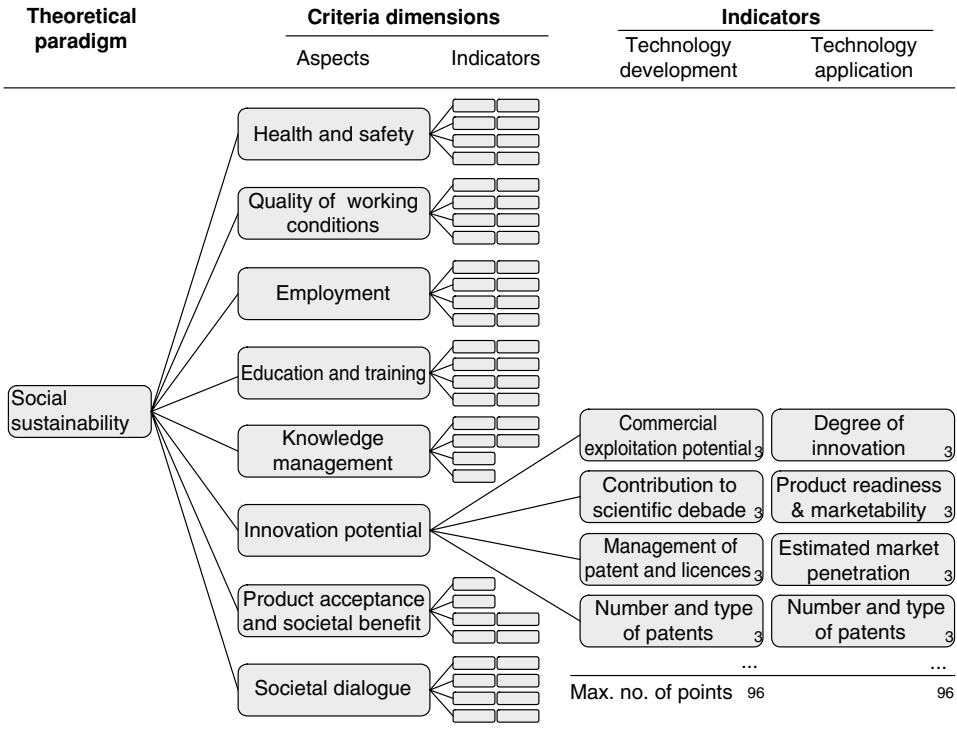


Figure 4.7 *Indicator set for the evaluation of social sustainability of biotechnological processes*

In fact it aims at enabling communicative cooperation with a large array of public actors, stakeholders, and political institutions [4.64, 4.65].

For each of the aspects eight indicators have been identified, covering two layers of evaluation: (i) The technology development and (ii) the technology application. This distinction has been made since the social context of the biotechnological processes (and other evolving technologies) varies between developing and applicative stages. For example, regarding the acceptance of a genetically engineered product, there is a difference in whether a biotechnological process is implemented in a secluded laboratory under controllable conditions or whether it is carried out on an agricultural area in a compound and more unpredictable ecosystem. Table 4.8 gives an overview of typical indicators for each aspect regarding the technology development and technology assessment. Figure 4.7 illustrates how these indicators can be merged for the assessment of social sustainability using a simple weighting method: A maximal three points for each indicator lead to a maximum of 96 points for each level. The indicators have been developed for the German context; in other regions other specific indicators might be more relevant.

The presented indicator set is being developed to support the assessment of social aspects in early stages of biotechnological process development. However, the single application of an assessment tool alone will not further sustainable development in the biotech sector. Along with internal evaluation and reporting tools it is necessary to develop a responsibly

Table 4.8 *Typical indicators used to describe and assess the different aspects of social sustainability*

Aspect	Social indicators	
	Technology development	Technology application
Health and safety	–Risk group of biological substances –Risk factors for health and safety –Voluntary health measures –Quality of health and safety management	–Job security levels –Amount of hazardous substances –Voluntary health measures during application –Voluntary health measures during usage
Quality of working conditions	–Working time arrangements –Degree of psychological strain –Percentage of women in leading positions –Measures taken to improve working conditions	–Working time arrangements –Degree of psychological strain –Percentage of women in leading positions –Measures taken to improve working conditions
Employment	–Safeguarding of jobs –Continuity of Job Creation Effects –Regions of Job Creation –Extent of Job Creation	–Safeguarding of jobs –Continuity of Job Creation Effects –Regions of Job Creation –Effects on related labor markets
Education and training	–Focus of employee training –Quality of human resource management –Identification of training needs –Incorporation of employee expectations	–Apprenticeship –Voluntary training offerings –Identification of training needs –Incorporation of employee expectations
Knowledge management	–Degree of knowledge exchange –Used information systems –Control of knowledge exchange –Employee involvement in decision-making	–Aspects of knowledge exchange –Used information systems –Control of knowledge exchange –Employee involvement in decision-making
Innovation potential	–Commercial exploitation potential –Contribution to scientific debate –Management of patents and licenses –Number and types of patents	–Degree of innovation –Product readiness and marketability –Estimated market penetration –Number and types of patents
Product acceptance and societal benefit	–Stakeholder involvement –Usage of genetic engineering methods –Social standards in supply chain –Societal benefits	–Product acceptance –Usage of genetic engineering methods –Social standards in supply chain –Societal benefits

Table 4.8 *cont.*

Societal dialogue	–Voluntary provision of information –Reporting of core activities to neighborhood –Stakeholder involvement in strategic decision making –Communication channels to political debates	–Used communication channels –Reporting of core activities to neighbors –Targeted dialogue partners –Measures taken to promote dialogue

minded culture [4.66]. A sustainability-oriented corporate culture promotes the 'ability to learn' – the central point in our ability to innovate for more sustainable production and consumption patterns.

4.5 Interactions between the Different Sustainability Dimensions

There exist manifold interactions between the three parts of sustainability, and it would need a large chapter to cover all of them. However, for the purpose of this book, it is important to recognize these interactions and how they may affect process assessment. Figure 4.8 gives an overview of the interactions.

The categories that are listed in Table 4.6 to assess the environmental sustainability are examples of such interactions. Almost all of them also affect the economic and social sustainability. The raw material availability considers the depletion of natural resources. This can cause price fluctuations or, in the long run, a strong increase in input material prices that affects the economic success of the process. The complexity of the synthesis or the agricultural area needed to produce a raw material also influences its price. The thermal

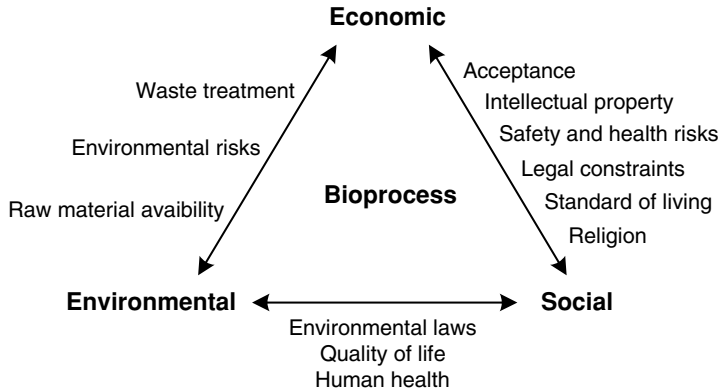


Figure 4.8 *Interaction between a process and the economic, environmental and social sustainability*

risks affect all three parts: They may cause an environmental damage as well as harm human life or cause injuries. From the economic point of view, they can result in the loss of product, parts of or even the complete plant, and lead to costs for compensation payments. The categories in the impact group organisms affect per se the environment and human health. However, by affecting human health, they also influence economic sustainability (less productivity of the workforce, more absence due to illness, etc.). Finally, the impact groups air and water/soil can affect indirectly human health (see e.g. [4.67]). Mainly, they influence the quality of life. Quality of life not only considers the economic standard of living but also demands an appropriate environmental quality and a social system that fulfills its functions.

The plant capacity is defined for an expected market demand and development that may be interpreted in a societal context and has a strong impact on the economic success of a process. The economic success is also influenced by the technological development of the company and its competitors. The general economic development influences product sales, which also has a strong social component. Furthermore, government policies and legal constraints have an effect on the process. This is particularly true for pharmaceutical processes. Religious beliefs may also influence the process. For example, to produce a kosher food or pharmaceutical can open a new market and might increase the achievable price.

A good example of these interactions is the use of genetically modified crops in agriculture. There has been a huge discussion in the literature covering this topic (e.g. [4.42, 4.68–4.71]). Within the environmental dimension there are two opposing aspects. On the one hand, the use of genetically modified (GM) crops might reduce the use of pesticides and increase the amount of food that can be produced per square meter. On the other hand, there is the risk that the GM plants might be distributed in the environment and may cause ecological damage. This risk is difficult to predict and quantify. The vagueness has led to fears and heated discussions in western societies [4.72]. However, the acceptance of a new technology can strongly affect its economic success. In the US the acceptance of GM crops is relatively high and GM crops are already widely used. Although the risks are the same, the acceptance in the EU is low. The fears of a possible direct impact on human health but also on the environmental quality as an aspect of the quality of life are an important reason for this low acceptance. This reduces the possible market size, probably also the price that can be achieved, and may cause additional costs to protect the crops in the field. Furthermore, the low acceptance has led to higher legal constraints for the use of GM crops. Owing to these social factors, the economic advantage of GM crops is substantially reduced and GM crops are used less in the EU compared with the US. This is a good example why one should consider all three dimensions of sustainability early in process development and be aware of the possible interactions between them.

References

[4.1] World Commission on Environment and Development (1987): Our common future. Oxford University Press, Oxford.

[4.2] Stiglitz, J. (1974): Growth with exhaustible natural resources: Efficient and optimal growth paths. *Review of Economic Studies Symposium*, 139–152.

[4.3] Peters, M., Timmerhaus, K., West, R. (2003): Plant design and economics for chemical engineers. McGraw-Hill, Boston.
[4.4] Ward, T. (2001): Economic evaluation. In: Kirk-Othmer Encyclopedia of Chemical Technology. Wiley-VCH, Weinheim.
[4.5] Vogel, H. (2002): Process development. In: Ullmann's Encyclopedia of Industrial Chemistry. Wiley-VCH, Weinheim.
[4.6] Mosberger, E. (2002): Chemical plant design and construction. In: Ullmann's Encyclopedia of Industrial Chemistry. Wiley-VCH, Weinheim, pp. 477–558.
[4.7] Perry, R., Green, D., Maloney, J. (1997): Perry's chemical engineers' handbook. McGraw-Hill. New York.
[4.8] Atkinson, B., Mavituna, F. (1991): Biochemical engineering and biotechnology handbook. Stockton Press, New York.
[4.9] Wheelwright, S. (1996): Economic and cost factors of bioprocess engineering. In: Avis, K., Wu, V.: Biotechnology and biopharmaceutical manufacturing, processing, and preservation. Interpharm Press, Buffalo Grove, pp. 333–354.
[4.10] Rathore, A., Latham, P., Levine, H., Curling, J., Kaltenbrunner, O. (2004): Costing issues in the production of biopharmaceuticals. *BioPharm Int.*, (Feb.) 46–55.
[4.11] Datar, R., Cartwright, T., Rosen, C. (1993): Process economics of animal cell and bacterial fermentations: A case study analysis of tissue plasminogen activator. *Bio/Technology*, **11**, 349–357.
[4.12] Brunt, J. van (1986): Fermentation economics. *Bio/Technology*, **4**, 395–401.
[4.13] US Energy Information Administration (2004): February 2004 monthly energy review; US EIA, Washington. Available at: http://www.eia.doe.gov
[4.14] Bundesverband der Deutschen Industrie e.V. (BDI) (2002): Industriestrompreisvergleich in der Europaeischen Union. *Circular Letter EP* 20/02.
[4.15] Bundesverband der Deutschen Gas und Wasserwirtschaft (2004): Marktdaten Abwasser 2003. BDGW, Berlin.
[4.16] Jia, X., Han, F., Tan, X. (2004): Integrated environmental performance assessment of chemical processes. *Comput. Chem. Eng.*, **29**, 243–247.
[4.17] Koller, G. (2000): Identification and assessment of relevant environmental, health and safety aspects during early phases of process development. PhD thesis, ETH, Zurich.
[4.18] Steinbach, A., Winkenbach, R. (2000): Choose processes for their productivity. *Chem. Eng.*, (April) 94–101.
[4.19] Young, D., Scharp, R., Cabezas, H. (2000): The waste reduction (WAR) algorithm: Environmental impacts, energy consumption and engineering economics. *Waste Management*, **20**, 605–615.
[4.20] Hendershot, D. (1997): Measuring inherent safety, health and environmental characteristics early in process development. *Proc. Safety Prog.*, **16**, 78–79.
[4.21] Turney, R., Mansfield, D., Malmen, Y., Royers, R.L., Verwoered, M., Sovkas, E., Plaisier, A. (1997): The inside project on inherent SHE in process development and design-The toolkit and its application. *I Chem E. Symp. Ser.*, **141**, 203–216.
[4.22] Elliott, A., Sowerby, B., Crittenden, B. (1996): Quantitative environmental impact analysis for clean design. *Comput. Chem. Eng. Suppl.*, **20**, 1377–1382.
[4.23] Goedkoop, M. (1995): The Eco-Indicator 95, Final Report. National Reuse of Waste Research Programme (NOH) Amersfoort.
[4.24] Thomas, S., Berger, S., Weber, V. (1994): Estimating the environmental cost of new processes in R&D. *AIChE Spring National Meeting Paper*, 1–12.
[4.25] Stephan, D., Knodel, R., Bridges, J. (1994): A 'Mark I' measurement methodology for pollution prevention progress occurring as a result of product design decisions. *Environ. Prog.*, **13**, 232–246.

[4.26] Schmidt-Bleek, F. (1993): MIPS. A universal ecological measure? *Fresenius Environ. Bull.*, **2**, 306–311.
[4.27] Biwer, A., Heinzle, E. (2004): Environmental assessment in early process development. *J. Chem. Technol. Biotechnol.*, **79**, 597–609.
[4.28] Morsey, D., Nishioka, M., Suter, G., Stahala, P. (1997): Improvements in waste minimization, process safety and running costs by integrated process development. *Chimia*, **51**, 207–210.
[4.29] Heinzle, E., Weirich, D., Brogli, F., Hoffmann, V., Koller, G., Verdyun, M., Hungerbuehler, K. (1998): Ecological and economic objective functions for screening in integrated development of fine chemical processes. 1. Flexible and expandable framework using indices. *Ind. Eng. Chem. Res.*, **37**, 3395–3407.
[4.30] OECD (2001): OECD Environmental Indicators: Towards sustainable development. OECD, Paris.
[4.31] Ullmann, F. (1985): Ullmann's Encyclopedia of Industrial Chemistry. Wiley-VCH, Weinheim.
[4.32] Kirk, R., Othmer, D. (1991): Encyclopedia of Chemical Technology. John Wiley & Sons, Inc., New York.
[4.33] Budavari, S., O'Neil, M., Smith, A. (1989): The Merck Index-An encyclopedia of chemicals, drugs, and biologicals. Merck & Co, Rahway.
[4.34] Lide, D.(editor) (2002): CRC Handbook of Chemistry and Physics. CRC Press, Boca Raton.
[4.35] Houghton, J., Ding, Y., Griggs, D., Noguer, M., van der Linden, P., Dai, X., Maskell, K., Johnson, C. (2001): Climate Change 2001: the scientific basis. IPCC, University Press, Cambridge.
[4.36] UNEP - Ozone Secretariat (Ed.) (2000): Handbook for the international treaties for the protection of the ozone layer, 5th edition. Unon press, Nairobi.
[4.37] Derwent, R., Jenkin, M., Saunders, S., Pilling, M. (1998): Photochemical ozone creation potentials for organic compounds in northwest Europe calculated with a master chemical mechanism. *Atmos. Environ.*, **32**, 2429–2441.
[4.38] Derwent, R., Jenkin, M., Saunders, S. (1996): Photochemical ozone creation potentials for a large number of reactive hydrocarbons under European conditions. *Atmos. Environ.*, **30**, 181–199.
[4.39] Heijungs, R., Guineé, J., Huppes, G. (1992): Environmental life cycle assessment of products: Guide. Center of Environmental Science, Leiden.
[4.40] Castells, F., Aelion, V., Abeliotis, K., Petrides, D. (1994): Life cycle inventory analysis of energy loads in chemical processes. In: El-Hawagi, M., Petrides, D.: Pollution prevention via process and product modifications. *American Institute of Chemical Engineers* New York, pp. 161–167.
[4.41] Glauser, M., Mueller, P. (1997): Eco-efficiency: a prerequisite for future success. *Chimia*, **51**, 201–206.
[4.42] Koenig, A., Cockburn, A., Crevel, R., Debruyne, E., Grafstroem, R., Hammerling, U., Kimber, I., Knudsen, I., Kuiper, H., Peijnenburg, A., Penninks, A., Poulsen, M., Schauzu, M., Wal, J. (2004): Assessment of safety of foods derived from genetically modified (GM) crops. *Food Chem. Toxicol.*, **42**, 1047–1088.
[4.43] Fent, K. (1998): Oekotoxikologie: Umweltchemie, Toxikologie, Oekologie. Thieme Verlag, Stuttgart.
[4.44] Jensen, A., Hoffman, L., Moller, B. et al. (1997): Life Cycle Assessment (LCA): A guide to approaches, experiences and information sources. European Environment Agency, Copenhagen.
[4.45] Molina, M., Rowland, F. (1974): Stratospheric sink for chlorofluoromethanes: Chlorine atom-catalysed destruction of ozone. *Nature*, **249**, 810–812.
[4.46] Wiesner, J., Christ, C., Fuehrer, W., Behre, H., Cuppen, H., Lumm, M., Mais, F., Schroeder, G., Senge, F., Stockburger, D., Schmidhammer, L., Lohrengel, G., Kerker, L., Regner, H., Rothe, U., Jordan, V. (1995): Production-integrated environmental protection. In:

Ullmann's Encyclopedia of Industrial Chemistry, Vol. B8. Wiley-VCH, Weinheim, pp. 213–309.

[4.47] Kuhndt, M., Liedtke, C. (1999): Die COMPASS-Methodik, COMPAnies and sectors path to sustainability. *Wuppertal Papers* No. 97. Wuppertal Institut, Wuppertal.

[4.48] Kuhndt, M., Geibler, J. v., Eckermann, A. (2004): Towards a sustainable aluminium industry: Stakeholder consultations; Final report. Wuppertal Institute for Climate, Environment and Energy and triple innova, Wuppertal.

[4.49] Kuhndt, M., Geibler, J. v., Eckermann, A., (2002): Developing a sectoral sustainability indicator set taking a stakeholder approach. 10th International Conference of the Greening of Industry Network, 23–26 June, 2002, Göteborg, Sweden.

[4.50] Federal German Government (2001): Perspektiven für Deutschland. Unsere Strategie für eine nachhaltige Entwicklung. Federal German Government, Berlin.

[4.51] European Commission (2001): Sustainable development in Europe for a better world. Strategy of the European Union for sustainable development. COM (2001) 264 final, Brussels.

[4.52] Koukoulaki, T., Boy, S. (2002): Globalizing technical standards. Impact and challenges for occupational health and safety. European Trade Union Technical Bureau for Health and Safety, Brussels.

[4.53] Adelmann, S., Schulze-Halberg, H. (2002): Arbeitsschutz in Biotechnologie und Gentechnik. Springer, Berlin.

[4.54] Reaser, A. (2002): Jobs in biotechnology. Applying old sciences to new discoveries. *Occupational Quarterly*, **48**, 25–35.

[4.55] DSM (2004): Industrial (white) biotechnology. An effective route to increase EU innovation and sustainable growth. In: Position Document on Industrial Biotechnology in Europe and the Netherlands. DSM, Heerlen, p. 7.

[4.56] Eurostat (2005): Betriebliche Weiterbildung in Europa. Ergebnis der zweiten europäischen Weiterbildungserhebung in Unternehmen;. Available at: http://europa.eu.int/comm/education/programmes/leonardo/new/leonardo2/cvts/cvts_de.pdf

[4.57] CRIS International (2001): Lebenslanges lernern. Best-Practices der betrieblichen Weiterbildung in führenden Hightech-Unternehmen der USA; Final Report. Federal Ministry of Economy and Technology, Berlin.

[4.58] Hodgson, J. (2001): The headache of knowledge management. Biotechnology companies face an elusive threat: How to handle what they know. *Nature Biotechnol.*, **19**, BE44–BE46.

[4.59] Henderson, S. (2001): Managing business risk. The commercial world abounds with risks, and life science companies can enhance their ability to manage them effectively. *Nature. Biotechnol.*, **19**, BE23–BE25.

[4.60] Argyris, C., Schoen, D. (1978): Organizational learning: A theory of action perspective. Addison-Wesley, Reading.

[4.61] Preskill, H., Torres, R. (1999): The role of evaluative enquiry in creating learning organizations. In: Easterby-Smith, M., Aaraujo, L., Burgoyne, J.: Organizational learning and the learning organization. Developments in theory and practice. Sage Publications, London, pp. 92–113.

[4.62] European Commission (2002): Innovation and competitiveness in European biotechnology; Papers No. 7/2002. EC, Brussels.

[4.63] Biotechnology Industry Organization (2004): New biotech tools for a cleaner environment: Industrial biotechnology for pollution prevention, resource conservation, and cost reduction; Final Report. BIO, Washington.

[4.64] European Commission (2002): Life sciences and biotechnology. A strategy for Europe. COM 27. EC, Brussels.

[4.65] Novozymes A/S (2002): The concise sustainability report 2002. Novozymes, Bagsvaerd.

[4.66] Hartmann, D., Brentel, H., Rohn, H. (2005): Lern- und Innovationsfaehigkeit von Organisationen und Unternehmen. Kriterien und Indikatoren zur Bewertung. Wuppertal Paper. Wuppertal Institute, Wuppertal (in preparation).

[4.67] Thanh, B., Lefevre, Th. (2001): Asseuing health benefits of controlling air pollution from power generation: The case of a lignite-fired power plant in Thailand *Envison. Manag*., **27**, 303–317.

[4.68] Stephenson, J., Warnes, A. (1996): Release of genetically modified Micro-organisms into the environment. *J. chem. Technol. Biotechnol.*, **65**, 5–142.

[4.69] Kaeppli, O., Auberson, L.(1998): Planned releases of genetically modified organisms into the environment: The evolution of sofety considerations. *Chinia,* **52**, 137–142.

[4.70] Losey, J., Rayor, L., carter, M. (1999): Transgenic Pollen harms monarch Larvae. *Nature*, **399** 214.

[4.71] Kok, E., Kuiper, H. (2003): comparative safety arrenment for biotech crops. *Trends Biotechnol.*, **21**, 439–444.

[4.72] Schurman, R. (2004): Fighting "frankenfoods": Industry opporntunity Structures and the effiency of the anti-biotech movement in Western Europe. *Social Problem*, **51**, 243–268.

Part II

Bioprocess Case Studies

Introduction to Case Studies

Sustainable bioprocesses should be: (i) commercially successful in both the short and long term, (ii) environmentally friendly using minimal and preferably renewable resources, while having minimal environmental burden, and (iii) contribute beneficially to the needs of society. The development of such processes is guided and supported by the systematic application of process modeling and sustainability assessment methods from the earliest phases of process development.

The inclusion of integrated methods for process development into the academic curricula, particularly in chemical and biochemical engineering, is greatly facilitated with the use of case studies. In Part II of this book we provide 11 case studies developed in our own research groups or supplied by experts all over the world.

These case studies are supplemented with fully operational models that are all supplied on the accompanying CD. The models are built using the software SuperPro Designer™ which is kindly supplied by Intelligen, Inc. (Scotch Plains, NJ, USA) in a version that allows running of all the examples. These examples are useful as classroom exercises as well as a platform for new case developments. Experienced practitioners might like to start modeling directly from an already well developed case to shorten model-development time. The necessary basic understanding of bioprocesses and of basic principles of assessment, the reader can obtain from studying Part I of this book, more detailed textbooks, or the primary literature. Online help and support is provided by SuperPro Designer™ (http://www.intelligen.com).

The 11 models were selected to cover examples of the major classes of bioprocesses that include: bulk biochemicals, fine chemicals, enzymes, and low- and high-molecular-weight biopharmaceuticals. This is illustrated in Figure I.1 below where all case studies are characterized in terms of their production volume and price.

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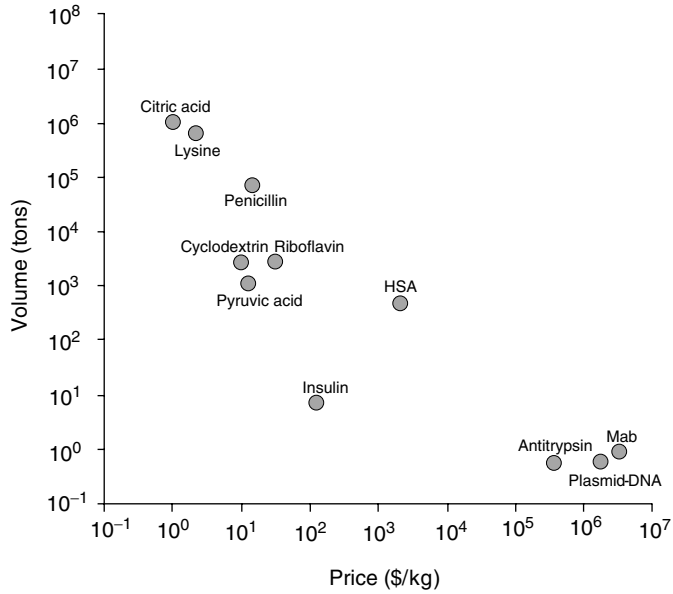


Figure I.1 *Production volume and price of case-study products*

Citric acid is a typical bulk chemical, having a price of about $ 1/kg, whereas some therapeutic proteins sell for more than $ 10^6/kg. A broad range of biocatalysts are applied; Isolated enzymes, wild-type and genetically engineered bacteria, yeasts, filamentous fungi, plant cells, and mammalian cells. Some of the case studies refer to existing, well established processes, whereas others have not yet been realized on a commercial scale, e.g. pyruvic acid. The case studies and some of their typical characteristics are summarized in Table I.1.

The version SuperPro Designer™ on the accompanying CD is free of charge and allows one to run all of the case studies. A fully functional program can be obtained from Intelligen, Inc. (http://www.intelligen.com; 2326 Morse Avenue, Scotch Plains, NJ 07076, USA). A second program, Crystal Ball™ (http://www.decisioneering.com/crystal_ball; Decisioneering, Inc., 1515 Arapahoe St., Suite 1311, Denver, CO 80202, USA) that is used for Monte Carlo simulation in some case studies for uncertainty analysis, is also provided. Details about integration of SuperPro Designer™, Excel, and Crystal Ball™ are provided in the help section (COM help/COM Application Examples/Risk Analysis using integration of SuperPro™, Excel, and Crystal Ball™) of SuperPro Designer™. The CD contains additional documentation about the program, tables for ecological and economic assessment, and process model case studies (Table I.2).

Note: The numerical values stored in the SuperPro Designer™ models on the CD do not always give results completely identical with those shown in tables and figures of the book.

Table I.1 *Case-study overview*

Product	Chapter	Biocatalyst	Special learning features
Citric acid	5	*Aspargillus niger* Filamentous fungus	Stoichiometric model
Pyruvic acid	6	*Escherichia coli* Bacterium	Detailed stoichiometric model, liquid–liquid extraction versus electrodialysis, scenario analysis
L-Lysine	7	*Corynebacterium glutamicum* Bacterium	Dynamic bioreaction model coupled to process model, sensitivity analysis
Riboflavin	8	*Eremothecium ashbyii* Filamentous fungus	Batch production
α-Cyclodextrin	9	Cyclodextrin glycosyl transferase Enzyme	Enzymatic conversion, scenario analysis
Penicillin V	10	*Penicillium chrysogenum* Filamentous fungus	Detailed process model, uncertainty analysis using Monte Carlo simulation
Recombinant human serum albumin	11	*Pichia pastoris* Yeast	New process, recombinant therapeutic protein from yeast, comparison of adsorption processes, scenario analysis
Recombinant human insulin	12	*Escherichia coli* Bacterium	Therapeutic protein from *E. coli*, protein processing and refolding, detailed model of complex process, scheduling
Monoclonal antibody	13	Chinese hamster ovary cells Mammalian cell	Animal cell culture, uncertainty analysis using scenarios, sensitivity analysis, and Monte Carlo simulation
α-1-Antitrypsin from transgenic plant cell suspension culture	14	Transgenic rice cells Plant cell	Plant cell culture, feasibility study
Plasmid DNA	15	Therapeutic DNA	DNA for gene therapy and gene vaccination

Table I.2 *Content of the accompanying CD*

Directory	Directory/File	Content
Demo_Version_ superPro_Designer		Installation software
Demo_Version_ Crystal_Ball		Installation software
Training_CaseCellulase	cellulase-base model.spf	SuperPro base model
	cellulase-base model-COM.spf	Base model Monte Carlo
	cellulase-scenario inoculum volume.spf	Model scenario
	cellulase-scenario ion exchange.spf	Model scenario
	Fermentation model cellulase production.xls	Basic calculations
	Model cellulase production-Monte Carlo Simulation.xls	Monte Carlo Simulation
Environmental_ Assessment_Case_ Studies	05_Env Assessment - citric acid.xls	Ecological assessment of case studies
	06_Env Assessment - pyruvic acid.xls	
	07_Env Assessment - lysine.xls	
	08_Env Assessment - riboflavin.xls	
	09_Env Assessment - cyclodextrin.xls	
	10_Env Assessment - penicillin.xls	
	11_Env Assessment - rHSA.xls	
	12_Env Assessment - insulin.xls	
	13_Env Assessment - monoclonal antibody.xls	
	14_Env Assessment - alpha-antitrypsin.xls	
	15_Env Assessment - DNA vaccine.xls	
Handbook_&_Tutorial_ Crystal_Ball		
Handbook_&_Tutorial_ SuperPro_Designer		
Process_Models_ Case_Studies	05_Citric_Acid	Case studies of the book
	06_Pyruvic_Acid	
	07_Lysine	
	08_Riboflavin	
	09_Cyclodextrin	
	10_Penicillin	
	11_Human_Serum_Albumin	
	12_Human_Insulin	
	13_Monoclonal_Antibody	
	14_Antitrypsin	
	15_Plasmid_DNA	

5

Citric Acid – Alternative Process using Starch

5.1 Introduction

Citric acid is one of the few commodity chemicals produced in a biotechnical process. The world production is approximately 1.1 million tons per year. Most of the production is used in beverages (45%) and foods (25%) as a flavor enhancer and a preservative. About 20% is used in soaps and detergents. In the chemical and pharmaceutical industry, citric acid is used in buffers, as an antioxidant, flavor additive, and for complexing metals.

A general introduction to citric acid production is given by Kristiansen et al. [5.1]. Citric acid has been produced for over 80 years using the filamentous fungus *Aspergillus niger*. More recently, yeast processes have been used as well. While molasses is a common raw material, in this case study we describe a process that uses pure starch as an alternative carbon source. Data were taken mainly from Marending [5.2]. There are other published citric acid processes starting from starch [5.3–5.5]. The process is described in greater detail by Biwer [5.6] and Biwer and Heinzle [5.7].

5.2 Fermentation Model

Figure 5.1 shows the reaction scheme for citric acid production. In the first step, α-amylase is added to hydrolyse the starch to dextrin. Complete starch hydrolysis cleaves starch into glucose monomers. In the citric acid case, starch is only hydrolysed to dextrin with five glucose units on average. Proteins and fats are common impurities in commercially available starch. We assume that the raw starch used is completely dry and that it contains approximately 1% proteins and 1% fats. The ash content of starch was not considered in this model. These facts have to be taken into account for the definition of the starch price.

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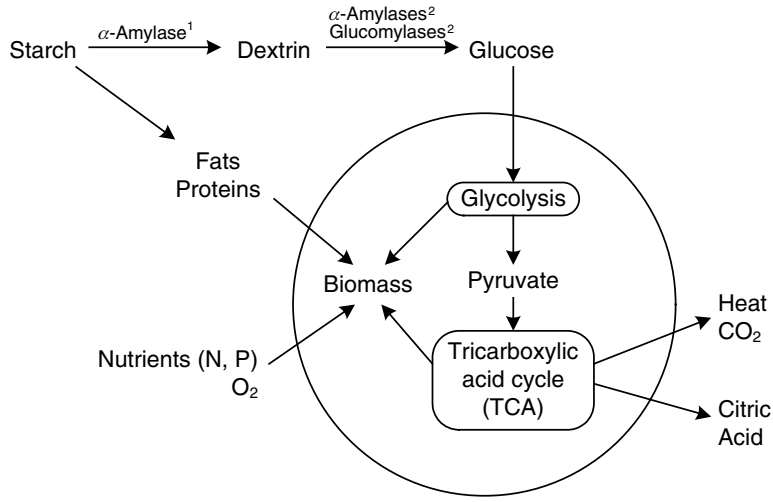


Figure 5.1 *Reaction scheme for citric acid production from starch using* Aspergillus niger. [1] *Added,* [2] *secreted to the medium by* A. niger

The molecular weight of dextrin is 828 g/mol with five glucose units. The degradation of a starch requires one molecule of water for each dextrin molecule formed:

$$(C_6H_{10}O_5)_x + \frac{x}{5} H_2O \rightarrow \frac{x}{5}(C_6H_{10}O_5)_5H_2O \tag{5.1}$$

Hence, 18 g of water (1 mol) are needed for 828 g dextrin, respectively 21.7 g for 1 kg. Thus, 978.3 g pure starch is necessary to obtain 1 kg dextrin. Including the impurities in the starch, the reaction equation is (in g):

$$998.26\,(C_6H_{10}O_5)_x + 21.74\,H_2O \rightarrow 1000.0\,(C_6H_{10}O_5)_5H_2O + 10.0\,\text{Proteins} + 10.0\,\text{Fats} \tag{5.2}$$

Since the exact elementary composition of the proteins and fats contained in raw starch is not known, they cannot be considered in detail for the equation. It is assumed that fats are not modified.

After the starch hydrolysis, temperature and pH are adjusted and the inoculum is added. During the fermentation several reactions run more or less in parallel. The fungus secretes α-amylases and glucoamylases into the media. These enzymes catalyse the degradation of dextrin to glucose that is consumed by *A. niger*. A molar yield of 100% of glucose from starch is assumed.

$$(C_6H_{10}O_5)_5H_2O + 4\,H_2O \rightarrow 5\,C_6H_{12}O_6 \tag{5.3}$$

and in grams:

$$828.7\,(C_6H_{10}O_5)_5 + 72.1\,H_2O \rightarrow 900.8\,C_6H_{12}O_6$$

The glucose is used to form biomass, produce citric acid, and provide energy via the degradation of glucose to carbon dioxide in the respiratory chain. Two phases of cultivation can be distinguished: (i) biomass formation and (ii) citric acid production. However, for the

process modeling, only the final concentration and productivity are relevant and, therefore, only one fermentation step is defined.

Starting concentrations of the raw materials were taken from Marending [5.2]. Ammonium nitrate and protein impurities are used as nitrogen source for the biomass formation. They are both assumed to be consumed completely. The amount of ammonium nitrate added in the medium is known [5.2]. The nitrogen content of the proteins is estimated. Nielsen and Villadsen [5.8] provided the average frequency of the different amino acids in yeast, and Creighton [5.9] calculated the average frequency of amino acids from 1000 known proteins. From the relative frequency and the elementary composition of the amino acids, an average composition was calculated. Thereby, it is assumed that during the polypeptide formation one mol of water is formed per mol of amino acid. The calculated average composition is very similar for the two literature sources. Their average is taken for the following calculations. Referred to one carbon atom, the elementary protein composition is $CH_{1.51}O_{0.3}N_{0.28}$, and the molecular weight is 22.23 g/C-mol. From this composition and the protein amount, the available nitrogen is calculated. In this case 25% of total nitrogen in the biomass is derived from proteins contained in starch and the rest from ammonium nitrate. The sulfur content of the proteins is neglected.

From the available amount of nitrogen and the amount of biomass formed, a nitrogen content of 5.5% is calculated for the biomass. This is lower than the 9.3% typically specified for *A. niger* in literature [5.8]. However, Schlieker [5.10] has shown that the nitrogen content of microbial biomass can substantially decrease under nitrogen limitation. The same is true for the phosphorus content. Here, also the calculated value is relatively small. The estimated elementary composition of the biomass used here is $CH_{1.72}O_{0.55}N_{0.09}P_{0.002}$ (MW = 23.89 g/C-mol).

The reaction equation for the biomass formation from ammonium nitrate is:

$$C_6H_{12}O_6 + 0.28\,NH_4NO_3 + 0.012\,KH_2PO_4 \rightarrow 6\,CH_{1.72}O_{0.55}N_{0.09}P_{0.002} + 1.412\,H_2O + 1.088\,O_2 + 0.012\,K^+ \quad (5.4)$$

The reaction equation for the biomass formation from the proteins is:

$$0.662\,C_6H_{12}O_6 + 2.026\,CH_{1.51}O_{0.3}N_{0.28} + 0.012\,KH_2PO_4 \rightarrow 6\,CH_{1.72}O_{0.55}N_{0.09}P_{0.002} + 0.356\,H_2O + 0.453\,O_2 + 0.012\,K^+ \quad (5.5)$$

For the product formation, glucose is degraded to pyruvate via glycolysis. Pyruvate enters the tricarboxylic acid cycle and is transformed to citric acid that is secreted to the media. The amount of citric acid is expressed as citric acid monohydrate which is the final product.

$$C_6H_{12}O_6 + 1.5\,O_2 \rightarrow C_6H_8O_7 \cdot H_2O + H_2O \quad (5.6)$$

The fermentation ends when the glucose concentration drops below 0.2 g/L. The amount of carbon dioxide produced is estimated via the carbon balance of the fermentation. Glucose start concentration, final citric acid, and start and final biomass concentrations were taken from Marending [5.2], and the overall amounts were calculated for a 210 m^3 working volume (see Table 5.1). 208 334 mol of CO_2 (= 9167 kg) are produced. The reaction equation is:

$$C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O \quad (5.7)$$

Table 5.1 *Estimation of the carbon dioxide produced in the model*

Component	Input (C-mol)	Output (C-mol)
Glucose	1 026 282	1398
Proteins	12 766	
Biomass	352	150 270
Citric acid		679 398
Difference (= CO_2)		208 334

A more detailed description of the citric acid biosynthesis is given in the literature [5.11, 5.12]. A more detailed model would have to consider all impurities as well as moisture content in the raw materials used.

5.3 Process Model

The citric acid production requires a couple of downstream steps following the fermentation. Figure 5.2 shows the process scheme. Based on this process scheme, a process model was developed. For the model, an annual production of 12 000 tons of citric acid is assumed and that is realized with 12 bioreactors, each with a volume of 240 m^3. The number of 12 bioreactors was chosen to facilitate scheduling optimization with a minimal idle time of the downstream equipment.

The corresponding process flow diagram is shown in Figure 5.3. The key process step is the bioreactor (P-6). Starch, water, and amylase (S-109 to S-111) are first added to the reactor, where starch is hydrolysed. Then the bioreactor is filled with medium (from tank P-1) and water (S-107). Both streams are sterilized in continuous heat sterilizers

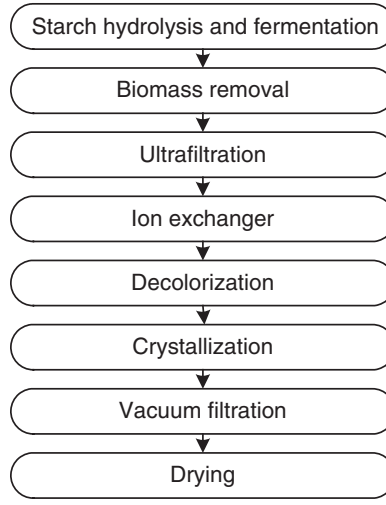


Figure 5.2 *Process scheme of the citric acid production (data taken from [5.2])*

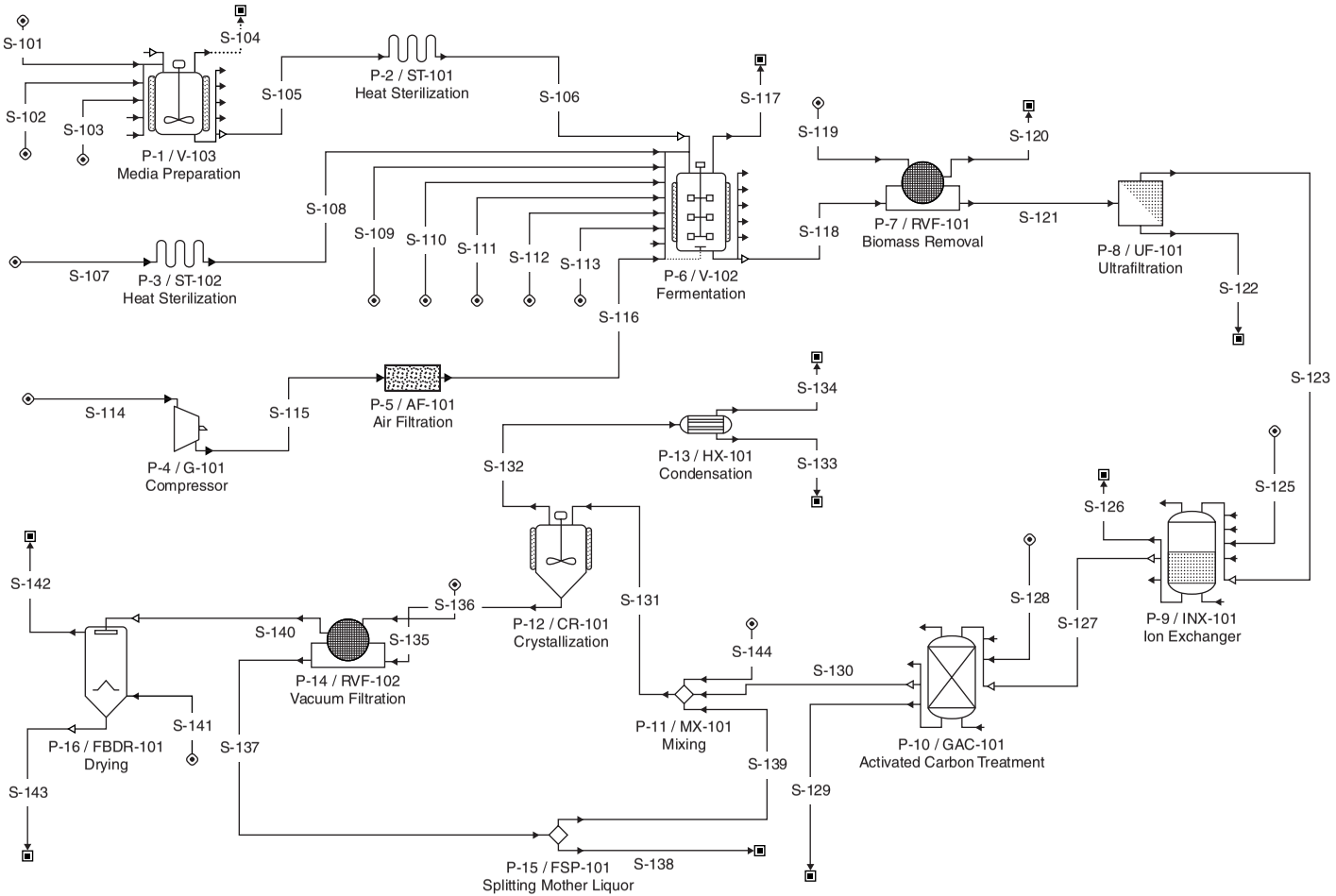


Figure 5.3 *Process flow diagram of the citric acid process*

(P-2, P-3). The pH is adjusted using hydrochloric acid (S-112) and the inoculum is added (S-113). The raw materials for starch hydrolysis do not need to be led through the sterilizer, because the temperature profile of the starch hydrolysis already meets sterilization requirements. During fermentation, the bioreactor is aerated (S-116). Air is supplied by the compressor P-4 and sterilized by the filter P-5.

After the end of the fermentation, proteases secreted to the media by the fungus are inactivated by heat. The bioreactor contents are led to the rotary vacuum filtration P-7 where most of the biomass is removed. The separated biomass is washed to reduce product loss (S-119). Remaining cells, cell debris, and proteins are retained in a subsequent ultrafiltration step (P-8). In the next step, magnesium and potassium ions are separated from the product stream in the ion-exchange column P-9. Cations are bound to the resin, then eluted using hydrochloric acid (S-125) and discharged (S-126), while the product and anions flow through the column. It is assumed that the anions do not affect the crystallization. The product solution is decolorized with activated carbon packed in column P-10. In P-11, caustic soda is added to prevent the evaporation of hydrogen chloride during crystallization.

In the crystallization tank P-12 most of the water is evaporated. Then the solution is cooled and citric acid crystallizes (data from [5.13]). The evaporated water is condensed in P-13. The citric acid crystals are separated and washed (S-136) in the rotary vacuum filter P-14. The mother liquor (S-137) is recycled to the crystallization tank to increase the recovery yield. A part of the mother liquor (S-138) is purged in P-15 to avoid the accumulation of undesired substances. Following Marending [5.2], a crystallization yield of 98% is assumed, although data from Gluszcz and Ledakowicz [5.14] indicate that the yield might be lower. Since citric acid has a high solubility (59% w/v), around 9 kg water/kg citric acid have to be evaporated to realize a high yield at the given product concentration of the feed. Glucose, fats, sodium, and chloride are the main impurities in the feed to the crystallizer. They all remain well below their maximum solubility, and it is assumed that they are separated with the bleed stream S-138.

The recovered crystals as citric acid monohydrate are dried in the fluid bed dryer P-16 using preheated air (S-141). For a 240 m^3 bioreactor, 28.3 tons/batch of starch are consumed and 22.4 tons/batch of citric acid monohydrate (= 20.4 tons pure citric acid) are obtained in the final product (S-143).

5.4 Inventory Analysis

One batch takes 189 h, with the bioreactor occupying 164 h and the downstream processing 34 h. A new batch is started every 14 h in one of the 12 bioreactors. The bioreactors are the bottleneck of the process. With 330 operating days, 12 630 tons of citric acid monohydrate are produced in 565 batches. This assumes that all batches are successful and meet the average target value.

From 100 kg starch, 79 kg of citric acid monohydrate are produced. The carbon yield of the process is 61% (C-mol citric acid/C-mol glucose). The respective yields of the bioreaction are 84% (kg/kg) and 65% (C-mol). The downstream processing yield is 94% with a product loss of 2% in the crystallization and 1% in the biomass removal, the ultrafiltration, and the two adsorption steps each.

Table 5.2 *Material balance of citric acid production. (kg/kg P) = kg component per kg final product (citric acid monohydrate)*

Component	Input (kg/kg P)	Output (kg/kg P)
α-Amylase	<0.01	<0.01
Ammonium nitrate	0.02	
Biomass	<0.01	0.16
Carbon dioxide		0.41
Chloride		<0.01
Citric acid monohydrate (product)		1.00
Citric acid (loss)		0.07
Fats		0.01
Glucose		<0.01
Hydrogen chloride	<0.01	
Sodium hydroxide	<0.01	
Sodium (dissolved)		<0.01
KH_2PO_4	<0.01	
Magnesium sulfate	<0.01	
Magnesium (dissolved)		<0.01
Oxygen	0.51	
Potassium (dissolved)		<0.01
Starch	1.27	
Sulfate		<0.01
Water	14.98	15.12
Mass Index (including water)	16.8	16.8
Mass Index (without water)	1.8	1.65

Table 5.2 shows the overall material balance for the process. The major input materials are typical for a bioprocess: A large amount of water, starch as the carbon source, oxygen for the respiration of the fungus, and a source of nitrogen to support biomass growth. All other compounds are needed in only small amounts. The overall material intensity is 16.8 kg/kg final product (see Table 5.2). Besides water and the product, biomass, carbon dioxide, and the product loss dominate the output. Additionally, several inorganic salts leave the process in smaller amounts.

The majority of the waste is accumulated as wastewater containing relatively low concentrations of organic materials and inorganic salts and must be led to a sewage-treatment plant. The chemical oxygen demand (COD) is around 65 g O_2/kg final product (kg P). Biomass is separated as solid waste in the first vacuum filtration (COD: 280 g O_2/kg P). It can be further used (e.g. as animal feed), added to a wastewater-treatment plant as carbon source, or else is disposed of. Gaseous emissions leave the process from the bioreactor (containing carbon dioxide) and from the dryer and the condenser (water-saturated air). They do not need further treatment. None of the waste streams contains critical materials in relevant amounts. We are assuming no need for odor control.

The process requires substantial amounts of steam, electricity, and cooling and chilled water. Around 10 MJ of electricity is needed per kg final product (2.65 kWh/kg P) with the main consumptions for the air compressor and the bioreactor agitation. The steam

requirement is 32 MJ/kg P (9 kWh/kg P), respectively 15 kg/kg P. Most of it is consumed for water evaporation in the crystallization step and also for the different heating operations in the bioreactor and the dryer. Cooling and chilled water are needed to condense the water vapor in P-13 and also to cool the bioreactors and the compressors, altogether 1.7 m^3/kg P. Air compression (P-4) and evaporation (P-12) are the dominating energy-consumption steps. The aeration rate and the fermentation duration determine the demands on the compressor. The amount of water that has to be evaporated in the crystallization step arises from the citric acid concentration in the feed after the adsorption steps (P-9, P-10).

5.5 Environmental Assessment

The environmental indices are summarized in Table 5.3. Four components are categorized at least once as class A (high environmental relevance). Concentrated hydrogen chloride and sodium hydroxide have a high acute toxicity, while the ammonium nitrate used as nitrogen source is classified A in the category Thermal Risk, because it can be explosive when mixed with flammable substances (R-code 9). A careful handling of these three substances in the process can minimize the risk. In the output, phosphate is classified A due to its importance for eutrophication processes. However, it leaves the process only in very small amounts.

When using the EF_{Mult}, the weighting factor for class C is one. This means the minimal possible $EI_{\text{Mult}} = MI$. When calculating EF_{MV}, class C is set to zero. Here, the minimal possible EI_{MV} is zero. The environmental indices, both for input and output, are all quite close to their minimum possible values (see Table 5.3); this indicates a generally low environmental relevance of the substances involved in the process.

The relative importance of the different components is shown in Figure 5.4 by the EI_{MV} and the EI_{Mult} of the process. Starch, oxygen, and ammonium nitrate are the three dominating components in the input EI_{Mult}. The EI_{MV} does not consider substances with a very low environmental relevance even if they are consumed in high amount. Therefore, starch and oxygen ($EF_{\text{MV}} = 0$) are not included in the input EI_{MV} and ammonium nitrate

Table 5.3 *Environmental assessment parameters of the citric acid process*

	Input		Output	
Assessment parameter	Including water	Without water	Including water	Without water
Mass Index *MI* (kg/kg P)	16.8	1.8	16.8	1.7
Number of A-components	3		1	
Environmental Index EI_{MW} (Index Points/kg P)	0.01		0.05	
Environmental Index EI_{Mult} (Index Points/kg P)	16.9	2.0	17.0	1.9
General Effect Index GEI_{MW} (0–1)	0.0006	0.005	0.003	0.032
General Effect Index GEI_{Mult}	1.01	1.09	1.01	1.13

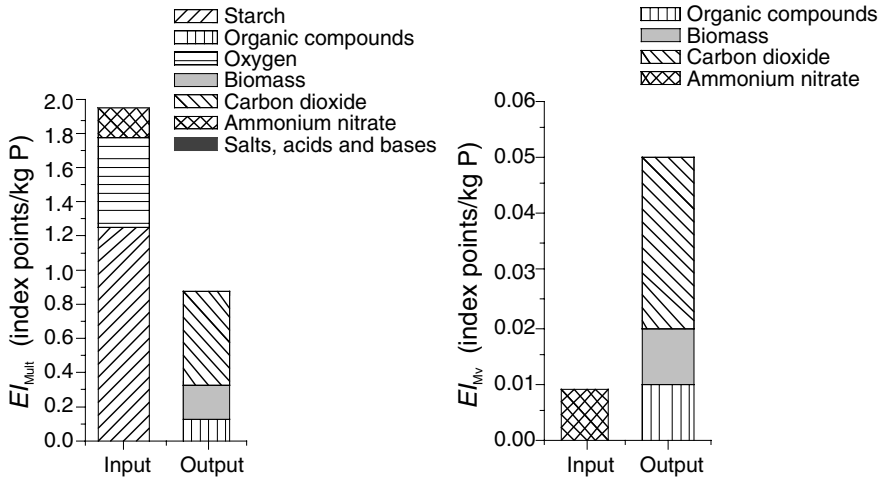


Figure 5.4 *Environmental Indices (EI_{Mult}, EI_{MV}) of the citric acid production. The final product is not considered in the graph*

is the only relevant component besides small amounts of acids and bases (HCl, NaOH). Carbon dioxide, biomass, and organic compounds (product loss, glucose, fats, etc.) are the dominating output components for both indices.

The impact group indices (*IGI*) show how strong the different impact groups contribute to the overall environmental impact of the process. The IG Risk dominates the *IGI* of the input components (see Figure 5.5), while the remaining three IGs (Organisms, Resources, Grey Inputs) are evenly weighted. The dominance of the risk IG is caused by ammonium nitrate (explosive, see above). The IG Risk does not contribute to the environmental relevance of the output components and the IG Organisms only to a small extent (no risk-relevant or toxic substances in the output). The impact groups Air and Water/Soil are relevant. The global warming potential of the carbon dioxide accounts for the importance of the IG Air,

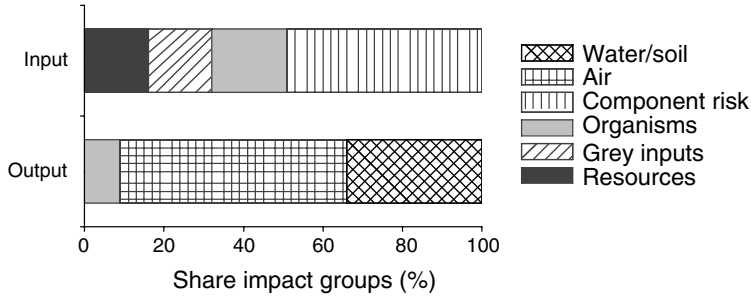


Figure 5.5 *Impact group indices of the input and output components of the citric acid production*

while the eutrophication and the organic carbon pollution potential of the biomass and the organic compounds are responsible for the importance of the IG Water/Soil.

5.6 Economic Assessment

With the model parameters chosen, the plant for the production 12 600 metric tons requires a direct fixed capital investment of $115 million and a total capital investment of $121 million. The most expensive pieces of equipment are the 12 bioreactors and the associated compressors.

The annual operating costs are $31 million. Figure 5.6 shows the allocation to the main cost categories. The facility-dependent costs are the dominating cost type, with the depreciation cost accounting for the largest part. Utilities, raw materials, and labor costs also contribute substantially, while the share of the waste treatment, consumables, and laboratory/QC/QA is small. The fermentation section accounts for 78% of the operating cost, mainly because of the high contribution to the facility-dependent cost (investment for bioreactors and compressors) and the raw material costs (starch).

Starch makes up almost 90% of the raw material costs. Additionally, the amylase, ammonium nitrate, and process water have a notable share. The labor costs are dominated by the fermentation section, almost exclusively caused by the labor demand to run the bioreactors. The waste-treatment costs are evenly divided between the solid waste and the wastewater.

In the model, the overall unit production costs are $ 2.5/kg, which is above the current citric acid selling price of around $ 1.8/kg (2005). Harrison et al. [5.15] found a slightly lower UPC for a citric acid process using molasses and year 2000 prices ($ 2.2/kg). Most citric acid plants operating today are already fully depreciated. Since the depreciation cost accounts for a substantial part of the operating cost, these plants can produce at lower costs, although their maintenance costs are surely higher than in a new plant. Furthermore, the model presented is a generalized representation of a citric acid process. Depending on the specific situation and location of a producer, the costs might be different. Access to inexpensive raw materials and inexpensive equipment is crucial. There is likely to be a strong competitive position for a manufacturer in an advantaged position with regard to raw materials.

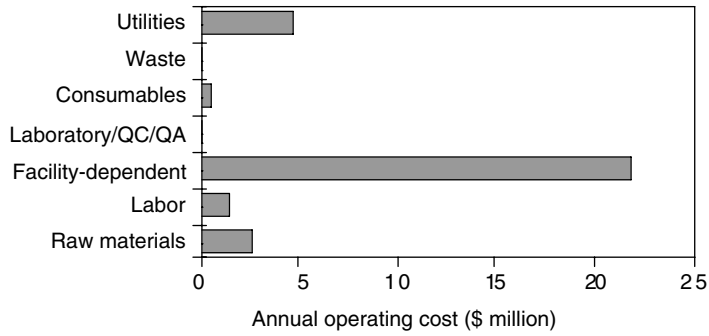


Figure 5.6 *Allocation of the annual operating costs of the citric acid production*

5.7 Conclusions

We modeled the production of citric acid with the use of starch instead of molasses. The use of starch necessitates a different downstream processing for the purification of the citric acid.

The process shows a relatively low potential environmental burden; the ammonium nitrate in the input, and the carbon dioxide emission, the biomass, and the organic compounds in the output, are the most relevant components. The energy consumption also substantially contributes to the environmental burden and to the operating costs. The key factors for a lower environmental impact and lower raw material and utility costs are a higher product concentration, a higher yield, and a shorter fermentation time; this is typical for a commodity process. For a new plant, low investment cost is crucial, e.g. through cheaper equipment acquired from an already depreciated plant or possibly by the replacement of stirred tank bioreactors with alternative reactor types, e.g. bubble columns.

Suggested Exercises

1. Study the influence of final product concentration. What is the influence of an increase in the final product concentration by 10%? Do this by increasing the feed starch concentration (S-109). First watch how biomass concentration, carbon dioxide production, and citric acid yield vary in the bioreactor (S-118). What is the impact on the size of the downstream equipment, unit production costs, and the environmental indices?
2. Assume the strain-development group has identified a new strain with higher productivity reaching the same product yield and concentration in a shorter time. The resulting process time for the fermentation (P-6) decreases from 145 h to 130 h. Examine the same parameters as in Exercise 1 and compare the results.

References

[5.1] Kristiansen, B., Mattey, M., Linden, J. (1999): Citric acid biotechnology. Taylor & Francis, London.
[5.2] Marending, T. (1992): Biotechnologische Herstellung von Zitronensaure aus Staerkehydrolysaten mit *Aspergillus niger*. Diploma thesis, ETH, Zurich.
[5.3] Lesniak, W. (1999): Fermentation substrates. In: Kristiansen, B., Mattey, M., Linden, J.: Citric acid biotechnology. Taylor & Francis, London, pp. 149–160.
[5.4] Sarangbin, S., Watanapokasin, Y. (1999): Yam bean starch: A novel substrate for citric acid production by the protease-negative mutant strain of *Aspergillus niger*. *Carbohydr. Polym.*, **38**, 219–224.
[5.5] Mourya, S., Jauhri, K. (2000): Production of citric acid from starch-hydrolysate by *Aspergillus niger*. *Microbiol. Res.*, **155**, 37–44.
[5.6] Biwer, A. (2003): Modellbildung, Simulation und oekologische Bewertung in der Entwicklung biotechnologischer Prozesse. PhD thesis, Universitaet des Saarlandes, Saarbruecken.
[5.7] Biwer, A., Heinzle, E. (2002): Early ecological evaluation in biotechnology through process simulation: case study citric acid. *Eng. Life Sci.*, **2**, 265–268.
[5.8] Nielsen, J., Villadsen, J. (1994): Bioreaction Engineering Principles. Plenum Press, New York.

[5.9] Creighton, T. (1993): Proteins: Structure and molecular properties. Freeman, New York.
[5.10] Schlieker, H. (1995): Stoechiometrie und Energetik des aeroben Wachstums von *Trichosporon cutaneum* und *Escherichia coli* TG1 unter Beruecksichtigung intrazellulaerer Intermediate. PhD thesis, TU Carola-Wilhelmina, Braunschweig.
[5.11] Wolschek, M., Kubicek, C. (1999): Biochemistry of citric acid accumulation by *Aspergillus niger*. In: Kristiansen, B., Mattey, M., Linden, J.: Citric acid biotechnology. Taylor & Francis, London, pp. 11–32.
[5.12] Karaffa, L., Kubicek, Ch. (2003): *Aspergillus niger* citric acid accumulation: do we understand this well working black box? *Appl. Microbiol. Biotechnol*., **61**, 189–196.
[5.13] Dorokhov, I., Gordeev, L., Vinarov, A., Leonteva, L., Bocharova, Y. (1997): Experimental and theoretical study of ion-exchange and crystallization operations in the production of citric acid. *Theor. Found. Chem. Eng*., **31**, 224–231.
[5.14] Gluszcz, P., Ledakowicz, S. (1999): Downstream processing in citric acid production. In: Kristiansen, B., Mattey, M., Linden, J.: Citric acid biotechnology. Taylor & Francis, London, pp. 135–148.
[5.15] Harrison, R., Todd, P., Rudge, S., Petrides, D. (2003): Bioseparations science and engineering. Oxford University Press, New York.

6

Pyruvic Acid – Fermentation with Alternative Downstream Processes

6.1 Introduction

Pyruvic acid (2-oxopropanoic acid, CAS Registry No. 127-17-3) is used as a raw material in the biosynthesis of pharmaceutically active ingredients, such as tryptophan, alanine, and L-DOPA, and also as a food additive [6.1]. The present world market is greater than 100 tons, with a market potential of approximately 1000 tons per year within the next decade. Pyruvic acid is a relatively low-priced biochemical. For demands in ton lots, the present market price is $15–25/kg (year 2005). However, with the increasing market demand, prices might decrease.

Pyruvic acid ($CH_3COCOOH$) is traditionally produced from tartaric acid via pyrolysis [6.2]. However, this process has both environmental and economic disadvantages. In recent years, a bioprocess has been developed that uses a genetically engineered strain of *Escherichia coli* [6.3–6.6]. In this case study we model such a process and compare solvent extraction and electrodialysis as alternative downstream processes for the separation of pyruvic acid from the fermentation broth.

6.2 Fermentation Model

Pyruvic acid is produced in a bioreactor from glucose using *Escherichia coli* YYC202 *ldhA::Kan* [6.3, 6.4]. A simplified reaction scheme is shown in Figure 6.1. The strain used is an acetate auxotroph [6.3, 6.4]. The bacteria consume glucose and acetate, as well as ammonia nitrogen, phosphorus, and oxygen. Glucose is mainly converted via glycolysis to yield pyruvic acid that is secreted into the medium. Glucose (via both glycolysis and pentose phosphate pathway) and acetate (tricarboxylic acid cycle) are converted into biomass, and additional acetate is oxidized to carbon dioxide to meet the energy demand of the cell. Parts

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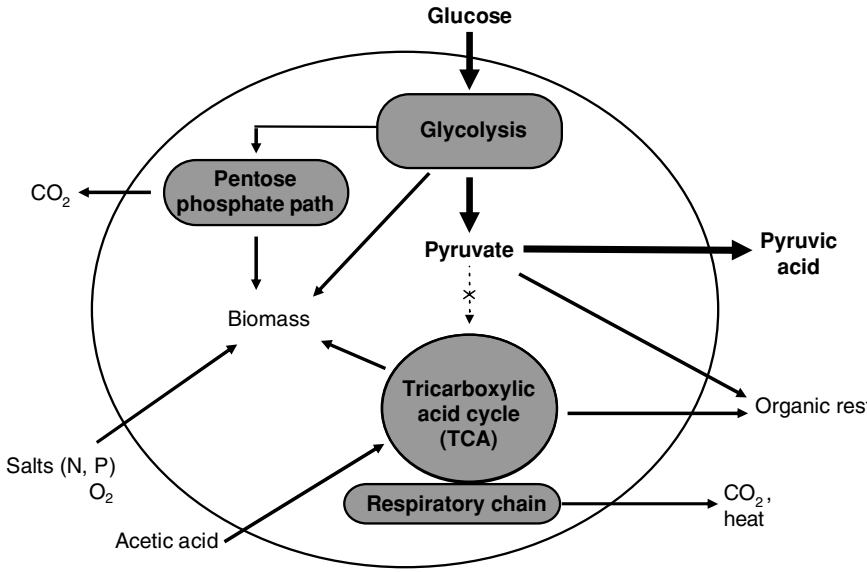


Figure 6.1 *Reaction scheme of the pyruvic acid production with* Escherichia coli *YYC202* ldhA::Kan. *The conversion of pyruvate into acetyl-CoA or acetate (dotted line) is completely blocked. Bold full lines = product-formation route. Reprinted with permission from* Ind. Eng. Chem. Res.*, Modeling and analysis of a new process for pyruvate production. Biwer, Zuber, Zelic, Gerharz, Bellman, and Heinzle,* ***44****, 3124–3133 (Figure 2). Copyright 2005 American Chemical Society*

of the glucose and acetate are converted into several soluble organic compounds that are not determined and are lumped collectively as 'organic rest'.

The fermentation is run as a repeated fed-batch with four cycles. The first cycle includes biomass growth and product formation. At the end of each cycle, a part of the fermentation broth is removed. The biomass remains in the bioreactor, which is refilled with fresh medium. In cycles 2–4 no further acetate is added and pyruvic acid is produced with resting (nongrowing) cells. The derivation of the bioreaction model is given in Appendix 1.

6.3 Process Model

6.3.1 Bioreaction and Upstream

Figures 6.2 and 6.3 show the process flow diagrams of the pyruvic acid process. The fermentation takes place in bioreactor P-20 (working volume: 50 m^3). Two reactors are run in a staggered mode to decrease the idle time of the downstream processing. In the tanks P-1 to P-5, solutions of carbon sources, mineral salts (N, P), and trace elements are prepared. These solutions are passed through the continuous heat sterilizer ST-101 (P-8, P-9) before entering the bioreactor. Trace elements are dissolved in (5N)-HCl solution (P-5). Owing to the high acid concentration additional heat sterilization is not necessary. Water used to fill up the bioreactor is provided in the streams S-120 and S-136.

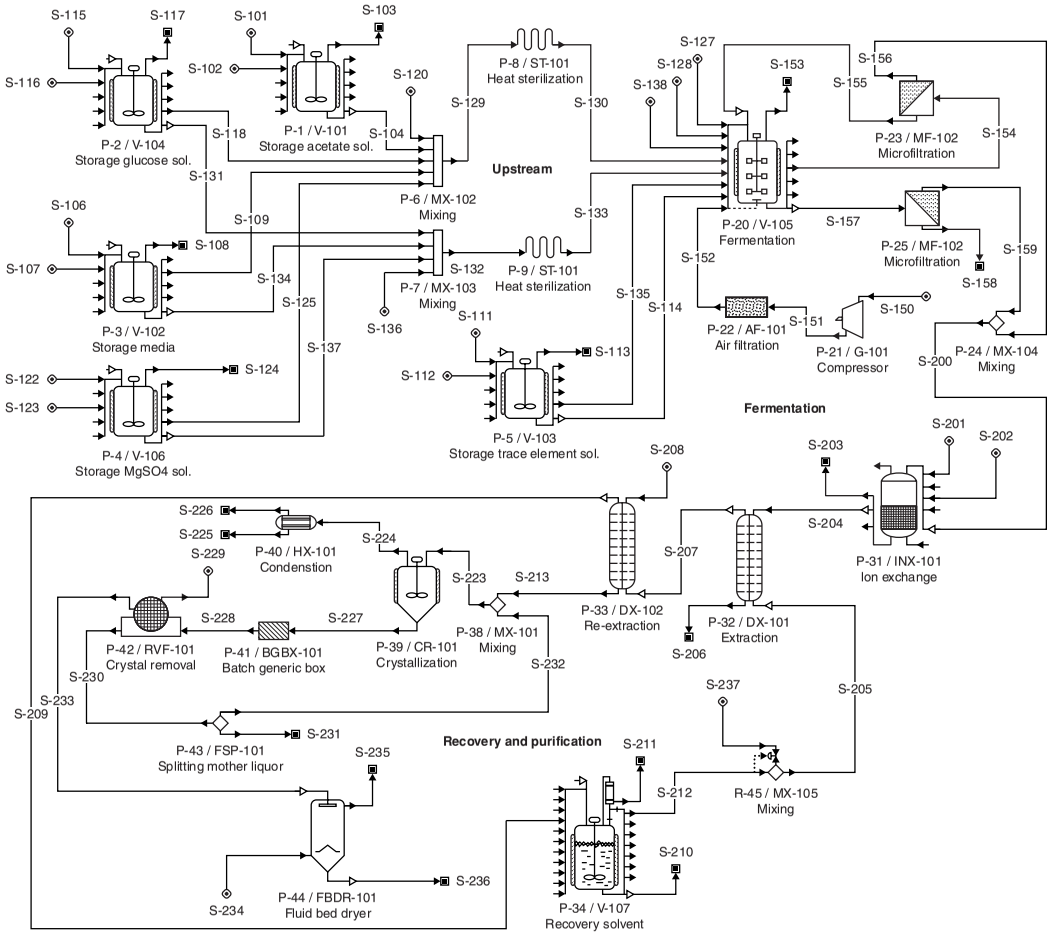


Figure 6.2 *Process flow diagram of the pyruvic acid product using solvent extraction in the downstream processing*

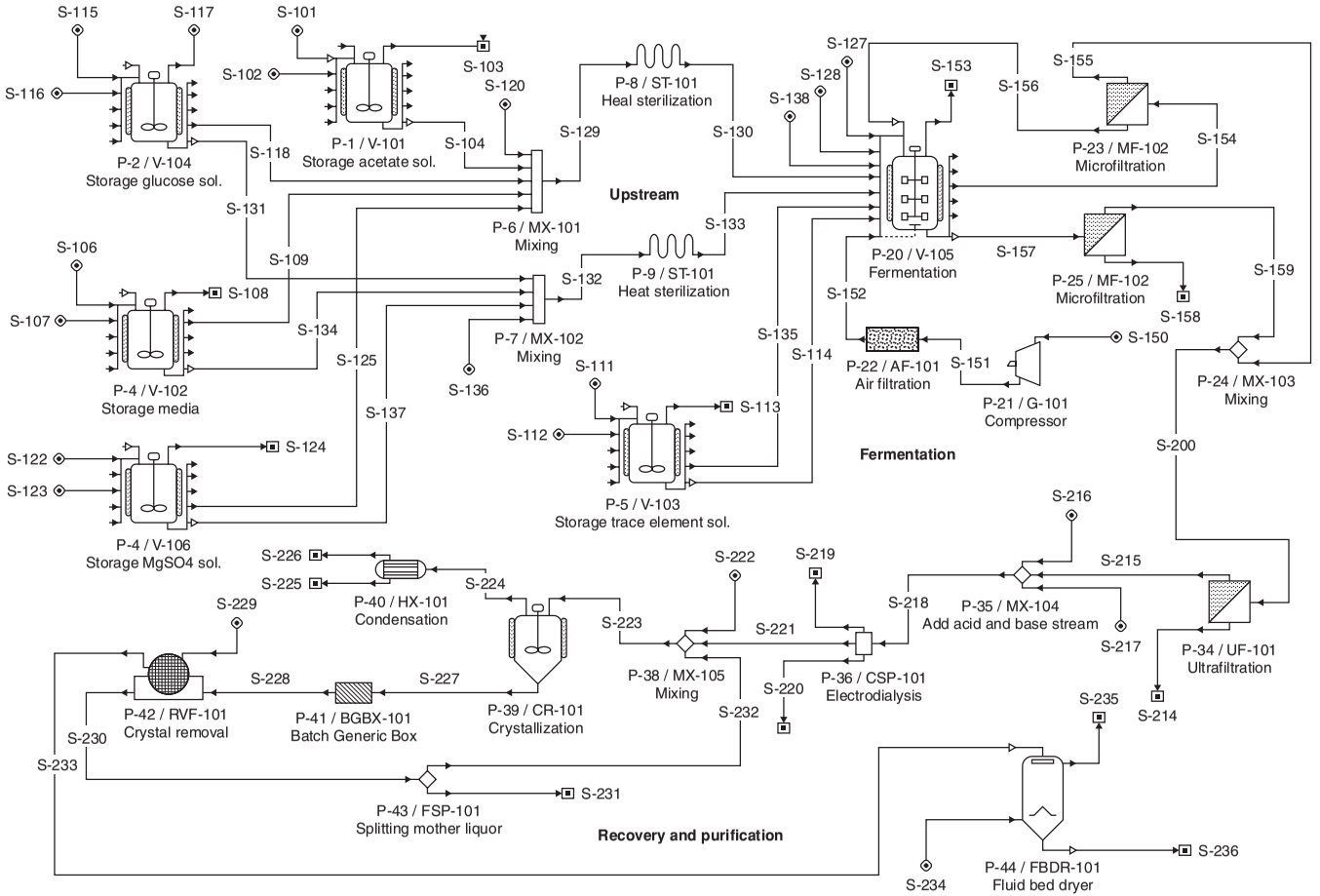


Figure 6.3 *Process flow diagram of the pyruvic acid product using electrodialysis in the downstream processing*

After the reactor is filled, pH is adjusted to 7 using an ammonia solution (S-128) and the inoculum is added (S-127). During the fermentation the reactor is agitated, aerated (0.5 vvm), and kept at 37 °C. Air (S-150) is supplied by compressor P-21 and sterilized by filter P-22. Ammonia is constantly added to the bioreactor to keep the pH constant while producing pyruvic acid. When a cycle is completed, the bioreactor content is passed through the microfiltration unit P-23, where part of the fermentation broth is removed while the biomass is returned to the bioreactor. Fresh medium is added, and the next cycle starts. The filtered solution containing pyruvic acid passes to the downstream processing. After the fourth cycle is completed, the reactor content is sterilized by heating to meet legal constraints for biomass inactivation. The biomass is separated from the product solution in the microfiltration P-25 and is disposed of as solid waste.

6.3.2 Downstream Processing

Solvent Extraction. After the removal of biomass, the product solution is led through the ion-exchange column P-31, where cations are adsorbed. The column is washed (S-201), and the cations are eluted with hydrochloric acid (S-202) and are discharged as wastewater (S-203). The separation of cations is necessary because they may cause problems in the following solvent-extraction steps.

Liquid–liquid extraction is applied on an industrial scale for the purification of a large group of acidic and basic compounds, e.g. carboxylic acids such as citric acid and acetic acid [6.7]. Various solvents are used for this purpose [6.8]. For pyruvic acid, the organic solvent to be used was not yet defined in the development process. The partition coefficient (K_i) between organic and aqueous phases is the key parameter. For pyruvic acid a K_i for the system water/diethyl ether is known [6.9, 6.10], but the coefficient is quite unfavorable ($K_i = 0.16$). However, there is a general rule for carboxylic acids, that alcohols and phosphorus compounds provide much better partition coefficients than do ethers and ketones [6.10]. Therefore, it is assumed, that a more suitable solvent with a higher partition coefficient can be found by sufficient solvent screening. For the initial model calculations, a hypothetical 'Solvent 1' with $K_i = 1$ was defined (price: \$ 0.85/kg). This is in the range of other carboxylic acids, for example citric acid ($K_i = 0.3$), lactic acid ($K_i = 0.75$), and propanoic acid ($K_i = 3.5$) [6.8, 6.10]. Further parameters for modeling the extraction columns, like pH control, contact time, temperature, yield, fluxes, specific mass transfer area, and others are taken from literature [6.7, 6.11–6.15].

A relevant mass transfer into the organic solvent occurs only for the nondissociated acids, while ions remain in the aqueous solution. After the removal of the cations in the ion exchanger, the pH of the product solution is well below the pK_a of pyruvic acid ($pK_a = 2.49$). Hence, in the following extraction (P-32) pyruvic acid is transferred into the organic solvent (S-205) and separated from the fermentation broth that is disposed of as wastewater (S-206). In the back-extraction (P-33) the organic solvent containing pyruvic acid is contacted with an aqueous sodium hydroxide solution (S-208). Owing to the high pH of this solution pyruvic acid is transferred into the aqueous solution and forms sodium pyruvate. The discharged organic solvent is led to unit P-34, where most of the solvent is recycled (S-212) but not shown in detail here. In industrial processes, organic solvents are normally recycled to a high extent. A recycling yield of 98% is assumed. In P-45, the necessary amount of fresh solvent is added.

In the crystallization step (P-39) most of the water is evaporated. Then the solution is cooled and sodium pyruvate precipitates. The resulting steam is condensed in P-40. The crystals are separated and washed (S-229) in the vacuum filtration unit (P-42). The mother liquor is led back to the crystallization tank to afford a higher recovery yield. To avoid the accumulation of undesired compounds a bleed stream is taken off in P-43. The sodium pyruvate separated is dried in the fluid bed dryer P-44 (S-236). SuperPro Designer® does not allow the modeling of the sodium pyruvate formation in the extraction or in the crystallization step in these units. To overcome this software limitation a virtual unit operation, P-41, is defined that does not represent any real apparatus. Therefore, in the model, pyruvate is first formally separated and crystallized as pyruvic acid and only then converted into crystalline sodium pyruvate in this virtual conversion unit P-41.

Electrodialysis. Pyruvic acid is the only organic acid secreted in relevant amounts in the fermentation broth. This allows the recovery of pyruvic acid by electrodialysis as an alternative to the solvent extraction. Electrodialysis is a separation operation in which electromotive force is used to transport ions through a semi-permeable, ion-selective membrane and thus to separate them from an aqueous solution. Preliminary experiments proved its applicability for pyruvic acid purification [6.16]. Besides the few experimental data, most of the data needed were taken from literature dealing with electrodialysis in the purification of other organic acids [6.17–6.23].

After biomass removal, the product solution passes through the ultrafiltration unit P-34, where remaining cells and proteins are separated to avoid fouling of the electrodialysis membranes. A water-splitting electrodialysis with bipolar membranes is assumed (P-35 and P-36). Since ammonium is used for pH control in the bioreactor, the product solution actually contains ammonium pyruvate when it enters the electrodialysis unit. Pyruvate and other monovalent anions pass through the anion membrane into the acid stream (S-221). Ammonium and other monovalent cations pass through the cation membrane into the base stream (S-220). Multivalent ions and uncharged molecules like glucose remain mainly in the solution, which is disposed as wastewater (diluate, S-219). Bipolar membranes located between the ion membranes split water into hydroxide ions and proton. In the acid stream pyruvic acid is formed, and in the base stream ammonium hydroxide is produced. It is assumed that the concentrations of pyruvic acid in the acid stream and ammonium hydroxide in the base stream reach 1.5 mol/L, though higher concentrations might be possible. The ammonia can be reused for pH control and as nitrogen source in the bioreactor. However, for the first analysis, it is assumed that S-220 is discharged as wastewater. After the electrodialysis, sodium hydroxide (P-38, S-222) is added to the product stream to crystallize sodium pyruvate in the subsequent crystallization step (P-39). The following process steps are identical to the alternative process using solvent extraction.

6.4 Inventory Analysis

Sodium pyruvate is the final product. Per batch, around 7.1 metric tons of sodium pyruvate are produced in both processes. Using two bioreactors, 2250 tons are produced annually in 316 batches, under the assumption of a new batch starting every 25 h. Total batch time is 64 h; the bioreactor with an occupation time of 50 h is the bottleneck in both processes. The yield of product separation and purification is 92%.

Table 6.1 provides the material balance of the process alternatives. The carbon sources glucose and acetic acid, water, hydrogen chloride [mainly used for the regeneration of the ion exchanger (only solvent extraction)], and sodium hydroxide (used for formation of sodium pyruvate) are the most important input materials. Additionally required are the organic solvent (solvent extraction), ammonia for pH control, mineral salts, and oxygen, consumed during bioreaction. Apart from water, organic solvent, and the product, dissolved inorganic salts are the most important output materials. The inorganic salts originate from the nutrients, hydrogen chloride, and sodium hydroxide. Further relevant output components are the bacterial biomass, the organic rest, and carbon dioxide produced during the fermentation, ammonium from the pH regulation, product loss, and unconsumed glucose.

The overall Mass Index is 53 kg/kg P (including water) and 3.1 kg/kg P (without water) for the solvent extraction process, and 34.5 kg/kg P or 2.2 kg/kg P for the electrodialysis process. Organic solvent is not needed when using electrodialysis. The additional removal of the ion exchanger further reduces the material intensity by decreasing the consumption of water and of hydrogen chloride (S-220). Furthermore, the ammonium in the base stream of the electrodialysis could be reused for pH control in the bioreactor. Then the specific consumption of ammonium decreases further from 0.19 to 0.04 kg/kg final product.

The energy consumption is notable form both economic (cost) and environmental aspects (depletion of fossil raw materials, air pollution, etc.). The steps with the highest energy consumption are the crystallization (P-39), the recycling of the solvent (P-34), the compressor (P-21), and the reactor (P-20). The specific electricity demand is 1.9 kWh/kg P for the solvent extraction process and 2.4 kWh/kg P for the electrodialysis process. The higher electricity demand is caused by the electrodialysis step. In contrast, the demand of steam, cooling, and chilled water is higher in the solvent-extraction process (50 kg/kg P,

Table 6.1 *Material balance of the pyruvic acid production. [kg/kg P] = kg component per kg sodium pyruvate*

	Input [kg/kg P]		Output [kg/kg P]	
Component	Extraction	Electrodialysis	Extraction	Electrodialysis
Acetic acid	0.09	0.09		
Ammonium sulfate	0.03	0.03		
Ammonium	0.19	0.19	0.18	0.18
Biomass	<0.01	<0.01	0.09	0.09
Carbon dioxide			0.05	0.05
Glucose	1.19	1.18	0.10	0.10
Hydrogen chloride	0.62	<0.01		
Solvent 1	0.31		−0.31	
Product loss			0.08	0.08
Organic rest			0.15	0.15
Oxygen	0.19	0.18		
Sodium hydroxide	0.37	0.38		
Mineral salts	0.14	0.14		
Inorganic salts			0.75	0.32
Water	46.9	32.3	47.3	32.7
Mass Index (*MI*)	52.6	34.5	51.6	33.5
MI without water	3.1	2.2	1.7	0.8

265 kg/kg P, 2.7 m^3/kg P, respectively) than in the electrodialysis process (15 kg/kg P, 50 kg/kg P, 1.5 m^3/kg P, respectively). It is assumed that the recycling of the organic solvent includes a distillation step, where most of the solvent is evaporated. The estimated energy consumption for this evaporation is considered in the analysis of the energy demand. The lower energy demand of the electrodialysis process is mainly due to the removal of the solvent recycling. Furthermore, the use of electrodialysis leads to an increased product concentration in S-223 and, thus, to a lower specific steam and cooling water demand in the crystallization.

6.5 Environmental Assessment

Waste is mainly produced as wastewater in both process alternatives. The separated biomass is the only solid waste. However, the water content of the resulting sludge is very high. Thus, it might be added to a wastewater treatment plant. Apart from air, the emissions produced contain only water and, in lesser amounts, carbon dioxide.

Figure 6.4 compares the Environmental Index (EI_{Mult}) of both processes. The most important input materials are the organic solvent, the carbon sources (glucose, acetate), hydrogen chloride (needed mainly for the regeneration of the ion exchanger), ammonia (consumed for pH control in the fermenter) and sodium hydroxide (needed in the re-extraction). The most important output components are mineral salts formed from salts, acids, and bases in the input and dissolved organic waste (by-products, unused carbon sources), the organic solvent, and biomass. Since organic solvent and HCl are not needed,

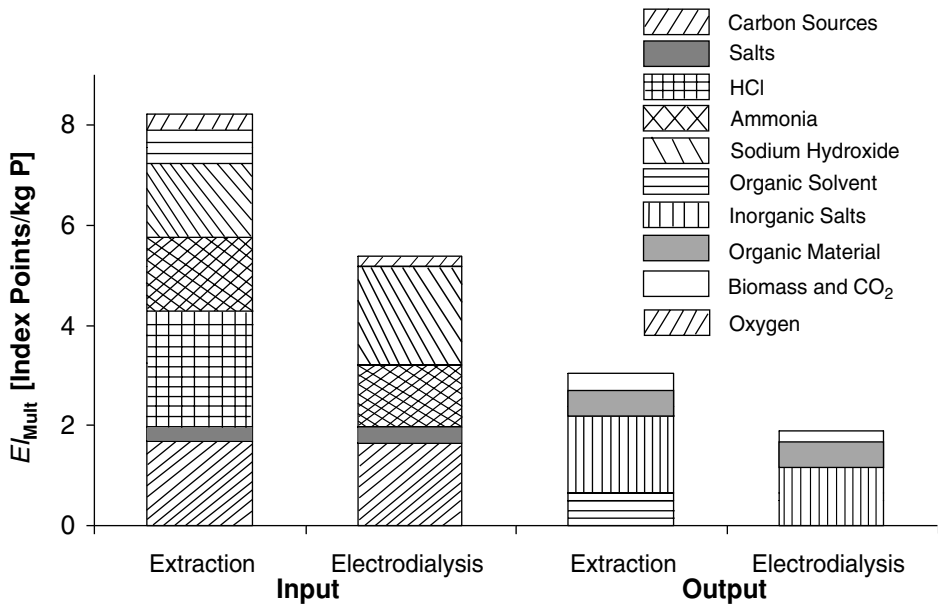


Figure 6.4 *EI_{Mult} of the output components of the two alternative processes for the production of pyruvic acid. BM = biomass (dry cell weight)*

the overall Environmental Index is lower for the electrodialysis process, indicating a lower environmental impact caused by this process alternative.

In the input, hydrogen chloride, ammonia, and sodium hydroxide are allocated to class A in the IC Acute Toxicity. In the output, phosphate and ammonium in the waste water are allocated to class A in the IC Eutrophication. At the input side, the IC Acute Toxicity (IG Organisms) dominates the *ICI*, due to the application of the A-components mentioned above. At the output side, the impact group Water/Soil is most important, originating from ammonium (IC Eutrophication) and biomass, organic compounds, and organic solvent in the waste water (IC Organic Carbon Pollution Potential). A sensitivity analysis concerning the uncertain environmental relevance of an unspecified solvent can be found in Biwer and Heinzle [6.24].

6.6 Economic Assessment

The solvent-extraction process requires a total capital investment (TCI) of $ 85 million, while the estimated TCI of the electrodialysis process is $ 60 million. The most expensive equipment units are the bioreactors, crystallizer, and electrodialysis unit (electrodialysis process), or the extractor P-32 and the solvent recycling (solvent-extraction process). The difference between the processes is caused by the lower cost of the electrodialysis and the ultrafiltration units in the electrodialysis process, compared with the cost for extractors, solvent recycling, and ion exchanger in the solvent-extraction process. Furthermore, the lower product concentration after the extraction steps (larger volume of the product stream) requires a larger crystallizer in the solvent-extraction process.

The annual operating costs of $ 20 million for the solvent-extraction-based process and $ 14.5 million when electrodialysis is used give a unit production cost of $ 9.0/kg P, or $ 6.5/kg P respectively. Figure 6.5 shows the allocation of the operating cost in the processes. The facility-dependent cost is the dominating cost parameter. Raw materials, labor, and utilities also contribute substantially, while waste treatment and consumables have only a small influence. Glucose, acetic acid, and solvent cause most of the raw material cost. The lower UPC of the electrodialysis process results from a lower facility-dependant cost (lower TCI), lower raw material cost (no solvent and HCl), and lower utility costs (see Chapter 4). The higher consumable cost for the membranes in the ultrafiltration and electrodialysis steps do not outweigh these savings. Although the investment and labor costs were estimated largely based on the default values of SuperPro Designer® and therefore involve some uncertainty, the electrodialysis process seems to be economically favorable.

The annual production and hence the annual revenue are identical in both processes. For an assumed selling price of $ 20/kg P, the revenue is $ 45 million. This results in a return on investment (ROI) of 28% for the solvent-extraction process and 42% for the electrodialysis process. However, depending on the application, the selling price might vary significantly and might decrease with increasing market size and therefore influence the ROI.

6.7 Conclusions

Based on process and literature data a new process for the production of pyruvic acid was modeled, and two alternative downstream options were compared. With the selected model

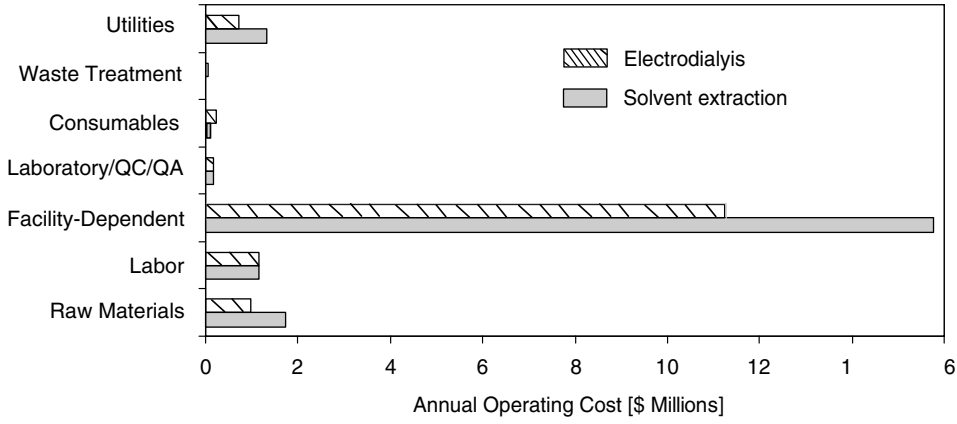


Figure 6.5 *Allocation of annual operating costs of solvent extraction and electrodialysis process*

parameters, the process using electrodialysis to separate the product shows a lower environmental impact of the components involved, lower energy consumption, lower capital investment, and lower unit production cost (UPC) compared with the process using two extraction steps. Although there are some uncertainties involved, one can expect that the electrodialysis process is economically and environmentally superior to the extraction process. However, independently of which process alternative is chosen, the biocatalyst needs further improvement to reduce the formation of unknown by-products. Together with further process optimization, a lower UPC may be obtained, enabling a long-term competitive process.

Suggested Exercises

1. The solvent used in the extraction process was not specified in the early phase of development and an average price for an organic solvent was assumed ($ 0.85/kg). Study the sensitivity of the unit production costs on the solvent price for a price range from $ 0.50/kg to $ 1.80/kg.
2. Assume that strain improvements resulted in an increase of final product yield in the production cycles 2–4 of the repeated fed-batch fermentation from 78 to 82% (P-20). Use the solvent-extraction model for this exercise. Make sure that the amount of sodium hydroxide in S-208 is sufficient to convert pyruvic acid into sodium pyruvate quantitatively. Watch stream compositions of bioreactor outlet (S-157) and re-extraction column effluent (S-213). What is the resulting overall product yield on glucose? How are the unit production costs affected? Is there a significant change in environmental performance?

References

[6.1] Li, Y., Chen, J., Lun, S. (2001): Biotechnological production of pyruvic acid. *Appl. Microbiol. Biotechnol.*, **120**, 451–459.
[6.2] Howard, J., Fraser, W. (1961): Pyruvic acid. *Org. Synth.*, 475–476.

[6.3] Bott, M.; Gerharz, T.; Takors, R.; Zelic, B. (2001): Process for pyruvate production by fermentation. German Patent Application 10 129 714.4.
[6.4] Gerharz, T.; Zelic, B.; Takors, R.; Bott, M. (2002): Processes and microorganisms for microbial production of pyruvate from carbohydrates and alcohols. German Patent Application 10 220 234.6.
[6.5] Zelic, B., Gostovic, S., Vuoriletho, K., Vasic-Racki, D., Takors, R. (2004): Process strategies to enhance pyruvate production by recombinant *Escherichia coli*: From repetitive fed-batch to ISPR with fully integrated electrodialysis. *Biotechnol. Bioeng.*, ***85***, 638–646.
[6.6] Biwer, A., Zuber, P., Zelic, B., Gerharz, T. Bellmann, K., Heinzle, E. (2005): Modeling and analysis of a new process for pyruvate production. *Ind. Eng. Chem. Res.*, **44**, 3124–3133.
[6.7] Sattler, K. (1995): Thermische Trennverfahren – Grundlagen, Auslegung, Apparate. VCH, Weinheim.
[6.8] Perry, R., Green, D., Maloney, J. (1997): Perry's chemical engineers' handbook. McGraw-Hill, New York.
[6.9] Uchio, R., Kikuchi, K., Enei, H.; Hirose, Y. (1976): Process for producing pyruvic acid by fermentation; US Patent 3 993 543.
[6.10] Kertes, A., King, C. (1986): Extraction chemistry of fermentation product carboxylic acids. *Biotechnol. Bioeng*., **28**, 269–282.
[6.11] McCabe, W., Smith, J., Harriott, P. (2001): Unit operations of chemical engineering. McGraw-Hill, New York.
[6.12] Katikaneni, S., Cheryan, M. (2002): Purification of fermentation-derived acetic acid by liquid-liquid extraction and esterification. *Ind. Eng. Chem. Res.*, **41**, 2745–2752.
[6.13] Prezhdo, O., Prezhdo, V., Nazarov, V. (1997): Effect of solvent nature on extraction of carboxylic acids. *Theor. Found. Chem. Eng*., **31**, 293–296.
[6.14] Benthin, S., Villadsen, J. (1995): Production of optically pure D-lactate by Lactobacillus bulgaricus and purification by crystallization and liquid/liquid extraction. *Appl. Microbiol. Biotechnol.*, **42**, 826–829.
[6.15] Weissermel, K., Arpe, H.-J. (1998): Industrielle organische Chemie – Bedeutende Vor- und Zwischenprodukte. Wiley-VCH, Weinheim.
[6.16] Zelic, B., Vasic-Racki, D. (2004): Process development and modeling of pyruvate recovery from model solution and fermentation broth. *Desalination*, **174**, 267–276.
[6.17] Pourcelly, G., Gavach, C. (2000): Electrodialysis water splitting – applications of electrodialysis with bipolar membranes (EDBM). In: Kemperman, A.: *Handbook of bipolar membrane technology*. Twente University Press, Twente, pp. 17–46.
[6.18] Kim, Y., Moon, S. (2001): Lactic acid recovery from fermentation broth using one-stage electrodialysis. *J. Chem. Technol. Biotechnol*., **76**, 169–178.
[6.19] Danner, H., Madzingaidzo, L., Thomasser, C., Neureiter, M., Biaun, R. (2002): Thermophilic production of lactic acid using integrated membrane bioreactor systems coupled with monopolar electrodialysis. *Appl. Microbiol. Biotechnol*., **59**, 160–169.
[6.20] Siebold, M., Rindfleisch, D., Schügerl, K., V. Frieling, P., Joppien, R., Röper, H. (1995): Comparison of the production of lactic acid by three different Lactobacilli and its recovery by extraction and electrodialysis. *Process Biochem*., **30**, 81–95.
[6.21] Novalic, S., Kulbe, K. (1998): Separation and concentration of citric acid by means of electrodialytic bipolar membrane technology. *Food Technol. Biotechnol.*, **36**, 193–195.
[6.22] Bauer, B., Holik, H., Velin, A. (2000): Cell equipment and plant design in bipolar membrane technology. In: Kemperman, A.: Handbook of bipolar membrane technology. Twente University Press, Twente, pp. 155–189.
[6.23] Min-tian, G., Hirata, M., Koide, M., Takanashi, H., Hano, T. (2004): Production of L-lactic acid by electrodialysis fermentation (EDF). *Process Biochem*., **39**, 1903–1907.
[6.24] Biwer, A., Heinzle, E. (2004): Environmental assessment in early process development. *J. Chem. Technol. Biotechnol*., **79**, 597–609.

Appendix 6.1 Bioreaction Model of the Pyruvic Acid Production

Pyruvic acid is produced in a bioreactor from glucose using *Escherichia coli* YYC202 *ldhA::Kan*. The fermentation is run as repeated fed-batch with four cycles. The first cycle includes biomass growth and product formation. At the end of each cycle, a part of the fermentation broth is removed. The biomass remains in the bioreactor, which is refilled with fresh medium. In cycles 2–4 no further acetate is added and pyruvic acid is produced with resting cells. For the bioreaction model the cycles 2–4 are lumped together.

1 Cycle 1

Known data from experiments:

- Start volume of laboratory bioreactor: 2.5 L, final volume: 3.78 L
- Volume of samples removed during cycle 1: 260 mL
- Bioreaction time: 13 h
- Acetate added: 2550 C-mmol
- CO_2 produced: 423 mmol
- Pyruvic acid: final concentration: 458 mmol/L; yield $Y_{\mathrm{Pyr/Gluc}} = 1.46$ mol/mol glucose consumed
- Biomass: Start concentration: 0.11 g/L; final concentration: 16.4 g/L
- Final concentrations: glucose: 3.5 g/L; acetate: 38 mmol (2.3 g/L)

The final concentrations of other media components are not known. The final concentrations and yields found in the experiments are used to estimate the performance of an industrial-scale bioreactor. Additionally, some assumptions are made:

- Ammonium, trace elements, and other media components are available in sufficient amounts during the bioreaction.
- The consumption of nitrogen and phosphorus is considered in the model, while the consumption of trace elements is excluded for simplification.
- The formation of fumarate, aspartate, glutamate, and alanine is neglected, because they account for less than 1% of the carbon balance.
- The final volume of the fermentation broth is 37.1 m^3 after cycle 1 (start volume: 24.6 m^3). For the given settings, this results in a final volume of 50 m^3 after cycle 4.
- The amount of glucose consumed is calculated from the final concentration of pyruvic acid and the yield coefficient. The amount of glucose added is derived from the amount of glucose consumed and the final glucose concentration after cycle 1.
- It is assumed that the acetate is completely consumed in an industrial fermentation (to avoid problems in the downstream processing). The final amount of acetate is set to zero. The amount actually remaining in the lab fermenter is subtracted from the starting amount: $2545 - 287 = 2258$ C-mol.
- The carbon balance is not completely closed. A number of by-products are obviously formed during the fermentation that are not measured and specified in detail. The carbon atoms that are not allocated to pyruvic acid, biomass, carbon dioxide, and unused glucose are summarized to the component ‘organic rest’. The average composition of the organic

Table 6.A1 *Input and output amounts for a bioreaction with a final volume of 37.1 m³*

Input (C-mol)		Output (C-mol)	
Glucose	74 217	Pyruvic acid	51 020
Acetate	22 177	CO_2	4155
Biomass	107	Biomass	24 029
Sum	96 501	Glucose	4328
		Sum	83 531
Difference = organic rest = 12 970		C-Balance (%)	86.6

rest is assumed to be identical to that of glucose. For the final volume of 37.1 m^3, the carbon balance is depicted in Table 6.A1.

The following amounts are added to the reactor: 2229 kg of glucose (12 370 mol), 666 kg of acetate (11 089 mol) and 2.7 kg of biomass (107 C-mol). The bioreaction results in the following output amounts: 1498 kg of pyruvic acid (17 006 mol), 609 kg of biomass (24 030 C-mol), 183 kg of carbon dioxide, and 130 kg of unused glucose (721 mol). The calculated amount of carbon dioxide is relatively small (4.3% of the carbon input). It might be that some of the carbon atoms that are allocated to the organic rest are actually converted into additional carbon dioxide.

1.1 Biomass Formation

Considering the input and output amounts of biomass (see above), 23 922 C-mol are formed in cycle 1. Table 6.A2 shows the contribution of the various precursors to the biomass formation of *Bacillus subtilis*. It is assumed that the data for *E. coli* are comparable. Furthermore, it is assumed that the precursors of glucose 6-phosphate to pyruvate (see Table 6.A2) are

Table 6.A2 *Contribution of the different precursor to the biomass of* Bacillus subtilis *[6.A1]. dcw = dry cell weight*

Precursor	Contribution (mmol/C-mol dcw)	Contribution (C-mmol/C-mol dcw)	Contribution (%)
Glucose 6-phosphate	3.9	23.6	2.3
Fructose 6-phosphate	4.9	29.1	2.8
Ribulose 5-phosphate	20.8	104	10.0
Erythrose 4-phosphate	7.9	31.4	3.0
Triose 3-phosphate	45.0	14.9	1.4
3-Phosphoglutarate	34.5	104	10.0
Phosphoenolpyruvate	18.2	54.6	5.3
Pyruvate	78.1	234	22.6
Sum (glucose)		595	57.4
Acetyl-CoA	54.4	109	10.5
Oxalacetate	49.1	196	18.9
α-Ketoglutarate	27.2	137	13.2
Sum (acetate)		442	42.6

derived from glucose (glycolysis; pentose phosphate path = PPP) while all other precursors are derived from acetate (tricarboxylic acid cycle). Thus, the contribution to the biomass formed is 57% (C-mol) for glucose and 43% for acetate.

The average biomass composition is initially assumed to be $CH_{1.8}O_{0.5}N_{0.2}$. The reaction equation for biomass formation from acetate or glucose is:

$$CH_2O + 0.2\ NH_3 + 0.1\ NADH_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + 0.5\ H_2O + 0.1\ NAD \tag{6.A1}$$

Considering the results from Table 6.A2, the equation is:

$$0.57\ CH_2O\ (Gluc.) + 0.43\ CH_2O\ (Ac.) + 0.2\ NH_3 + 0.1\ NADH_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + 0.5\ H_2O + 0.1\ NAD \tag{6.A2}$$

which is equivalent to:

$$0.095\ C_6H_{12}O_6 + 0.215\ C_2H_4O_2 + 0.2\ NH_3 + 0.1\ NADH_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + 0.5\ H_2O + 0.1\ NAD \tag{6.A3}$$

During the formation of ribulose 5-phosphate (R5P) and erythrose 4-phosphate (E4P) in the PPP, one mole of CO_2 is formed per mole of precursor. For all other precursors derived from glucose, there is no CO_2 formation. For the PPP the following, simplified equation is assumed:

$$C_6H_{12}O_6 + H_2O\ +\ 2\ NAD \rightarrow C_5H_{10}O_5 + CO_2 + 2\ NADH_2 \tag{6.A4}$$

For simplification, $NADH_2$ and $NADPH_2$ are lumped together. The carbon dioxide formation in the PPP can be described as following:

$$C_6H_{12}O_6 +\ 6\ H_2O +\ 12\ NAD \rightarrow 6\ CO_2 + 12\ NADH_2 \tag{6.A5}$$

which is equivalent to:

$$CH_2O +\ H_2O +\ 2\ NAD \rightarrow CO_2 + 2\ NADH_2 \tag{6.A6}$$

R5P and E4P account for 13% of the biomass (see Table 6.A2). Hence, 0.13 mol of CO_2 are formed per mol of biomass. Considering Equations (6.A4) and (6.A6), the reaction equation is:

$$0.7\ CH_2O\ (Gluc.) + 0.43\ CH_2O\ (Ac.) + 0.2\ NH_3 + 0.16\ NAD \rightarrow CH_{1.8}O_{0.5}N_{0.2} + 0.13\ CO_2 + 0.37\ H_2O + 0.16\ NADH_2 \tag{6.A7}$$

In the initial equation 0.1 mol $NADH_2$/mol biomass are consumed. However, in the PPP 0.26 mol/mol are formed, resulting in an overall formation of 0.16 mol/mol.

The phosphorus content of *E. coli* is 3% [6.A2]. To consider the phosphorus, a modified biomass composition is assumed: $CH_{1.8}O_{0.5}N_{0.2}P_{0.024}$. Potassium dihydrogen phosphate is used as P-source. For the phosphorus consumption a simplified equation is assumed (P-BM = phosphorus in biomass)

$$KH_2PO_4 \rightarrow KOH + \text{P-BM} + 0.5\ H_2O + 1.25\ O_2 \tag{6.A8}$$

$$0.024\ KH_2PO_4 \rightarrow 0.024\ KOH + 0.024\ \text{P-BM} + 0.012\ H_2O + 0.03\ O_2 \tag{6.A9}$$

Equations (6.A7) and (6.A9) are merged to a new biomass-formation equation:

$$0.7\ CH_2O\ (Gluc.) + 0.43\ CH_2O\ (Ac.) + 0.2\ NH_3 + 0.16\ NAD + 0.024\ KH_2PO_4 \rightarrow$$
$$CH_{1.8}O_{0.5}N_{0.2}P_{0.024} + 0.13\ CO_2 + 0.382\ H_2O + 0.16\ NADH_2 + 0.024\ KOH + 0.03\ O_2 \quad (6.A10)$$

In this equation, 0.16 mol of $NADH_2$ are produced per C-mol of biomass. It is assumed that it is oxidized in the respiratory chain. There, usually 2 mol of ATP are formed per mol of $NADH_2$; hence 0.32 mol/mol biomass. The reaction equation is:

$$NADH_2 + {}^1/_2\ O_2 \rightarrow H_2O + NAD \quad (6.A11)$$

Equations (6.A10) and (6.A11) are merged to a new biomass-formation equation:

$$0.7\ CH_2O\ (Gluc.) + 0.43\ CH_2O\ (Ac.) + 0.2\ NH_3 + 0.024\ KH_2PO_4 + 0.05\ O_2 \rightarrow$$
$$CH_{1.8}O_{0.5}N_{0.2}P_{0.024} + 0.13\ CO_2 + 0.542\ H_2O + 0.024\ KOH \quad (6.A12)$$

which is equivalent to:

$$0.116\ C_6H_{12}O_6 + 0.215\ C_2H_4O_2 + 0.2\ NH_3 + 0.024\ KH_2PO_4 + 0.05\ O_2 \rightarrow$$
$$CH_{1.8}O_{0.5}N_{0.2}P_{0.024} + 0.13\ CO_2 + 0.542\ H_2O + 0.024\ KOH \quad (6.A13)$$

Using Equation (6.A13), the biomass formation in cycle 1 is:

$$2775\ C_6H_{12}O_6 + 5143\ C_2H_4O_2 + 4784\ NH_3 + 574\ KH_2PO_4 + 1100\ O_2 \rightarrow$$
$$23\,922\ CH_{1.8}O_{0.5}N_{0.2}P_{0.024} + 3014\ CO_2 + 12\,966\ H_2O + 574\ KOH \quad (6.A14)$$

During biomass formation, 23 922 mol × 0.32 mol = 7250 mol ATP are formed. The reaction extent considered in the process model is 46% referred to the acetate added.

1.2 Energy Recovery and CO_2 Formation

The yield coefficient for energy consumption during biomass formation is assumed to be $Y_{X/ATP} = 0.35$ C-mol/mol [6.A3]. In cycle 1, 68 349 mol of ATP are required. 7655 mol of ATP are produced (see above). Thus, an additional amount of 60 694 mol is needed.

The overall formation of carbon dioxide is 4155 mol. 3014 mol are already considered in the biomass formation [Equation (6.A14)]. The remaining 1141 mol are formed during the oxidation of the acetate:

$$C_2H_4O_2 + 2\ O_2 \rightarrow 2\ CO_2 + 2\ H_2O \quad (6.A15)$$

$$570\ C_2H_4O_2 + 1141\ O_2 \rightarrow 1141\ CO_2 + 1141\ H_2O \quad (6.A16)$$

In the process model, the reaction extent is 8.7% referred to acetate. In the respiratory chain, 3 mol of $NADH_2$ and 1 mol of $FADH_2$ are formed per mol of acetate [6.A4]. Their oxidation results in the formation of 8 mol of ATP. In the tricarboxylic acid cycle, an

additional mol of ATP is formed per mol of acetate. Thus, 9 mol of ATP are formed per mol of acetate, in cycle 1 1141 × 9 = 10 270 mol of ATP overall.

1.3 Product Formation

Glucose is converted into pyruvic acid via glycolysis:

$$C_6H_{12}O_6 + O_2 \rightarrow 2\,C_3H_4O_3 + 2\,H_2O \tag{6.A17}$$

Overall, 17 006 mol of pyruvic acid are formed and 8503 mol of glucose are consumed. This equals a yield of 69% (C-mol/C-mol glucose), or 53% (C-mol/C-mol glucose + acetate). In the glycolysis, 2 mol of ATP are formed per mol of glucose (in cycle 1: 8503 × 2 = 17 006 mol) and also 2 mol of $NADH_2$ per mol of glucose [6.A4] (17 006 mol). Per mol of $NADH_2$, 2 mol of ATP are formed in the respiratory chain (in cycle 1: 17 006 × 2 = 34 013 mol of ATP). Thus, the overall ATP formation is 51 019 mol. The reaction extent of the product formation in the process model is 89% referred to the remaining glucose.

1.4 Organic Rest

Organic rest formed from glucose:

Added:	74 217 C-mol
Converted:	69 889 C-mol
Into pyruvic acid:	−51 019 C-mol
Into biomass/CO_2:	−16 650 C-mol
Into organic rest:	2220 C-mol

Since the average composition of the organic rest was assumed to be identical to that of glucose, the reaction equation is:

$$C_6H_{12}O_6 \rightarrow C_6H_{12}O_6 \tag{6.A18}$$

Organic rest formed from acetate

Added/consumed:	22 177 C-mol
Into biomass:	−10 287 C-mol
Into CO_2:	−1141 C-mol
Into organic rest:	10 749 C-mol

The reaction equation is:

$$3\,C_2H_4O_2 \rightarrow C_6H_{12}O_6 \tag{6.A19}$$

Considering Equations (6.A18) and (6.A19) and the amounts formed, the reaction equation is:

$$370\,C_6H_{12}O_6 + 5375\,C_2H_4O_2 \rightarrow 2162\,C_6H_{12}O_6 \tag{6.A20}$$

1.5 Maintenance

For the estimation of the energy demand for maintenance, the following parameter values are considered: $m_{ATP} = 0.002$ mol/g biomass h; amount of biomass at the beginning: 2.7 kg (0.11 g/L); at the end: 609 kg; arithmetic mean: 306 kg; fermentation time: 13 h. Hence, the ATP demand is: 305 850 × 13 × 0002 = 7952 mol.

1.6 ATP Balance

Reaction	ATP (mol)
Glycolysis	51 019
Acetate oxidation	10 270
Biomass formation	−60 694
Maintenance	−7952
Difference	−7357

This equals a shortage of 0.01 mol ATP/g biomass.

1.7 Oxygen Balance

Reaction	O_2 (kg)
Glycolysis	−72
Acetate oxidation	−272
Total	−344

In the model, 344 kg of oxygen are consumed (10 744 mol). Dunn et al. [6.A3] state that the average biomass yield coefficient is $Y_{x/o2} = 1$–2 C-mol/mol O_2. For the given biomass formation of 23 922 mol, the coefficient is $Y_{x/o2} = 2.2$ in the model. However, if one excludes the oxygen consumption in the product formation, the coefficient is $Y_{x/o2} = 11$. This very high value would be reduced if more acetate were oxidized to carbon dioxide.

2 Cycles 2–4

Cycles 2–4 are calculated separately, but summarized to one step in the process model. No further acetate is added in these cycles. Hence, further biomass formation does not take place and pyruvic acid is produced with resting cells. The model of cycles 2–4 is based on experiments in lab-scale fermenters. The basic data are given in Table 6.A3. Using the equations in Chapter 1 and the data in Table 6.A3, the reaction parameters are calculated for a 50 m^3 working volume (see Table 6.A4).

Table 6.A3 *Basic data for cycles 2–4 derived from experiments in lab fermenters*

Parameter	Cycle 2	Cycle 3	Cycle 4
Start volume (L)	3.8	4.6	4.9
Final volume (L)	4.1	4.9	5.1
Sample removal (L)	2.0	2.5	5.1
Medium added (L)	2.5	2.5	2.5
Final conc. pyruvic acid (mmol/L)	670	660	600
Yield $Y_{P/G}$ (mol/mol)	1.7	1.7	1.8
Final conc. glucose (g/L)	8.0	5.9	6.0
CO_2 produced (mmol)	152	101	84

Table 6.A4 *Reaction parameters of cycles 2–4 for the industrial-scale process*

Reaction balance	Cycle 2	Cycle 3	Cycle 4	Cycles 2–4
Glucose consumption (mol)	12 676	10 209	8 275	31 161
Glucose consumption (C-mol)	76 057	61 255	49 652	186 964
Pyruvic acid formation (mol)	21 676	17 560	14 730	53 966
Pyruvic acid formation (C-mol)	65 029	52 679	44 190	161 898
CO_2 formation (mol)	1493	992	825	3310
Organic rest (mol)	1589	1264	773	3626
Organic rest (C-mol)	9535	7583	4637	21 755
$Y_{(P/G)}$ (mol/mol)	1.71	1.72	1.78	1.73

2.1 Energy Recovery/CO_2 Formation

In cycle 1 it is assumed that carbon dioxide is formed during biomass formation and oxidation of acetate. Neither process takes place in the cycles 2–4. Instead, the carbon dioxide must originate from a glucose oxidation as described in Equation (6.A5). Usually, the oxidation of glucose in *E. coli* proceeds via glycolysis and the tricarboxylic acid cycle. However, in the strain used, the transfer of pyruvate to the tricarboxylic acid cycle is blocked. It is assumed that the CO_2 is mainly produced in the PPP and that the $NADPH_2$ formed can be oxidized in the respiratory chain.

References

[6.A1] Sauer, U., Hatzimanikatis, V., Hohmann, H., Manneberg, M. van Loon, A., Bailey, J. (1996): Physiology and metabolic fluxes of wild-type and riboflavin-producing *Bacillus subtilis*. *Appl. Environ. Microbiol.*, **62**, 3687–3696.

[6.A2] Doran, P. (1995): Bioprocess engineering principles. Academic Press, London.

[6.A3] Dunn, J., Heinzle, E., Ingham, J., Prenosil, J. (2003): Biological reaction engineering. Wiley-VCH, Weinheim.

[6.A4] Schlegel, H. (1995): Allgemeine Mikrobiologie. Thieme, Stuttgart.

7

L-Lysine – Coupling of Bioreaction and Process Model

*Arnd Knoll and Jochen Buechs**

7.1 Introduction

To date, optimization of chemical and biochemical processes is achieved by empirical or intuitive methods. Industrial research in particular is characterized by enormous pressure of time and costs. There is a need for a systematic method to evaluate whether or not expense of research in process optimization will be compensated for by a possible reduction in production costs. This creates a demand for methods that enable an estimation of the essential economic data of a process to be made in order to develop decision criteria for the general research strategy. Additionally, there is a need for fast and unerring methods for the determination of process optima.

Biochemical processes in particular are characterized by a large number of influential parameters and their variations. Empirical, not fully systematic, strategies are time- and cost-intensive. In this context, mathematical modeling of biotechnological systems has proven to be an efficient and modern tool to evaluate these bioprocesses. This case study presents methods for bioprocess development by modeling in an industrial environment and examines the advantages of this approach. It combines a dynamic model of the bioconversion, the core part of each bioprocess, with a model of the overall production process.

The production of L-lysine, a well established bioprocess, is used as a case study. Lysine is mainly used as an animal feed additive. The annual production is approximately 700 000 tons/year. A comprehensive overview of lysine production is given by Pfefferle et al. [7.1].

* Corresponding author: buechs@biovt.rwth-aachen.de, ++49/241/80-25546

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7.2 Basic Strategy

The strategy is based on the idea of coupling a dynamic model describing the bioreaction to a process model of the complete production plant. Because the reaction is integrated in a complex network of unit procedures, the model must consequently be capable of calculating the energy and mass balances as well as the cost factors of the entire process.

Whereas the first part can be performed by almost any numerical simulation tool, the second part is ideally carried out using a process-simulation tool. For the bioreaction model, considering the kinetics of the lysine production, Modelmaker® (Modelmaker Tools BV, Arnhem, The Netherlands) was used, while the process model was built using SuperPro Designer®. The aim of this strategy is to achieve information about what impact a process parameter of the bioreaction has on the economic or environmental objective functions of the complete production plant. This overall information is of greater interest than the optimization of a single unit procedure. The numerical simulation of the bioreaction alone gives no information about the impact that parameter variations have on the nature and economic efficiency of the overall process. The process model, however, requires data about the time and stoichiometry of the bioreaction. This information can be derived from experimental results or, as in this study, from simulation results.

7.3 Bioreaction Model

The first fundamental step is a mathematical description of the biological activities. Basic data usually emerge from a few preliminary experiments. The resulting model should be able to reflect the influence of relevant process parameters with adequate accuracy. A procedure serving to establish such a model is described by Büchs [7.2]. The present study makes use of an existing biological model [7.2] that is described in detail in the Appendix, and model parameters are presented in Table 7.A1. This model describes the production of the amino acid lysine in a fed-batch process by a mutant of the bacterium *Corynebacterium glutamicum*.

In wild type strains of *C. glutamicum*, lysine is usually feedback regulated by the amino acid threonine. Therefore it is assumed that the applied mutant is disrupted in threonine metabolism, resulting in a threonine auxotrophy. Even though such amino acid-auxotrophic mutants have disappeared from industrial lysine production [7.1], this simplified study is based on such a strain. The reason is that this particular strain characteristic is very suitable to explain the introduced optimization approach. Threonine is an essential component of proteins and has to be supplemented to the culture media in case of auxotrophic strains. Therefore the carbon flux into cell growth and lysine production, and further the stoichiometry of the applied microorganism's metabolism, can be controlled by the medium-specific process parameter 'initial threonine concentration' ($c_{\mathrm{Thr_IN}}$) [7.2]. Figure 7.1 shows three different pathways which carbon may take within general metabolism. Variation of these three carbon fluxes will alter the substrate yield.

In Equation (7.1) the glucose-consumption rates for lysine, growth, and cell maintenance are taken into account. The fluxes are characterized by the corresponding coefficients and

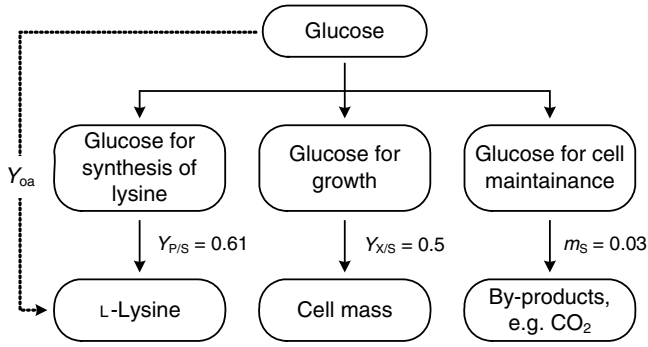


Figure 7.1 *Stoichiometry of glucose consumption for a lysine-producing strain of* C. glutamicum *[7.2]. For abbreviations see Nomenclature section. Reproduced by permission of Wiley-VCH*

describe the stoichiometry of the glucose consumption of lysine-producing microorganisms. For abbreviations see the Nomenclature section.

$$\frac{dc_S}{dt} = -\frac{1}{Y_{X/S}} \cdot \mu \cdot c_X - \frac{1}{Y_{P/S}} \cdot r_P \cdot c_X - m_S \cdot c_X \tag{7.1}$$

Once the biological model is established and validated, it can be used to simulate and optimize the bioreaction. Figure 7.2 shows a simulation of a fed-batch process of lysine production with an initial threonine concentration of 1.6 g/L as an example. After depletion of the glucose in the initial batch phase, a fed-batch phase is initiated where the feed rate is controlled by the dissolved oxygen value (DO_2). A DO_2 set point of 25% of air saturation was chosen. Owing to the limited oxygen-transfer capacities of conventional fermenters, the oxygen transfer limits the feed rate of fed-batch fermentations and therefore might extend the fermentation time [7.3, 7.4].

To account for a limited oxygen transfer in the fermenter, the start k_La value was assumed to be 1000 h^{-1}. The maximal oxygen-transfer rate at a dissolved-oxygen value of 25% is limited to a value of around 0.2 mol/L h. Figure 7.2 depicts the course of threonine, biomass, and lysine concentration during the process. As long as threonine is present in the medium, cell growth is favored, while only small amounts of lysine are produced. After the complete consumption of threonine, cell growth stops and lysine synthesis is enhanced. The biomass concentration decreases due to the dilution of the fermentation broth by the substrate feed stream. An important problem is specifying the optimization function. Figure 7.3 recapitulates the most influential input (left) and output parameters (right) for this process. The input parameters can be classified into three categories:

1. Strain-specific parameter, e.g. maximum specific growth rate, specific productivity, etc.
2. Medium-specific parameter, e.g. concentration of the main carbon source glucose and the initial threonine concentration.
3. Operation-specific parameter, e.g. dilution rate of a continuous fermentation, feed rate of a fed-batch fermentation.

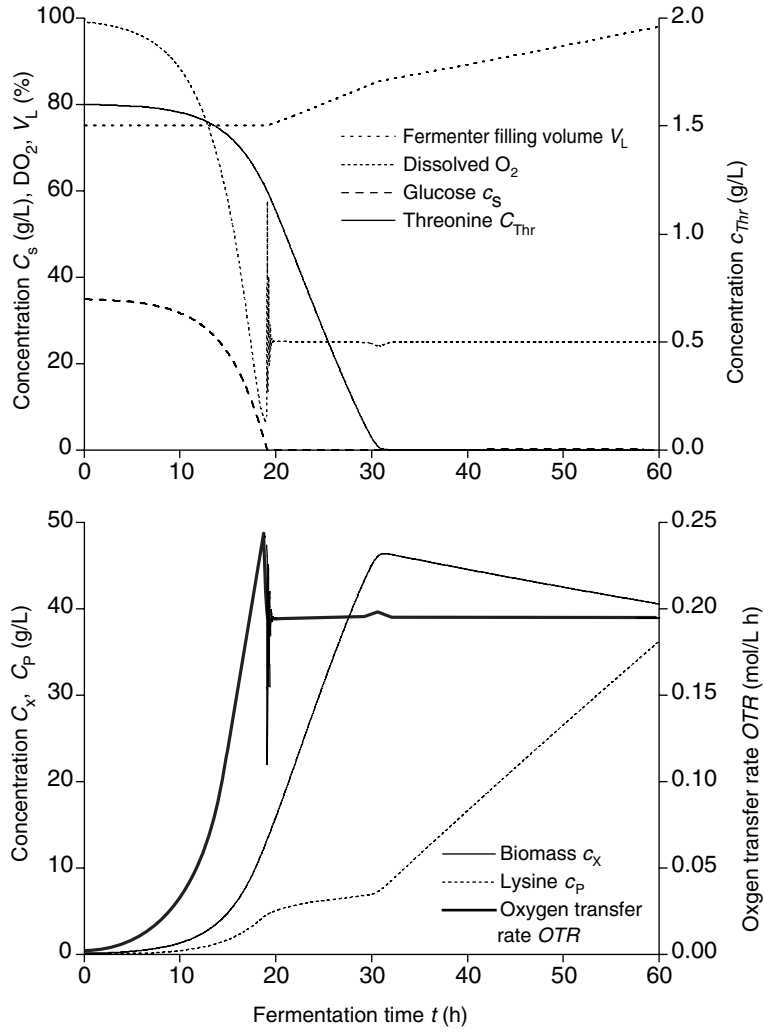


Figure 7.2 *Simulated kinetics of c_S, V_L, DO_2, OTR, c_P, c_X, c_{Thr} versus fermentation time for an example of an initial threonine concentration of 1.6 g/L for a process of lysine production in a glucose-fed-batch of 700 g/L substrate concentration in the feed*

The most important output parameters are the space-time yield *STY*, the overall yield Y_{oa}, and the product concentration c_P. These parameters can be obtained from the simulations of the bioreaction model. In this work, only three optimization parameters are considered. The question arises as to which of these parameters should be optimized. A purely intuitive optimization of the complete system is extremely difficult if not impossible to achieve.

To demonstrate this challenge, the initial threonine concentration c_{Thr_IN} as a medium-specific parameter may be chosen for variation. Figure 7.4a shows the final biomass concentration c_{X_END}, the final lysine concentration c_{P_END}, and the total fermentation time t_{END}

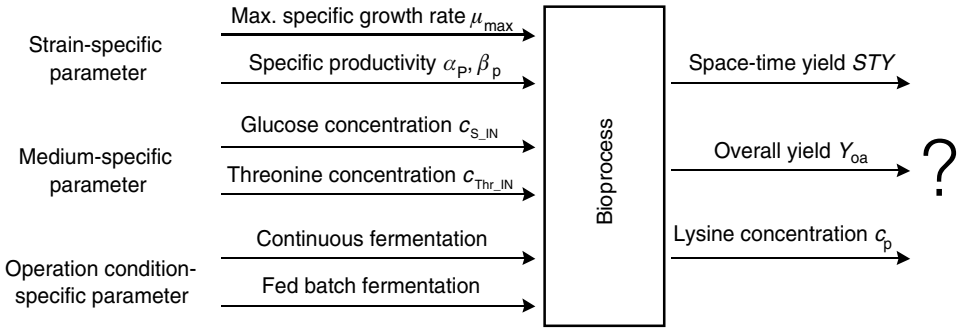


Figure 7.3 *Problem of assessment criteria for bioprocess optimization for the example of a lysine-producing strain of* C. glutamicum. *Reproduced by permission of Wiley-VCH*

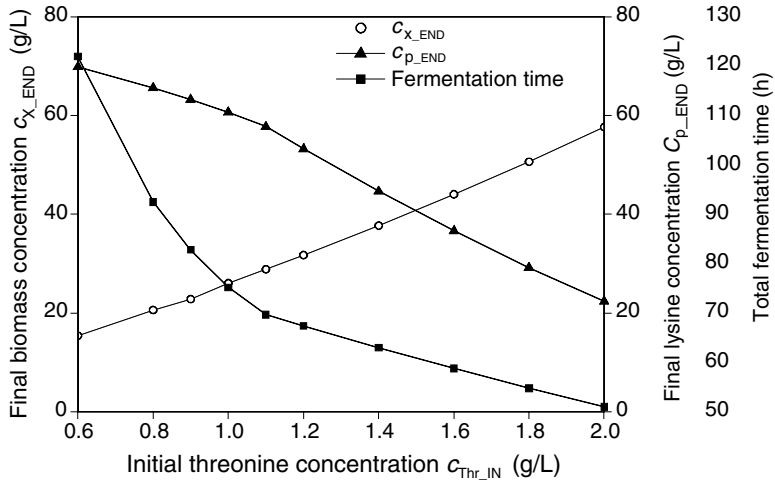
in the bioreaction model versus initial threonine concentration c_{Thr_IN} from 0.6 to 2.0 g/L. The final biomass concentration increases with increasing c_{Thr_IN}. The lysine concentration decreases because more glucose is consumed for the growth of biomass. Finally, the total fermentation time, being a very important process parameter with a strong influence on the process economy, decreases with increasing initial threonine concentrations.

In Figure 7.4b, the output parameters space-time yield and overall yield are plotted versus initial threonine concentration. The c_{Thr_IN} values for maximal values of these two parameters are not at all equal. This means that the highest Y_{oa} cannot be achieved together with the highest final space-time yield.

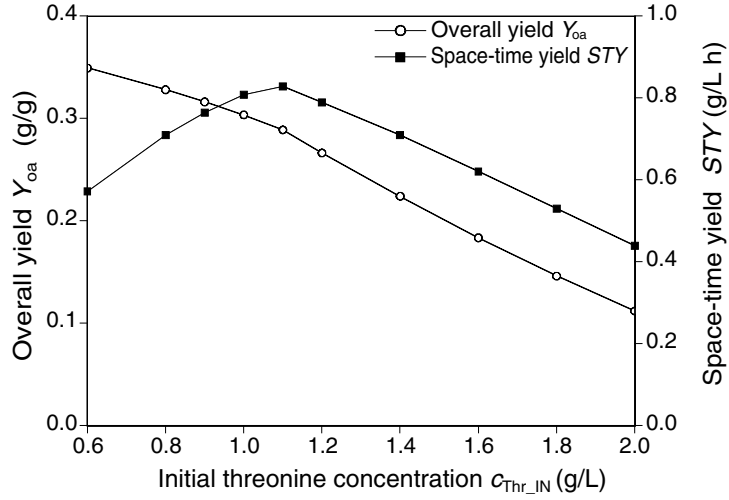
7.4 Process Model

The process model of an industrial L-lysine process is derived from Stevens [7.5]. It should be noted that details of recent lysine-production processes are neither published nor distributed by the lysine-producing companies. Therefore, this example represents neither the actual sequence of unit procedures nor the real manufacturing costs of the final product. The process flow diagram is shown in Figure 7.5.

The flow sheet consists of the fermentation and the downstream section. Since the desired final product is supposed to be used as a feed additive, the downstream section is kept very simple [7.1, 7.5]. In the first part of the process the required amounts of water, threonine, and nutrients are mixed in a blending tank (P-1, units of stirred tanks of maximum capacity 80 m^3). Additional water is added separately by a mixing unit (P-8) to lower the cost for P-1. As nutrients, only the C-source (glucose) and the P-source (KH_2PO_4) are considered. The N-source, NH_4OH, will be titrated into the fermenter during the reaction. The culture medium is sterilized through a heat sterilizer (P-2) passing the medium on to the fermenter unit (P-20) (units of stirred tank fermenters of maximum capacity 300 m^3). Since SuperPro Designer® accounts for a fixed stoichiometry of a reaction and a fixed total fermentation time, the fed-batch process is modeled as a batch process with an initial glucose concentration of 200 g/L. The mass balance of the elements C, O, P, and N in the bioreactor is derived from the stoichiometry from the bioreaction model and the elemental analysis of



(a)



(b)

Figure 7.4 *(a) Final biomass concentration and lysine concentration, and (b) overall yield and final space-time yield for initial threonine concentrations from 0.6 to 2.0 g/L for a process of lysine production with* C. glutamicum *in a fed-batch mode*

biomass and lysine. After the fermentation, the fermentation broth is transferred to a storage vessel (P-6). From this vessel, the broth is transferred to a rotary vacuum filter (P-10) to separate the biomass from the liquid. The permeate is transferred to an evaporation unit (P-23) that removes around 80% of the water content of the liquid stream. After storage in a vessel (P-12) the broth is spray dried and processed to granules in P-15. These granules are transferred to a storage tank (P-17).

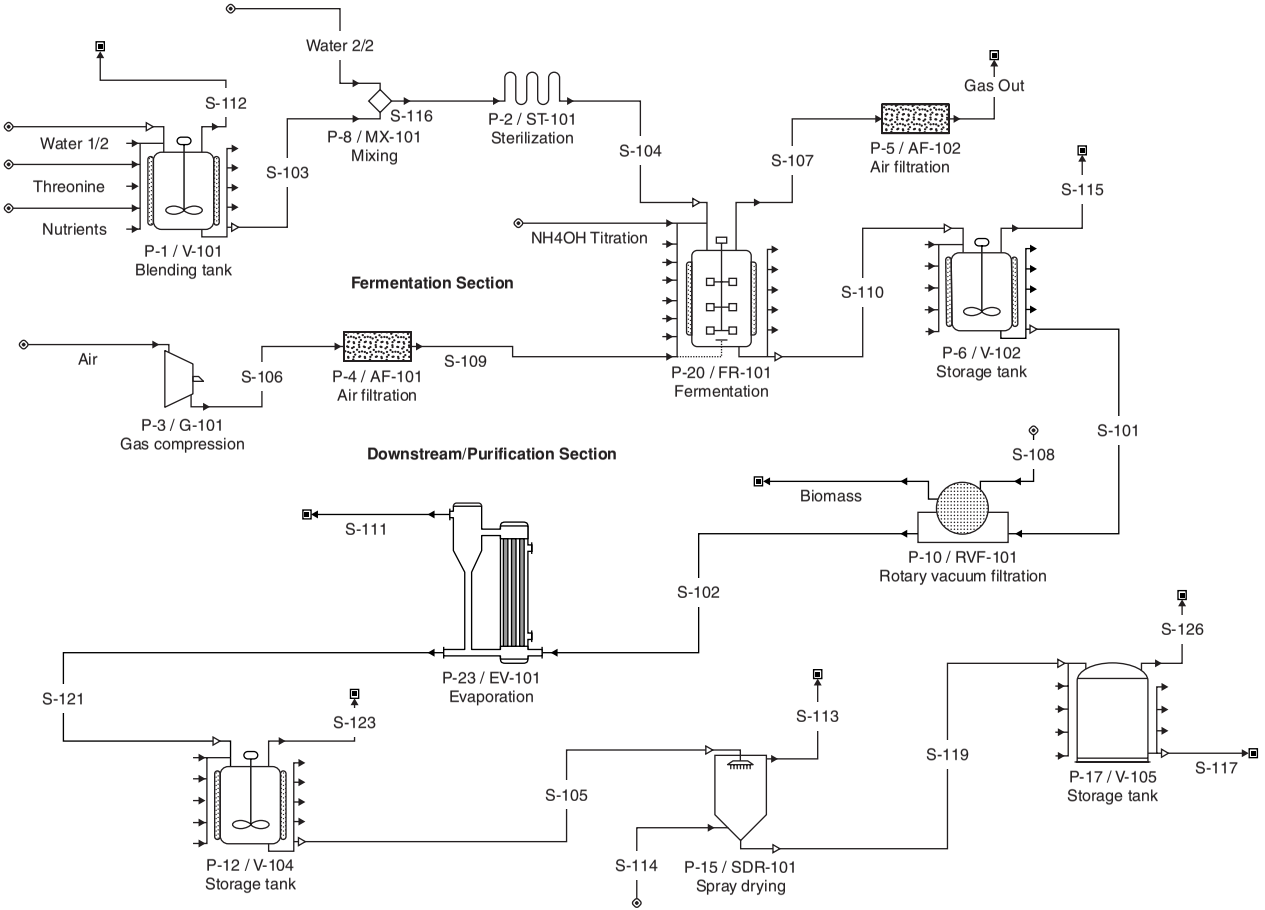


Figure 7.5 *Process flow diagram of a lysine-production plant*

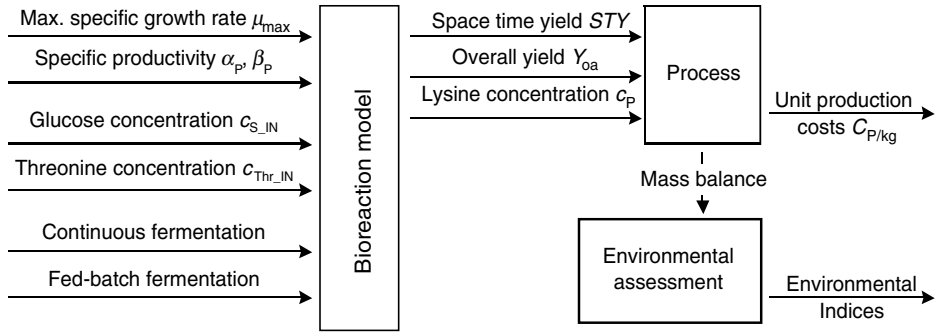


Figure 7.6 *Assessment criterion for bioprocess optimization derived from coupling the bioreaction model and process model of a lysine-producing strain of* C. glutamicum. *The reaction model is simulated with Modelmaker®-software; the process model is implemented in the simulation software SuperPro Designer®*

7.5 Coupling of Bioreaction and Process Model

An optimization of the complete process based only on the biochemical reaction model is insufficient. Figure 7.6 represents a general method to solve the problem of optimizing the complete process. The bioreaction model is coupled to the process model created in SuperPro Designer®.

In this case study, the bioreaction model as well as the process model are used as standalone systems with different features. The bioreaction model contributes information about the stoichiometry and time behavior of the biological reactions. The process model is able to calculate the energy and mass balances and the economic behavior (among many other parameters) for a complete production plant of a fixed reaction stoichiometry and fixed reaction time. Subsequently, the mass balance that is calculated in the process model can also be used for the environmental assessment. To investigate the influence of different input parameters of the bioreaction model on the process economic behavior (i.e. unit production costs) or on the environmental impact, the output parameters of the bioreaction model are transferred to the process model. For every c_{Thr_IN}, the stoichiometry and total fermentation time (t_{END}) are calculated with the bioreaction model. A mass-based stoichiometry of the presented process of lysine production derived from the bioreaction model is demonstrated in Equation (7.2) for the example of $c_{Thr_IN} = 1.1$ g/L.

$$\begin{aligned} &200\text{ g Glucose} + 3.8\text{ g } KH_2PO_4 + 37.8\text{ g } NH_4OH + 1.1\text{ g Threonine} \rightarrow \\ &28.9\text{ g Biomass} + 57.7\text{ g Lysine} + 30\text{ g Water} + 126.1\text{g } CO_2 \end{aligned} \tag{7.2}$$

These parameters are used in the process model of the production plant. From this, production costs or environmental impact can be calculated. The calculations are conducted following a user-defined stoichiometry of the biochemical or chemical reactions as well as a user defined total fermentation time. A variation of a process parameter (such as c_{Thr_IN}) that effects the stoichiometry and therefore the amount of product per batch, as well as the

total fermentation time, will determine the mass output of the plant. Since a production plant is usually designed for a certain amount of product per year, this has to be considered for comparative studies or scenario analysis. If the equipment size of the unit procedures is not adapted to stoichiometric changes of the process, the annual process throughput will change. To maintain a constant annual production, the throughput-adjustment tool of SuperPro Designer® is applied to automatically re-adapt the annual production with varying bioreaction variables. A higher product titer, for example, would result in a smaller size and, therefore, lower cost for the fermentation unit. For the present example, the annual process throughput was fixed to 6000 metric tons of lysine. This value represents around 1% of the estimated worldwide demand of lysine per year [7.6]. The optimization procedure is now demonstrated for the above discussed process model of a lysine-production plant.

7.5.1 Assumptions

(i) Type and sequence of unit procedures are considered as known and constant in this example. The process simulator allows this to be altered if desired.
(ii) For simplification, trace elements and other nutrients are not considered, but only C, N, and P assimilation for biomass and lysine synthesis are taken into account.
(iii) Product losses of the different unit procedures in the downstream processing are considered as constant.

For the biomass a C-content of 50% w/w is assumed. The P- and N-content of biomass of *C. glutamicum* is around 3% w/w and 14% w/w, respectively [7.7]. For N consumption, the N-content of lysine is also considered. One mole of lysine contains two moles of N. Nitrogen is provided as NH_4OH with a molecular weight of 35 g/mol. The molecular weight of lysine is 146 g/mol. Hence, for the synthesis of 1 g of lysine, 0.48 g of NH_4OH is required.

As a result of the bioreaction model, the reaction stoichiometry for a certain initial threonine concentration, as demonstrated in Equation (7.2), has to be adjusted in the Reaction Stoichiometry module of the process model. This stoichiometric equation is part of the reaction operation within the unit procedure Fermentation. Starting from a base case process model, several new scenarios were created, each including the resulting reaction stoichiometry for initial threonine concentrations from 0.6 to 2.0 g/L in steps of 0.2 g/L. This led to eight different scenarios of the same plant, which were used for the economic and environmental assessment. Having adjusted the annual process throughput to 6000 metric tons per year, the process model can be solved. Mass and energy balances for the process are solved and the required equipment size of each unit procedure is calculated.

Before deriving the information about economic and environmental impacts of each scenario, another aspect has to be considered. Regardless of the amount of lysine or biomass to be produced, glucose is assumed to be completely consumed during fermentation. However, the amount of lysine and biomass determines the amount of N- and P-source consumed during fermentation. The amount of N- and P-source influences both the economic calculations and the environmental assessment. The addition of NH_4OH and KH_2PO_4 was adjusted such that a residual amount of 5 kg per fermentation was registered in every batch. To enable the comparison of each scenario, the residual values have to be equalized. After properly adjusting these parameters, the mass and energy balances can be solved again.

7.6 Results and Discussion

For the environmental assessment, only the Environmental index EI_{Mult} is applied for the output components in this study. Components under consideration are glucose, threonine, NH_4OH, KH_2PO_4, biomass, lysine loss, and CO_2. The EI_{Mult}, $C_{P/kg}$, *STY* and Y_{oa} values are plotted versus the optimization parameter c_{Thr_IN} in Figure 7.7.

Figure 7.7 represents the results of coupling the bioreaction model to the process model. In addition to overall yield and space-time yield, the production costs and EI_{Mult} Output is plotted versus initial threonine concentration. The unit-production cost decreases to a minimum of \$ 4.9/kg L-lysine at an initial threonine concentration of 0.9 g/L and rises with increasing initial threonine concentrations up to \$ 12/kg L-lysine at c_{Thr_IN} of 2.0 g/L.

Within the investigated range of initial threonine concentrations, the EI_{Mult} rises continuously from 4.1 index points/kg P to 13.1 index points/kg P. The increasing *EI* with increasing c_{Thr_IN} is mainly caused by an increase of biomass and CO_2 at higher c_{Thr_IN}, which are considered as waste compounds. The more that glucose is converted into biomass and CO_2 instead of lysine, the higher is the EI_{Mult}. The difference between the minimum EI_{Mult} Output at 0.6 g/L threonine and the EI_{Mult} Output at 0.9 g/L where the unit-production cost shows its minimum is very small. Thus, running the process under the economically optimal conditions also results in an environmental impact near the minimum.

As seen in Figure 7.7, the minimal production costs are not achieved at the same c_{Thr_IN} as the highest final space-time yield (*STY*) or overall lysine yield (Y_{oa}). It can be concluded that overall yield or space-time yield are not the most appropriate process parameters to be optimized in this case study.

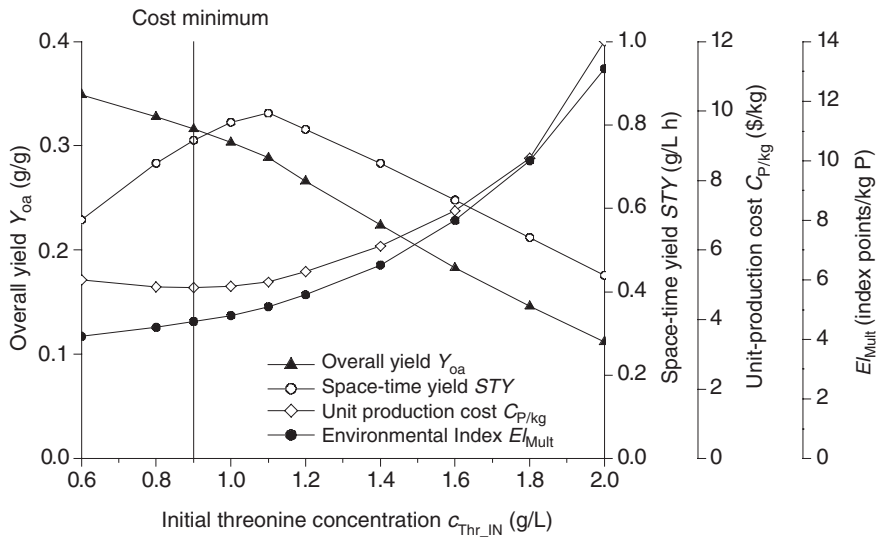


Figure 7.7 *Overall yield, final space-time yield, unit-production cost, and environmental index (EI_{Mult} Output) versus the initial threonine concentration from 0.6 to 2.0 g/L for a process of lysine production with* C. glutamicum *in a fed-batch mode*

Owing to the application of a threonine auxotroph mutant of *C. glutamicum,* the achieved minimal production costs of $ 4.9/kg L-lysine might no longer be the state of the art. Regarding the limited knowledge used in this case equivalent to an early stage of process development, however, the achieved production costs coincide quite well with real industrial prices. While Festel et al. [7.8] published production cost of $ 1–2/kg L-lysine for an unspecified sequence of unit procedures and quality of product, the calculated value from this example is in the same order of magnitude.

Suggested Exercises

1. The purchase equipment cost of the process is calculated using the built-in model of SuperPro Designer®. Naturally, this provides only a first estimate. For a bioreactor (P-20), you have obtained a quotation of $ 2 million (FoB) for the desired volume of 550 m^3. Assume that the transport of a reactor to the facility costs $ 150 000. Set the new price (Equipment Data of P-20) and study how capital investment and operating costs vary.
2. The first step of the downstream processing is the removal of the biomass. In the base case, rotary vacuum filtration is assumed. Model an alternative process setting with a disk-stack centrifuge that removes 98% of the biomass and a subsequent ultrafiltration that retains the remaining biomass. First, use the default values for the parameters in the newly added unit operations. Compare the economic performance with the base case. Then, go through the two unit models and check if the default values are realistic for this case. Make reasonable changes and watch the effect on the process performance.

References

[7.1] Pfefferle, W., Moeckel, B., Bathe, B., Marx, A. (2003): Biotechnological manufacture of lysine. *Adv. Biochem. Eng. Biotechnol.*, **79**, 59–112.

[7.2] Büchs, J. (1994): Precise optimization of fermentation processes through integration of bioreaction and cost models. In: Ghose T: Process computations in biotechnology. McGraw-Hill, New Delhi, pp. 194–237.

[7.3] Riesenberg, D., Guthke, R. (1999): High-cell-density cultivation of microorganisms. *Appl. Microbiol. Biotechnol.*, **51**, 422–430.

[7.4] van Hoek, P., de Hulster, E., van Dijken, J., Pronk, J. (2000): Fermentative capacity in high-cell-density fed-batch cultures of baker's yeast. *Biotechnol. Bioeng.*, **68**, 517–523.

[7.5] Stevens, J., Binder, T. (1999): Process for making granular L-lysine. US Patent US 005 990 350A.

[7.6] Kennerknecht, N., Peters-Wendisch, P., Eggeling, L., Sahm, H. (2003): Metabolic Engineering: Entwicklung von Bakterienstraengen zur Lysinproduktion. *BIOspektrum*, **5**, 582–585.

[7.7] Büchs, J. (1988): Immobilisierung von aeroben Mikroorganismen an Gassintermaterial am Beispiel der L-Leucin-Produktion von *Corynebacterium glutamicum*. PhD thesis, TU Hamburg-Harburg.

[7.8] Festel, G., Knoell, J., Goetz, H., Zinke, H. (2004): Impact of biotechnology production processes in the chemical industry. *Chem. Ing. Technol.*, **76**, 307–312.

Nomenclature

c_L = oxygen concentration (g/L)
c_P = product concentration (g/L)
c_{P_END} = final product concentration (g/L)
$C_{P/kg}$ = unit-production cost ($/kg)
c_S = substrate concentration (g/L)
c_{SF} = substrate concentration in the feed (g/L)
c_{S_IN} = initial substrate concentration (g/L)
c_{Thr} = threonine concentration (g/L)
c_{Thr_IN} = initial threonine concentration (g/L)
c_X = biomass concentration (g/L)
c_{X_END} = final biomass concentration (g/L)
c_{X_IN} = initial biomass concentration (g/L)
DO_2 = dissolved oxygen (%)
EI_{Mult} = Environmental Index (index points/kg P)
F = rate of feed (feed rate) (L/h) or (m^3/h)
K_La = specific mass transfer coefficient (1/h)
K_{IP} = product inhibition constant (g/L)
K_{IThr} = threonine inhibition constant (g/L)
K_O = substrate oxygen affinity constant (g/L)
K_{PS} = product affinity constant (g/L)
K_S = substrate carbon source affinity constant (g/L)
K_{Thr} = substrate threonine affinity constant (g/L)
L_{O_2} = oxygen solubility (mol/L/bar)
m_O = specific oxygen consumption for maintenance (g/L)
m_S = specific substrate consumption for maintenance (g/L)
OTR = oxygen transfer rate (mol/L h)
P_R = reactor pressure (bar)
r_P = rate of lysine production (g/L h)
STY = space time yield (g/L h)
t = time
t_{END} = total fermentation time (h)
V = fermenter filling volume (m^3)
V_L = fermenter filling volume (%)
y_L = mole fraction of oxygen in the liquid phase (mol/mol)
y_{O_2} = mole fraction of oxygen in the gas phase (mol/mol)
Y_{oa} = overall yield (g/g)
$Y_{P/O}$ = product yield per amount of oxygen (g/g)
$Y_{P/S}$ = product yield per amount of substrate (g/g)
$Y_{X/S}$ = biomass yield per amount of substrate (g/g)
$Y_{X/O}$ = biomass yield per amount of oxygen (g/g)
$Y_{X/Thr}$ = biomass yield per amount of threonine (g/g)
α_P = growth-associated coefficient for product synthesis (g/g)
β_P = non-growth-associated coefficient for product synthesis (g/g h)
μ = specific growth rate (1/h)
μ_{max} = maximum specific growth rate (1/h)

Appendix 7.1 Biochemical Reaction Model for a Fed-Batch Fermentation to Produce Lysine

Mass balance for glucose

$$\frac{dc_S}{dt} = -\frac{1}{Y_{X/S}} \cdot \mu \cdot c_X - \frac{1}{Y_{P/S}} \cdot r_P \cdot c_X - m_S \cdot c_X + \frac{F}{V}(c_{SF} - c_S)$$

Mass balance for oxygen $\frac{dc_L}{dt} = -\frac{1}{Y_{X/O}} \cdot \mu \cdot c_X - \frac{1}{Y_{P/O}} \cdot r_P \cdot c_X - m_O \cdot c_X$

Mass balance for threonine $\frac{dc_{Thr}}{dt} = -\frac{1}{Y_{X/Thr}} \cdot \mu \cdot c_X - \frac{F}{V} \cdot c_{Thr}$

Mass balance for biomass $\frac{dc_X}{dt} = \mu \cdot c_X - \frac{F}{V} \cdot c_X$

Mass balance for lysine $\frac{dc_P}{dt} = r_P \cdot c_X - \frac{F}{V} \cdot c_P$

Mass balance for the fermenter volume $\frac{dV}{dt} = F$

Kinetic model for oxygen transfer $OTR = k_L a \cdot L_{O_2} \cdot p_R \cdot (y_{O_2} - y_L)$

Kinetic model for growth $\mu = \mu_{max} \frac{c_S}{c_S + K_S} \cdot \frac{c_L}{c_L + K_O} \cdot \frac{c_{Thr}}{c_{Thr} + K_{Thr}}$

Kinetic model for lysine formation

$$r_P = (\alpha_P \cdot \mu + \beta_P) \cdot \frac{c_S}{c_S + K_{PS}} \cdot \frac{c_L}{c_L + K_O} \cdot \frac{K_{IThr}}{c_{Thr} + K_{IThr}} \cdot \frac{K_{IP}}{c_P + K_{IP}}$$

Overall yield $Y_{oa} = \frac{c_P}{c_{S_IN}}$

Space-time yield $STY = \frac{c_P}{t}$

Table 7.A1 *Parameters of the biological model taken from [7.2]*

Stoichiometric constants	Value	Unit
$Y_{X/S}$	0.52	(g/g)
$Y_{P/S}$	0.6	(g/g)
$Y_{X/Thr}$	33	(g/g)
$Y_{P/O}$	4.11	(g/g)
$Y_{X/O}$	1.29	(g/g)
Monod constants		
μ_{max}	0.28	(1/h)
K_O	6.4×10^{-6}	(g/L)
K_S	0.1	(g/L)
K_{Thr}	0.1	(g/L)
K_{PS}	0.072	(g/L)
L_{O_2}	0.00118	(mol/L/bar)
m_O	0.036	(g/L)
m_S	0.034	(g/L)
α	0.2	(g/g)
β	0.043	(g/g h)
c_{S_IN}	50	(g/L)
c_{X_IN}	0.1	(g/L)

8

Riboflavin – Vitamin B_2

Wilfried Storhas and Rolf Metz*

8.1 Introduction

Riboflavin was first isolated by Blyth in 1879 from whey, and the water-soluble, yellow, fluorescent material was named lactochrome. Kuhn and Karrer first synthesized riboflavin in 1935 [8.1]. According to IUPAC rules, riboflavin [83-88-5] is called 7,8-dimethyl-10-(D-1′-ribityl)isoalloxazine, also known as vitamin B_2 or lactoflavin. Riboflavin is an essential vitamin required for the synthesis of flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD) which are essential coenzymes required for the functioning of more than 100 flavoproteins. Riboflavin is thus involved in various redox and energy-delivering oxidation processes in cells. The daily human demand for riboflavin is around 1.7 mg, and deficiencies lead to various symptoms such as, e.g., versions of dermatitis. The vitamin cannot be stored in the body and a constant intake is required. Green plants, most bacteria, and moulds, however, can produce their own riboflavin. Riboflavin is used as an additive to soft drinks and yogurt, but 80% of the worldwide annual production of more than 3000 t/year is used in animal feed, mainly for poultry and pigs [8.2].

Chemical synthesis was the first production method to be established and is still dominating, but in recent years the production is shifting more and more to fermentation [8.2]. At present, three organisms are used for the industrial production of riboflavin by fermentation: the filamentous fungus *Ashbya gossypii* (BASF, Germany), the yeast *Candida famata* (ADM, USA), and a genetically engineered strain of *Bacillus subtilis* (DSM, Germany).

* Corresponding author: w.storhas@t-online.de, ++49/621/292 6494

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8.2 Biosynthesis and Fermentation

The biosynthesis of one riboflavin molecule requires one molecule of guanosine triphosphate (GTP) and two molecules of ribulose 5-phosphate [8.3]. The biosynthesis starts with GTP as depicted in Figure 8.1. In the first step GTP cyclohydrolase II converts GTP (**1**) into 2,5-diamino-6-ribosylamino-4(3*H*)-pyrimidinone 5′-phosphate (**2**), and simultaneously formate and pyrophosphate are released. 5-Amino-6-(5′-phosphoribitylamino)uracil (**5**) is produced via alternative pathways by deamination and side-chain reduction. The route proceeds via (**4**) in fungi and via (**3**) in bacteria. The dephosphorylated compound 5-amino-6-ribitylamino-2,4(1*H*,3*H*)-pyrimidinedione (**6**) is converted into 6,7-dimethyl-8-(1-D-ribityl)lumazine (**9**) by condensation with 3,4-dihydroxy-2-butanone 4-phosphate (**8**) derived from ribulose 5-phosphate (**7**). Dismutation of the lumazine derivative yields

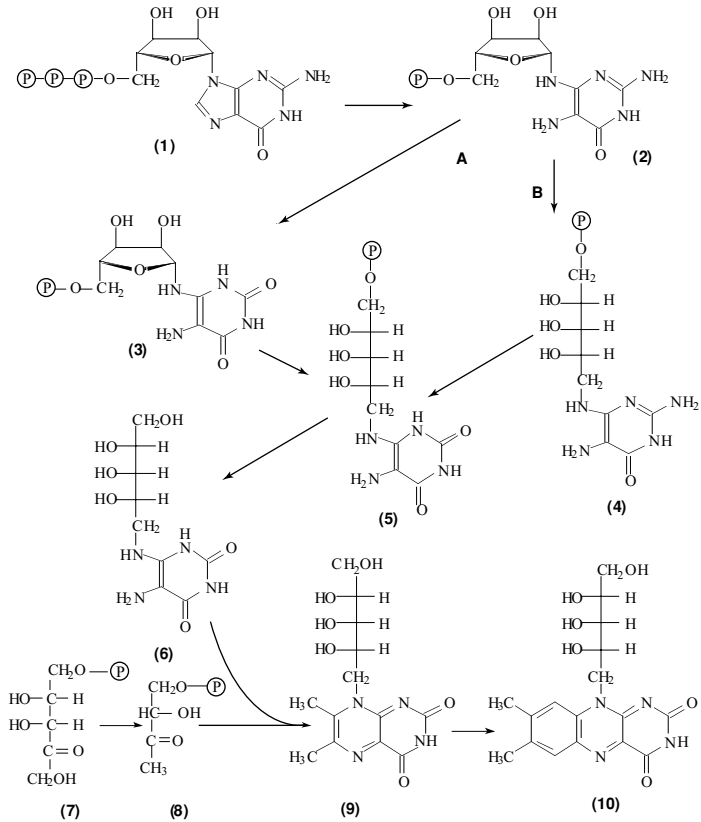


Figure 8.1 *Biosynthesis of riboflavin.* ***1*** *– guanosine 5′-triphosphate (GTP);* ***2*** *– 2,5-diamino-6-ribosylamino-4(3H)-pyrimidinone 5′-phosphate;* ***3*** *– 5-amino-6-ribosylamino-2,4(1H,3H)-pyrimidinedione 5′-phosphate;* ***4*** *– 2,5-diamino-6-ribitylamino-4(3H)-pyrimidinone 5′-phosphate;* ***5*** *– 5-amino-6-ribitylamino-2,4(1H,3H)-pyrimidinedione 5′-phosphate;* ***6*** *– 5-amino-6-ribitylamino-2,4(1H,3H)-pyrimidinedione;* ***7*** *– ribulose 5-phosphate;* ***8*** *– 3,4-dihydroxy-2-butanone 4-phosphate;* ***9*** *– 6,7-dimethyl-8-(1-D-ribityl)lumazine;* ***10*** *– riboflavin*

riboflavin (**10**) and 5-amino-6-ribitylamino-2,4(1*H*,3*H*)-pyrimidinedione (**6**), which is recycled in the biosynthetic pathway.

Production of riboflavin by *A. gossypii* and *C. famata* can be stimulated by feeding precursors, especially purine derivates, e.g. hypoxanthine. Fermentation media consist of glucose, corn steep liquor, saccharose, and maltose as carbon and nitrogen sources. The use of lipids in the media increases the yield. A further increase of riboflavin productivity can be achieved by adding peptone, glycine, and yeast extract. Important for riboflavin production is an optimized sterilization procedure, which is best carried out continuously [8.4].

The organism used in this case study is *Eremothecium ashbyii*, a strain closely related to *A. gossypii*. *E. ashbyii* is, however, genetically not very stable. Therefore fermentation has to be carried out batch-wise starting from fresh stock each time. An additional reason for batch operation is that riboflavin production occurs only in the stationary phase after growth of *E. ashbyii* has slowed down.

8.3 Production Process and Process Model

In this case study, a batch process using *E. ashbyii* with a capacity of around 1000 metric tons a year is analysed [8.5]. Smaller-scale production is very unlikely to be economically competitive. Important reaction parameters and the medium composition are listed in Table 8.1. The data originate from laboratory experiments and pilot plant data, where the complete process was developed and tested. Upstream processing consists of preparation of medium and associated continuous counter-current sterilization (Figure 8.2). Feed components are: 70% glucose syrup, yeast and malt extract, sunflower oil, sulfuric acid, and concentrated salt solution at room temperature. Fermentation is operated batch-wise with

Table 8.1 *Data for process design, reaction parameters, and medium [8.1]*

Parameter	Value	Component	(g/L)	(€/kg)
Temperature	30 °C	Glucose	5.0	1.00
pH, controlled (PIC)	6.5	Peptone	5.0	9.00
Pressure, uncontrolled	1 bar	Yeast extract	5.0	4.00
DO, dissolved oxygen	free	Malt extract	5.0	4.00
Aeration rate (FIC)	0.30 vvm	$MgSO_4 \cdot 7H_2O$	0.2	1.50
Power input (NIC)	0.80 W/kg	K_2HPO_4	0.2	1.50
Product concentration	27 g/L	Sunflower oil	15.0	0.55
Cell concentration	22 g/L	H_2SO_4 (1N*)	2.0	0.10
Fermentation time	500 h	Methionine	0.4	35.00
Preparation time	15 h			
Inoculum ratio	10%			
Reaction rate (netto)	54 mg/L h			
Yield (downstream)	0.80			
Reaction heat	2 W/L			
Growth rate	0.28 h^{-1}			

*For neutralization, pH-control.

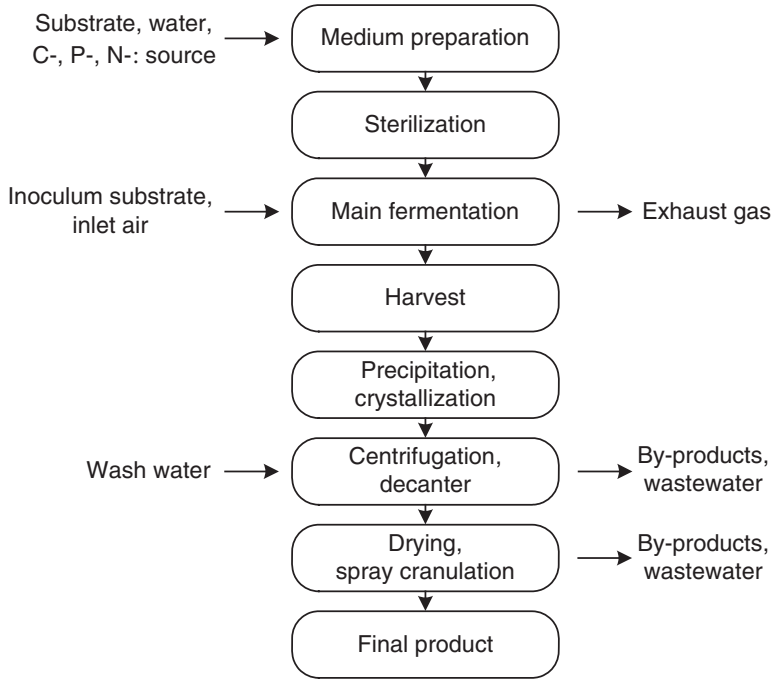


Figure 8.2 *Process flow diagram for riboflavin production by fermentation [8.2]*

10% inoculum ratios (Table 8.1). Downstream processing starts with harvesting followed by crystallization, centrifugation (decanter), and final drying (spray dryer) [8.4]. The requested purity of riboflavin is 70%. The residual 30% consists of salts and biomass. The product is obtained as dry powder or as granulate.

8.3.1 Upstream Processing

The upstream processes include preparation and sterilization of the medium (see Figure 8.3). The medium's composition, as specified in Table 8.1, does not allow sterilization of all components mixed together and using classical batch conditions (121 °C, 20 minutes). Under these conditions carbon and nitrogen sources would cause Maillard reactions, which destroy media components and produce side products, which may be strong inhibitors. Therefore, the medium would be divided into several groups, i.e. (i) glucose and sunflower oil, (ii) peptone, yeast and malt extracts, (iii) salts in water, and (iv) methionine. The latter is sterilized by filtration. Sulfuric acid does not require sterilization. Sterilization is drastically improved by applying continuous operation as is used in the case study (Figure 8.3). Only two separate solutions have to be prepared, 70% glucose (P-1) and other nutrients (Nutrients Tank / P-4). These are sterilized continuously and pass directly into the fermenter (P-9). Optimal conditions with respect to temperature and time could be chosen. These are generally combinations of high temperature and short incubation, i.e. $T = 140$ to 150 °C, respectively, with a holding time of down to 10 s. Heating and cooling phases

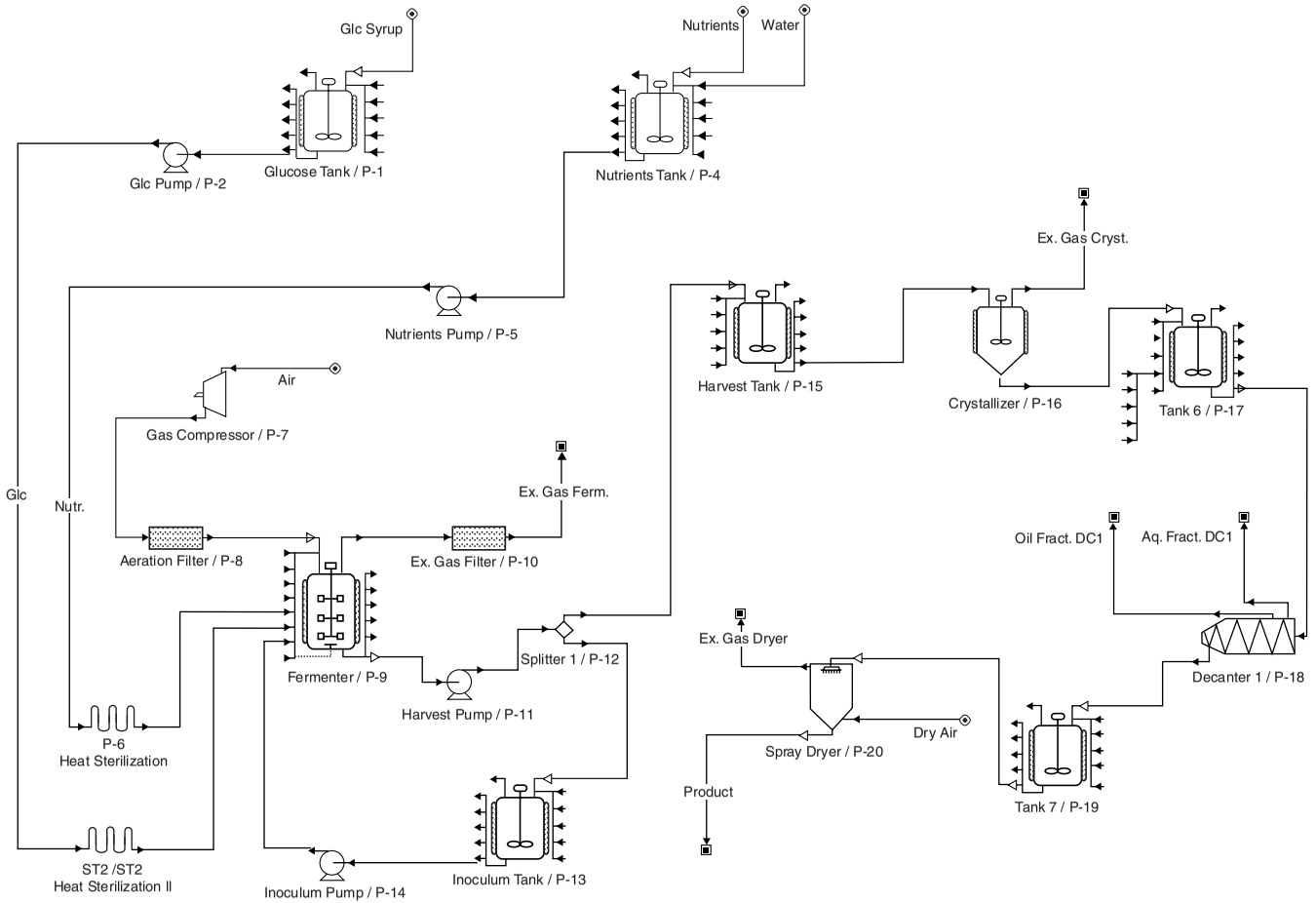


Figure 8.3 *Flowsheet of the process model for riboflavin production using* E. ashbyii

are kept as short as possible. On a production scale, heating and cooling periods of 30 to 50 seconds are possible for temperature increases of 100 °C. Optimal sterilization helps increasing yield and therefore also increases profit.

8.3.2 Fermentation

In several steps the necessary seed cultures are prepared in different seed fermenters, increasing in capacity by a ratio of 10:1. The last seed culture is the start inoculum for the main fermentation. The duration of a seed-fermentation is around 50 hours, while the main fermentation lasts about 500 hours. During this time the strain produces 27 g/L riboflavin. Fermentation requires aeration accomplished by a gas compressor (P-7) and a sterile filter (P-8). Exhaust gases are filtered by a second filter (P-10). In the simulation model, a small fraction of the harvested broth is put into another tank and is used as inoculum for the next batch (P-13).

8.3.3 Downstream Processing

After fermentation the broth is harvested into the harvest tank (P-15). Part of the product crystallizes in the fermenter and also in the harvesting tank. Crystallization is completed in the crystallizer (P-16) by evaporation of some of the water. Afterwards the suspension is stored in tank P-17. If the riboflavin crystals have a needle structure, they are recrystallized in this vessel into cubic particles, which show better separation behavior in the decanter (P-18). From the decanter three streams are harvested, two liquid phases and the cell/crystal suspension. To achieve higher purity, a washing step is used with a second separation. The last step is drying, either using a spray dryer to obtain a powdered product or applying a spray granulation to obtain granulate. Granulate can be dosed more precisely, e.g. when used in food or feed, which increases the quality of the final product. Processing of powder is much more difficult because of the electrostatic charge. The powder sticks to all vessel walls, dosing becomes difficult, and, during product finishing, losses of product can be significant. In the model presented here, a spray dryer (P-20) is used to yield the final product and exhaust gas.

8.4 Inventory Analysis

The mass balance is dominated by air and water flows as seen in Table 8.2. Air is used for aeration but only a little oxygen is actually consumed (<1%). Nitrogen is inert in this process and passes through the system unchanged. At first glance, it might be surprising that air is dominating the raw material flows. This can be explained by the long fermentation of 500 h with continuous aeration at 0.3 vvm during the entire process. This points to a first possibility for improvement. Much less aeration should suffice to supply sufficient oxygen and remove carbon dioxide. Water is used in the fermentation and in downstream processing. Major streams are received from the decanter and from partial condensation of the gas from the spray dryes. Major organic materials used are sunflower oil, glucose, malt and yeast extracts, and peptone. Additional feeding of racemic methionine is required for growth of the organism. Only small amounts of inorganic salts, namely potassium hydrogen sulfate,

Table 8.2 *Material balance of the riboflavin production. (kg/batch) = kg component per batch*

Component	Input (kg/batch)	Output (kg/batch)
Biomass	0	46 240
Carbon dioxide	0	126 880
DL-Methionine	2400	22
Glucose	30 000	279
K_2HPO_4	1200	11
Manganese sulfate	1200	11
Malt extract	30 000	270
Nitrogen	30 581 000	30 581 000
Oxygen	9 284 000	9 195 000
Peptone	30 000	270
Riboflavin	0	9420
Riboflavin (crystals)	0	84 800
Sunflower oil	90 000	811
Water	3 078 000	3 112 000
Yeast extract	30 000	270
Sum	43 157 800	43 157 284

and manganese sulfate, are needed. The major products of the fermentation are riboflavin, biomass, carbon dioxide, and water. Roughly 50% of the carbon used is converted into product, 25% to biomass, and 25% to carbon dioxide. Typically less than 1% of the total substrate is not consumed and remains in the fermentation broth after fermentation. During downstream processing slightly more than 10% of the product is lost. Expenses for utilities are dominated by the high consumption of electric energy, mainly used for air compression, bioreactor stirring, and centrifugation (see Table 8.3).

8.5 Ecological Assessment

In this process there are no environmentally critical components used. Most compounds used are from biological origin, e.g. glucose, malt and yeast extracts, peptone, and sunflower oil. Only for the chemical production of DL-methionine are hazardous chemicals used. The

Table 8.3 *Utility requirements per batch*

Utility	Annual amount	Annual cost ($)	Cost share (%)
Electricity	64 803 000 kWh	6 480 000	84
Steam	26 398 000 kg	528 000	6.9
Cooling water	3 010 857 000 kg	75 000	1.0
Chilled water	3 441 616 000 kg	602 000	7.8
Sum		7 685 000	100

Table 8.4 *Emissions per unit product amount*

Component	Total waste (g/kg)
Carbon dioxide	1530
Biomass	445
Glucose	3.00
Malt extract	3.00
Peptone	3.00
Riboflavin	113
Sunflower oil	10.0
Water	36 600
Yeast extract	3.00

major mass flow in this process is caused by aeration. However, only a very small fraction of the oxygen supplied to the fermenter (<1%) is actually consumed. This very large air flow does not directly cause any environmental pollution but the necessary electric energy consumption by the compressor is very high (Table 8.3), causing increased costs and indirect environmental pollution during production of electric energy. Water is used in large amounts and primarily converted to wastewater, which has to be treated before release to a receiving water body. All components contained in the wastewater are, however, readily biodegradable. More details about the environmental assessment can be found on the accompanying CD-ROM.

The overall mass index is 39 kg/kg of product excluding air, and 2.6 kg/kg, excluding water and air. The emissions to the environment are summarized in Table 8.4. There are no solid wastes produced in this process. As in almost any bioprocess, water is dominating. Water pollutants are dominated by product losses. Incomplete consumption of substrates is much less significant. The amount of carbon dioxide produced is low, with only about 0.6 kg/kg product. If odor problems become significant in a particular environment, the waste gas can be treated, e.g., by a so-called biofilter. A chamber filled with peat adsorbs the organic components from the gas, and microorganisms transform carbons into biomass, carbon dioxide, and water. After a long period (some years) the peat will be rotted and has to be exchanged. The rotted peat can be used as a fertilizer.

8.6 Economic Assessment

The economic calculations are based on a facility using twelve bioreactors with about 500 m^3 total volume. This corresponds to a typical maximum size of stirred tank fermenters. The total plant direct costs are then $ 66 million and total plant indirect costs $ 26 million (summing to total plant costs of $ 92 million). With contractors fees and contingencies the resulting direct fixed capital cost is $ 106 million, and further total capital investment is $ 119 million (see Table 8.5). The annual operating costs are $ 45 million for about 1000 metric tons of product. The total amount of annual raw material cost is around $ 13.6 million, where peptone dominates with 40%, followed by yeast extract and malt extract

Table 8.5 *Executive summary. 2004 Prices*

Total capital investment ($)	118 800 000
Operating cost ($/year)	45 460
Production rate (kg/year)	997 000
Unit production cost ($/kg P)	45.6
Total revenues ($/year)	49 869 000
Gross margin (%)	8.84
Return on investment (%)	11.6
Payback time (years)	8.6
Net present value (at 7.0% interest) ($)	−9 980 000

with 18% each, and DL-methionine with 7%. In this model, facility-dependent (49%), raw material (30%), and utility costs (17%) dominate the annual operating costs. Labor costs are very small (<3%). The payback time is about 8.5 years. Costs for waste treatment and disposal are about $ 588 000, which amounts to only 1.5% of the annual operating costs and is therefore almost negligible compared with the total annual costs of $ 45.5 million.

8.7 Discussion and Concluding Remarks

Today, large-scale riboflavin production by fermentation is state-of-the-art. There is a lot of experience with moulds like *Ashbya gossypii* as well as with recombinant *Bacillus subtilis* strains. While in older plants contamination rates up to 15% and more occurred, in modern plants this rate has been reduced to less than 5%. This encourages building plants with a reactor volume of several hundred cubic meters.

The total required reaction volume is calculated by:

$$V_{R,L} = \frac{CAP \cdot (1 + COR)}{APT \cdot RRG \cdot EFF} \tag{8.1}$$

CAP = annual capacity (kg/yr)
COR = contamination rate
APT = annual production time including idle time (h/yr)
RRG = gross reaction rate (kg/m^{-3} h)
EFF = downstream efficiency factor

If the total required fermenter volume is divided into several reactors, the investment will increase. The influence of the reactor number can be calculated as follows:

$$\frac{INV(n)}{INV(1)} = n^{0.35} \tag{8.2}$$

This means that if the reaction volume is installed in 10 reactors instead of one, the investment (purchase cost) will increase by a factor of 2.5 for the reactors. The installation costs would increase, too, but by a factor 10.

There are only a few uncertainties associated with riboflavin production. The genetic stability of the production organism is essential and realized to a satisfying degree with modern strains of, e.g., *A. gossypii* or *B. subtilis*. Upstream and downstream processing use well established techniques and are therefore very reliable.

There are several possibilities for improvement. Generally, modern metabolic engineering techniques offer further improvement of strains, thereby increasing yields, and production rate and permitting the use of cheaper raw materials. In the prevailing case, the use of peptone is a major cost driver. Less expensive nitrogen and carbon sources could be applied, e.g. glucose derived from starch. Operating costs are dominated by the air supply. As stated already, only a small fraction of the oxygen present is actually consumed. Therefore, aeration rates could be reduced drastically. In fed-batch processes it is usually possible to reach higher final product concentrations, which reduces costs for downstream processing. Generally, the increase in plant size offers further reduction in production costs by the economy-of-scale.

The chosen process data were documented in the literature 20 years ago. These original process results do not lead to an economically feasible process. Therefore, these data were adapted to a more realistic level to achieve meaningful results. The assumed productivities are, however, still below what is possible today. Up to now the riboflavin market has been strongly controlled by a few producers, and it will be very difficult to increase the market volume. The market is also not too attractive, with decreasing price levels below $ 30 per kg (reference year 2005). The riboflavin market seems quite stable with constant demand but also limited growth potential.

Suggested Exercises

1. Reactor volume: The largest bioreactor that can be operated under sterile conditions has a volume of 3000 m^3. Compare the resulting process with one having 12 equally sized bioreactors and the same total volume. What consequences can you find for investment and total plant direct cost? Observe costs for installation, instrumentation, and unit production.
2. Replace the decanter centrifuge with a cross-flow filtration module using a microfiltration membrane. Observe changes in energy consumption, investment, and production cost.
3. Exhaust-gas treatment: Implement exhaust-gas treatment to reduce the odor problems by selecting a biofilter (absorption unit). Watch the investment and production costs and its impact on product cost.
4. Replace the batch-fermentation with a continuous fermentation. Use the following assumptions to simulate a continuous process: Genetic stability of the organism, outlet product concentration 27 g/L, dilution rate 0.05 h^{-1}, same medium composition as batch process, idle time reduction to less than 0.5% of annual production time.
5. Apply batch fermentation and continuous downstream processing. The interface is the harvest vessel. Check and improve the process. Make it function with proper definition of timetable, harvest time (duration), number of bioreactors and harvest vessels, and distribution to downstream operation modules.

References

[8.1] Kurth, R., Paust, J., Hähnlein, W. (2002): Vitamins – Riboflavin. In: *Ullmann's encyclopedia of industrial chemistry*. Wiley-VCH, Weinheim. DOI: 10.1002/14356007.a27_443.

[8.2] Stahmann, K., Revuelta, J., Seulberger, H. (2000): Three biotechnical processes using *Ashbya gossypii*, *Candida famata*, or *Bacillus subtilis* compete with chemical riboflavin production. *Appl. Microbiol. Biotechnol.*, **53**, 509–516.

[8.3] Fischer, M., Bacher, A. (2005): Biosynthesis of flavocoenzymes. *Nat. Prod. Rep.*, **22**, 324–350.

[8.4] Storhas, W. (2003): Bioverfahrensentwicklung, Wiley-VCH, Weinheim.

[8.5] Özbas, T., Kutsal, T., Caglar, A. (1984): The Production of riboflavin by *Eremothecium asgbyii*, 3rd European Congress on Biotechnology, Munich, Sept. 10–14, 1984. ISBN 0-89573-414-1.

9

α-Cyclodextrin

9.1 Introduction

Cyclodextrins (CD) are cyclic oligosaccharides composed of α-1, 4-glycosidic-linked glucosyl residues. Cyclodextrin glycosyl transferase [EC 2.4.1.19, CGTase] is used to produce α-CD from starch or starch derivates. There are three different types of CDs, according to the number of glucosyl residues in the molecule: α-, β-, and γ-CDs consisting of 6, 7, or 8 glucose units, respectively. Each type is produced industrially today. In 1998, global consumption was around 6000 metric tons, with a high annual growth rate [9.1]. Owing to its easier purification, the price for β-CD went significantly down in the past, whereas α- and γ-CD are still more expensive. For industrial application, β-CD costs around \$ 3–4/kg, α-CD \$ 20–25/kg, and γ-CD \$80–100/kg (reference prices from 2002) [9.2].

CDs have a cylindrical shape with a hydrophobic inside and a hydrophilic outside. They are able to form inclusion complexes with many hydrophobic molecules, thus changing their physical and chemical properties. These and other properties make CDs attractive for various applications in the food, chemical, pharmaceutical, and textile industries [9.2, 9.3].

Today, two types of production processes are applied: In the solvent process, an organic complexing agent precipitates α-CD selectively and, thus, directs the enzyme reaction to produce mainly α-cyclodextrin. Here, 1-decanol is often used as a complexing agent. The nonsolvent process does not use any complexing agent. The proportions of the different cyclodextrins produced depends only on the CGTase used and on the reaction conditions. A mixture of all three cyclodextrin types is usually produced. However, new or genetically improved CGTases may be able to form α-cyclodextrin with similar yields and selectivity, as reached in solvent processes.

In this chapter, we model and assess both a solvent and a nonsolvent process for the production of α-cyclodextrin. Similar models have been previously described by Biwer and Heinzle [9.4]. A general description of cyclodextrin production and applications is given by Biwer et al. [9.2], Schmid [9.5], and Bender [9.6].

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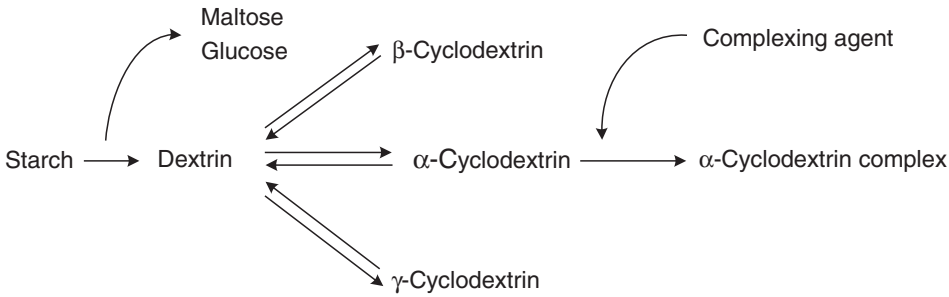


Figure 9.1 *Reaction scheme of cyclodextrin formation using a complexing agent. Reprinted from* Appl. Microbiol. Biotechnol. ***59****, 2002, 609–617, Enzymatic production of cyclodextrins, Biwer, Antranikian, and Heinzle, Figure 3. With kind permission of Springer Science and Business Media*

9.2 Reaction Model

Figure 9.1 shows the reaction scheme. Starch is the starting material for the α-CD production. First, it is liquefied and partially hydrolysed to dextrin by added α-amylase. Thereby, also glucose and maltose are formed. After liquefaction, α-amylase is inactivated by heat and the dextrin solution is cooled down to the optimum temperature of the CGTase.

CGTase and decanol are added, and α-CD is enzymatically produced, also small amounts of β-CD and partly some γ-CD. The exact proportions of CDs produced depends on the CGTase, the complexing agent used, and on reaction conditions. Typical α-CD : β-CD : γ-CD proportions in the precipitate are 96.5 : 3.5 : 0 [9.5], which are used in the model. Main characteristics of the enzymatic conversion have been defined as follows: Yield: 50%; reaction temperature: 40 °C; reaction time: 6 h; starting concentration of starch: 30%; working volume reactor: 10 m^3. Two bioreactors are operated in staggered mode to minimize the idle time of the downstream equipment.

In the solvent process, α-cyclodextrin forms a complex with decanol and precipitates. It is assumed that one mol of decanol is needed to precipitate one mol of α-CD. A small part of the cyclodextrins remains dissolved.

9.3 Process Model

The available process data were collected from the literature and patents (major sources are [9.5–9.8]).

9.3.1 Solvent Process

The process flow diagram of the solvent process is shown is Figure 9.2. The enzymatic conversion takes place in reactor P-1. After the bioreaction, the CD–agent complex is removed in centrifuge P-2 and is washed (P-4, P-5). The supernatant (S-109) contains unused starch, linear dextrins, glucose, maltose, the enzymes used (α-amylase, CGTase), unused decanol, some other by-products, and water. It is transferred to the decanter tank P-3 where decanol is separated and recycled to the reactor. In the next step, the complex is cleaved and decanol is removed in the steam distillation P-7. The necessary steam is supplied by the generator P-6.

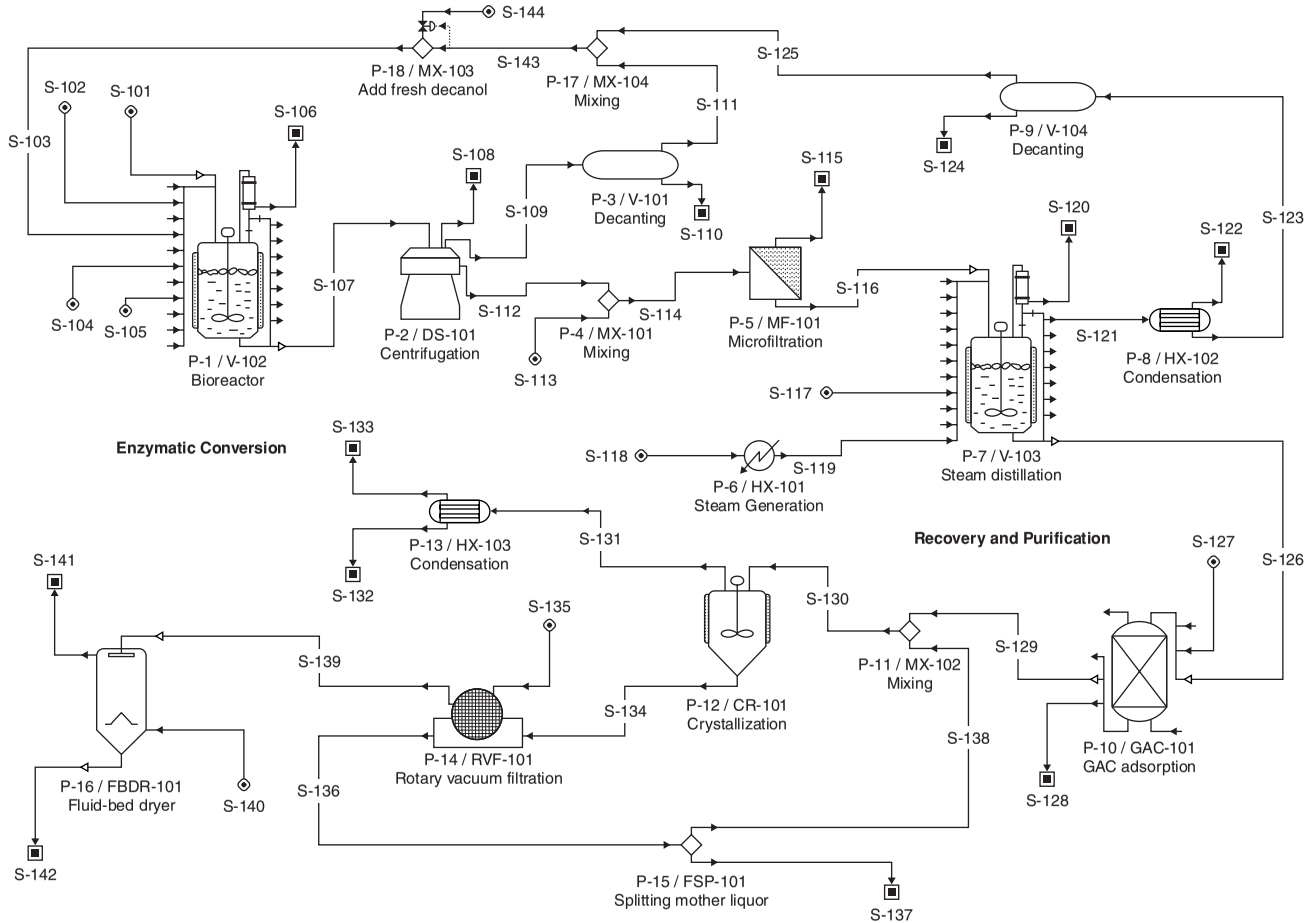


Figure 9.2 *Process flow diagram of the solvent process. Reprinted from* Enz. Microbiol. Technol. ***34****, Biwer and Heinzle, Process modeling and simulation can guide process development: Case study alpha-cyclodextrin, 642–650, Figure 2, 2004. With kind permission of Elsevier*

The modeling of the steam distillation is used to illustrate how the performance of a unit procedure can be estimated based on material properties, thermodynamic data, and engineering principles. In the first step, decanol and CD are cleaved by heating, and CD is dissolved. In a second step decanol is removed with steam. The amount of steam needed is estimated from the vapor pressure of water and decanol. Since the two liquids are immiscible the total vapor pressure is the sum of the vapor pressures of both pure components. Total pressure is assumed to be 1013 hPa (1 atm). In the gas phase, the ratio of the molar fraction of decanol (n_D) and the molar fraction of water (n_W) is equal to the ratio of their vapor pressures (p) at a given temperature:

$$\frac{n_D}{n_W} = \frac{p_D}{p_W} \tag{9.1}$$

Using the Antoine equation, the curves of vapor pressure are derived by regression using data from Jordan [9.9] for decanol and data from Atkins and de Paula [9.10] for water. The boiling temperature of the mixture can be calculated by using a Newton algorithm to solve the nonlinear equation. The calculated boiling point is 99.7 °C at atmospheric pressure. According to the estimated vapor/liquid-equilibrium curves, the vapor pressures at this temperature are $p_W = 1005.1$ hPa and $p_D = 8.1$ hPa. After resolving the complex, there are 1.03 mol decanol per mol α-CD in the vessel. Putting these values in the above equation, the amount of steam necessary to remove the decanol completely is 2.3 kg/mol α-CD (= 2.4 kg/kg α-CD), which is equal to 14.1 kg/kg decanol.

The decanol–water gas phase (S-121) is condensed in P-8. Organic and aqueous phase are separated in decanter P-9. The decanol from the two decanters is recycled to the bioreactor. In P-18, fresh decanol (S-144) is added to supply the necessary amount to the bioreactor (S-103). After steam distillation, the product solution passes through an activated carbon filter for decolorization (P-10). Most of the water is evaporated in the crystallization step P-12. Then the solution is cooled and α-CD precipitates. The generated steam is condensed in P-13. In the vacuum filtration (P-14) the CD crystals are separated and finally dried to a water content of around 5% in the fluid-bed dryer P-16. To ensure a high yield, the mother liquor (S-136) is recycled to the crystallization unit. A part of the mother liquor (1%) is discharged to prevent accumulation of undesired substances (P-15).

9.3.2 Non-solvent Process

In the solvent process, a complexing agent is required to direct the enzymatic conversion to α-CD to reach a high selectivity. Potential new CGTases may be able to form α-CD selectively without any complexing agent that forms a solid complex. Without such an agent the product remains dissolved in the solvent. Thus, a different separation approach is necessary. Here, the use of an adsorption column seems to be the most likely approach.

In the model, it is assumed that the same yield and selectivity of the enzymatic conversion can be realized with an improved CGTase. In the bioreactor, the solution is cooled to adsorption temperature after the enzymatic conversion. Centrifugation (P-2), complex washing (P-4, P-5), and steam distillation (P-6, P-7, and P-8) are no longer required and also both decanters (P-3, P-9) can be removed. They are replaced by an adsorption column that is packed with a polymeric resin (named P-20, for details of the flowsheet refer to the non-solvent process model on the CD-Rom). Thus the number of downstream steps is reduced. The second part of the downstream processing, including activated carbon treatment, crystallization, vacuum filtration and drying, remains unchanged.

Process data to model the adsorption column were taken from the literature [9.11–9.14]. α-CD is retained selectively while all other components flow through the column and become waste. The resin capacity is set to be 200 g α-CD/L and a yield of 95% is assumed. Also, β-CD is partly retained. After loading, the column is washed with water to remove impurities. The α-CD adsorption is highly temperature dependent. The product can be eluted by passing hot water (85 °C), supplied by a heat exchanger, through the column.

9.4 Inventory Analysis

The final product consists of 91.5% α-CD, 3.5% β-CD, and 5% water. Per batch, 1.52 tons of final product are produced in both processes. Using two bioreactors, 1480 tons are produced annually in 969 batches under the assumption of 330 operating days and a new batch starting every 8 h. Total batch time is 26 h in the solvent process and 21 h in the nonsolvent process; the bioreactor with an occupation time of 14 h is the bottleneck in both processes. The shorter downstream time (due to the lower number of steps) of the nonsolvent process would enable the use of a third bioreactor without adding a second downstream train. However, process-time calculations in the models are only a rough estimate and depend strongly on the capacity of the downstream units.

Bioreaction yield is 50% (g α-CD/g starch); downstream yield is 92%. In the solvent process, most of the product is lost by the incomplete precipitation in the bioreactor (4%), in the centrifuge (2%), and the crystallization (2%). The adsorption (5%) and the crystallization (2%) show the largest losses in the nonsolvent process.

Table 9.1 shows the material balance. The input consists of starch (raw material), decanol (complexing agent), the enzymes needed, and water (solvent). Besides water and the other input materials, the output includes fats, proteins (starch impurities), dextrin, glucose,

Table 9.1 *Material balances of solvent and nonsolvent process. The recycling of decanol in the solvent process is already considered in the table. CD = Cyclodextrin. Data taken from Elsevier*

	Solvent process		Nonsolvent process	
Component	Input (kg/kg P)	Output (kg/kg P)	Input (kg/kg P)	Output (kg/kg P)
α-Amylase	<0.01	<0.01	<0.01	<0.01
CGTase	0.01	0.01	0.01	0.01
α-CD (final product)		0.91		0.91
α-CD (loss)		0.08		0.08
β-CD in product		0.12		0.12
in waste		0.04		0.04
γ-CD		0.03		0.03
Decanol	0.04	0.04		
Dextrin		0.24		0.24
Fats and proteins		0.04		0.04
Glucose		0.26		0.26
Maltose		0.24		0.24
Starch	1.97	0.06	1.97	0.06
Water	12.7	12.7	19.3	19.3
Sum	14.7	14.7	21.3	21.3

maltose, β-CD, and γ-CD as by-products of the enzymatic conversion. Except from the decanol consumption in the solvent process, specific material consumptions are practically the same. The material intensity (Mass Index) without water is in both cases 2 kg/kg P. However, water consumption is higher in the nonsolvent process. In the solvent process, 79% of the decanol is recycled.

Most of the energy consumed is used for heating and for vaporization. The allocation to the process steps shows that steam distillation, crystallization, and reactor have the highest energy consumption. The specific steam demand is identical in both processes (9 kg/kg P). Significant differences are shown in the specific demand of electricity and cooling water. The removal of the centrifuge reduces the electricity consumption of the nonsolvent process (0.6 kWh/kg P compared with 0.75 kWh/kg P). The reduction of the cooling water demand is mainly caused by the removal of the steam distillation and the condensation P-8 (0.8 m^3/kg P compared with 1 m^3/kg P). The energy demand of the crystallization unit is, however, increased in the nonsolvent process, because product concentration is lower after the adsorption column than after the steam distillation. Thus, during crystallization more water has to be evaporated.

9.5 Environmental Assessment

The results of the environmental evaluation are summarized in Table 9.2. In general, the environmental impact of both alternatives and the environmental relevance of the compounds involved are relatively small. None of the substances is allocated to class A in any impact category. Owing to the additional use of decanol, the solvent process has a slightly higher potential environmental impact expressed by the different *EIs* and *GEI*. Considering the uncertainties involved in process modeling and in the assessment, a significant difference between the two alternatives cannot be identified.

Figure 9.3 compares the Environmental Index (EI_{Mult}) of the output components of both processes. The EI_{Mult} shows the most significant outputs: (i) starch and dextrin not consumed, (ii) glucose and maltose as by-products, (iii) product loss and other cyclodextrin types produced, and (iv) decanol. The input includes mainly water, starch (raw material), enzymes and, in the solvent process, additional decanol. The use of decanol and the organic load of the waste streams are the most relevant points.

The Impact Group Index of the input is dominated by decanol. Decanol is allocated to class B in the impact group Resources (based on oil or natural gas), Component Risk (low flashpoint), and Organisms (eye irritating). Consequently these three groups dominate the Impact Group Index of the input. At the output the IG Water/Soil is most affected by the COD of the several organic compounds in the waste streams.

9.6 Economic Assessment

A plant with two 10 m^3 bioreactors requires a fixed capital investment of $ 33 million and total capital investment of around $ 35 million in both process alternatives. The most expensive equipment includes the reactors, the crystallizer, and additionally the tank

Table 9.2 *Environmental performance of solvent and nonsolvent processes. incl. w. = with water, excl. w. = without water. Reproduced by permission of John Wiley & Sons, Ltd*

	Solvent process				Nonsolvent process			
	Input		Output		Input		Output	
Parameter	incl. w.	excl. w.	incl. w.	excl. w.	incl. w.	excl. w.	incl. w.	excl. w.
Mass Index *MI* (kg/kg P)	14.7	2.0	14.7	2.1	21.3	2.0	21.3	2.0
Number A-components	0		0		0		0	
Environmental Index EI_{Mw} (index points/kg P)	0.01		0.10		0.00		0.09	
Environmental Index EI_{Mult} (index points/kg P)	14.7	2.0	15.1	2.5	21.3	2.0	21.6	2.4
General Effect Index GEI_{Mw} (0–1)	0.0004	0.003	0.006	0.046	0.0	0.0	0.004	0.043
General Effect Index GEI_{Mult} (1–256)	1.00	1.01	1.03	1.19	1.0	1.0	1.02	1.17

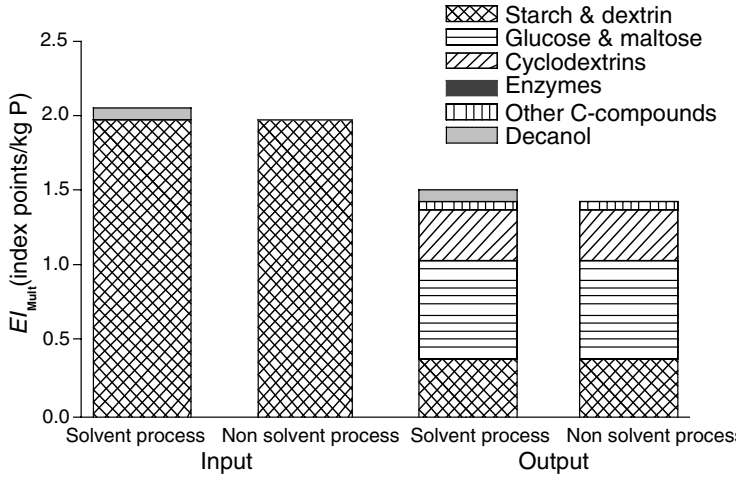


Figure 9.3 *Environmental Index (EI_{Mult}) of solvent and nonsolvent process*

for steam distillation in the solvent process and the adsorption column in the nonsolvent process.

Annual operating costs are $ 11 million in the solvent process and $ 10 million in the nonsolvent process. In both cases, facility-dependent expenditures and labor are responsible for most of the operating cost (see Figure 9.4). Raw material, mainly starch, and utility cost play a significant role, while consumables and waste treatment costs are relatively small. The higher operating cost for the solvent process is mainly caused by higher labor costs due to the higher number of downstream steps.

The unit-production costs are $ 7.40/kg in the solvent process and $ 6.70/kg in the nonsolvent process. With an annual revenue of $ 27 million this results in an ROI of 36% (solvent process), 38% (nonsolvent process) respectively. The two major cost factors, equipment and the labor demand, were estimated largely based on the SuperPro Designer® default values. In the specific situation, and at the specific location of a manufacturer, these costs

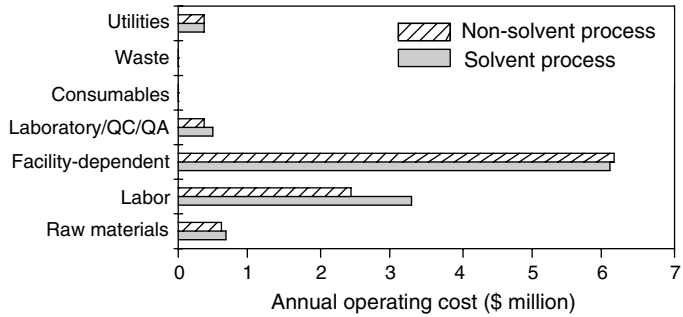


Figure 9.4 *Allocation of annual operating costs of solvent and nonsolvent process*

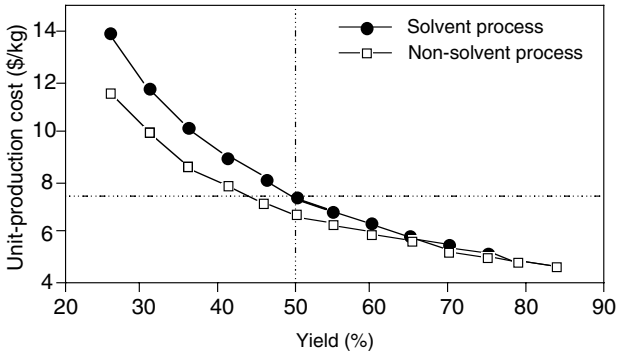


Figure 9.5 *Unit-production cost (UPC) of the two processes at different yields of the enzymatic conversion. The dotted lines indicate yield and UPC of the standard model of the solvent process*

can vary substantially. Therefore, a significant economic difference between the two alternatives cannot be used to select definitively the best process with only the available data basis. However, lower UPC in the model, shorter downstream time, and a smaller number of downstream steps might indicate some advantage of the nonsolvent process, on condition that the same yields are reached in the bioreactor.

Model yields were estimated based on literature data and can vary, especially in the nonsolvent process. Figure 9.5 shows the UPC tested against different yield values. As one could expect, the UPC is highly sensitive to the yield of the enzymatic conversion. For a competitive nonsolvent process, it is crucial to reach similar or higher yields than obtained in the solvent process. Using the model settings, the UPC of a nonsolvent process with 40–45% yield (see Figure 9.5) are similar to those for the standard solvent process with a yield of 50%. Below that yield, the standard solvent process is superior. Very similar results have been found for the environmental impact of the processes [9.4]. The integrated use of the adsorption column to remove the product during the enzymatic conversion could help to realize higher yields in the nonsolvent process.

9.7 Conclusions

The two processes regularly used for the production of α-cyclodextrin have been modeled and assessed. The environmental evaluation shows a low potential environmental impact for both alternatives. Considering the uncertainties involved in the process modeling and assessment, a significant difference cannot to be stated. The economic assessment also does not show any significant difference in the competitiveness of the processes, although the nonsolvent process seems to have some advantage due to less complex downstream processing and a shorter process time. However, the yield that can be realized in the bioreaction is the key parameter. A competitive nonsolvent process requires high yields that might be realized with improved enzymes or the integrated use of an adsorption column. Finally, the

specific situation and location of a manufacturer will determine which alternative should be chosen.

Suggested Exercises

1. In the nonsolvent process the effluent concentration of the product after the adsorption column (P-20) seems to be important for further processing. Check its influence by varying the eluent volume, which is defined relative to the bed volume (bv). Suggested variations are between 0.3 and 1.5 bv. How do the total operating costs change? Which of the contributing costs cause the variation of the total operating costs? What is the underlying reason for this?
2. In the solvent process two reactors are used for the enzymatic conversion (P-1). Since the downstream equipment has substantial idle time, as can be observed in the Gantt charts, there is a potential to enlarge the annual production. Study the influence of the addition of another reactor in the staggered mode (in Equipment Data in P-1). Observe the increase in annual production and the change of the unit-production cost. Then, add a fourth reactor and check the annual production and the unit-production cost. Why does the unit-production cost not decrease further?

References

[9.1] McCoy, M. (1999): Cyclodextrins: Great product seeks a market. *Chem. Eng. News*, **77** (9), 25–27.
[9.2] Biwer, A., Antranikian, G., Heinzle, E. (2002): Enzymatic production of cyclodextrins. *Appl. Microbiol. Biotechnol*., **59**, 609–617.
[9.3] Atwood, J., Davies, J., MacNicol, D., Voegtle, F. (1996): Comprehensive supramolecular chemistry. Volume 3: Cyclodextrins, Pergamon, Oxford.
[9.4] Biwer, A., Heinzle, E. (2004): Process modeling and simulation can guide process development: Case study α-cyclodextrin. *Enzyme Microb. Technol.,* **34**, 642–650.
[9.5] Schmid, G. (1996): Preparation and industrial production of cyclodextrins. In: Atwood, J., Davies, J., MacNicol, D., Voegtle, F.: Comprehensive supramolecular chemistry. Volume 3: Cyclodextrins. Pergamon, Oxford, pp. 41–56.
[9.6] Bender, H. (1986): Production, characterization, and application of cyclodextrins. *Adv. Biotech. Proc*., **6**, 31–71.
[9.7] Hedges, A. (1992): Cyclodextrin: Production, properties, and applications. In: Schenk, F., Habeda, R.: Starch hydrolysis products. VCH, New York, pp. 319–333.
[9.8] Ammeraal, R. (1988): Process for producing and separating cyclodextrins; US Patent 4 738 923.
[9.9] Jordan E. (1954): Vapor pressure of organic compounds. Interscience Publishers, New York.
[9.10] Atkins, P., de Paula, J. (2004): Atkins' physical chemistry. Oxford University Press, Oxford.
[9.11] Tsuchiyama, Y., Nomura, H., Okabe, M., Okamoto, R. (1991): A novel process of cyclodextrin production by the use of specific adsorbents: Part II. A new reactor system for selective production of α-cyclodextrin with specific adsorbents. *J. Ferment. Bioeng*., **71**, 413–417.
[9.12] Okabe, M., Tsuchiyama, Y., Okamoto, R. (1993): Development of a cyclodextrin production process using specific adsorbents. *Bioprocess Technol*., **16**, 109–130.

[9.13] Maekelae, M., Mattsson, P., Korpela, T. (1989): Specific adsorbents in isolation and purification of cyclodextrins. *Biotechnol. Appl. Biochem*., **11**, 193–200.
[9.14] Yamamoto, M., Horikoshi, K. (1981): Isolation and purification of α-cyclodextrin by synthetic adsorption polymer. *Starch/Staerke*, **33**, 244–246.

10
Penicillin V

10.1 Introduction

Penicillins belong to a family of hydrophobic β-lactams and are still among the most important antibiotics. They are produced by *Penicillium chrysogenum*. Lowe [10.1] estimates the world production of penicillin to be 65 000 metric tons in 2001. Besides penicillin G, penicillin V (phenoxymethylpenicillin) is the commercially most important penicillin. It is mainly converted to in 6-aminopenicillanic acid (6-APA), which in turn is used to make amoxicillin and ampicillin [10.2]. Furthermore, it is used directly as an antibiotic and ranks in the 100 top prescribed drugs in the US [10.3].

In this case study, we place emphasis on Monte Carlo simulations (MCS) to investigate the effect of parameter uncertainty on overall process performance. First, we develop the base-case model, which is later used for the MCS. The base-case model and the uncertainty analysis have been described in more detail in Biwer et al. [10.4].

10.2 Modeling Base Case

10.2.1 Fermentation Model

Penicillin V is a secondary metabolite produced at low growth rates and its syntheses have been described extensively in the literature [e.g. 10.5, 10.6]. Penicillin formation starts from three activated amino acids, and involves several enzymes and isopenicillin N as a major intermediate [10.7]. A typical medium consists of glucose, corn steep liquor, mineral salts, and phenoxyacetic acid as precursor for penicillin V [10.8–10.10]. *P. chrysogenum* has difficulty synthesizing the phenolic side chain for penicillin. Therefore, phenoxyacetic acid is added continuously to the culture medium as precursor.

In this case study, we use a simplified bioreaction model to describe the dependence of final product and biomass concentrations on the cell yield and maintenance coefficient

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Table 10.1 *Parameter values of the bioreaction model. dcw = dry cell weight. Reproduced by permission of John Wiley & Sons Inc.*

Parameter	Value	Yield coefficients	Value
t_{exp} (time of exponential growth) (h)	50	$Y_{X/pharmamedia}$ (g/g)	2.14
t_{prod} (time of production) (h)	106	$Y_{X/gluc.}$ (g/g)	0.45
X_f (biomass concentration at t_{exp}) (g/L)	30	$Y_{pen./gluc.}$ (g/g)	0.81
X_{nl} (final biomass concentration) (g/L)	45	$Y_{pen./phenoxyacetic\ acid}$ (g/g)	2.00
V_{in} (initial volume) (L)	55 000	$Y_{X/O2}$ (g/g)	1.56
V_{final} (final volume) (L)	75 000	$m_{gluc.}$ (maintenance coefficient) (g glucose/g dcw h)	0.022
P_{final} (final product concentration) (g/L)	63.3	m_{O2} (maintenance coefficient) (g/g dcw h)	0.023

and the specific product-formation rate and yield coefficient. The values for the model parameters that are derived from a combination of literature and process data are shown in Table 10.1. Two bioreaction phases (growth and production) are assumed. The first (primary) phase lasts about 50 h and during this time mainly biomass is produced in a batch culture. After the biomass formation slows down, penicillin V is produced over 106 h in the secondary phase. Glucose is fed continuously during the secondary phase.

10.2.2 Process Model

The process model is based on available literature [10.1, 10.9–10.11]. We assume a facility with 11 fermenters, each with a volume of 100 m^3, thus optimizing the usage of the downstream equipment. Final product is penicillin V sodium salt.

Figure 10.1 shows the process flow diagram. Medium (pharmamedia, trace metals, phenoxyacetate) is prepared in tank P-1, the glucose solution in tank P-2. They are sterilized in the continuous heat sterilizer P-4 and fed to the fermenter P-7. Air (S-113) is compressed (P-5) and filter sterilized (P-6). The exhaust air containing mainly carbon dioxide is filtered in P-8. In the bioreactor P-7, biomass and penicillin V are produced, consuming the carbon sources, the precursor, and the mineral salts. After the fermentation, the bioreactor content is transferred to the harvest tank P-9.

Biomass is removed in the rotary vacuum filter P-20 and discharged (S-151). In the centrifugal extractor P-23, penicillin is extracted into butyl acetate (S-156). Prior to extraction, the cell-free broth has to be acidified to a pH of around 3 using sulfuric acid (P-22) and cooled (P-21) to minimize degradation during acid extraction. After the extraction, the remaining aqueous solution is neutralized with sodium hydroxide (P-24) and discharged. Penicillin is re-extracted (P-25) into acetone/water (S-162). Sodium acetate is added (S-163) and penicillin V sodium salt precipitates. The crystals are separated and washed in the basket centrifugation P-26. In the fluid-bed dryer P-30, the penicillin is dried with air (S-175) and the final product stored in tank P-31.

The mother liquor is led to P-27, where most of the butyl acetate is recovered in a recycling step (not shown in detail). The rest is discharged and neutralized in P-28 (NaOH, 10% w/w). The butyl acetate is reused in the extraction. In P-29 fresh butyl acetate is added (S-172).

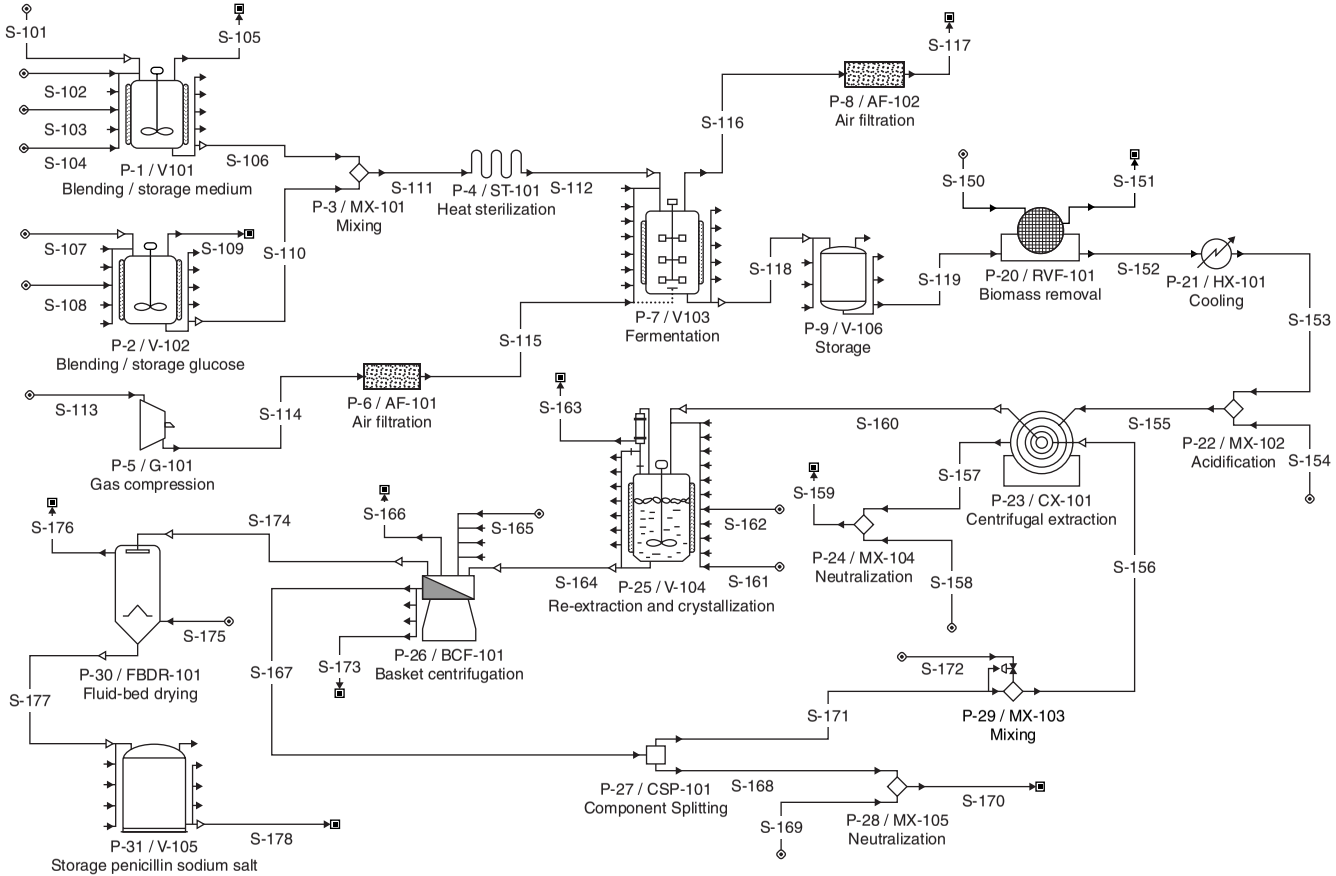


Figure 10.1 *Process flow diagram of the penicillin V production model. Reproduced by permission of John Wiley & Sons Inc.*

10.3 Inventory Analysis

The average production rate from the facility is approximately 263 kg of penicillin V sodium salt per hour. This results in an annual production of 2090 tons with the assumption of 330 operating days. Initial fermenter volume is 55 m^3, and 20 m^3 are added as nutrient and precursor feeds (36%). The volume added in the model is in the range given by Lowe [10.1]. Annual production is 546 batches, and it is assumed that 16 fail (3%). The overall yield of the fermentation is 0.21 g penicillin/g glucose. The yield across downstream recovery is 90%. The carbon balance shows that around 25% of the carbon is converted into penicillin, 17% into biomass, and 60% into carbon dioxide.

Table 10.2 shows the summary material balance for the base-case process. Altogether, there are 30 kg of raw materials per kg of final product (kg/kg P). The input includes a number of materials that are typical for fermentation processes: a high amount of water, glucose as carbon source, oxygen, medium, and trace metals. Specific to the penicillin production is the demand for phenoxyacetic acid. Furthermore, relevant amounts of the solvents, butyl acetate and acetone, are needed for extraction and a smaller amount of sodium acetate that forms the final product with the penicillin in the crystallization step.

Besides the product, the fermentation output consists of large amounts of carbon dioxide and biomass. Furthermore, significant amounts of unused raw materials and unrecovered product leave the process. An 80% recycling of butyl acetate is assumed in this model (see also Chang et al., 2002). Acetone (S-167, S-173) is also recycled (70%) (not shown in Figure 10.1).

Table 10.2 *Material balance of the penicillin V production. Recycling of butyl acetate and acetone is already considered. From the air transported through the bioreactor, only the consumed oxygen is considered in the table. [kg/kg P] = kg component per kg penicillin V sodium salt; dcw = dry cell weight. Reproduced by permission of John Wiley & Sons Inc.*

Component	Input (kg/kg P)	Output (kg/kg P)
Acetic acid		0.17
Acetone	0.12	0.12
Biomass (dcw)		0.88
Butyl acetate	0.32	0.32
Carbon dioxide		5.47
Glucose	5.10	0.10
Oxygen	2.56	
Penicillin V (loss)		0.10
Penicillin V sodium salt		1.00
Pharmamedia	0.47	0.06
Phenoxyacetic acid	0.60	0.01
Sodium acetate	0.23	0.01
Sulfuric acid	0.01	0.01
Trace metals	0.77	0.10
Sodium hydroxide	0.12	0.12
Water	19.2	21.1
Sum	29.8	29.8

10.4 Environmental Assessment

Most of the wastewater produced is discharged from the extraction step (S-158; remaining broth after penicillin removal) and after the separation of the crystals in P-26 (S-173, S-168; mixture of butyl acetate, acetone, water, and some impurities). Butyl acetate (P-27) and acetone (not shown in the flowsheet) are partially recycled. The remaining streams are led to a biological wastewater-treatment plant. Solid waste is produced in the biomass removal. The only relevant emission is the exhaust air of the fermenter, which includes a large amount of carbon dioxide (S-117). We have not attempted to assess fugitive emissions from the process.

The EI_{MV} is shown in Figure 10.2. Media components (mainly ammonium sulfate), the precursor phenoxyacetic acid, butyl acetate (extraction), acetone (re-extraction), and auxiliary materials, mainly acids and bases used for pH control of the extraction and neutralization of waste streams, are the most relevant input components. Although glucose and pharmamedia are used in large amounts, they are not relevant in any of the input impact categories, and their environmental factor is $EF_{MV} = 0$. Hence, they do not appear in the evaluation of the input. Carbon dioxide produced during the fermentation strongly dominates the output *EI*. Furthermore, the biomass, the butyl acetate, unused raw materials, and acetic acid formed in the re-extraction step (P-25) have some impact.

The overall EI_{MV} for the input is $EI_{In} = 0.46$ Index Points/kg P (= IP/kg P), for the output $EI_{out} = 0.74$ IP/kg P. In addition to the compounds involved, the energy consumption also contributes significantly to the environmental impact of the process [10.12]. The supply of energy affects the input side by consuming fossil energy sources and the output side by generating air pollution (e.g. carbon dioxide, sulfur dioxide).

10.5 Economic Assessment

The base-case model provides an estimate of the costs involved in penicillin manufacture. The estimated total purchased equipment cost is $ 9 million, which leads to a fixed capital

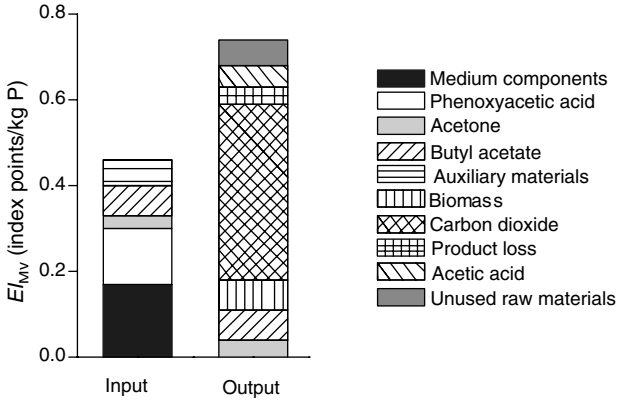


Figure 10.2 *Environmental Indices (EI_{MV}) of the input and output components of the penicillin V production model. Reproduced by permission of John Wiley & Sons Inc.*

investment of $ 44 million and a total capital investment (TCI) of $ 51 million. The bioreactors dominate the equipment costs with their purchase price of $ 5.5 million, which is consistent with results from Swartz [10.13].

Annual operating costs are $ 31.5 million. The biggest cost is raw material costs (40%), mainly glucose, phenoxyacetic acid, and butyl acetate (including recycling costs); this is in agreement with the analysis of Lowe [10.1]. They are followed by equipment-dependant costs (29%), mainly depreciation and maintenance. Labor (15%) and utility costs (12%, mainly electricity) also play a role, while the impact of laboratory/QC/QA, waste treatment, and consumables (altogether 4%) is small. Seven single operating-cost parameters capture each by themselves more than 2% of the total operating costs. The bioreactor-related costs of glucose (6.8%), phenoxyacetic acid (13.9%), and electricity for bioreactor (2.2%) and compressor (3.2%) constitute 26% of the annual operating costs. Furthermore, basic labor costs (12.3%), butyl acetate (9.9%, including recycling cost), and chilled water demand (3.3%) contribute considerably to the operating cost. This shows that the price of glucose and assumed hourly labor rates play an important role. This explains why today most penicillin-producing plants are located in countries where sugar and labor costs are low but are capable of supplying a stable source of energy given the high energy requirements of the process.

At the calculated annual production and operating costs, the unit-production costs are $ 15/kg final product. Based on an assumed selling price of $ 17.3/kg [10.14], the annual revenue is $ 36 million. This results in a return on investment (ROI) of 14%. Note that the ROI number assumes a 35% tax rate and no financial leverage for the project (i.e. no interest payments).

10.6 Monte Carlo Simulations

10.6.1 Objective Functions, Variables, and Probability Distributions

Monte Carlo simulations (MCS) can be used to explore how variance propagates through the entire process to impact both economic and environmental results. A crucial step in this analysis is selecting the objective functions, the input variables and their probability distributions. Several output parameters can be useful as objective functions. Here, we study the unit-production costs (UPC) and the input and output environmental index (EI_{MV}) of the process. For the analysis of profitability measurements such as earnings before interest and taxes (EBIT), earnings before interest, taxes, depreciation, and amortization (EBITDA) and return on investment (ROI) see Biwer et al. [10.4]. In our analysis, the capital investment is kept constant to represent an existing plant.

From the process model, a number of technical, supply chain, and market parameters routinely exhibit uncertainty. These parameters and their probability distribution are summarized in Table 10.3. Their probability distributions are derived from experimental and statistical data and are assumed to reflect the expected uncertainty in a process. For more details see Biwer et al. [10.4].

Technical parameters are all process parameters that affect the performance of the unit procedures in the process. In our analysis of technical parameter variability, we take the perspective of product development and assume that the true mean of each parameter is unknown but described by a distribution. This allows us to calculate economic parameters, such as *UPC*, for each Monte Carlo trial in a meaningful way. We recognize, however, that

Table 10.3 *Parameters used for Monte Carlo simulation and their variation and probability distribution chosen. SD = Standard deviation; V = Coefficient of variance. Reproduced by permission of John Wiley & Sons Inc.*

Parameter	Base-case value	Source	Probability distribution	Variation data	Source
1. Technical Parameters					
Yield biomass on glucose (g/g)	0.45	Own estimate, based on fermentation data	Normal	$V = 17.5\%$; min: 0.2	Industry data
Maintenance coefficient (mg glucose/g dcw h)	22	Own estimate, based on fermentation data	Normal	$V = 17.5\%$; min: 10	Industry data
Precursor utilization efficiency (%)	92	[10.15]	Normal	$V = 15.0$; 70–100 (min, max)	Industry data
Final biomass concentration (g/L)	45.0	Own estimate, based on fermentation data	Normal	$V = 17.5\%$; min: 25	Industry data
Final product concentration (g/L)	63.6	[10.8, 10.9]	Normal	$V = 10\%$; 20–100 (min, max)	Industry data
Aeration rate (vvm)	0.8	[10.1, 10.10]	Normal	$V = 10\%$; 0.5–1.0 (min, max)	[10.10]; Own estimate
Agitator power (kW/m^3)	2.5	[10.1, 10.10]	Normal	$V = 20\%$; 1.5–3.5 (min, max)	[10.10]; Own estimate
Yield downstream recovery (%)	90	[10.1, 10.9]	Normal	Calculated for single-step yields	

(*Continued*)

Table 10.3 *Parameters used for Monte Carlo simulation and their variation and probability distribution chosen. SD = Standard deviation; V = Coefficient of variance. Reproduced by permission of John Wiley & Sons Inc. (continued)*

Parameter	Base-case value	Source	Probability distribution	Variation data	Source
Yield biomass removal (%)	97	Own estimate	Normal	±2 (SD)	
K_{Pen} extraction	60	[10.16]	Uniform	60 – 80	Industry data
Yield crystallization (%)	97	Own estimate	Normal	±2 (SD)	(overall yield)
Yield basket centrifuge (%)	99	Own estimate	Normal	±1 (SD)	
Yield fluid-bed dryer (%)	99	Own estimate	Normal	±1 (SD)	
Yield butyl acetate recycling (%)	80	Own estimate	Normal	±5 (SD)	Own estimate
Yield acetone recycling (%)	70	Own estimate	Normal	±5 (SD)	Own estimate
2. Supply-chain parameters					
Price glucose ($/kg)	0.216	[10.17]	Beta	$\alpha = 3.49$; $\beta = 1.2$; Scale = 29.1 (for a normal distribution: $V = 25\%$)	[10.17]
Price phenoxyacetic acid ($/kg)	3.80	Own estimate; supplier data	Normal	$V = \pm 10\%$	Own estimate
Electricity cost ($/kW h)	0.0468	[10.18, 10.19]	Weibull	Loc: 4.13; Scale: 0.61; Shape: 1.96 (for a normal distr.: $V = 6\%$)	[10.18, 10.19]
3. Market parameters					
Selling price final product ($/kg)	17.30	[10.14]	Normal	$V = \pm 10\%$	Own estimate

the penicillin process is quite well characterized, and so we could have performed technical parameter uncertainty analysis with regard to process capabilities, which are defined by operating specifications, means, and standard deviations.

For this process, variability is described for parameters that determine biomass and product formation. Fermentation time and initial and final broth volumes are assumed to be deterministic. Fermentation conditions do vary as represented in the MCS by the aeration rate and the power consumption of the stirrer. In the base case, overall yield of the downstream section is 90%. In the MCS, variation in overall separation and purification is achieved by varying the yield of individual steps (P-20, P-25, P-26, and P-30) and the partition coefficient (K_{Pen}) of the extraction step (P-23). With regard to environmental and economic aspects the recycling of butyl acetate and acetone is crucial. Mean values and standard deviations are defined based on yields and variability usually occurring in the recycling of organic solvent.

The technical parameters are largely defined by the process and are under the control of the manufacturer (i.e. strain used, fermentation or purification conditions, etc.). Supply-chain and market parameters are not affected by the process conditions, but exhibit variance that influences the economics of the process. Raw material costs account for a large part of the operating costs. They are dominated by the costs for glucose and phenoxyacetic acid. Therefore, the prices of these materials are considered in the MCS. For phenoxyacetic acid an average price is chosen that is realistic for the annual demand of 1600 tons. The energy costs are dominated by the costs for electricity that is therefore considered in the MCS.

The price for penicillin V and penicillin in general has varied dramatically over the last few years. As the mean value, the current (2003) price stated by Milmo [10.14] is used, and a coefficient of variation of 10% is assumed.

10.6.2 Results

In this case study we did not use the COM function of SuperPro but transferred the model from SuperPro Designer® to MS Excel. The MCS were run in Excel using Crystal Ball 2000 as a random-number generator (Not contained in the CD). In the first MCS only the technical parameters are varied (MCS-TP), followed by a variation of the supply chain and market parameters (MCS-SCMP). In the next step, Monte Carlo simulations are done for all parameters defined in Table 10.3 (MCS-AP). The first results showed that the final penicillin concentration of the fermentation is the dominant technical parameter. To study its influence separately, additional MCS are run, one simulation varying the technical parameters without the final penicillin concentration (MCS-TPW) and another varying only the final penicillin concentration (MCS-Pen). For all parameter sets, 100 000 trials are run to ensure a low mean standard error for all objective functions (<1%). All distribution curves are more or less normally distributed. The results of the MCS are summarized in Appendix 1.

Unit-Production Cost. Figure 10.3 shows, as an example, the probability distribution of the unit-production cost (UPC) on a batch-to-batch basis for the MCS-TP. All supply-chain variables have distributions balanced around their base-case values. Therefore, the mean value in the MCS-SCMP is equal to the UPC of the base case. However, the mean UPC is significantly higher for the MCS-TP, MCS-TPW, and MCS-Pen. For several technical parameters the definition of a minimum or maximum value results in an unbalanced distribution, e.g. the downstream yield and the precursor-utilization efficiency are truncated at

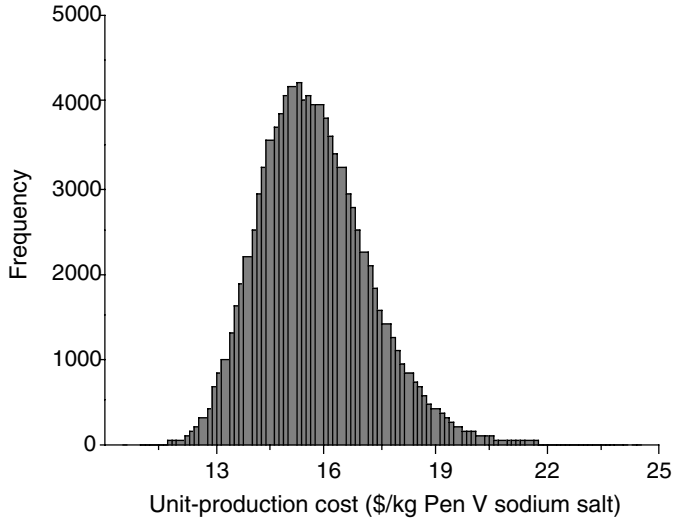


Figure 10.3 *Probability distribution of the unit-production costs in the MCS-TP (100 000 trials; 100 groups in the graph). Reproduced by permission of John Wiley & Sons Inc.*

100%. The average of these parameters in the MCS is therefore less than their base-case values. This leads to a higher mean UPC in the MCS. Since the supply-chain parameters do not have such an effect, the MCS-AP also shows a higher mean UPC of $ 15.6/kg.

The technical parameters cover a much broader range of values than do the supply chain parameters. The same tendency is shown by the standard deviation. The MCS-TP has a standard deviation (*SD*) of $ 1.5/kg, equal to a coefficient of variation (V) of 9.5%. The coefficient of variation of the MCS-SCMP is only $V = 2\%$. Thus, the variance of the MCS-AP is dominated by the technical parameters and its coefficient of variation (10%) is almost identical to the value of the MCS-TP. In the MCS-AP, the *UPC* is less than $ 17.7/kg with a probability of 90% and below $ 16.3/kg with a probability of 70%.

Figure 10.4 shows the parameters that drive the variance of the UPC. The final penicillin concentration dominates the variation in the MCS-TP. The concentration defines the amount of final product per batch and thus the percentage of raw materials converted into

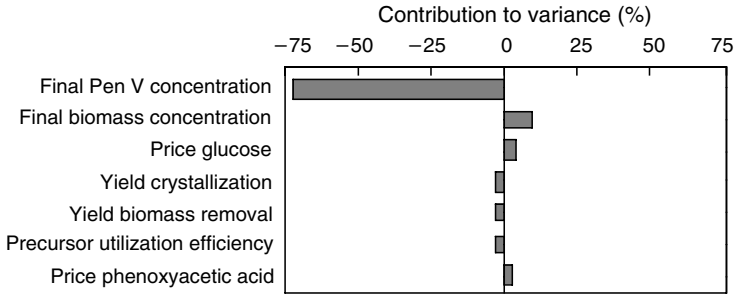


Figure 10.4 *Contribution of the parameters to the variance of the unit-production costs in the MCS-AP. Only parameters with more than 1% contribution to the variance are included. Negative values represent negative correlations*

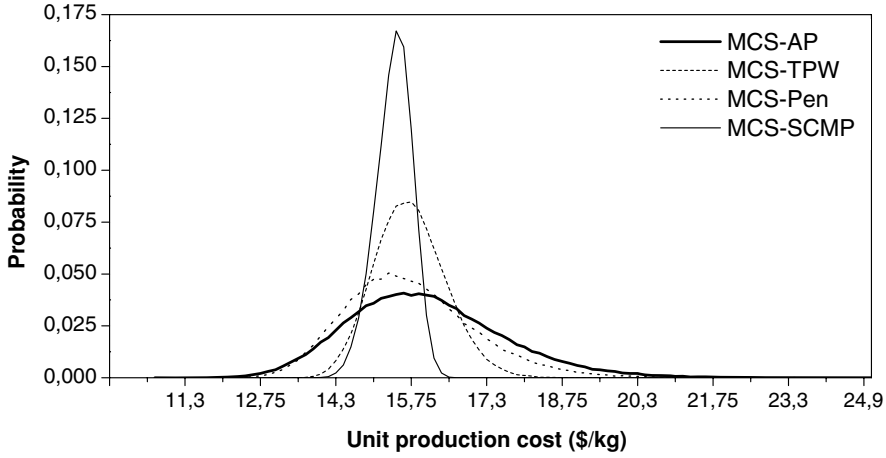


Figure 10.5 *Probability distribution of the unit-production costs in MCS-AP, MCS-TPW, MCS-Pen, and MCS-SCMP (100 000 trials; 100 groups in each graph). The curve for MCS-TP (not shown) is very similar that curve for MCS-AP. The area under the curves is always the same. For abbreviations see Notation section. Reproduced by permission of John Wiley & Sons Inc.*

biomass and carbon dioxide. Additionally, the relative amount (and cost) of butyl acetate necessary in the extraction stage decreases with increasing product concentration (as long as the solvent/broth ratio remains unchanged). The second driver is the final biomass concentration. Higher biomass concentration increases the diversion of C-atoms to cell growth and respiration (i.e. CO_2) and increases the raw materials requirements to produce a specific amount of penicillin. Besides these factors, the different recovery yields in the downstream process contribute to the variation because they determine the amount of final product that is ultimately recovered. Furthermore, the precursor-utilization efficiency influences the phenoxyacetic acid demand. In the MCS-SCMP the variance is mostly caused by the variation of the glucose and phenoxyacetic acid prices. With the probability distribution used in the MCS, the impact of the electricity cost is small.

The parameter contribution shown in Figure 10.4 explains why the additional MCS-TPW and MCS-Pen simulations were performed. The high impact of the final penicillin concentration is reaffirmed in the MCS-Pen. The penicillin concentration alone causes a variation of $V = 8.5\%$, while all other technical parameters (MCS-TPW) result in a coefficient of variation of $V = 4.5\%$.

Figure 10.5 compares the different probability distributions for the UPC. The MCS-SCMP shows the smallest variation. As one might expect, the MCS-AP displays the broadest variation. The MCS-Pen, which includes substantial variation contributed by penicillin concentration, is only slightly smaller; the MCS-TPW distribution lies between those of MCS-SCMP and MCS-Pen.

Environmental Index Input and Output. The variation of the EIs is determined only by the technical parameters. Hence, the results of the MCS-TP and MCS-AP are identical. The mean values for all parameter sets are more or less identical to their base-case values. The variation of the *EI* Input is significantly lower than for the *EI* Output. The specific amount of carbon dioxide, environmentally the most relevant output component, varies more than the specific amount of phenoxyacetic acid, the most relevant input component.

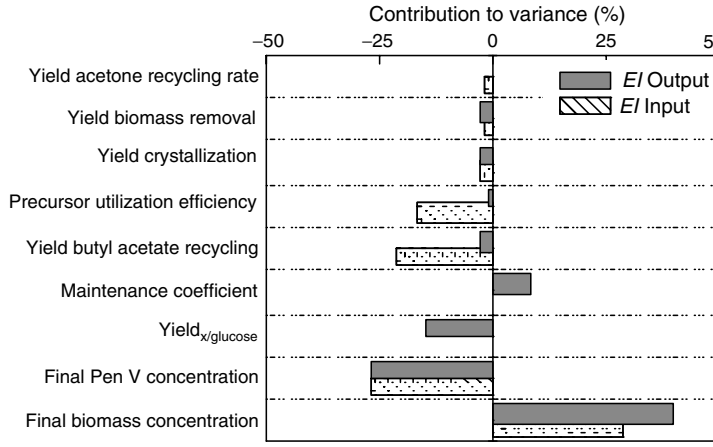


Figure 10.6 *Contribution of the technical parameters to the variance of the EI Input and Output in the MCS-TP. Negative values represent negative correlations*

Figure 10.6 shows the contribution of the technical parameters to the variance of the *EI* Input (MCS-TP). Medium, butyl acetate, acetone, and phenoxyacetic acid have the highest input *EF*s, and *EI*s and this influences the variance. The final biomass concentration shows the strongest contribution. It defines the amount of medium that must be added to the bioreactor. In contrast to the UPC, the penicillin concentration is only the second relevant factor. It determines the total amount of final product and the specific consumption of raw materials and solvents. Furthermore, the butyl acetate recycling rate and, to a smaller extent, the acetone recycling contribute to the variation by defining the amount of butyl acetate and acetone in the waste. However, they do not contribute significantly to the economic uncertainty. As with the UPC variance, the precursor (phenoxyacetic acid) utilization efficiency and the recovery yields (amount of final product) contribute substantially to the EI variance.

The contribution of the parameters to the variance of the *EI* Output is also shown in Figure 10.6. Carbon dioxide, biomass, and butyl acetate have the highest output *EI*s, which again affects the *EI* variance. The final biomass concentration determines the amount of biomass in the waste and by association the amount of carbon dioxide formed. The maintenance coefficient for glucose and the yield coefficient of biomass on glucose also influence the CO_2 amount. Neither parameter has any significant impact on the economic uncertainty.

The reduced impact of the final penicillin concentration compared with the economic objective function is shown clearly in Figure 10.7 by the smaller variance of the MCS-Pen curve. The MCS-TPW is wider and lies nearer to the MCS-AP distribution curve.

Sensitivity Analysis Penicillin Concentration. The final penicillin concentration is the most important technical parameter in the model. Therefore, it is interesting to see how the variation of the objective function changes when the coefficient of variation of the penicillin concentration varies. In general, it can be expected that the higher the coefficient of variation of the penicillin concentration, the higher is the variation of the objective function since each draw of the MCS will assess a different mean concentration. Figure 10.8(a) shows the probability distribution of the *UPC* at different coefficients of variation of the

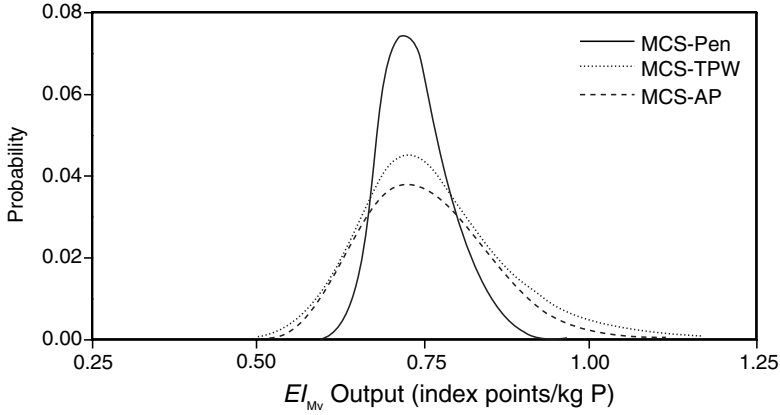


Figure 10.7 *Probability distribution of the Environmental Index Output (EI_{MV}) in the MCS-AP, MCS-TPW, and the MCS-Pen (100 000 trials; 100 groups in each graph). The area under the curves is always the same. For abbreviations see Notation section. Reproduced by permission of John Wiley & Sons Inc.*

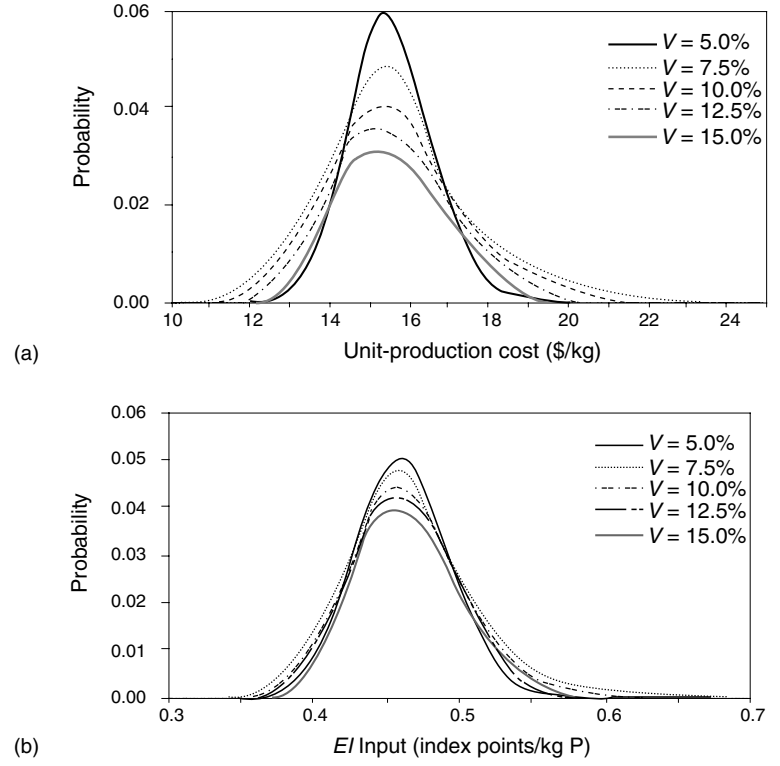


Figure 10.8 *Probability distribution of the UPC (a) and the EI Input (b) at different coefficients of variance (V) of the final penicillin concentration. The area under the curves is always the same. Reproduced by permission of John Wiley & Sons Inc.*

penicillin concentration. The strong impact of this variable on the *UPC* results in significant change of the curve shape and a higher standard deviation of the objective function (V_{UPC} = 7–14%). Figure 10.8(b) shows the *EI* Input for the same sets of coefficients of variation. Here, the variation of the penicillin concentration also leads to a broader variance of the *EI* Input ($V_{\text{EI Input}}$ = 7–9%). However, the effect is much smaller than for the UPC, based on the smaller impact of the penicillin concentration.

10.7 Conclusions

The development of the base model and the use of Monte Carlo simulations have led to a better understanding of penicillin V production and the influence of both technical and market variance. The most relevant stochastic variables are identified and proposed as parameters that are critical to an efficient process-control strategy, as well as for starting points for potential process improvements. Final penicillin and biomass concentrations in the fermenter have the highest contribution to the uncertainty of unit-production cost and environmental impact. Fermentation parameters such as yield, maintenance coefficient, and precursor utilization also have a high impact on the variance of the environmental impact, as well as the recycling rate of the organic solvents. The production costs are significantly affected by downstream yield and raw material costs.

The results show that the relevant parameters, and how strongly they contribute to the uncertainty, differ to some extent between the economic and environmental indicators. However, the direction of change is the same for all relevant parameters. The contributions of the variables to the overall uncertainty reflect the sensitivity of the process to these variables. Thus, there are parameters that can be changed to improve the economic performance without affecting the environmental performance and vice versa, while, for other parameters, an economic improvement leads directly to an environmental improvement. This represents an economic and environmental (eco-efficiency) win-win scenario that is in contrast to the use of end-of-pipe technologies for environmental pollution control that involve additional costs.

We note that the case presented is limited by the fact that the base model is a generalized model of the penicillin V product process. Depending on the location, the cost structure for a manufacturer might vary.

Suggested Exercises

1. The manufacturer gets an offer from a contract research organization to develop a new production strain. New metabolic engineering methods promise development of a new recombinant production strain requiring a shorter time to reach the same product concentration with identical yields. It is expected that the required fermentation time will be drastically reduced from 142 to 100 h (P-7). The offered research costs are $ 2 million. How long would it take to amortize this investment neglecting interests and the time-value of money? Assume that the additional annual amount of product can be sold at the same selling price.
2. Unions are successfully forcing the company to increase salaries by 10%. What is the impact on operating costs and unit-production costs?

Nomenclature

EBITDA = Earnings before interests, taxes, depreciation, and amortization
EBIT = Earnings before interests and taxes
EF = Environmental factors
EI = Environmental index, input = *EI* of the input components, Output = *EI* of the output components
FCI = Fixed capital investment
IP/kg P = Index points per kg final product
MCS = Monte carlo simulation
MCS-TP = Monte Carlo simulation using technical parameters
MCS-TPW = Monte Carlo simulation using technical parameters without final penicillin concentration in the fermenter
MCS-Pen = Monte Carlo simulation using final penicillin concentration in the fermenter
MCS-AP = Monte Carlo simulation using all parameters
MCS-S/MP = Monte Carlo simulation using supply chain and market parameters
ROI = Return on investment
SD = Standard deviation
TCI = Total capital investment
TOC = Total operating cost
UPC = Unit-production costs
V = Coefficient of variation

References

[10.1] Lowe, D. (2001): Antibiotics. In: Ratledge, C., Kristiansen, B.: Basic biotechnology. University Press, Cambridge, pp. 349–375.
[10.2] McCoy, M. (2000): Antibiotic restructuring follows pricing woes. *Chem. Eng. News* (4), 21–25.
[10.3] American Druggist (2005): http://www.rxlist.com/top200a.htm
[10.4] Biwer, A., Griffith, S., Cooney, C. (2005): Uncertainty analysis of penicillin V production using Monte Carlo simulation. *Biotechnol. Bioeng*, **90**, 167–179.
[10.5] Paradkar, A., Jensen, S. Mosher, R. (1997): Comparative genetics and molecular biology of ß-lactam biosynthesis. In: Strohl, W.: Biotechnology of antibiotics. Dekker, New York, pp. 241–277.
[10.6] Strohl, W. (1997): Biotechnology of antibiotics. Dekker, New York.
[10.7] Strohl, W. (1999): Secondary metabolites, antibiotics. In: Flickinger, M., Drew, St.: Encyclopedia of Bioprocess Technology Fermentation, Biocatalysis, and Bioseparation. Wiley-VCH, Weinheim.
[10.8] Demain, A., Elander, R. (1999): The β-lactam antibiotics: past, present, and future. *Antonie van Leeuwenhoek*, **75**, 5–19.
[10.9] van Nistelrooij, H., Krijgsman, J., de Vroom, E., Oldenhof, C. (1998): Penicillin update: Industrial. In: Mateles, R.: Penicillin: A paradigm for biotechnology. Candida Cooperation, Chicago, pp. 85–91.
[10.10] Perry, R., Green, D., Maloney, J. (1997): Perry's chemical engineers' handbook. McGraw-Hill, New York.

[10.11] Ohno, M., Otsuka, M., Yagisawa, M., Kondo, S., Oppinger, H., Hoffman, H., Sakutsch, D., Hepner, L., Male, C. (2002): Antibiotics. In: Ullmann's encyclopedia of industrial chemistry. Wiley-VCH, Weinheim.

[10.12] Chang Z., Wei X., Chen J. (2002): Simulated foam separation of butyl acetate from wastewater discharged by solvent extraction operation in penicillin production. *Separation Sc Tech* **37**: 981–991.

[10.13] Castells, F., Aelion, V., Abeliotis, K., Petrides, D. (1994): Life cycle inventory analysis of energy loads in chemical processes. In: El-Hawagi, M., Petrides, D.: Pollution prevention via process and product modifications. AIChE, New York, pp. 161–167.

[10.14] Swartz, R.W. (1979): The use of economic analysis of penicillin G manufacturing costs in establishing priorities for fermentation process improvements. *Ann. Rep. Ferment. Proc*, **3**, 75–110.

[10.15] Milmo, S. (2003): Challenges for European antibiotics producers: competition from China and a soaring euro are just two of the factors making business more difficult. *Chem. Market Reporter*, **263** (24).

[10.16] DeTilly, G., Mou, D., Cooney, C. (1982): Optimization and economics of antibiotic production. In: Smith, J.: Filamentous Fungi. Edward Arnold Publishers, London, pp. 190–209.

[10.17] McCabe, W., Smith, J., Harriott, P. (2001): Unit operations of chemical engineering. McGraw-Hill, New York.

[10.18] Foreign Agricultural Service (2001): World and U.S. raw and refined sugar prices. Available at U.S. Dept. of Agriculture, Foreign Agricultural Service: http://www.fas.usda.gov/htp/sugar/2000/November/prices.pdf

[10.19] U.S. Energy Information Administration (2004): February 2004 Monthly Energy Review. Available at: http://www.eia.doe.gov.

[10.20] Peters, M., Timmerhaus, K., West, R. (2003): Plant design and economics for chemical engineers, McGraw-Hill, Boston.

Appendix 10.1 Results of Monte Carlo Simulations of the Penicillin Production Process

Monte Carlo simulation	Parameter	Mean	Median	Standard deviation	Variance	Skewness	Kurtosis	Coeff. of variability %	Range minimum	Range maximum	Range width	Mean std. error
Technical parameters	*UPC*	15.63	15.50	1.50	2.24	0.58	3.71	10	10.46	25.57	15.11	0.47
	EI Input	0.46	0.46	0.04	0.00	0.37	3.27	8	0.34	0.66	0.31	0.01
	EI Output	0.76	0.75	0.11	0.01	0.67	3.91	15	0.44	1.56	1.12	0.04
Technical parameters without Pen concentration	*UPC*	15.30	15.27	0.70	0.50	0.29	3.12	5	12.71	19.02	6.31	0.22
	EI Input	0.46	0.46	0.03	0.00	0.16	2.97	7	0.35	0.60	0.25	0.01
	EI Output	0.75	0.74	0.09	0.01	0.57	3.71	12	0.46	1.37	0.91	0.03
Supply chain/ market parameters	*UPC*	14.98	15.00	0.36	0.13	−0.34	3.01	2	13.33	16.19	2.86	0.11
All parameters	*UPC*	15.62	15.49	1.55	2.39	0.58	3.74	10	10.83	25.62	14.80	0.49
	EI Input	0.46	0.46	0.04	0.00	0.40	3.35	8	0.34	0.68	0.34	0.01
	EI Output	0.76	0.75	0.11	0.01	0.69	4.00	15	0.45	1.83	1.39	0.04

(*Continued*)

Monte Carlo simulation	Parameter	Mean	Median	Standard deviation	Variance	Skewness	Kurtosis	Coeff. of variability %	Range minimum	Range maximum	Range width	Mean std. error
Penicillin concentration only	*UPC*	15.30	15.17	1.29	1.66	0.66	3.93	8	11.62	25.27	13.65	0.41
	EI Input	0.45	0.45	0.02	0.00	0.66	3.93	4	0.40	0.60	0.20	0.01
	EI Output	0.74	0.73	0.06	0.00	0.66	3.93	8	0.57	1.17	0.60	0.02
Pen V conc. = 5%	*UPC*	15.53	15.49	1.03	1.06	0.28	3.12	7	11.94	20.75	8.81	0.33
	EI Input	0.46	0.46	0.03	0.00	0.19	3.02	7	0.35	0.64	0.29	0.01
	EI Output	0.76	0.75	0.10	0.01	0.58	3.72	13	0.46	1.36	0.90	0.03
Pen V conc. = 7.5%	*UPC*	15.57	15.49	1.26	1.59	0.40	3.31	8	11.66	22.52	10.86	0.40
	EI Input	0.46	0.46	0.03	0.00	0.27	3.12	7	0.34	0.64	0.30	0.01
	EI Output	0.76	0.75	0.10	0.01	0.60	3.74	14	0.46	1.42	0.97	0.03
Pen V conc. = 10%	*UPC*	15.62	15.49	1.55	2.39	0.58	3.74	10	10.83	25.62	14.80	0.49
	EI Input	0.46	0.46	0.04	0.00	0.40	3.35	8	0.34	0.68	0.34	0.01
	EI Output	0.76	0.75	0.11	0.01	0.69	4.00	15	0.45	1.83	1.39	0.04
Pen V conc. = 12.5%	*UPC*	15.71	15.50	1.87	3.50	0.77	4.32	12	9.61	29.82	20.20	0.59
	EI Input	0.47	0.46	0.04	0.00	0.51	3.59	8	0.34	0.73	0.39	0.01
	EI Output	0.77	0.75	0.12	0.01	0.79	4.29	16	0.44	1.75	1.31	0.04
Pen V conc. = 15%	*UPC*	15.80	15.49	2.24	5.01	1.04	5.70	14	9.72	41.57	31.84	0.71
	EI Input	0.47	0.46	0.04	0.00	0.73	4.36	9	0.33	0.84	0.51	0.01
	EI Output	0.77	0.75	0.13	0.02	0.97	5.08	17	0.44	2.05	1.61	0.04

11

Recombinant Human Serum Albumin

*M. Abdul Kholiq and Elmar Heinzle**

11.1 Introduction

Human serum albumin (HSA) is applied to stabilize blood volume during surgery and during shock or burn cases. It is also used for the formulation of protein therapeutics, for vaccine formulation and manufacturing, for coating of medical devices, for drug delivery, etc. The worldwide sales of HSA from human blood are approximately $ 1–1.5 billion, requiring roughly 400–500 tons of HSA per year [11.1, 11.2]. One gram of HSA derived from human blood costs about $ 2–3.5 [11.3, 11.4].

HSA is currently extracted from human plasma by fractionation based on the method of Cohn originating from 1946, which is often combined with chromatography steps or other purification techniques [11.5–11.7]. However, collected blood sometimes contains undesirable substances, e.g. hepatitis viruses. Another disadvantage is the varying, uncertain blood supply. Therefore, it is desireable to develop a bioprocess to produce recombinant HSA (rHSA).

Potential expression systems for the production of recombinant human serum albumin are yeasts (*Saccharomyces cerevisiae, Kluyveromyces sp., Pichia pastoris*), bacteria (*Escherichia coli, Bacillus subtilis*), and also transgenic plants and animals [11.8]. Sijmons et al. [11.9] reported the expression of human serum albumin in transgenic plants (in tobacco and potato). GTC Biotherapeutics [11.1] has developed an rHSA production process using transgenic cows.

For these new processes, the crucial question is, whether the large-scale production of rHSA can be more economical than the fractionation of human blood plasma. This requires

* Corresponding author: e.heinzle@mx.uni-saarland.de, ++49/681/302-2905

Development of Sustainable Bioprocesses E. Heinzle, A. Biwer and C. Cooney
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high expression levels of the recombinant protein, the use of inexpensive media, and an efficient downstream processing. The methylotrophic yeast *P. pastoris* can provide such high expression levels of heterologous proteins. Reported expression rates of heterologous proteins range from several μg up to 22 g/L [11.10, 11.11]. The expression rate of rHSA in *P. pastoris* exceeded 10 g/L [11.2]. Cultivation can be carried out using inexpensive and defined media, consisting usually of carbon sources (glycerol and methanol), mineral salts, trace elements, biotin, and water [11.12].

Two industrial production plants, each with a capacity of 12.5 metric tons/year, have been constructed by Mitsubishi Pharma Corporation (Osaka, Japan) and by Kaketsuken (Kumamoto, Japan) using *P. pastoris* and *S. cerevisiae*, respectively. In 1998, Mitsubishi Pharmaceuticals announced the construction of the world's first plant for the production of rHSA using *P. pastoris*, and is currently awaiting the approval of its rHSA-manufacturing facility [11.12]. Kaketsuken is constructing a production plant for rHSA to be used in therapeutic applications using *S. cerevisiae*. Licensed from Delta Biotechnology (Nottingham, UK), commercial production is expected to start in 2008 [11.13, 11.14].

This case study concentrates on the rHSA production using *P. pastoris*. The data were taken from patent and scientific literature. It is an example of a large-scale bioproduction of a recombinant protein for pharmaceutical applications, whose selling price is relatively low. The case illustrates the important role of the expression rate of the recombinant proteins in the chosen host cell and the downstream processing strategy. The downstream processing has to have a high yield due to the low selling price, and, at the same time, provide a high purity due to the pharmaceutical use. Here, we focus on the application of expanded-bed adsorption (EBA) and compare it with the conventional purification method of proteins based on filtration and packed-bed adsorption (PBA).

11.2 Bioreaction Model

The carbon sources and media components are converted into cell biomass, rHSA, and by-products. The bioreactor size is derived from the desired annual production, the overall downstream yield, the possible number of batches per year, and the final product concentration and recovery yield.

11.2.1 Stoichiometry

The stoichiometry representing the conversion of glycerol into cell biomass and of methanol into cell biomass and product is derived from a simplified elemental formula ($C_pH_qO_rN_s$) of the cell biomass and of the product. Ammonia is assumed to be the only nitrogen source. The elemental formula of the cell biomass was set to be $CH_{1.67}O_{0.50}N_{0.17}$, which is close to values reported by Nielsen et al. [11.15] for *S. cerevisiae*. The cell biomass has a molar mass of 24 g/C-mol. The simplified elemental formula of HSA is determined using the ProtParam tool (Swiss Institute of Bioinformatics, Basle), a computation tool to identify physical and chemical characteristics of a given protein (*http://au.expasy.org/tools/protparam.html*) as $CH_{1.57}O_{0.30}N_{0.27}$, having a molar mass of 22.2 g/C-mol. The carbon sources used in this case study are glycerol ($C_3H_8O_3$) and methanol (CH_4O).

The stoichiometry for the conversion of glycerol into cell biomass is:

$$C_3H_8O_3 + a_g O_2 + b_g\,NH_3 \rightarrow \hat{Y}_{x/s,g}\,CH_{1.67}O_{0.5}N_{0.17} + c_g\,H_2O + d_g\,CO_2 \qquad (11.1)$$

Methanol is converted into cell biomass and into rHSA:

$$CH_4O + a_m\,O_2 + b_m\,NH_3 \rightarrow \hat{Y}_{x/s,m}\,CH_{1.67}O_{0.5}N_{0.17} + \hat{Y}_{P/s,m}\,CH_{1.57}O_{0.30}N_{0.27} + c_m\,H_2O + d_m\,CO_2 \quad (11.2)$$

$\hat{Y}_{x/s}$ (mol/mol) is the molar biomass yield, which can be calculated from the biomass yield $Y_{x/s}$ (g/g) using the molar mass of substrates (M_s) and cell biomass (M_c).

$$\hat{Y}_{x/s} = Y_{x/s} \cdot M_s/M_c \quad (11.3)$$

The molar product yield $\hat{Y}_{P/s}$ (mol/mol) is determined analogously. For known $Y_{x/s,g}$, $Y_{x/s,m}$, and $Y_{P/s,m}$, the stoichiometric coefficients a_g, b_g, c_g, d_g, a_m, b_m, c_m, and d_m can be determined based on the elemental balances for C, H, O, and N.

Reported cell yields of glycerol are about 0.40–0.45 g/g [11.16, 11.17], while cell yields of methanol vary from 0.15 [11.17], 0.4 [11.18], to 0.61 and 1.73 g/g [11.16]. In the model, the cell yields of glycerol and methanol are set to be 0.45 and 0.25 g/g, which is equal to 1.73 and 0.33 mol/mol, respectively. The product yield of methanol is assumed to be 0.05 g/g.

For these assumptions, the reaction stoichiometry is [using Equations (11.1) and (11.2)]:

$$C_3H_8O_3 + 1.7\,O_2 + 0.29\,NH_3 \rightarrow 1.73\,CH_{1.67}O_{0.5}N_{0.17} + 2.99\,H_2O + 1.28\,CO_2 \quad (11.4)$$

for using glycerol, and:

$$CH_4O + 1.08\,O_2 + 0.08\,NH_3 \rightarrow 0.33\,CH_{1.67}O_{0.5}N_{0.17} + 0.07\,CH_{1.57}O_{0.30}N_{0.27} + 1.78\,H_2O + 0.60\,CO_2 \quad (11.5)$$

for using methanol.

In the process model the calculations are defined on a mass basis. Then, the reaction equations (in grams) are:

$$C_3H_8O_3 + 0.59\,O_2 + 0.05\,NH_3 \rightarrow 0.45\,CH_{1.67}O_{0.5}N_{0.17} + 0.58\,H_2O + 0.61\,CO_2 \quad (11.6)$$

for using glycerol, and:

$$CH_4O + 1.08\,O_2 + 0.04\,NH_3 \rightarrow 0.25\,CH_{1.67}O_{0.5}N_{0.17} + 0.05\,CH_{1.57}O_{0.30}N_{0.27} + 1.00\,H_2O + 0.82\,CO_2 \quad (11.7)$$

for using methanol.

11.2.2 Multi-stage Fermentation and Feeding Plan

The fermentation is usually carried out in a fed-batch modus. First, the cells are grown in a batch fermentation with glycerol as carbon source. After glycerol is consumed, feeding of a medium containing methanol as carbon source is started. The feeding rate is controlled to avoid the accumulation of toxic methanol in the culture medium [11.19, 11.20].

To achieve high cell and product concentrations, a so-called multi-stage fermentation was proposed [11.21, 11.22]. After the batch growth on glycerol in the first stage, maximum cell density is achieved in the second stage by feeding further glycerol (see Table 11.1). The

Table 11.1 *Feeding plan of a multi-stage fermentation for the production of rHSA (based on data from [11.21, 11.22])*

Stage	Modus	Aim	Carbon source	Initial concentration or feed	Flow rate (mL/L h)	Time (h)
1	Batch	Growth	Glycerol	50 g/L	–	23
2	Fed-batch	Growth	Glycerol	50%	24	24
3	Batch	Starvation			–	1
4	Fed-batch	Induction	Methanol	50%	7.5	12
5	Fed-batch	Production	Methanol	50%	15	84

third stage is a starvation phase in which the glycerol feeding is stopped. After the glycerol in the medium is totally consumed, the rHSA production is induced by feeding methanol at a very low flow rate. The last stage is the production stage with an increased feeding rate of methanol. Part of the methanol is also converted into cell biomass [see Equation (11.5)].

Starting from a given batch volume, the amount of substrates and the final broth volume can be estimated using the feeding plan of Table 11.1. The density of glycerol and methanol used for this estimation is 1.26 and 0.79 g/L, respectively. For example, starting from a batch volume of 1 L containing 50 g of glycerol, 363 g of glycerol (in 0.3 L of water) and 716 g of methanol (in 1 L of water) are fed to the bioreactor. The estimated final broth volume is about 3.9 L. The ammonia consumption is determined by the stoichiometric equations [Equations (11.6) and (11.7)].

11.2.3 Total Broth Volume in Production Scale and Raw Material Consumption

The total broth volume needed for a desired annual production can be estimated from the product concentration, the overall downstream yield, and the possible number of batches per year, as shown in Table 11.2. For an annual production target of 12.5 metric tons/year and an

Table 11.2 *Estimation of the total broth volume and the necessary bioreactor volume for an annual production of 12.5 tons rHSA*

Description	Value	Source
Production target	12.5 tons/year	[11.23, 11.24]
Overall purification yield	60%	[11.2]
Total amount of product in the fermentation broth	21 tons/year	calculated
Working days	338 days/year	estimated
Process time per batch	6.5 days	[11.2]
Number of batches	52	calculated
Total amount of product in the fermentation broth	0.40 tons/batch	calculated
Product concentration	10 g/L	[11.25]
Broth volume	40 m^3	calculated
Volume of fermenter	50 m^3	[11.2]
Working volume	80%	estimated

overall total purification yield of 60%, the total amount of product in the fermentation broth should be about 21 metric tons per year. For the calculated number of batches and the product concentration of 10 g/L, a broth volume of about 40 m^3 (50 m^3 working volume) is needed for the production of 12.5 tons of rHSA per year. This value agrees with literature data [11.2].

From the data shown in Table 11.2 and the calculated working volume of 40 m^3, the necessary amount of raw materials can be calculated for the industrial process: 4.2 tons of glycerol (in 13 m^3 of water) and 7.4 tons of methanol (in 9.4 m^3 of water) as carbon sources, and 0.5 tons of ammonia.

11.3 Process Model

11.3.1 Bioreaction

Figure 11.1 shows the process flow diagram of the rHSA production. In the first medium tank (P-1), 4.27 tons of glycerol and 13 m^3 of water are mixed; in the second tank (P-2), 7.4 tons of methanol and 9.4 m^3 of water. Other media components, such as salts, vitamins, and trace elements, are also added to these tanks. A 20% (v/v) ammonium solution containing 500 kg of ammonia is fed in during the fermentation in the bioreactor (P-3).

The cell and product formation in the bioreactor is described by the Equations (11.6) and (11.7). The first step [Equation (11.6)] represents the bioconversion of glycerol into biomass with a reaction extent of 100% referred to glycerol (stages 1–3 in Table 11.1). The conversion of methanol into cell biomass and product in the second step is aimed to achieve a product concentration of 10 g/L (stages 4 and 5 in Table 11.1). The total amount of cell biomass in the fermentation broth is about 3.8 tons (S-007), corresponding to a cell density of about 100 g/L. Reported cell densities for yeast are in the range of 100 to 160 g/L [11.17, 11.26–11.29].

11.3.2 Downstream Processing

Ohmura et al. [11.20] described the application of packed-bed adsorption (PBA) for the purification of rHSA requiring filtration and chromatography steps. Alternatively, using expanded-bed adsorption (EBA) the product can be captured directly from the cell-containing fermentation broth [11.2, 11.30]. Sumi et al. [11.2] reported significant improvements by a 50% reduction of the downstream time and a 45% increase of the overall yield compared with the results using the conventional purification method. Some cost comparisons of EBA and PBA processes have been published, e.g. by Curbelo et al. [11.31] for bovine serum albumin from *P. pastoris* broth and by Amersham Biosciences [11.32] for monoclonal antibody.

In this case study, we compare the use of EBA and PBA as alternative downstream routes in the production of rHSA (EBA process, PBA process). Figure 11.2 shows the flow diagrams of the recovery section for the EBA and PBA process. The application of the EBA process allows the removal of at least three downstream processing steps (microfiltration and two ultrafiltrations) that is expected to lead to a better purification yield and a reduction in the purification time. The process flow diagrams of the bioreaction section and the downstream section following the EBA step and the PBA step, respectively, are identical in both cases (as shown in Figure 11.1).

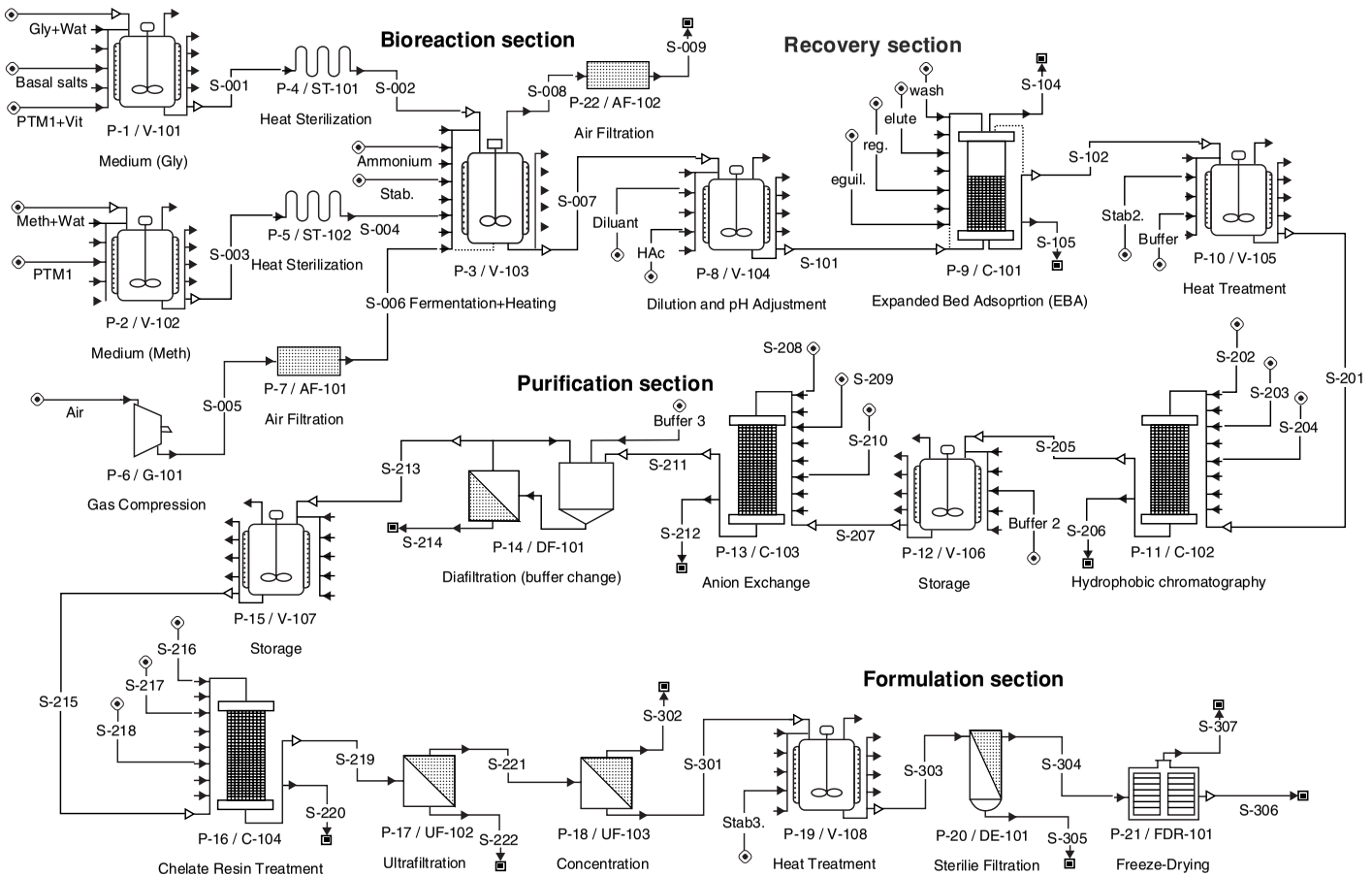


Figure 11.1 *Process flow diagram of the production and purification of rHSA using expanded-bed adsorption (EBA)*

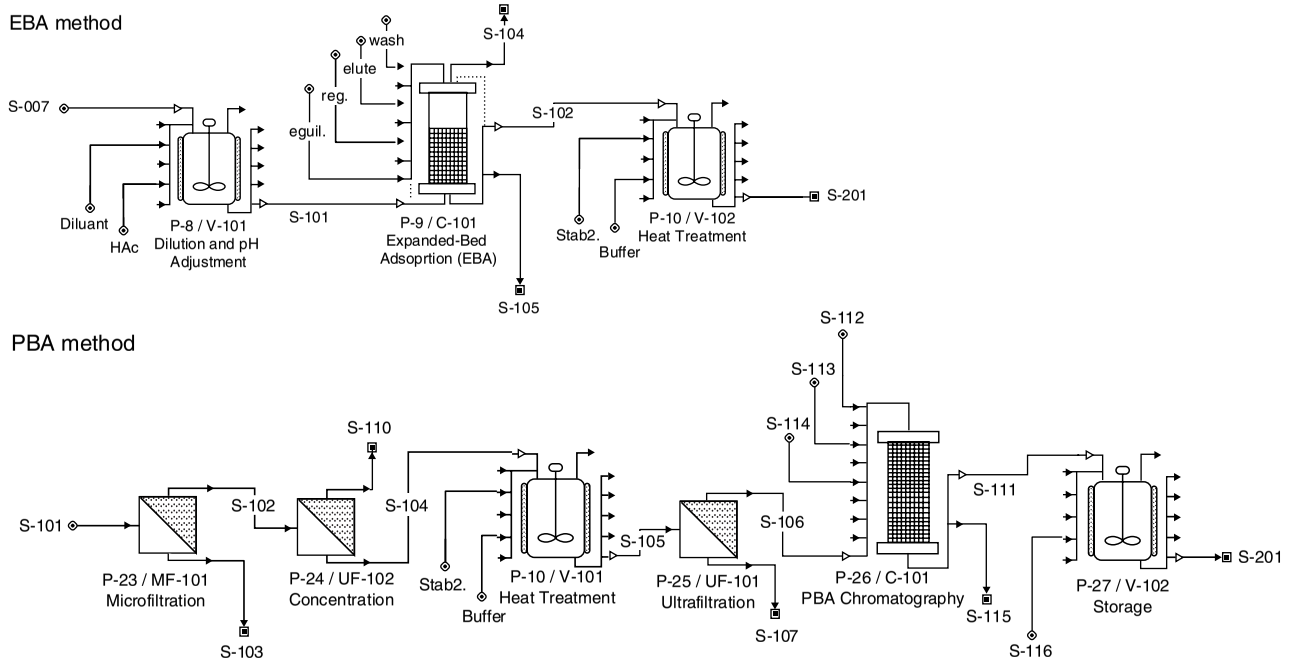


Figure 11.2 *Comparison of the recovery section of the expanded-bed (EBA) process and the packed-bed adsorption (PBA) process. Entering stream S-101 is equivalent to stream S-007 of the EBA-method and to the corresponding stream in Figure 11.1*

EBA Process. Under the acidic conditions necessary for the EBA that uses strongly cationic adsorbent particles, rHSA would be rapidly degraded by proteases contained in the culture medium. Therefore, the proteases are inactivated by heating the fermentation broth to 68 °C for 30 minutes in the presence of 10 mM sodium caprylate as stabilizer and 10 mM cysteine and 100 mM aminoguanidine hydrochloride to suppress the coloration caused by heating. The heat treatment is done in the bioreactor (P-3), saving an additional tank. To represent the rHSA denaturation due to the heat treatment, a reaction operation is used with a reaction extend of 4% converting rHSA into waste proteins.

The heated solution has an electric conductivity of about 20 mS/cm. An optimum binding of rHSA to the strongly cationic adsorbent particles is at an electrical conductivity of 8–12 mS/cm. For a better adsorption, the solution is diluted (1:1) in the vessel P-8 using acetate buffer (50 mM) and distilled water. The pH value is adjusted to 4.5 using acetic acid.

The diluted solution containing the cells is then loaded upwardly into the EBA column (P-9), which has been equilibrated with an acetate buffer. The target rHSA binds to the adsorbent particles, while impurities are discarded. The equilibration buffer is used for washing. rHSA is recovered by feeding downwardly a phosphate buffer (100 mM, pH 9).

For decolorization, the rHSA solution is then heat-treated again in P-10 at 60 °C for 1 hour in the presence of 10 mM cysteine, 5 mM sodium caprylate, and 100 mM aminoguanidine hydrochloride at pH 7.5. The solution is then adjusted to pH 6.8 using phosphoric acid and the salt concentration is reduced to 200 mM by adding water. The adjusted solution is loaded onto the hydrophobic interaction chromatography (HIC) column (P-11), where impurities are retained.

In next step, the salt concentration of the solution is again reduced to about 100 mM (P-12), and loaded onto the anion exchanger (P-13). rHSA flows through the column, while coloring matters and trace impurities are removed. In the diafiltration step (P-14) the phosphate buffer (pH 6.8) is replaced by an acetate buffer (pH 4.5), which is required for the chelate resin treatment (P-16). The chelate resin treatment again retains coloring matter. Pyrogens are removed by ultrafiltration (P-17, molecular weight cut-off: 100 kDa). In a further ultrafiltration step (P-18), the solution is concentrated and then freeze-dried (P-21).

PBA Process. In the PBA process, the cell biomass is separated from the fermentation broth by microfiltration (P-23). The solution is then further concentrated by ultrafiltration (P-24). The proteases in the solution are inactivated by heat in P-9 at 60 °C for 1 hour (also for decolorization). The heat-treated solution is adjusted to pH 4.5 using acetic acid. Polymerized high-molecular-weight contaminants are removed by ultrafiltration (P-25).

The rHSA solution is then loaded onto the PBA column (P-26), where the product is retained. Acetate buffer is used for washing and for equilibration of the column. Similar to the EBA process, rHSA is recovered by feeding in a phosphate buffer. The further purification steps (P-10 and subsequent units) are identical to the EBA process.

11.4 Economic Assessment

Table 11.3 compares key economic metrics of both the EBA process and the PBA process. The selling price of the product (rHSA) was set as $ 3000/kg. At the same bioreactor size and productivity, there is no significant difference in the overall capital investment. The higher

Table 11.3 *Comparison of EBA and PBA processes and the influence of the number of bioreactors*

Parameter	Unit	EBA process		PBA process	
Number of bioreactors		1	2	1	2
Investment (TCI)	($ million)	92	125	91	125
Revenue	($ million/yr)	37	73	31	60
Operating cost	($ million/yr)	23	36	20	30
Annual production	(tons/yr)	12	24	10	20
Unit-production cost	($/g)	1.90	1.50	1.95	1.50
Gross margin	(%)	37	50	35	49
Return of investment	(%)	18	27	16	24
Payback time	(years)	5.6	3.8	6.1	4.3

cost due to the larger equipment size caused by the necessary dilution (P-8) and the higher equipment cost of the EBA column outweigh the savings due to the lower number of downstream steps. However, owing to the better yield, the annual production in the EBA process is about 12.2 tons rHSA, and thus about 1.7 tons higher than in the PBA process (additional revenue: $ 5 million/year). Therefore, the specific investment cost of the EBA process is lower ($ 7500/kg annual production) compared with the PBA process ($ 9000/kg annual production). The annual operating costs in the EBA process are about $ 3 million higher, mainly caused by the dilution at the beginning of the downstream process. Thus, the gross margin of the EBA process is only slightly better (37%) than that for the PBA process (35%).

In the base-case model of the EBA process, the fermentation time is about three times longer than the duration of the downstream processing. To increase the usage of the downstream equipment, an additional bioreactor can be added, which runs in stagger mode. Table 11.3 shows the influence of a second bioreactor on both process alternatives.

After addition of one extra bioreactor set, the total capital investment cost in the EBA process increases from $ 92 to 125 million and the operating cost from $ 23 to 36 million/year. However, the production rate nearly doubles. Therefore, the unit-production cost lowers from about $ 2000 to $ 1500/kg rHSA. Accordingly, the values of the gross margin, return on investment, and payback time are clearly improved. Similar improvements can be obtained also for the PBA process (see Table 11.3).

The product concentration plays an important role in the economic success of the process. Figure 11.3 shows the influence of the product concentration on both unit-production cost and gross margin. To enable a unit-production cost lower than the expected selling price ($ 3/g) to be obtained, the product concentration has to be above 6 g/L. At a product concentration of 10 g/L (base case), the unit-production cost is $ 2/g. For product concentrations higher than 15 g/L, the unit-production cost drops below $ 1.5/g and the gross margin rises above 50%.

11.5 Ecological Assessment

The EBA process has some ecological drawbacks compared with the PBA process. The total Mass Index of the PBA and the EBA process is about 1250 and 1930 kg/kg P, respectively,

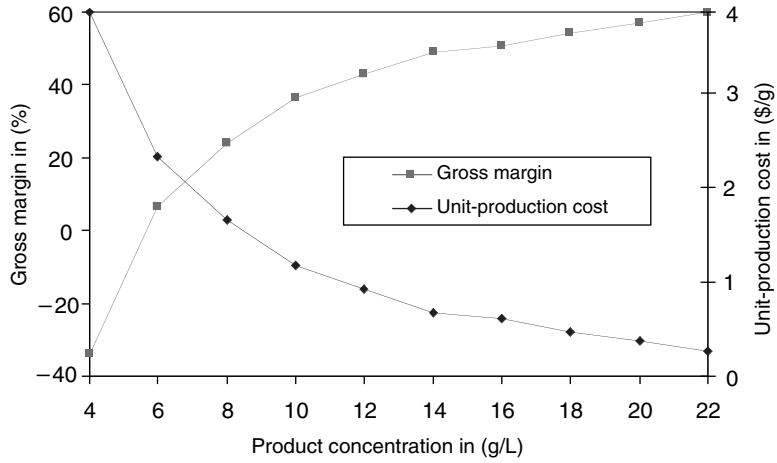


Figure 11.3 *Influence of product concentration on gross margin and unit-production cost of the EBA process (using one bioreactor)*

giving a difference of more than 50%. In the case of the EBA process, a 1:1 dilution of the fermentation broth with acetate buffer is necessary, whereas in the case of the PBA process the fermentation broth is even concentrated after cell separation. Furthermore, the consumption of z-propanol and sodium hydroxide is about 3 times higher in the EBA process, while the consumption of glycerin and methanol is slightly lower due to the better yield of the EBA process.

Figure 11.4 shows a comparison of the Environmental Index (EI_{Mult}) of the PBA and EBA methods. The materials used in the EBA and PBA methods are largely the same. Therefore, the differences of the Environmental Indices are mainly caused by the differences of the Mass Indices described above.

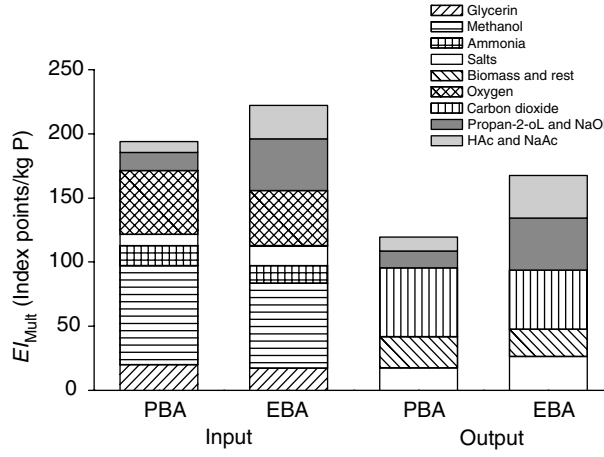


Figure 11.4 *Comparison of the Environmental Index (EI_{Mult}) of the PBA and EBA processes (without water)*

11.6 Conclusions

At the same bioreactor size and productivity, the EBA process and the PBA process have more or less the same overall capital investment. However, owing to the better purification yield, the EBA process shows a higher annual production and revenue. This results in lower specific investment cost, lower unit-production cost, and a higher gross margin for the EBA process, although the annual operating costs are higher in the EBA process.

Overall, the EBA process shows some economic advantages compared with the PBA process. However, particularly owing to the dilution of the fermentation broth and higher consumption of washing buffer, the EBA process shows some ecological drawbacks.

The production process with the EBA method could be improved by achieving a lower level of electric conductivity of the fermentation broth by reducing salt concentrations in the medium to avoid or reduce the necessary dilution. Another interesting option is the use of alternative fermenter operation such as repeated fed-batch fermentation to increase yield and productivity [11.33, 11.34]. For example, Ohya et al. [11.33] reported that with repeated fed-batch fermentation a 47% increase in annual rHSA production could be achieved.

Suggested Exercises

1. The usage of the downstream equipment can be increased by addition of one or more bioreactors, which run in stagger mode. The impact of the addition of one additional bioreactor is described in the book text. What is the impact of the addition of two bioreactors (total three bioreactors) on total capital investment costs, operating costs, and return on investment? What is the bottleneck in this case? Does the addition of a third bioreactor give the same improvement as the addition of the second bioreactor? Does an addition of a fourth bioreactor bring any process improvements? Why/Why not?
2. The EBA process could be improved by achieving a lower level of the electrical conductivity of the fermentation broth to reduce the necessary dilution before the EBA step. Observe the impact of the dilution degree of the heated fermentation broth in P-8.

References

[11.1] GTC Biotherapeutics (2002): Annual Report 2002. Framingham, MA. Available at: http://www.transgenics.com/

[11.2] Sumi, A., Okuyama, K., Kobayashi, K., Ohtani, W., Ohmura, T., Yokoyama, K. (1999): Purification of recombinant human serum albumin – efficient purification using STREAMLINE. *Bioseparation*, **8**, 195–200.

[11.3] Kostandini, G. (2004): Potential impacts of pharmaceutical uses of transgenic tobacco - the case of human serum albumin and Gaucher's Disease treatment. Master thesis. Virginia Polytechnic Institute and State University, Blacksburg.

[11.4] Flesland, O., Seghatchian, J., Solheim, B. (2003): The Norwegian plasma fractionation project – a 12 year clinical and economic success story. *Transfus. Apheresis. Sci.*, **28**, 93–100.

[11.5] Burnouf, T. (1995): Chromatography in plasma fractionation: Benefits and future trends. *J. Chromatogr., B: Biomed. Appl.*, **664**, 3–15.

[11.6] Tanaka, K., Shigueoka, E., Sawatani, E., Dias, G., Arashiro, F., Campos, T., Nakao, H. (1998): Purification of human albumin by the combination of the method of Cohn with liquid chromatography. *Br. J. Med. Biol. Res.*, **31**, 1383–1388.

[11.7] Adcock, W., MacGregor, A., Davies, J., Hattarki, M., Anderson, D., Goss, N. (1998): Chromatographic removal and heat inactivation of hepatitis A virus during manufacture of human albumin. *Biotechnol. Appl. Biochem.*, **28**, 85–94.

[11.8] Ballance, J. (2001): Characterization of yeast-derived recombinant human albumin. IBC Well Characterized Biologicals Conference, Seattle. Available at: http://www.transgenics.com/

[11.9] Sijmons, P., Dekker, B., Schrammeijer, B., Verwoerd, T., van den Elzen, P., Hoekema, A. (1990): Production of correctly processed human serum albumin in transgenic plants. *Biotechnology* (*New York*), **8**, 217–221.

[11.10] Cereghino, J., Cregg J. (2000): Heterologous protein expression in the methylotrophic yeast *Pichia pastoris*. *FEMS Microbiol. Rev.*, **24**, 45–66.

[11.11] Cregg, J. (2005): Heterologous proteins expressed in *Pichia pastoris*. Available at: http://faculty.kgi.edu/cregg/Pichia2004.htm

[11.12] Ilgen, C., Cereghino, J., Cregg J. (2005): *Pichia pastoris*. In: Gellissen, G.: Production of recombinant proteins. Wiley-VCH, Weinheim, pp. 143–162.

[11.13] Kaketsuken (2000): Human blood serum albumen. Press release translation from Nikkei Sangyo Shinbun. 15 December 2000. Available at: http://www.deltabiotechnology.com/

[11.14] Delta Technology (2003): Delta technology goes into another large scale manufacturing facility. Available at: http://www.deltabiotechnology.com/

[11.15] Nielsen, J., Villadsen, J., Liden, G. (2003): Bioreaction engineering principles. Kluwer Academic, New York.

[11.16] d'Anjou, M., Daugulis, A. (2000): Mixed-feed exponential feeding for fed-batch culture of recombinant methylotrophic yeast. *Biotechnol. Lett.*, **22**, 341–346.

[11.17] Ren H., Yuan J., Bellgardt K. (2003): Macrokinetic model for methylotrophic *Pichia pastoris* based on stoichiometric balance. *J. Biotechnol.*, **106**, 53–68.

[11.18] Siegel, R., Brierley, R. (1989): Methylotrophic yeast *Pichia pastoris* produced in high-cell-density fermentations with high cell yields as vehicle for recombinant protein production. *Biotechnol. Bioeng.*, **34**, 403–404.

[11.19] Sreekrishna, K., Tschopp, J., Thill, G., Brierley, R., Barr, K. (1998): Expression of human serum albumin in *Pichia pastoris*. US Patent 5 707 828.

[11.20] Ohmura, T., Sumi, A., Ohtani, W., Furuhata, N., Takeshima, K., Kamide, K., Noda, M., Kondo, M., Ishikawa, S., Oohara, K., Yokoyama, K., Fujiwara, N. (1999): Recombinant human serum albumin, process for producing the same and pharmaceutical preparation containing the same. US Patent 5 986 062.

[11.21] Wallman, S. (2003): Process controlled fed-batch fermentation on recombinant HSA secreting *Pichia pastoris* – A standard operating procedure. Available at: http://biotech.nhctc.edu/BT220/SOP/SOP3Obj.html

[11.22] Cino, J. (1999): High yield protein production from *Pichia pastoris* yeast - A protocol for benchtop fermentation. *Am. Biotechnol. Lab.* May edition. Available at: http://www.nbsc.com/files/papers/ABL_Pichia.pdf

[11.23] Amersham Pharmacia (1998): Amersham Pharmacia Biotech chosen to supply equipment for recombinant human serum albumin production. *Downstream*, **27**, 23.

[11.24] Pharmaceutical-Technology.com (2001): Bipha human serum albumin plant. Available at: http://www.pharmaceutical-technology.com/projects/chitose/

[11.25] Kobayashi, K. (2000): Production of recombinant human serum albumin from the methylotrophic yeast *Pichia pastoris*. *Downstream*, **31**, 5.

[11.26] Schilling, B., Goodrick, J., Wan, N. (2001): Scale-up of a high cell density continuous culture with *Pichia pastoris* X-33 for the constitutive expression of rh-Chitinase. *Biotechnol. Prog.*, **17**, 629–633.

[11.27] Jahic, M., Rotticci-Mulder, J., Martinelle, M., Hult, K., Enfors, S. (2002): Modeling of growth and energy metabolism of *Pichia pastoris* producing a fusion protein. *Bioprocess Biosyst. Eng.*, **24**, 385–393.

[11.28] Cornelissen, G., Bertelsen, H., Hahn, B., Schultz, M., Scheffler, U., Werner, E., Leptien, H., Krüß, S., Jansen, A., Gliem, T., Hielscher, M., Wilhelm, B., Sowa, E., Radeke, H., Luttmann, R. (2003): Herstellung rekombinanter Proteine mit *Pichia pastoris* in Integrierter Prozessführung, *Chem.-Ing.-Tech.*, **75**, 281–290.

[11.29] Trinh, L., Phue, J., Shiloach, J. (2003): Effect of methanol feeding strategies on production and yield of recombinant mouse endostatin from *Pichia pastoris*. *Biotechnol. Bioeng.*, **82**, 438–444.

[11.30] Noda, M., Sumi, A., Ohmura, T., Yokoyama, K. (1999): Process for purifying recombinant human serum albumin. US Patent 5 962 649.

[11.31] Curbelo, D., Gahrke, G., Anspach, F., Deckwer, W. (2003): Cost comparison of protein capture from cultivation broths by expanded and packed bed adsorption. *Eng. Life Sci.*, **3**, 406–415.

[11.32] Amersham Biosciences (2002): Cost analysis study favours Streamline for capture. *Downstream*, **34**, 16–18.

[11.33] Ohya, T., Ohyama, M., Kobayashi, K. (2005): Optimization of human serum albumin production in methylotrophic yeast *Pichia pastoris* by repeated fed-batch fermentation. *Biotechnol. Bioeng.*, **90**, 876–887.

[11.34] Bushell, M., Rowe, M., Avignone-Rossa, C., Wardell, J. (2003): Cyclic fed-batch culture for production of human serum albumin in *Pichia pastoris*. *Biotechnol. Bioeng.*, **82**, 679–683.

12

Recombinant Human Insulin

*Demetri Petrides**

12.1 Introduction

Insulin facilitates the metabolism of carbohydrates and is essential for the supply of energy to the cells of the body. Impaired insulin production leads to the disease diabetes mellitus, which is the third largest cause of death in industrialized countries, after cardiovascular diseases and cancer [12.1]. Approximately 18 million people suffer from diabetes in the US [12.2]. Worldwide, the total number of diabetics is estimated to be between 150 and 200 million [12.3] and it is growing at an annual rate of 3–4% [12.4, 12.5].

Human insulin is a polypeptide consisting of 51 amino acids arranged in two chains: A with 21 amino acids, and B consisting of 30 amino acids. The A and B chains are connected by two disulfide bonds. Human insulin has a molecular weight of 5734 g/mol and an isoelectric point of 5.4. Human insulin has historically been produced by four different methods:

- Extraction from human pancreas
- Chemical synthesis via individual amino acids
- Conversion of pork insulin or 'semi-synthesis'
- Fermentation of genetically engineered microorganisms

Extraction from the human pancreas cannot be practiced because the availability of raw material is so limited and there are concerns with propagation of infectious agents. Total synthesis, while technically feasible, is not economically viable because the yield is very low. Production based on pork insulin, also known as 'semi-synthesis,' transforms the

* Corresponding author: ++1/908/654-0088; dpetrides@intelligen.com

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porcine insulin molecule into an exact replica of the human insulin molecule by substituting a single amino acid, threonine, for alanine in the G-30 position. This technology has been developed and implemented by Novo Nordisk A/S (Denmark). However, this option is also quite expensive because it requires the collection and processing of large amounts of porcine pancreases. In addition, its supply is limited by the availability of porcine pancreas.

At least three alternative technologies have been developed for producing human insulin based on fermentation and utilizing recombinant DNA technology [12.6].

12.1.1 Two-chain Method

The first successful technique of biosynthetic human insulin (BHI) production based on recombinant DNA technology was the two-chain method. This technique was developed by Genentech, Inc. (South San Francisco) and scaled up by Eli Lilly and Co. (Indianapolis). Each insulin chain is produced as a β-galactosidase fusion protein in *Escherichia coli,* forming inclusion bodies. The two peptide chains are recovered from the inclusion bodies, purified, and combined to yield human insulin. Later, the β-galactosidase operon was replaced with the tryptophan (Trp) operon, resulting in a substantial yield increase.

12.1.2 Proinsulin Method

The so-called intracellular method of making proinsulin eliminates the need for the separate fermentation and purification trains required by the two-chain method. Intact proinsulin is produced instead. The proinsulin route has been commercialized by Eli Lilly and Co. [12.7]. Figure 12.1 shows the key transformation steps. The *E. coli* cells overproduce Trp-LE'-Met-proinsulin (Trp-LE'-Met-proinsulin is a 121 amino acid peptide signal sequence; proinsulin, with 82 amino acids, is a precursor to insulin) in the form of inclusion bodies, which are recovered and solubilized. Proinsulin is released by cleaving the methionine linker using CNBr. The proinsulin chain is subjected to a folding process to allow intermolecular disulfide bonds to form; and the C peptide, which connects the A and B chains in proinsulin, is then cleaved with enzymes to yield human insulin. A number of chromatography and membrane-filtration steps are utilized to purify the product.

A second method of producing proinsulin was developed by Novo Nordisk A/S. It is based on yeast cells that secrete insulin as a single-chain insulin precursor [12.1]. Secretion simplifies product isolation and purification. The precursor contains the correct disulfide bridges, identical to those of insulin. It is converted into human insulin by transpeptidation in organic solvent in the presence of a threonine ester and trypsin followed by de-esterification. Another advantage of this technology is the ability to reuse the cells by employing a continuous bioreactor-cell separator loop.

In this case example we analyse a process based on the intracellular proinsulin method.

12.2 Market Analysis and Design Basis

The worldwide market for synthetic insulin is estimated to be $ 3.5–4.0 billion and the major players include Novo Nordisk, Eli Lilly, and Sanofi Aventis [12.8]. The market for insulin products is higher because it also includes the cost of the delivery devices and packaging. Treatment with insulin requires on average 0.5 g/patient/year of purified product. Considering the total number of diabetics (150 to 200 million), that corresponds

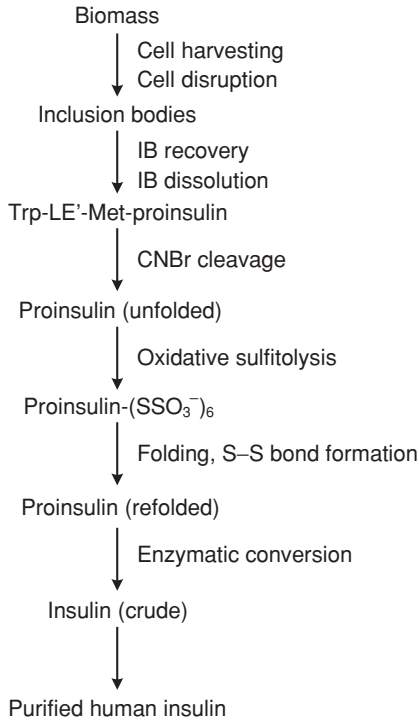


Figure 12.1 *Human insulin from proinsulin fusion protein*

to an annual demand of 75 000 to 100 000 kg of purified insulin. However, the current worldwide production is only 20 000 to 30 000 kg/year because most patients in the developing countries cannot afford to pay the $ 250–750/year required for purchasing the medicine. There is a great need for additional capacity and improved processes that can manufacture the product at a lower cost to satisfy the demand in the developing nations. The plant analysed in this example has a capacity of around 1800 kg of purified biosynthetic human insulin (BHI) per year. This is a relatively large plant for producing polypeptide-based biopharmaceuticals. It can satisfy the demand of around 3.5 million patients or roughly 25% of the US market. The plant operates around the clock for 330 days a year. A new batch is initiated every 48 hours, resulting in 160 batches per year. The fermentation broth volume per batch is approximately 37.5 m^3.

12.2.1 Process Description

The entire flowsheet for the production of BHI is shown in Figure 12.2. It is divided into four sections: (i) Fermentation, (ii) Primary Recovery, (iii) Reactions, and (iv) Final Purification. A section in SuperPro Designer® is simply a set of unit procedures (processing steps).

Fermentation Section. Fermentation medium is prepared in a stainless steel tank (V-101) and sterilized in a continuous heat sterilizer (ST-101). The axial compressor (G-101) and the absolute filter (AF-101) provide sterile air and ammonia to the fermenter at an average

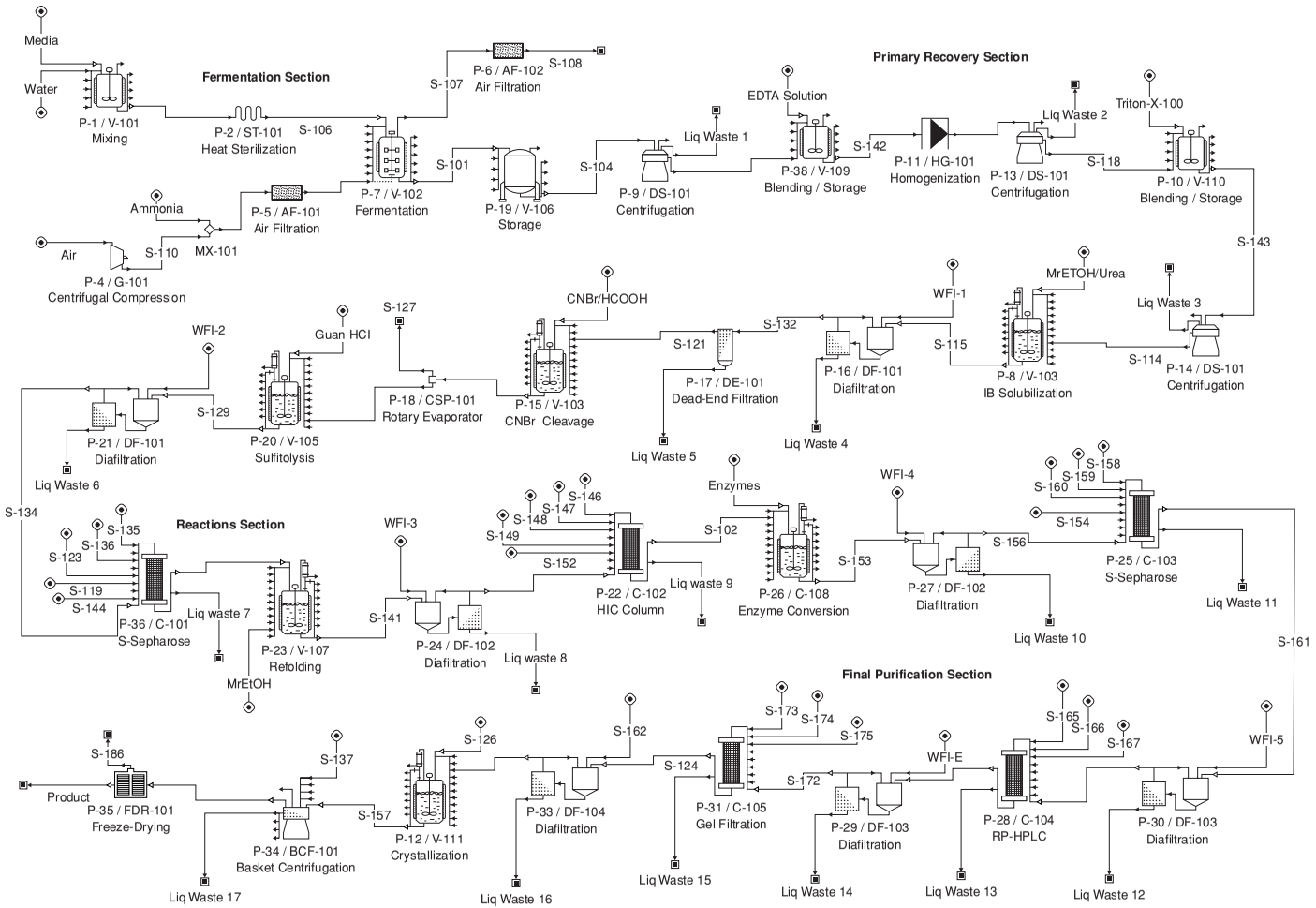


Figure 12.2 *Insulin production flowsheet*

rate of 0.5 VVM. A two-step seed fermenter train (not shown in the flowsheet) is used to inoculate the 50 m^3 production fermenter (V-102) with transformed *E. coli* cells. These cells are used to produce the Trp-LE'-MET-proinsulin precursor of insulin, which is retained in the cellular biomass. The fermentation time in the production fermenter is about 18 hours, and the fermentation temperature is 37 °C. The final concentration of *E. coli* in the production fermenter is about 30 g/L (dry cell weight). The Trp operon is turned on when the *E. coli* fermentation runs out of tryptophan. The chimeric protein Trp-LE'-MET-proinsulin accumulates intracellularly as insoluble aggregates (inclusion bodies), and this decreases the rate at which the protein is degraded by proteolytic enzymes. In the base case, it was assumed that the inclusion bodies (IBs) constitute 20% of total dry cell mass. At the end of fermentation, the broth is cooled to 10 °C to minimize cell lysis. After completing each processing step in the Fermentation Section (and subsequent sections), the equipment is cleaned thoroughly in order to prepare for the next batch of product.

Downstream Sections

(i) *Primary recovery section*
After the end of fermentation, the broth is transferred into a surge tank (V-106), which isolates the upstream from the downstream section of the plant. Three disk-stack centrifuges (DS-101) operating in parallel are used for cell harvesting. Note that a single unit-procedure icon on the screen of SuperPro Designer may represent multiple equipment items operating in parallel. During centrifugation, the broth is concentrated from 37 000 L to 9165 L, and most of the extracellular impurities are removed. The cell-recovery yield is 98%. The cell sludge is diluted with an equal volume of buffer solution [buffer composition: 96.4% w/w WFI (water for injection), 0.7% EDTA, and 2.9% TRIS-base] using a blending tank (V-109). The buffer facilitates the separation of the cell debris particles from inclusion bodies. Next, a high-pressure homogenizer (HG-101) is used to break the cells and release the inclusion bodies. The broth undergoes three passes under a pressure drop of 800 bar. The exit temperature is maintained at around 10 °C. The same centrifuges as before (DS-101) are used for inclusion-body recovery (P-13). The reuse of these centrifuges can be seen by noting that procedures P-9 and P-13 have the same equipment name, DS-101. The IBs are recovered in the heavy phase (with a yield of 98%) while most of the cell debris particles remain in the light phase. This is possible because the density (1.3 g/cm^3) and size (diameter about 1 μm) of the IBs are significantly greater than those of the cell debris particles. The IB sludge, which contains approximately 20% solids w/w, is washed with WFI containing 0.66% w/w Triton-X100 detergent (the volume of solution is twice the volume of inclusion body sludge) and recentrifuged (P-14) using the same centrifuges as before (DS-101). The detergent solution facilitates purification (dissociation of debris and soluble proteins from inclusion bodies). The exit temperature is maintained at 10 °C. The slurry volume at the end of the primary recovery section is around 1400 L.

(ii) *Reactions section*
Inclusion body solubilization. The inclusion-body suspension is transferred to a glass-lined reaction tank (V-103) and is mixed with urea and 2-mercaptoethanol to final concentrations of 300 g/L (5 M) and 40 g/L, respectively. Urea is a chaotropic agent

that dissolves the denatured protein in the inclusion bodies and 2-mercaptoethanol is a reductant that reduces disulfide bonds. A reaction time of 8 hours is required to reach a solubilization yield of 95%. The inclusion bodies are composed of 80% w/w Trp-LE′-Met-proinsulin, with the remainder being other (contaminant) proteins. At the end of the solubilization reaction, a diafiltration unit (DF-101) is used to replace urea and 2-mercaptoethanol with WFI and to concentrate the solution. This operation is performed in 6 h with a recovery yield of 98%. All remaining fine particles (biomass, debris, and inclusion bodies) are removed using a polishing dead-end filter (DE-101). This polishing filter protects the chromatographic units that are used further downstream. The solution volume at this point is around 5200 L.

CNBr cleavage. The chimeric protein is cleaved with CNBr (cyanogen bromide) into the signal sequence Trp-LE′-Met, which contains 121 amino acids, and the denatured proinsulin (82 amino acids) in the same reactor (V-103) that was used for IB solubilization (procedure P-15). The reaction is carried out in a 70% formic acid solution containing 30-fold molar excess of CNBr (stoichiometrically, one mole of CNBr is required per mole of Trp-LE′-Met-proinsulin). The reaction takes 12 h at 20 °C and reaches a yield of 95%. The mass of the released proinsulin is approximately 30% of the mass of Trp-LE′-Met-proinsulin. A small amount of cyanide gas is formed as a by-product of the cleavage reaction. Detailed information on CNBr cleavage is available in the patent literature [12.9]. The formic acid, unchanged CNBr, and generated cyanide gas are removed by applying vacuum and raising the temperature to around 35 °C (the boiling point of CNBr). This operation is carried out in a rotary vacuum evaporator (CSP-101) and takes 1 h. Since cyanide gas is toxic, all air exhausted from the vessels is scrubbed with a solution of hypochlorite, which is prepared and maintained in situ [12.7].

Sulfitolysis. Sulfitolysis of the denatured proinsulin takes place in a reaction tank (V-105) under alkaline conditions (pH 9–11). This operation is designed to unfold proinsulin, break any disulfide bonds, and add SO_3 moieties to all sulfur residues on the cysteines. The product of interest is human proinsulin$(S\text{-}SO_3^-)_6$ (protein-S-sulfonate). The sulfitolysis step is necessary for two reasons: (1) the proinsulin probably is not folded in the correct configuration when expressed in *E. coli* as part of a fusion protein, and (2) the cyanogen bromide treatment tends to break existing disulfide bonds. The final sulfitolysis mixture contains 50% w/w guanidine•HCl (6 M), 0.35% ammonium bicarbonate (NH_4HCO_3), 3% Na_2SO_3, and 1.5% $Na_2S_4O_6$ [12.10]. A reaction time of 12 h is required to reach a yield of 95%. The presence of the denaturing reagent (guanidine•HCl) prevents refolding and cross-folding of the same protein molecule onto itself or two separate protein molecules onto each other. Urea also may be used as a denaturing reagent. Upon completion of the sulfitolysis reaction, the sulfitolysis solution is exchanged with WFI to a final guanidine•HCl concentration of 20% w/w. This procedure, P-21, utilizes the DF-101 diafilter that also handles buffer exchange after IB solubilization. The human proinsulin$(S\text{-}SO_3^-)_6$ is then chromatographically purified using three ion-exchange columns (C-101) operating in parallel. Each column has a diameter of 140 cm and a bed height of 25 cm. A cation-exchange resin is used (SP Sepharose Fast Flow from GE Healthcare) operating at pH 4. The eluant solution contains 69.5% w/w WFI, 29% urea, and 1.5% NaCl. Urea, a denaturing agent, is used to prevent incorrect refolding and cross-folding of proinsulin$(S\text{-}SO_3^-)_6$. The following operating assumptions are made: (1) the column is equilibrated for

30 minutes prior to loading, (2) the total resin binding capacity is 20 mg/mL, (3) the eluant volume is equal to 5 column volumes (CVs), (4) the total volume of the solutions for column wash, regeneration, and storage is 15 CVs, and (5) the protein of interest is recovered in 1.5 CVs of eluant buffer with a recovery yield of 90%.

Refolding. This operation catalyses the removal of the SO_3^{2-} moiety and then allows disulfide-bond formation and correct refolding of the proinsulin to its native form. It takes place in a reaction tank (V-107). This process step involves treatment with mercaptoethanol (MrEtOH), a reductant that facilitates the disulfide interchange reaction. It is added at a ratio of 1.5 mol of mercaptoethanol to 1 mol of SO_3^{2-}. Dilution to a proinsulin$(S\text{-}SO_3^-)_6$ concentration of less than 1 g/L is required to prevent cross-folding of proinsulin molecules. The reaction is carried out at 8 °C for 12 h and reaches a yield of 85%. After completion of the refolding step, the refolding reagents are replaced with WFI and the protein solution is concentrated using a diafiltration unit (DF-102), which has a product recovery yield of 95% (5% of the protein denatures). The volume of the solution at this point is around 5000 L. Next, the human proinsulin is chromatographically purified in a hydrophobic interaction chromatography (HIC) column (C-102). The following operating assumptions were made: (1) the column is equilibrated for 30 minutes prior to loading, (2) the total resin binding capacity is 20 mg/mL, (3) the eluant volume is equal to 6 column volumes (CVs), (4) the total volume of the solutions for column wash, regeneration, and storage is 15 CVs, (5) the protein of interest is recovered in 1 CV of eluant buffer with a recovery yield of 90%, and (6) the material of a batch is handled in three cycles.

Enzymatic conversion. The removal of the C-peptide from human proinsulin is carried out enzymatically (using trypsin and carboxypeptidase B) in a reaction tank (V-108). Trypsin cleaves at the carboxy terminal of internal lysine and arginine residues, and carboxypeptidase B removes terminal amino acids. The amount of trypsin used is rate-limiting and allows intact human insulin to be formed. Carboxipeptidase is added to a final concentration of 4 mg/L, while trypsin is added to a final concentration of 1 mg/L. The reaction takes place at 30 °C for 4 h and reaches a conversion yield of 95%. The volume of the solution at this point is around 4300 L. A schematic representation of the various transformation steps required to convert fusion protein into active insulin is available in the literature [12.3].

(iii) *Final purification section*

A purification sequence based on multimodal chromatography, which exploits differences in molecular charge, size, and hydrophobicity, is used to isolate biosynthetic human insulin. A description of all the purification steps follows.

The enzymatic conversion solution is exchanged with WFI and concentrated by a factor of four in a diafilter (DF-102). An ion-exchange column (C-103) is used to purify the insulin solution. The following operating assumptions were made: (1) the column is equilibrated for 30 minutes prior to loading, (2) the total resin binding capacity is 20 mg/mL, (3) the eluant volume is equal to 8 CVs and the eluant is a 11.5% w/w solution of NaCl in WFI, (4) the total volume of the solutions for column wash, regeneration, and storage is 14 CVs, (5) the protein of interest is recovered in 1.5 CV of eluant buffer with a recovery yield of 95%, and (6) the material from each batch is handled in four cycles. The liquid volume at this point is around 1100 L.

Next, the ion-exchange eluant solution is exchanged with WFI in a diafilter (DF-103) and is concentrated by a factor of 2.0. A recovery yield of 98% was assumed for this step (2% denatures).

The purification of the insulin solution proceeds with a reversed-phase high-performance liquid chromatography (RP-HPLC) step (C-104). Detailed information on the use of RP-HPLC for insulin purification is available in the literature. Analytical studies with a variety of reversed-phase systems have shown that an acidic mobile phase can provide excellent resolution of insulin from structurally similar insulin-like components. Minor modifications in the insulin molecule resulting in monodesamido formation at the 21st amino acid of the A chain, or derivatization of amines via carbamoylation or formylation, result in insulin derivatives having significantly increased retention. Derivatives of this nature are typical of the kind of insulin-like components that are found in the charge stream going into the reversed-phase purification. The use of an acidic mobile phase results in elution of all the derivatives after the insulin peak, while the use of mildly alkaline pH results in derivatives eluting on either side of the parent insulin peak. An ideal pH for insulin purification is in the region of 3.0–4.0, since this pH range is far enough below the isoelectric pH of 5.4 to provide for good insulin solubility. An eluant buffer with an acetic acid concentration of 0.25 M meets these operational criteria because it is compatible with the chromatography and provides good insulin solubility. A 90% insulin yield was assumed in the RP-HPLC step with the following operating conditions: (1) the column is equilibrated for 30 minutes prior to loading, (2) the total resin binding capacity is 15 mg/mL, (3) the column height is 25 cm, (4) the eluant volume is equal to 6 CV and its composition is 25% w/w acetonitrile, 1.5% w/w acetic acid, and 73.5% w/w WFI, (5) the total volume of the solutions for column wash, equilibration, regeneration, and storage is 6 CVs, and (6) the protein of interest is recovered in 1 CV of eluant buffer with a recovery yield of 90%.

The RP-HPLC buffer is exchanged with WFI and concentrated by a factor of 2.0 in a diafilter (DF-103) that has a product recovery yield of 98% (2% denatures). Purification is completed by a gel-filtration chromatography column (C-105). The following operating assumptions were made: (1) the column is equilibrated for 30 minutes prior to loading, (2) the sample volume is equal to 5% of the column volume, (3) the eluant volume is equal to 4 CVs, (4) the total volume of the solutions for column wash, depyrogenation, stripping, and storage is 6 CVs, and (5) the protein of interest is recovered in 0.5 CV of eluant buffer with a recovery yield of 90%. The mobile phase is a solution of acetic acid.

Next, the same diafilter (DF-103) is used to concentrate the purified insulin solution by a factor of ten. The liquid volume at this point is around 500 L, which contains approximately 12.8 kg of insulin. This material is pumped into a jacketed and agitated reaction tank (V-111). Ammonium acetate and zinc chloride are added to the protein solution until each reaches a final concentration of 0.02 M [12.4]. The pH is adjusted to between 5.4 and 6.2. The crystallization is carried out at 5 °C for 12 h. Insulin crystallizes with zinc with the following stoichiometry: $insulin_6$-Zn_2. Step recovery on insulin is around 90%.

The crystals are recovered with a basket centrifuge (BCF-101) with a yield of 95%. Finally, the crystals are freeze-dried (FDR-101). The purity of the crystallized end product is between 99.5 and 99.9% measured by analytical high-pressure liquid chromatography (HPLC). Approximately 11.5 kg of product is recovered per batch. The overall recovery yield is around 32%.

12.2.2 Inventory Analysis and Environmental Assessment

Table 12.1 displays the raw material requirements in tons per year, kg per batch, and kg per kg of main product (MP = purified insulin crystals). Note the huge amounts of WFI, water, NaOH (0.5 M), H_3PO_4 (20% w/w), urea, acetic acid, formic acid, guanidine hydrochloride, and acetonitrile required per kg of final product. All of these materials end up in waste streams. The total waste-to-product ratio is 55 000:1 and 12 600:1 without considering water.

Figure 12.3 shows the EI_{MV} of the process. Acetic acid, phosphoric acid, formic acid, acetonitrile, urea, and guanidine • HCl dominate the input side. They are all based on oil and have some acute toxicity. Therefore, the impact groups Availability and Organisms contribute most to the overall environmental impact. The same substances also dominate the output EI. Their nitrogen and phosphorus content and their chemical oxygen demand leads to the dominance of the impact group Water/Soil on the output side of the overall environmental impact.

In the base case, it was assumed that this waste is treated and disposed of. However, opportunities may exist for recycling some chemicals for in-process use and recovering others for off-site use. For instance, formic acid (HCOOH), acetonitrile, and urea are good candidates for recycling and recovery. Formic acid is used in large quantities (11 tons/batch)

Table 12.1 *Raw material requirements (1 batch = 11.5 kg MP)*

Raw material	(metric tons/yr)	(kg/batch)	(kg/kg P)
Glucose	782.2	4889	432
Salts	71.43	446	39.5
Air	3647	22 800	2020
Ammonia	75.69	473	41.8
Water	9854	61 590	5450
Water for injection (WFI)	67 030	418 900	37 000
NaOH (0.5 M)	3991	24 940	2210
H_3PO_4 (20% w/w)	4405	27 530	2430
TRIS Base	43.20	270	23.9
EDTA	10.43	65.2	5.8
Triton-X-100	3.035	19.0	1.7
Cyanogen bromide (CNBr)	15.27	95.4	8.44
Formic acid	1752	10 950	968
Urea	3062	19 140	1690
Mercaptoethanol	98.66	617	54.5
NH_4HCO_3	5.551	34.7	3.07
$Na_2S_4O_6$	24.16	151	13.4
Sodium sulfite	48.32	302	26.7
Guanidine HCl	805.6	5034	445
Sodium chloride	778	4862	423
Sodium hydroxide	137.7	860	76.1
Acetic acid	2262	14 140	1250
Enzymes	0.003	0.021	0.002
Acetonitrile	767.2	4794	424
Ammonium acetate	0.181	1.133	0.100
Zinc chloride	0.320	2.000	0.177
Sum	99 670	622 905	55 037

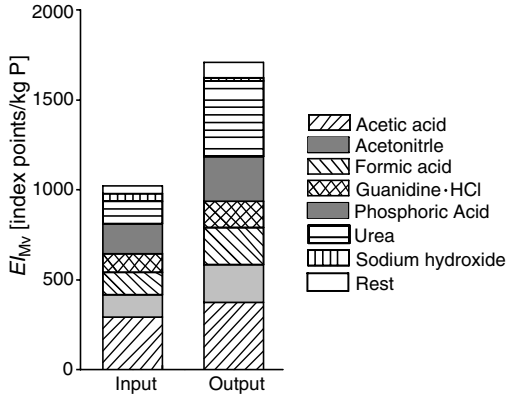


Figure 12.3 *Environmental indices (EI_{MV}) of the process*

in the CNBr cleavage step (V-103) and it is removed using a rotary vacuum evaporator (CSP-101), along with small quantities of CNBr, water, and urea. The recovered formic acid can be readily purified by distillation and recycled in the process. Around 2 tons per batch of urea is used for the dissolution of inclusion bodies (V-103) and 17 tons per batch is used in the first chromatography step (C-101) to purify proinsulin$(S\text{-}SO_3^-)_6$ before its refolding. Approximately 90% of the urea appears in just two waste streams (Liquid Wastes 4 and 7). It is unlikely that these urea-containing streams can be purified economically for in-process recycling. However, these solutions can be concentrated, neutralized, and shipped off site for further processing and utilization as a nitrogen fertilizer.

Approximately 4.8 tons per batch of acetonitrile is used in the reversed-phase HPLC column (C-104), and most of it ends up in the waste stream of the column (Liquid Waste 13) along with 6.8 tons of water, 1.85 tons of acetic acid, and small amounts of NaCl and other impurities. It is unlikely that acetonitrile can be recovered economically to meet the high purity specifications for a step so close to the end of the purification train. However, there may be a market for off-site use.

12.2.3 Production Scheduling

Figure 12.4 displays the equipment occupancy chart for six consecutive batches. The process batch time is approximately 12 days. This is the time required to go from the preparation of raw materials to final product for a single batch. However, since most of the equipment items are utilized for much shorter periods within a batch, a new batch is initiated every 2 days. Multiple bars on the same line within a batch (e.g., for DS-101, V-103, DF-101, DF-102, and DF-103) represent reuse (sharing) of equipment by multiple procedures. White space represents idle time. The equipment with the least idle time between consecutive batches is the *time (or scheduling) bottleneck* (DF-101 in this case) that determines the maximum number of batches per year. Its cycle time (approximately 41.5 h) is the minimum possible time between consecutive batches. This plant operates around the clock and processes 160 batches per year. The top six lines of Figure 12.4 correspond to cleaning-in-place (CIP) skids utilized to thoroughly clean the equipment. CIP skids are common bottlenecks in biopharmaceutical manufacturing facilities.

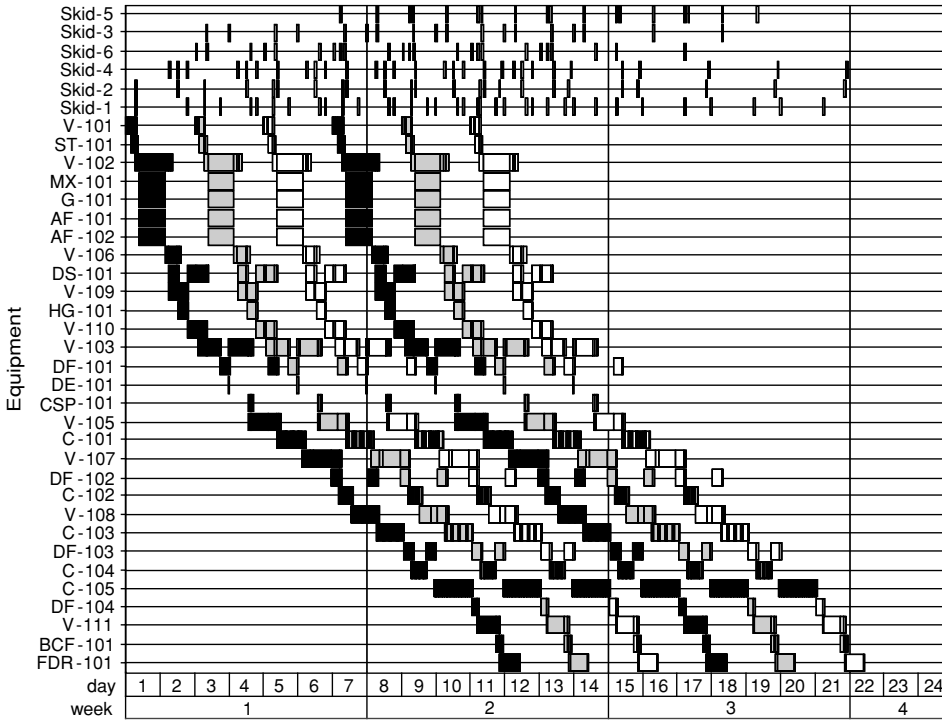


Figure 12.4 *Equipment occupancy as a function of time for six consecutive batches*

12.3 Economic Assessment

Table 12.2 shows the key economic evaluation results generated by using the built-in cost functions of SuperPro Designer®. For a plant of this capacity, the total capital investment is $ 145 million. The unit-production cost is $ 67.2/g of purified insulin crystals. Assuming a selling price of $ 120/g, the project yields an after-tax internal rate of return (IRR) of 63.5% and a net present value (NPV) of $ 397 million (assuming a discount interest of 7%). In the US, the retail price of vials that contain 40 mg of insulin is around $ 25 [12.3], which is equivalent to $ 625/g of active insulin. Therefore, a selling price of $ 120/g of bulk insulin corresponds to around 20% of the retail selling price of the final product, which is reasonable considering the additional cost and profit margins for formulation, packaging, distribution, etc. Based on these results, this project represents a very attractive investment. However, if amortization of up-front R&D costs is considered in the economic evaluation, the numbers change drastically. For instance, a modest amount of $ 150 million for up-front R&D cost amortized over a period of 10 years reduces the IRR to 21%.

Figure 12.5 breaks down the operating cost. The cost of consumables is the most important, accounting for 38% of the overall manufacturing cost. This represents the expense for periodically replacing the resins of the chromatography columns and the membranes of the membrane filters. The cost of raw materials lies in the second position, accounting

Table 12.2 *Key economic evaluation results*

Parameter	Value
Direct fixed capital	$ 117 million
Total capital investment	$ 145 million
Plant throughput	1810 kg/year
Manufacturing cost	$ 122 million/year
Unit-production cost	$ 67.2/g
Selling price	$ 120/g
Revenues	$ 217 million/year
Gross profit	$ 95.4 million/year
Taxes (40%)	$ 38.2 million/year
Net profit	$ 68.4 million/year
Internal rate of return (after taxes)	63.5%
Net present value (for 7% discount interest)	$ 397 million

for 29% of the overall cost. The facility overhead accounts for 18% of the total cost. This mainly represents the depreciation and maintenance of the facility. Treatment and disposal of waste materials account for 7% of the total cost. As mentioned in the material-balance section, recycling and reuse of some of the waste materials may reduce this cost. Labor lies in the fifth position, accounting for 6% of the total cost. Approximately 60 operators are required to run the plant around the clock, supported by 12 scientists for QC/QA work. The cost of utilities is only 0.2% because it comprises only electricity and the small amounts of heating and cooling required. The cost of purified water is treated as a raw material and not as a utility.

Figure 12.6 displays the cost distribution per flowsheet section. Only 6% of the overall cost is associated with fermentation. The other 94% is associated with the recovery and purification sections. This is common for high-value biopharmaceuticals that are produced from recombinant *E. coli*. Most of the cost is associated with the reactions section because of the large amounts of expensive raw materials and consumables that are utilized in that section.

Table 12.3 for each consumable displays its annual amount, unit cost, annual cost, and contribution to the overall consumables cost. The gel filtration resin is the most expensive consumable, followed by the first S-Sepharose resin and the HIC resin. Gel filtration accounts for 10% of the overall manufacturing cost. Replacement of the gel-filtration step with an alternative and more efficient chromatography step can have a significant impact on the manufacturing cost and should be considered in future versions of this process.

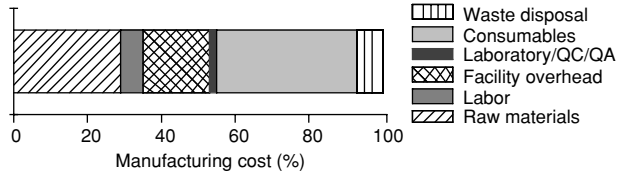


Figure 12.5 *Breakdown of manufacturing cost*

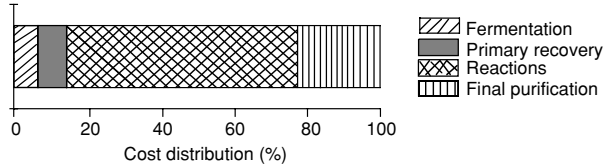


Figure 12.6 *Cost distribution per flowsheet section*

Finally, Table 12.4 for each raw material displays its price, annual amount, annual cost, and contribution to the overall raw materials cost. WFI, acetic acid, urea, and H_3PO_4 (20% w/w) are the major contributors to the raw materials cost. The solution of H_3PO_4 is used for equipment cleaning.

12.4 Throughput-Increase Options

In the base case, a new batch is initiated every 48 h. Most of the equipment items, however, are utilized for less than 24 h per batch (see Figure 12.4). If the market demand for insulin grows, this provides an opportunity for increasing plant throughput without increasing major capital expenditures. A realistic improvement is to initiate a batch every 24 h. This will require a new fermenter of the same size whose operation will be staggered relative to the existing unit so that one fermenter is ready for harvesting every day. Such a production change will also require additional equipment of the following types: (1) disk-stack centrifuges to reduce the occupancy of DS-101 to less than 24 h; (2) two new reactors to reduce the occupancy of V-103 and V-107; a new gel-filtration chromatography column, and (3) membrane filters to reduce the occupancy of DF-101, DF-102, and DF-103.

The additional capital investment for such a retrofit is around $ 30–40 million. This additional investment will allow the plant's capacity to be doubled, and the new unit-production cost will be around $ 62/g. The reduction in the unit-production cost is rather small because the majority of the manufacturing cost is associated with raw materials, consumables, and waste disposal that scale approximately linearly with production.

Table 12.3 *Cost of consumables*

Consumable	Annual amount	Unit cost ($/unit)	Annual cost ($ million)	Share (%)
UF membrane	4790 m^2	800	3.83	8.5
HIC resin	4310 L	2000	8.62	19.0
Gel-filtration resin	16 200 L	800	13.0	28.7
DEF cartridge	3840 items	800	3.07	6.8
S-Seph-1 resin	8290 L	1200	9.94	22.0
S-Seph-2 resin	2220 L	1500	3.33	7.4
RP-HPLC-resin	1750 L	2000	3.49	7.7
Sum			45.3	100.0

Table 12.4 *Cost of raw materials*

Raw Material	Unit Cost ($/kg)	Annual amount (tons)	Annual cost (ths. $)	Share (%)
Glucose	0.60	782.2	469	1.3
Salts	1.00	71.43	71	0.2
Ammonia	0.70	75.69	53	0.15
Water	0.05	9854	493	1.4
WFI	0.10	67 030	6703	18.9
NaOH (0.5 M)	0.50	3991	1995	5.6
H_3PO_4 (20% w/w)	1.00	4405	4405	12.4
TRIS base	6.00	43.20	259	0.73
EDTA	18.50	10.43	193	0.54
Triton-X-100	1.50	3.035	5	0.01
Cyanogen bromide (CNBr)	11.00	15.27	168	0.47
Formic acid	1.60	1752	2802	7.9
Urea	1.52	3063	4655	13.1
Mercaptoethanol	3.00	98.66	296	0.84
NH_4HCO_3	1.00	5.551	6	0.02
$Na_2S_4O_6$	0.60	24.16	14	0.04
Sodium sulfite	0.40	48.32	19	0.05
Guanidine HCl	2.15	805.6	1732	4.9
Sodium chloride	1.23	778.0	957	2.7
Sodium hydroxide	3.50	137.7	482	1.4
Acetic acid	2.50	2262	5656	16.0
Enzymes	500 000	0.003	1691	4.8
Acetonitrile	3.00	767.2	2302	6.5
Ammonium acetate	15.00	0.181	3	0.01
Zinc chloride	12.00	0.320	4	0.01
Sum		95 218	35 433	100.0

12.5 Conclusions

In this chapter, we have analysed the production of biosynthetic human insulin from recombinant *E. coli*. The development of the process was based on information available in the literature. The work was facilitated using SuperPro Designer®, a comprehensive process simulator. The analysis has clearly shown that most of the cost for manufacturing high-value biopharmaceuticals with recombinant *E. coli* is associated with the recovery and purification of the product. The large number of conversion and separation steps required to recover and purify the product lead to a low recovery yield of 32% and a huge waste-to-product ratio (55 000:1). Improved processes that result in reduced manufacturing cost can greatly contribute towards the effort of making insulin accessible to diabetics in the developing nations.

References

[12.1] Barfoed, H. (1987): Insulin production technology. *Chem. Eng. Prog.*, **83**, 49–54.
[12.2] Dalton, L. (2004): Drugs for diabetes. *Chem. Eng. News*, **82**, October 25, 59–67.

[12.3] Klyushnichenko, V., Bruch, R., Bulychev, A., Ditsch, A., Pernenkil, L., Tao, F. (2004): Feasibility of international technology transfer for the production of recombinant human insulin. *BioProcess Int.*, **2** (8), 48–59.

[12.4] Datar, R., Rosen C. (1990): Downstream process economics. In: Asenjo, J.: Separation processes in biotechnology. Marcel Dekker, New York, pp. 741–793.

[12.5] Petrides, D., Sapidou, E., Calandranis, J. (1995): Computer-aided process analysis and economic evaluation for biosynthetic human insulin production – A case study. *Biotechnol. Bioeng.*, **5**, 529–541.

[12.6] Ladisch, M., Kohlmann, K. (1992): Recombinant human insulin. *Biotechnol. Prog.*, **8**, 469–478.

[12.7] Kehoe, J. (1989): The story of biosynthetic human insulin. In: Sikdar, S., Bier, M., Todd, P.: Frontiers in bioprocessing. CRC Press, Boca Raton, pp. 45–49.

[12.8] Ainsworth, S. (2005): Biopharmaceuticals. *Chem. Eng. News*, **83**, June 6, 21–29.

[12.9] Di Marchi, R. (1984): Process for inhibiting undesired thiol reactions during cyanogen bromide cleavage of peptides. US Patent 4 451 396.

[12.10] Bobbitt, J., Manetta, J. (1990): Purification and refolding of recombinant proteins. US Patent 4 923 967.

13
Monoclonal Antibodies

13.1 Introduction

Monoclonal antibodies (Mabs) are an important class of bioproducts. Their application includes diagnostic as well as therapeutic use [13.1–13.4]. The number of therapeutic Mabs in production is expected to rise in the coming years and the annual demand of many Mab will exceed 100 kg/yr [13.5]. Antibody-based therapeutics are emerging as an important segment of biopharmaceuticals, representing $ 5.2 billion or 22% of total sales and growing 38% in 2002 [13.6]. However, production capacities are limited today [13.7]. Therefore, new plants will be built and existing facilities will be optimized to increase production. Hence, there is a strong need for a better understanding of the Mab processes and the uncertainties that influence them.

13.2 Process Model

Figure 13.1 shows the process-flow diagram used in this model. Today, the method of choice for the production of Mabs is animal cell culture techniques [13.3, 13.8–13.11]. In addition to the production fermenter, a seed train is necessary to provide the needed amount of cells. In the model, the seed train includes the T-flasks P-1, the roller bottle P-2, the bag bioreactor P-3 (5 L), the 40 L bag bioreactor P-4, the first seed bioreactor P-5, and as the last step the second seed reactor P-10 (2 m^3). The volume is increased by a factor 7.5 in each step and the whole seed train takes 24 days. The cells are first grown in serum-containing medium. In the second seed reactor they are adapted to serum-free medium (following [13.11–13.13]). The medium for the two seed reactors is prepared in the tanks P-6 and P-11, respectively, and filter-sterilized (P-7, P-12).

In tank P-21 and filter P-22, the serum-free production medium is prepared and sterilized. At a concentration of 25 g of media powder per liter of fermenter volume, the cost of the bioreactor solution is $ 5/L. Units P-23 and P-24 supply air to the bioreactor (P-20), and

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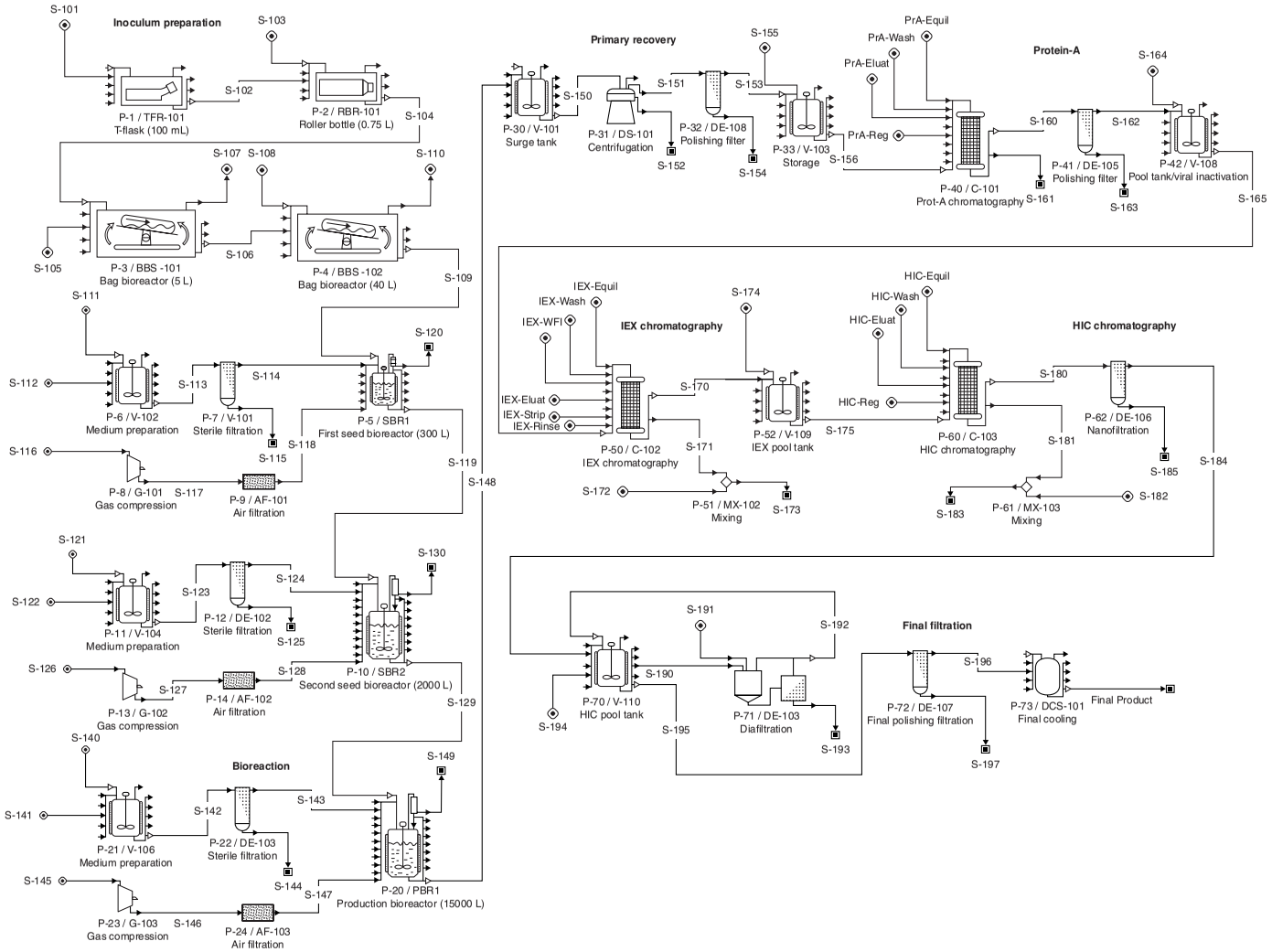


Figure 13.1 *Process-flow diagram for the base-case model of the monoclonal antibody production*

the spent air is discharged in stream S-149. The fermentation is run as a fed batch [13.14, 13.15]. In the base case, two fermenters, each with a working volume of 15 m^3, were assumed to work in staggered mode. 5% of the batches are expected to fail. Over a period of 14 days, the media's components are converted into biomass (11 g/L), carbon dioxide, monoclonal antibodies (1 g/L), and several organic components (proteins, peptides, organic acids, and others) that are described as 'impurities' (1 g/L). After the bioreaction stops, the reactor content is transferred to the harvest (surge) tank P-30.

The exact structure of the product separation and purification depends on the class and subclass of the Mab, as well as the experience of the design engineer, and varies from company to company. However, the general structure is usually very similar for most Mabs, and the process-flow diagram presented here should give a realistic representation for Mab processes.

In the primary recovery section the biomass is separated by centrifugation (P-31), and the remaining cell debris in the depth (polishing) filter P-32. The product solution is stored in tank P-33. The next step in purification is usually carried out in several chromatography steps [13.10, 13.11, 13.16]. The chromatography has certain sensitivity for ionic strength, salt concentration, and pH of the feed [13.16]. In the model, the product solution is first passed through a protein A chromatography column (P-40). The Mab is retained, then the column is washed and the product is eluted with a sodium citrate buffer. The product concentration is increased by a factor of 7.5 in this step. Four cycles are needed to process the product solution of one fed-batch fermentation. The eluant is filtered in P-41 and stored in tank P-42. There, acetic acid is added (S-164) and the solution is held for 1.5 h to inactivate possible viral contaminants.

In the next step, the solution is processed through ion-exchange chromatography P-50. After the load, the column is washed with a buffer solution and the Mab is eluted with a gradient elution (sodium chloride concentration). The serum proteins are eluted in the order of their isoelectric point. Since immunoglobulins are the most basic of the major serum proteins, they are eluted first [13.17]. Three cycles are needed to process the product solution. In tank P-52, ammonium sulfate is added to increase the ionic strength of the solution. How exactly the ionic strength is varied before the hydrophobic-interaction chromatography (HIC) depends on the kind of Mab produced. Ammonium sulfate is one of several possibilities. HIC is used to remove protein A that might be leached from the protein A column, as well as antibody aggregates and DNA [13.9]. The Mab is retained in the column, then eluted, and filtered in P-62. The liquid waste is neutralized in P-61 (HCl).

In the final filtration section the volume of the product solution is reduced by diafiltration in P-71 where the product is transferred to phosphate-buffered saline (PBS) buffer. Glycine is added to stabilize the product (S-194). After a final polishing filtration (P-72), the product is cooled before final formulation and packaging (P-73). In the downstream processing, a 90% yield in each step was assumed for the centrifugation and the three chromatography steps.

13.3 Inventory Analysis

The final product contains 9.5 kg of Mab per batch with one 15 m^3 fermenter. For the complete recipe, a duration of 41 days was calculated where a fermentation takes 14 days,

Table 13.1 *Material balance for the model of monoclonal antibody production*

Component	Input (kg/kg P)	Output (kg/kg P)
Acetic acid	1.14	1.14
Ammonium sulfate	11.4	11.4
Biomass		19.1
Carbon dioxide		3.82
Glycine	0.11	0.11
Hydrogen chloride	61.4	
Inorganic salts		214
Organic impurities		1.9
Mab (final product)		1.0
Mab (loss)		0.66
Medium (inoculum prep.)	0.66	0.30
Na_2HPO_4/NaH_2PO_4	3.4	3.1
Serum-free medium	42.3	4.5
Sodium chloride	97.2	
Sodium citrate	1.8	1.8
Sodium hydroxide	61.1	
RO water/water (CIP)	6060	
Water for injection (WFI)	8510	
Water (in output)		14 600
Sum	14 830	14 830

and an additional lag time of 1.5 days was expected between consecutive batches. With 326 operating days assumed, the annual production is 307 kg produced in 34 batches. The yield of the downstream recovery is 63%, which lies in the range stated by Chovav et al. [13.6].

Table 13.1 shows the material balance for the process. Altogether, there are 4560 tons of raw materials needed per year, which equals 14 800 kg per kg of final product (kg/kg P), or 260 kg/kg P without considering water. Most of this is water for injection (WFI), of which around 75% is needed in the inoculum preparation and bioreaction sections and 25% in the downstream sections. Besides the WFI, less purified water (called RO water in the model) is used in cleaning procedures. In the bioreactions a large amount of serum-free medium is needed and also some serum-containing medium for the inoculum preparation. All other input materials are mainly consumed in the downstream process: Sodium chloride is used in the different buffers for chromatography and diafiltration steps. Sodium hydroxide is needed in the cleaning-in-place procedures (CIP) and for the regeneration of the HIC and ion-exchange chromatography (IXC) columns. Hydrogen chloride is mainly consumed to neutralize the waste streams containing higher concentrations of NaOH that result from the chromatography steps (S-171, S-181) and the cleaning procedures. Ammonium sulfate is used before the HIC to increase the ionic strength of the product solution, and acetic acid is needed for viral inactivation. Additionally, some other compounds are part of the buffers used in the chromatography and diafiltration steps (see Table 13.1).

Besides water, the different components used in the buffers, the CIP, and the viral inactivation dominate the output. The output of the fermentation includes unused raw materials

Table 13.2 *Energy demand for the base-case model*

Energy source	Annual demand	Specific demand (per kg P)
Electricity	185 MWh	600 kWh
Steam	14 metric tons	45 kg
Cooling water	7300 m^3	24 m^3
Chilled water	9250 m^3	30 m^3

(medium, serum-free medium), biomass, carbon dioxide, impurities, inorganic salts, and the product. The demand for the different energy sources is shown in Table 13.2.

13.4 Economic Assessment

In the model, the total purchased equipment cost (PEC) is $ 9.3 million, leading to a total capital investment (TCI) of $ 133 million for a plant with a bioreactor capacity of 30 m^3 (reference year 2005). This lies in the range given in literature [13.5, 13.6]. The most expensive equipment items are the production bioreactors (P-20, 30% of PEC), the centrifuge (P-31, 7%), and the two seed reactors (P-5, P-10; respectively 4%, 5%).

Annual operating costs are $ 44 million. They are dominated by the facility-dependent costs (70%). Furthermore, the consumables (13%), the raw-material cost (7%), and the labor cost (6%) play an important role. The facility-dependent cost is mostly depreciation cost. The serum-free medium dominates the raw-material cost (81%). Furthermore, the caustic soda (4%), the water for injection (4%), and the serum-containing medium used in the inoculum preparation (2%) have a relevant impact. The cost for the Protein A resin (75%) and also the resin costs for the HIC (11%) and IXC (9%) mainly account for the consumables cost. Most labor is needed in the inoculum preparation (43%) and in the bioreaction section (39%). Laboratory/QC/QA estimated at 4% plays a notable role in the operating costs, while waste treatment and utilities have only a very small impact. The QC/QA expense as estimated here may be on the low side.

The bioreaction and upstream sections contribute to 55% of the operating cost, while all the downstream sections amount to 45%. This compares well with data from Chovav et al. [13.6]. Six operating-cost parameters contribute more than 1% to the operating cost: protein A resin (10%), media powder (6.4%), labor cost for inoculum preparation (2.8%) and for the production fermenters (2.6%), and resin cost for HIC (1.5%) and IXC (1.2%).

Based on an annual production of 307 kg, the unit-production cost (UPC) in the model is $ 143/g Mab. The selling price is assumed to be $ 800/g. The actual price depends very much on the specific Mab and the kind of application it is used for. From data published by Chovav et al. [13.6], an average price for Mabs of $ 4500/g can be calculated. In the long run, however, prices will go down significantly. Furthermore, the product described by this model lacks the final formulation and packaging, and further marketing and transportation costs also are not considered. Therefore, a relatively low selling price is assumed. It results in annual revenues of $ 246 million, leading to a gross profit of $ 202 million per year. With an assumed income tax of 35% and $ 12 million depreciation (10 years, linear), the

net profit is $ 143 million, and the return on investment (ROI) is 108% (payback time = 1 year). However, initial R&D costs that can vary significantly from project to project are not considered in this study.

13.5 Environmental Assessment

The Environmental Indices (EI_{MV}) are shown in Figure 13.2. Sodium hydroxide and hydrogen chloride that are used in the downstream processing dominate the input side. Both substances are considered environmentally relevant due to their acute toxicity (e.g. they can cause severe chemical burns). Ammonium sulfate that is used before the HIC step also has some impact at the input side because it is based on gas/oil (raw-material availability), and it has the highest EI at the output side due to its high nitrogen content (eutrophication). All other input components have only a small EI. At the output side, the biomass produced (organic carbon, eutrophication) and sodium phosphate used in the buffers (eutrophication) have some impact. Here, the 'Rest' includes unused medium components, carbon dioxide, organic by-products, product loss, and inorganic salts.

The overall EI_{MV} of the input is $EI_{In} = 43$ Index Points/kg Product (= IP/kg P), of the output $EI_{out} = 7$ IP/kg P. From an environmental point of view, the input is more relevant than the output. NaOH and HCl are the most important materials of the input. The neutralization turns them into environmentally far less harmful salts (mainly NaCl) in the output, causing the main differences between input and output. In general, the components involved in the process show only a low or medium environmental relevance, respectively their negative potential can be handled in the process (e.g. acids and bases by simple safety measures).

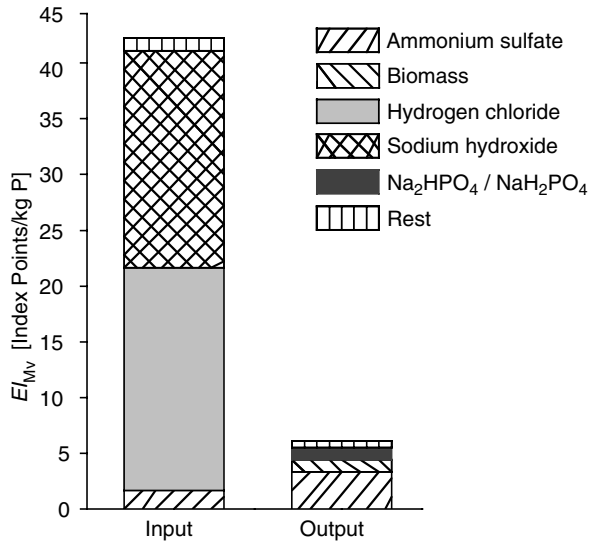


Figure 13.2 *EI_{MV} of the input and output components for the Mab production model. [IP/kg P] = Index points per kg of final product*

13.6 Uncertainty Analysis

The analysis of the base case leads to a better understanding of the process. However, to appreciate the risks and needs, a deeper understanding of how design and operating parameters impact the results is needed. In the next section, the uncertainty in this design is studied. First, alternative process configurations are examined in a scenario analysis. The impact of key technical and market parameters is assessed in a sensitivity analysis and the overall uncertainty is quantified in a number of Monte Carlo simulations.

13.6.1 Scenarios

There are a number of different process configurations, including changes in the process-flow diagram or changes in the number or size of units. As an example, scenarios that consider alternative chromatography steps are modeled. The simulation results are summarized in Table 13.3.

Chromatography steps are the core purification for Mab recovery. The number and kind of chromatography steps varies between different Mab processes. In the first scenario a gel filtration is added in the final purification as a final polishing step (and an additional pool tank). The second scenario adds an additional anion-exchange step that retains DNA and other impurities. Compared with the base case, both scenarios cause an expected increase of the total and unit cost (per kg product). Owing to the larger column volume necessary for the gel filtration the increase of the TCI is higher than for anion exchange. Downstream yield and annual amount of product decreases (a yield of 90% was assumed for both). This and the higher investment cause an increase of UPC values by 18% (IXC) and 33% (gel filtration), respectively. The use of additional buffers and the lower amount of product also result in less favorable environmental indices.

In the third downstream scenario, the Protein A chromatography is replaced by a second ion-exchange column. For simplification, the same buffers, binding capacity, loading flowrate, etc. were assumed for both ion-exchange chromatography steps. Also, the same yield was presumed, resulting in an identical annual production. Since the environmental differences between the buffers used in the IXC and the Protein A chromatography are small, the environmental impacts are almost identical. However, the lower cost of the ion-exchange resin leads to a lower UPC. Thus, if the same yield and degree of purification

Table 13.3 *Key results for different scenarios of the downstream processing. IXC = Ion-exchange chromatography; HIC = Hydrophobic-interaction chromatography*

Scenario	Amount Mab (kg)	*TCI* ($ million)	*UPC* ($/g)	*ROI* (%)	EI_{MV} Input (IP/kg P)	EI_{MV} Output (IP/kg P)
Base case	307	133	143	108	43	7.0
Gel filtration	277	149	188	83	50	8.6
2nd IXC	277	140	167	91	51	8.0
IXC instead of Protein A chromotography	307	132	131	111	44	7.0
Without HIC	341	125	120	130	32	2.4

can actually be achieved with two ion-exchange columns, and the same resin replacement frequency assumed, this approach is superior to the base case.

There are also Mab processes reported that only use two chromatography steps. Therefore, the HIC section is removed in the fourth downstream scenario. Here, the removal of the HIC results in higher annual production, lower UPC (16%), and smaller environmental impact. The reduction of the environmental indices is mainly caused by the removal of the ammonium sulfate (to increase ionic strength before the HIC) and the reduction of the sodium hydroxide consumption (regeneration of HIC column).

13.6.2 Sensitivity Analysis

Fermentation time and product concentration are usually the most crucial fermentation parameters. Therefore, they are presented here as an example for sensitivity analysis. In the base case, a fermentation time of 14 days is assumed. Following the literature [13.5, 13.6] the fermentation duration is varied between 8 and 20 days. All other parameters, especially the product concentration, are kept constant (i.e. productivity varies as well). The fermentation duration neither influences the overall investment cost nor the environmental impact since neither the equipment needed nor the mass balance of the process varies. However, the fermentation time defines the number of batches per year and consequently the annual amount of product. Hence, specific investment cost, UPC, and annual revenue vary. They show a more or less linear dependency on the fermentation time. The annual amount of product varies between 362 kg (8 days) and 226 kg (20 days), resulting in annual revenues of $ 290 million and $ 181 million, respectively. The UPC rises from $ 126/g (8 days) to $ 181/g (20 days) mainly because the constant facility-dependent cost (investment) is allocated to less product per year. Additionally, the longer fermentation time causes higher labor costs. The UPC increases by $ 4.5/g or 3.5% with every additional fermentation day.

In the next analysis the product concentration is varied between 0.5 and 3 g/L. Yield and reaction stoichiometry remain unchanged. Thus, the overall amounts of medium added and biomass and carbon dioxide produced rise, but their relative amounts (per kg product) are constant. The annual production varies enormously, from 150 kg (0.5 g/L) to 960 kg (3 g/L). To handle the linearly increasing amount of product, the downstream equipment must be resized, resulting in an also linear rise of the overall investment cost ($ 129–147 million). Figure 13.3 shows the UPC values at different concentrations. From 0.5 to around 2 g/L, the UPC values decrease strongly. At higher concentrations, the further reduction is relatively small.

Owing to the constant fermentation and downstream yields, annual raw material and consumable costs rise linearly with increasing Mab concentration (increasing amount of product), while their costs per kg of product hardly change. However, labor costs are largely independent of the amount of product, and the higher downstream capacity causes only a moderate increase of the TCI. Therefore, the specific facility dependent cost (or depreciation of TCI), and to a lesser extent the specific labor cost, decrease with increasing Mab concentration, causing the exponential shape of the UPC curve. The ROI remains clearly positive for all cases. Even at a concentration of 0.5 g/L, the ROI is still 51%.

The curve of the EI Input has a very similar shape to the UPC (see Figure 13.3). The consumption of the two components that dominate the EI Input (NaOH, HCl) is mainly defined by the volume of the fermentation broth, rather than the amount of product. In

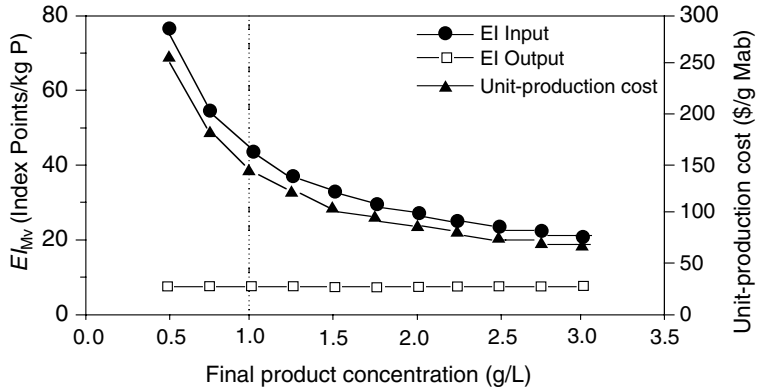


Figure 13.3 *UPC, EI Input and Output for the Mab process model at different Mab concentrations in the bioreaction. The vertical dotted line indicates the base-case value*

contrast, the EI Output more or less does not change. Here, the three environmentally most important output components (ammonium sulfate, sodium phosphate, and biomass) are largely defined in the model by the amount of product, and therefore their specific consumption per kg of product hardly changes. Under the conditions of this sensitivity analysis, the biomass concentration rises with the product concentration. In reality, this does not have to be the case, but even if a higher product concentration is reached with the same amount of biomass, the impact on the EI Output remains small.

These results indicate that process improvements should be targeted to reach a product concentration of around 2 g/L. At higher concentrations, significant improvements can only be expected if the higher concentration leads to a simplification of the separation and purification section.

13.6.3 Monte Carlo Simulations

Variables and Objective Functions. One of the specific problems in cell-culture process development is the variability associated with both the biology and the process itself. Using the process model as the basis for a Monte Carlo simulation (MCS) we can explore how variance propagates through the entire process to impact both economic and environmental results. The variables selected for this case study are presented in Table 13.4. The probability distributions defined for these variables are derived from available process and statistical data, supplier information, expert opinions, and internal estimates. The bioreaction parameters for process duration, final concentration, and yield are chosen to be examined. Additionally the aeration rate is selected as a typical fermentation condition parameter. The different chromatography steps are the core steps of the downstream process. Their yields, replacement frequencies, and unit cost of the resin show variation. The replacement frequency of the membranes also is considered.

As supply-chain parameters, the cost for medium powder and electricity are chosen as they dominate raw material and utility costs, respectively. Depending on the application, the selling price varies strongly and is therefore considered as a market parameter. With this selection of sources of variance, uncertainties affecting raw material, consumables, and

Table 13.4 *Parameters used in the Monte Carlo simulation and their chosen probability distributions. V = Coefficient of variance*

Parameter	Base-case value	Source	Probability distribution	Variation data	Source
1. Technical parameters					
Fermentation time (days)	14	[13.6]; Own estimate	Triangular	10–20; base-case value as the most likeliest	[13.6]; Own estimate
Final product concentration (g/L)	1	[13.12], [13.6]	Triangular	0.5–2; base-case value as the most likeliest	[13.6]; Own estimate
Yield (g media powder/g Mab)	25	Own estimate	Triangular	15–35; base-case value as the most likeliest	Own estimate
Aeration rate (vvm)	0.1	Own estimate	Normal	$V = 20\%$; 0.05–0.2 (min, max)	Own estimate
Replacement frequency resin Protein A chromatography (cycles)	50	Own estimate; supplier data	Triangular	20–100; base-case value as the most likeliest	Own estimate
Replacement frequency resin IXC (cycles)	50	Own estimate; supplier data	Triangular	20–100; base-case value as the most likeliest	Own estimate
Replacement frequency resin HIC (cycles)	50	Own estimate; supplier data	Triangular	20–100; base-case value as the most likeliest	Own estimate
Replacement frequency diafiltration membranes (cycles)	25	Own estimate; supplier data	Triangular	10–40; base-case value as the most likeliest	Own estimate
Yield Protein A chromatography (%)	90	Own estimate; supplier data	Normal	$V = 10\%$; max: 100	Own estimate
Yield IXC (%)	90	Own estimate; supplier data	Normal	$V = 10\%$; max: 100	Own estimate
Yield HIC (%)	90	Own estimate; supplier data	Normal	$V = 10\%$; max: 100	Own estimate
2. Supply chain parameters					
Price of medium ($/kg)	200	Own estimate; supplier data	Triangular	100–300; base-case value as the most likeliest	Own estimate
Resin cost of Protein A Chromatography ($/L)	9000	Own estimate; supplier data	Triangular	7000–11 000; base-case value as the most likeliest	Own estimate
Resin cost IXC ($/L)	1500	Own estimate; supplier data	Triangular	500–2,500; base-case value as the most likeliest	Own estimate
Resin cost HIC ($/L)	2000	Own estimate; supplier data	Normal	1,000–3,000; base-case value as the most likeliest	Own estimate
Electricity cost [$/kW h]	0.0468	[13.18, 13.19]	Weibull	Loc: 4.13; Scale: 0.61; Shape: 1.96 (for a normal distribution: $V = 6\%$)	[13.19]
3. Market parameters					
Selling price of final product [$/g]	800	Own estimate	Normal	$V = 20\%$	Own estimate

Table 13.5 *Mean value, median, and standard deviation of the objective functions for the Monte Carlo simulation using all parameter (MCP-AP) defined in Table 13.4*

Objective function	Abbreviation	Mean value	Median	Standard deviation
Annual production (kg/year)	*AP*	322	307	109
Unit-production cost ($/g)	*UPC*	153	145	45
Total capital investment ($ million)	*TCI*	134	133	2.2
Gross profit ($ million)	*GP*	213	197	101
Net profit ($ million)	*NP*	154	143	66
Return on investment (%)	*ROI*	112	105	48
Environmental Index input (IP/kg P)	EI_{In}	45	43	12
Environmental Index output (IP/kg P)	EI_{Out}	7.6	7.4	1.0

utility costs, and plant capacity are covered. Uncertainties concerning the PEC, which might be also relevant, are not included.

The annual amount of product is used as an objective function to document the technical performance of the process. While unit-production cost, gross and net profit, ROI, and capital investment are used as objective functions that describe the economic performance of the process, the EIs cover the variation of environmental impact.

Results. Five different sets of parameters are used: Monte Carlo simulations using the technical parameters (MCS-TP), and simulations using the supply-chain and market parameters (MCS-SCMP). Then the influence of these parameters is studied in a simulation using the parameters affecting the upstream and fermentation sections of the model (MCS-FP) and in a simulation considering the parameters affecting the downstream sections (MCS-DSP). Finally one simulation uses all parameter defined in Table 13.4 (MCS-AP). For each simulation, 10 000 trials were chosen. At this number of runs, the mean standard error stays below 1% for all objective functions. Table 13.5 highlights the most important results of the MCS-AP that describes the overall uncertainty in the process. The complete simulation results are given in Appendix 1. In the following text, we discuss only the annual product and the economic objective functions in detail.

There are some general trends that apply to all objective functions. The technical parameters contribute much more to the existing uncertainty than do the supply-chain parameters. For objective functions that are affected by the selling price of the final product, the selling price, not surprisingly, plays a dominant role and increases the variability substantially. All objective functions show in general a relatively high variability. This is caused by the broad range of values (probability distributions) that are defined for key parameters. The contributions of the DSP and the FP to the overall uncertainty are in the same range, whereby the FP usually cause a higher variability, which is originated mainly by the wide range of final product concentrations possible in the fermentation. One could take this analysis further and look at the impact of varying not only the mean values, as is done in sensitivity testing, but also look at the width and shape of the variance.

(i) *Annual amount of product*
We discuss the variation of the annual amount of product first because it influences all other objective functions. The possible range of values is quite broad, with values in

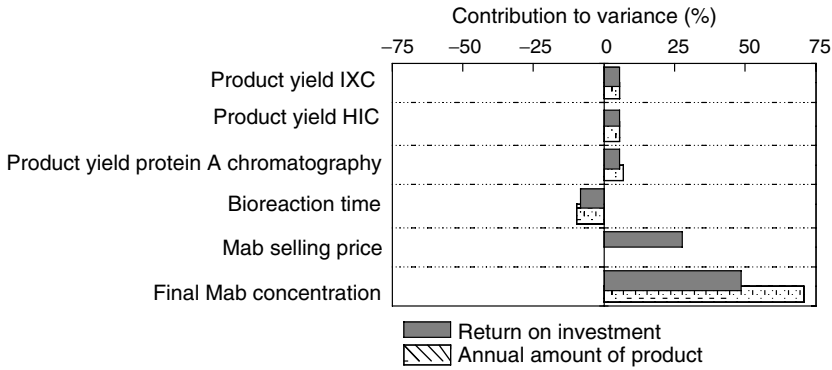


Figure 13.4 *Contribution of the variables to the variance of the annual amount of product and the ROI in the Monte Carlo simulation using all parameters in Table 13.4. Only parameters with more than 1% contribution to the variance are shown. Negative values represent negative correlations. HIC = Hydrophobic-interaction chromatography; IXC = Ion-exchange chromatography*

the range 95–805 kg/year (MCS-AP). The variability is higher for the FP (30%) than for the DSP (14%), with an overall variability of 34% (AP). Figure 13.4 shows the parameters that contribute to the variation of the annual amount of product. The most important parameter is the Mab concentration. It defines the total amount per batch. On the one hand, this is very typical for fermentation processes. On the other hand, it is amplified by the wide range of possible concentrations defined for this parameter (see Table 13.4). Other relevant parameters are the fermentation time that defines the number of batches per year and in this way the annual amount of product, and the yields of the chromatography steps (amount per batch).

The MCS-DSP shows a lower mean than the base case. The variation is dominated by the chromatography yields (base case: 90%). For them, a variability of 10% is defined with an upper limit of 100%. Therefore, the actual mean in the MCS is slightly lower than in the base case, resulting in higher product loss and a lower annual production. The MCS-FP shows a higher mean. Here, the Mab concentration dominates the variation for which a range from 0.5 g/L to 2 g/L was defined with the base-case value as the likeliest in a triangular distribution. Since the base-case value is not in the middle of the range, this also results in an actual mean different from the base-case value, leading to more product per batch. In the MCS-TP and MCS-AP, these effects almost offset each other. Mean and median are only slightly higher than in the base case.

(ii) *Unit-production cost*

The UPC is a key measurement to evaluate the economy of a process. The contribution of the variables to its variation is given in Figure 13.5. In the MCS-SCMP the medium powder's price and the unit costs for the Protein A resin have the highest contribution. The resin unit costs for the other two chromatography steps also play a significant role, while the impact of the electricity price is negligible. The variation of the TP is dominated by the same variables that influence the amount of product (see Figure 13.4).

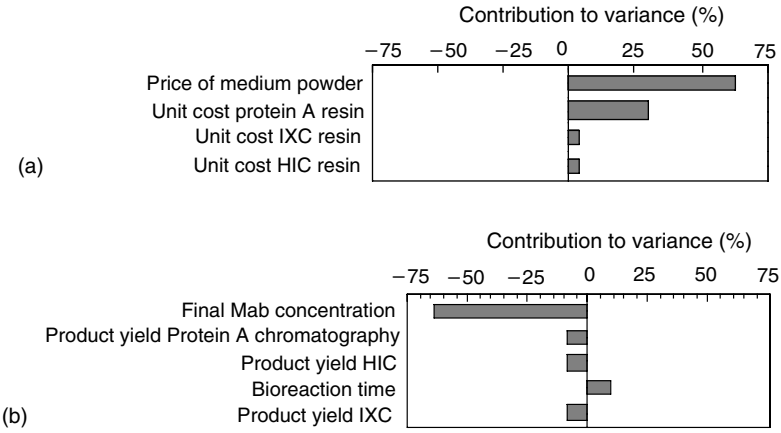


Figure 13.5 *Contribution of the variables to the variance of the unit-production costs in MCS-SCMP (a) and MCS-AP (b). Shown are only those parameters with more than 1% contribution to the variance. Negative values represent negative correlations. HIC = Hydrophobic-interaction chromatography; IXC = Ion-exchange chromatography*

Besides the annual amount of product, the Mab concentration affects also the TCI (higher capacity needed) and, thus, the annual operating costs. The fermentation time also influences the annual operating cost via the electricity and the cooling-water consumption. The contribution of all other technical parameters is clearly lower than 1%.

The UPC values for the MCS-AP lie in a wide range from $ 64/g to $ 411/g. However, with a 90% probability the UPC is between $ 103/g and $ 213/g. A UPC of below $ 170/g has a probability of 70%. The probability distributions of the UPC for the different sets of parameters are compared in Figure 13.6(a). The MCS-TP shows a variability of 30% (standard deviation = $ 45/g) while the variability in the MCS-SCMP is only 2%. This explains why the impact of the SCMP on the overall variation is negligible and why none of them contributes more than 1% to the overall uncertainty [see Figure 13.5(b)]. Owing to the domination of the technical parameters, the MCS-AP has almost the same distribution as the MCS-TP. Similarly to the annual amount of product, the MCS-DSP has a smaller variation (16%) than does the MCS-FP (25%).

The relative variation of mean values compared with the base case is identical to the variation of the annual production.

(iii) *Gross profit, net profit, and return on investment*

The variability of these parameters is very similar due to the same parameters that influence their calculation. Figure 13.4 shows the contribution of the variables to the variation of the ROI. In addition to the variables dominating the UPC variance, the selling price of the final product as a key market parameter contributes strongly to the overall variance. However, the Mab concentration remains the most important parameter.

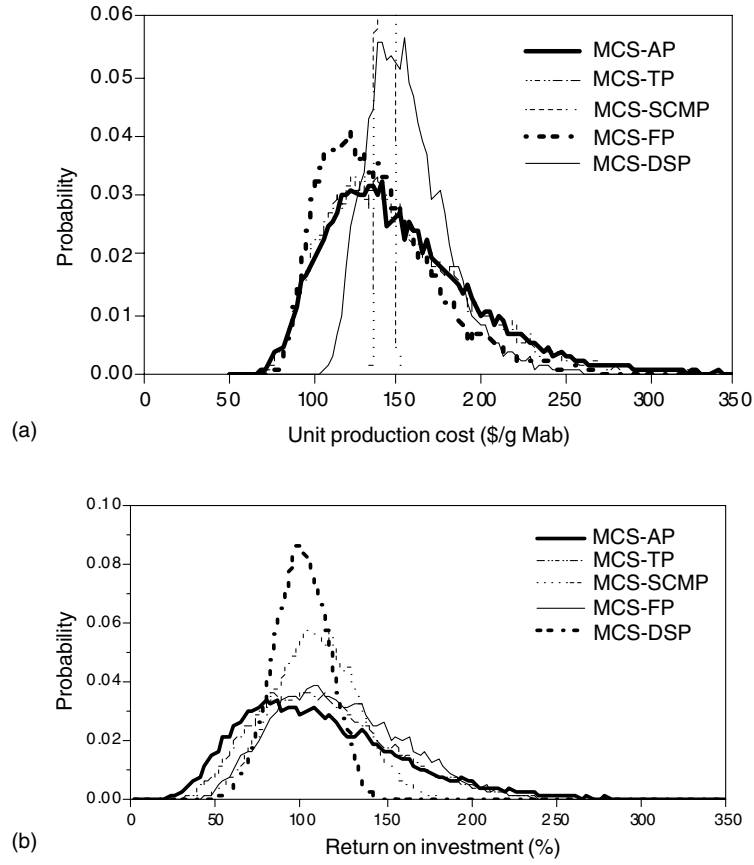


Figure 13.6 *Probability distributions of the unit-production costs (a) and the return on investment (b) for the different parameter sets (10 000 trials, 100 groups in each graph). The area under the curves is always the same. The peak of the MCS-SCMP in (a) is at a probability of* $p = 0.4$ *(outside the scale of the graph). For abbreviations see Nomenclature section*

The ROI values range from 16% to 388% (MCS-AP). However, with a 90% probability the ROI is above 58%. For the net profit, the range is $ 24–540 million. The standard deviation is $ 66 million. The net profit is above $ 80 million with a 90% probability and above $ 97 million with an 80% probability. The variation of the gross profit is even higher. It ranges from $ 13 to $ 807 million with a 90% probability of a gross profit higher than $ 99 million.

Figure 13.6(b) compares the probability distributions of the ROI for the different parameter sets. Compared with the UPC, the selling price substantially broadens the distribution of the MCS-SCMP, but is still shows a smaller variability (22%) than does the MCS-TP. The variation of the MCS-FP (30%) is again higher than the impact of the DSP (16%). Similarly to the UPC, the MCS-TP remains at a variability of 34%, while the additional impact of the selling prices causes a higher variability of 42% for the MCS-AP.

13.7 Conclusions

In this case study, alternative process-flow diagrams for the production of monoclonal antibodies are compared, and the impact of cultivation parameters was quantified. The final Mab concentration of the fermentation and the yields of the chromatography steps contribute most to the economic uncertainty and, together with the fermentation yield, also to the environmental variability. The objective functions of ROI and net profit are additionally affected by the selling price. The environmental performance is characterized by a large volume of waste with a relatively low pollution potential.

Based on the process-uncertainty analysis, the most promising approach for improvements is an increase of the productivity in the fermenter, especially when a higher Mab concentration of around 2 g/L can be reached. If impurities were to be reduced such that only two chromatography steps were necessary then this would lead to a substantially better process. This might be achieved if strain requirements allow a change in the medium.

Suggested Exercises

1. The resin cost in the Protein A chromatography plays an important role. In the case provided on the CD the price is set to $ 9000/L. How does a price of $ 12 000/L affect operating costs and unit-production costs? The supplier guarantees an increased operating time of the new resin ($ 12 000/L). Assume that the replacement frequency rises from 50 to 80 cycles. Is it worth trying the new product?
2. In the supplied model two bioreactors are used with 15 m^3 working volume each. Develop a scenario where only one bioreactor with a working volume of 30 m^3 is used. Since the units in the model are in 'design mode', they are resized automatically when the volume of an input stream changes. To model the use of only one bioreactor, double the size of the input streams S-140 and S-141 and remove the one extra (second) unit for steps P-20 to P-24 (via the Equipment Data of these units). A larger bioreactor requires a resizing of the inoculum train. Do this also by doubling the volume of all input streams in this section. How is the economic performance affected?

Nomenclature

CIP = Cleaning-in-place
DSP = Downstream-section parameters
EF = Environmental Factor
EI = Environmental Index
FCI = Fixed capital investment
FP = Fermentation Parameters
HIC = Hydrophobic-interaction chromatography
IXC = Ion-exchange chromatography
Mab = Monoclonal antibody
MCS = Monte Carlo simulation

MCS-AP = Monte Carlo simulations using all parameters
MCS-FP = Monte Carlo simulations using fermentation parameters
MCS-DSP = Monte Carlo simulations using downstream parameters
MCS-SCMP = Monte Carlo simulations using supply-chain and market parameters
MCS-TP = Monte Carlo simulations using technical parameters
PEC = Purchased equipment cost
ROI = Return on investment
TCI = Total capital investment
TP = Technical parameters
UPC = Unit-production cost
WFI = Water for injection
VBA = Visual Basic for Applications
(IP/kg P) = Index Points/kg of product
(kg/kg P) = kg per kg of final product

References

[13.1] King, D. Use of antibodies for immunopurification. In: *Biotechnology – Vol. 5a: Recombinant proteins, monoclonal antibodies and therapeutic genes*. Mountain, A., Ney, U., Schomburg, D., Eds.; Wiley-VCH, Weinheim, 1999, pp. 276–287.

[13.2] Sopwith, M. Therapeutic applications of monoclonal antibodies: A clinical overview. In: *Biotechnology – Vol. 5a: Recombinant proteins, monoclonal antibodies and therapeutic genes*. Mountain, A., Ney, U., Schomburg, D., Eds.; Wiley-VCH, Weinheim, 1999, pp. 312–324.

[13.3] Clark, M. Immunochemical applications. In: *Basic biotechnology*. Ratledge, C., Kristiansen, B., Eds.; University Press, Cambridge, 2001, pp. 503–530.

[13.4] Albert, C., Patel, P., Rho, J. Monoclonal antibodies. In: *Handbook of pharmaceutical biotechnology*. Rho, J., Louie, S., Eds.; Haworth Press, New York, 2003, pp. 15–42.

[13.5] Harrison, R., Todd, P., Rudge, S., Petrides, D. Bioseparation science and engineering. Oxford University Press, New York, 2003.

[13.6] Chovav, M., Wales, M., De Bruin, D., Samimy, A., Meacham, G., Kim, K., Farhadu, D. The state of biomanufacturing. UBS's Q-series, London, 2003.

[13.7] Jones, D., Kroos, N., Anema, R., Montfort, B., Vooys, A., Kraats, S., Helm, E., Smits, S., Schouten, J., Brouwer, K., Lagerwerf, F., Berkel, P., Opstelten, D., Logtenberg, T., Bout, A. High-level expression of recombinant IgG in the human cell line PER.C6. *Biotechnol. Prog.*, 2003, 19, 163–168.

[13.8] Gerber, R., McAllister, P., Smith, C., Simth, T., Zabriskie, D., Gardner, A. Establishment of proven acceptable process control ranges for production of a monoclonal antibody by cultures of recombinant CHO cells. In: Validation of biopharmaceutical manufacturing processes. Kelley, B., Ramelmeier, A., Eds., ACS Symposium Series 698, ACS, Washington, 1998, pp. 44–54.

[13.9] Smith, T., Wilson, E., Scott, R., Misczak, J., Bodek, J., Zabriskie, D. Establishment of operating ranges in a purification process for monoclonal antibody. In: Validation of biopharmaceutical manufacturing processes. Kelley, B., Ramelmeier, A., Eds., ACS Symposium Series 698, ACS, Washington, 1998, pp. 80–92.

[13.10] Walsh, G. Biopharmaceuticals: Biochemistry and biotechnology. John Wiley & Sons, Ltd, Chichester, 2003.

[13.11] Racher, A., Tong, J., Bonnerjea, J. Manufacture of therapeutic antibodies. In: Biotechnology – Vol. 5a: Recombinant proteins, monoclonal antibodies and therapeutic genes. Mountain, A., Ney, U., Schomburg, D., Eds., Wiley-VCH, Weinheim, 1999, pp. 247–274.

[13.12] Dean, C. Monoclonal antibodies. In: Molecular biology and biotechnology. Walker, J., Rapley, R., Eds., Royal Society of Chemistry, Cambridge, 2000, pp. 497–520.

[13.13] Muething, J., Kemminer, S., Conradt, H., Sagi, D., Nimtz, M., Kaerst, U., Peter-Katalinic, J. Effects of buffering conditions and culture pH on production rates and glycosylation of clinical phase I anti-melanoma mouse IgG3 monoclonal antibody R24. *Biotechnol. Bioeng.*, 2003, 83, 321–334.

[13.14] Zhou, W., Chen, C., Buckland, B., Aunins, J. Fed-batch culture of recombinant NSO myeloma cells with high monoclonal antibody production. *Biotechnol. Bioeng.*, 1997, 55, 783–792.

[13.15] Sanfeliu, A., Cairo, J., Casas, C., Sola, C., Godia, F. Analysis of nutritional factors and physical conditions affecting growth and monoclonal antibody production of hybridoma KB-26.5 cell line. *Biotechnol. Prog.*, 1996, 12, 209–216.

[13.16] Necina, R., Amatschek, K., Jungbauer, A. Capture of human monoclonal antibodies from cell culture supernatant by ion-exchange media exhibiting high charge density. *Biotechnol. Bioeng.*, 1998, 60, 689–698.

[13.17] Goding, J. Monoclonal antibodies: Principles and practice. Academic Press, London, 1996.

[13.18] Peters, M., Timmerhaus, K., West, R. Plant design and economics for chemical engineers. McGraw-Hill, Boston, 2003.

[13.19] US Energy Information Administration February 2004 Monthly Energy Review, 2004. Available at: http://www.eia.doe.gov.

Appendix 13.1 Results Monte Carlo Simulations

The following parameter sets were used:

Technical parameters (TP): Fermentation time, fermentation yield, fermentation final MAb concentration, aeration rate, replacement frequencies of chromatography resins (Protein A, IXC, HIC) and diafiltration membranes, chromatography yields (Protein A, IXC, HIC)

Supply-chain and market parameters (SCMP): Price of medium powder, resin cost of Protein A, IXC, and HIC, electricity price, selling price of final product

Fermentation parameters (FP): Fermentation time, fermentation yield, fermentation final MAb concentration, aeration rate, price of medium powder, electricity price (inoculum and bioreaction section)

Downstream parameters (DSP): Replacement frequencies of chromatography resins (Protein A, IXC, HIC) and diafiltration membranes, chromatography yields (Protein A, IXC, HIC), resin cost of Protein A, IXC, and HIC, electricity price (downstream sections)

All parameters (AP): All parameters listed above.

Attribute	TP	SC/MP	DSP	FP	AP
Trials	10 000	10 000	10 000	10 000	10 000
Annual amount of product (kg)			base case: 307 kg		
Mean	326		286	349	322
Median	313		286	337	307
Standard deviation	108		41	104	109
Skewness	0.61		−0.08	0.5	0.7
Kurtosis	3.2		2.8	2.8	3.2
Coefficient of variability (%)	33		14	30	34
Range minimum	75		115	120	95
Range maximum	822		410	719	804
Range width	747		296	600	709
Mean standard error (%)	0.3		0.1	0.02	0.3
UPC ($/g)			base case: 143 $/g		
Mean	151	143	156	139	153
Median	143	143	153	133	144
Standard deviation	44.8	2.2	24.4	34.2	45.1
Skewness	1.19	0.02	1.04	0.98	1.05
Kurtosis	5.42	2.74	5.14	4.12	4.62
Coefficient of variability (%)	30	2	16	25	30
Range minimum	64	136	105	78	64
Range maximum	495	150	359	317	411
Range width	432	15	254	239	347
Mean standard error (%)	0.3	0.0	0.2	0.2	0.3

Attribute	TP	SC/MP	DSP	FP	AP
TCI ($ million)			base case: $ 133 million		
Mean	134	133	132	134	134
Median	133	133	133	134	133
Standard deviation	2.2	0.0	0.4	2.2	2.2
Skewness	0.252	0.014	−0.37	0.26	0.30
Kurtosis	2.50	2.41	2.95	2.47	2.52
Coefficient of variability (%)	1.62	0.04	0.3	1.6	1.6
Range minimum	128	133	131	129	129
Range maximum	140	133	133	140	140
Range width	12	0	3	11	12
Mean standard error (%)	0.0162	0.0004	0.003	0.02	0.02
Gross Profit ($ million)			base case: $ 202 million		
Mean	216	203	185	234	213
Median	206	202	185	225	197
Standard deviation	83.5	49.1	32.9	80.1	101.0
Skewness	0.61	0.05	−0.08	0.48	0.96
Kurtosis	3.18	2.87	2.81	2.78	4.25
Coefficient of variability (%)	39	24	18	34	47
Range minimum	23	55	51	58	13
Range maximum	598	387	285	518	807
Range width	575	332	235	460	793
Mean standard error (%)	0.4	0.2	0.2	0.3	0.5
Net Profit ($ million)			base case: $ 143 million		
Mean	155	147	135	167	154
Median	149	146	135	161	143
Standard deviation	54.4	31.9	21.4	52.2	65.8
Skewness	0.61	0.05	−0.08	0.48	0.96
Kurtosis	3.18	2.87	2.81	2.78	4.24
Coefficient of variability (%)	35	22	16	31	43
Range minimum	29	51	48	52	24
Range maximum	404	267	200	352	540
Range width	375	216	153	300	516
Mean standard error (%)	0.4	0.2	0.2	0.3	0.4
ROI (%)			base case: 108%		
Mean	113	108	100	122	112
Median	109	108	100	118	105
Standard deviation	38.9	24.1	15.9	37.0	47.5
Skewness	0.56	0.05	−0.09	0.43	0.92
Kurtosis	3.10	2.87	2.82	2.73	4.14
Coefficient of variability (%)	34	22	16	30	42
Range minimum	21	36	34	38	16
Range maximum	286	199	148	250	388
Range width	266	163	114	212	372
Mean standard error (%)	0.3	0.2	0.2	0.3	0.4
EI Input (IP/kg P)			base case: 43 IP/kg P		
Mean	44		47	41	45
Median	42		46	39	43
Standard deviation	11.9		6.9	9.0	12.1

(*Continued*)

Attribute	TP	SC/MP	DSP	FP	AP
Skewness	1.13		1.04	0.89	0.99
Kurtosis	5.03		5.15	3.65	4.24
Coefficient of variability (%)	27		15	22	27
Range minimum	22		33	26	22
Range maximum	122		108	75	115
Range width	99		75	49	92
Mean standard error (%)	0.3		0.1	0.2	0.3
EI Output (IP/kg P)			base case: 7.0 IP/kg P		
Mean	7.6		7.6	7.0	7.6
Median	7.4		7.4	7.0	7.4
Standard deviation	1.0		1.0	0.3	1.0
Skewness	0.88		0.97	−0.02	0.84
Kurtosis	4.19		4.55	2.40	4.04
Coefficient of variability (%)	13		13	4	13
Range minimum	5.2		5.6	6.2	5.3
Range maximum	13.6		14.4	7.8	12.7
Range width	8.5		8.8	1.5	7.4
Mean standard error (%)	0.1		0.1	0.04	0.1

14

α-1-Antitrypsin from Transgenic Plant Cell Suspension Cultures

*Elizabeth Zapalac and Karen McDonald**

14.1 Introduction

Human α-1-antitrypsin (AAT) (also known as α-1-proteinase inhibitor) is a 52 kDa glycoprotein present in fairly high levels ($\sim$2 mg/mL) in the blood of healthy individuals [14.1, 14.2]. It acts as a serine protease inhibitor that helps maintain appropriate levels of neutrophil elastase and other proteinases in humans. Its structure is shown in Figure 14.1.

Patients with genetic disorders resulting in limited AAT functionality require augmentation therapy via intravenous weekly administration and currently there are three FDA-approved sources of AAT (Prolastin®, Bayer Corporation; Aralast™, Baxter; Zemaira®, Aventis). The recommended dosage is typically around 60 mg/kg body weight administered weekly. The selling price of these therapeutics ranges from $ 280/g to $ 390/g [14.4], resulting in annual drug costs of over $ 60 000/year if administered weekly. A recent study illustrates the high cost of treating AAT deficiency and finds that the largest direct medical cost is the cost of the therapeutic agent itself. All three products are purified from pooled human serum. Owing to the limited blood supply, shortages of human AAT have been a problem in the past. The potential combined markets of emphysema and dermatological disorders are estimated at 1.5 million g/year while the current annual US production (isolated from donated human blood) is only 250 000 g/year [14.5]. Owing to both supply

* Corresponding author: kamcdonald@ucdavis.edu; ++1/530/752-0559

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Figure 14.1 *A picture of the three-dimensional structure (ribbon representation) of human α-1-antitrypsin where the black lines indicates the α-helices, the light grey indicates the β-sheets. The Met358 residue of the active site is denoted [14.1, 14.3]*

and safety concerns associated with human blood, alternative recombinant sources for AAT are being investigated. In this case study, we consider the purification of recombinant AAT (rAAT) from the broth of transgenic rice-cell suspension culture that has been genetically engineered to produce and excrete rAAT [14.6–14.13].

14.2 Process Description

Recovery and purification processes for recombinant proteins depend strongly on the production host, since unique impurities exist in each system and it is from these impurities that the target protein must be separated. It is assumed that the purified product containing rAAT consists of active rAAT 8.3 mg/mL and glycerine in WFI (water for injection) 100 mg/mL. Since the focus of this case study is on the recovery and purification of rAAT from transgenic rice-cell suspension culture as well as the influence of downstream-process parameters on process economics, the total production costs associated with preparing the rice-cell culture harvest (e.g. the upstream processing costs) were not considered in this study. Instead, the cost of the starting material for the entire purification process was estimated as the sum of the weighted individual material costs and equaled $ 5.80/L. In the case study presented, a bioreactor with a harvest batch size of 10 000 kg, a maximum annual operating time of 7920 h, and an annual production of active AAT of 32.7 kg are assumed.

In this case study, rAAT is purified according to the laboratory-scale process developed by Huang and co-workers [14.10]. In this process, the plant-cell culture broth containing rAAT is clarified, and then immobilized-lectin affinity chromatography using Concavalin A (ConA) resin is used to separate proteins that are not glycosylated, particularly native secreted α-amylases. Active and inactive forms of rAAT, as well as other secreted glycosylated proteins are retained on the resin. Next, anion-exchange chromatography using DEAE [2-(diethylamino)ethyl-protected] resin further removes low-molecular-weight impurities, and, finally, active rAAT is isolated from inactive rAAT via octyl hydrophobic-interaction chromatography. In this case study, we aim to determine which factors significantly impact the cost and productivity of the three-step packed-bed adsorption process to purify rAAT from plant-cell culture. Such an analysis is useful to define the research and development agenda for improved production.

14.3 Model Description

Figure 14.2 shows the process flowsheet for the recovery and purification of rAAT from transgenic rice-cell suspension culture broth. The starting material for the purification process is a 10 000 kg batch of rice-cell culture harvest separated from the rice cells using gravitational settling. Since rAAT is secreted into the cell culture fluid after it is produced, no cell disruption is required. It is assumed that the culture broth contains 0.008 wt% active rAAT, 0.007 wt% inactive rAAT, 0.01 wt% α-amylase (a naturally secreted native rice protein), 0.005 wt% low-molecular-weight proteins, 1.0 wt% residual nutrient medium components, 2 wt% biomass, and 96.97 wt% water for injection. This corresponds to rAAT levels of 80 μg/mL and a total protein concentration of 300 μg/mL obtained experimentally [14.11]. The clarified liquid broth that is recovered following gravity sedimentation of the cell aggregates is filtered to remove fine cell debris using a normal-flow (dead-end) filter train and is stored in a holding tank. Overall, essentially 100% of the initial cell debris is removed in the sequential filtration by 8 μm, 0.45 μm, and 0.22 μm filters. These filters are each used for only one batch to prevent the potential for cross-contamination between batches. The filtered harvest fluid is concentrated 10 times by ultrafiltration in a

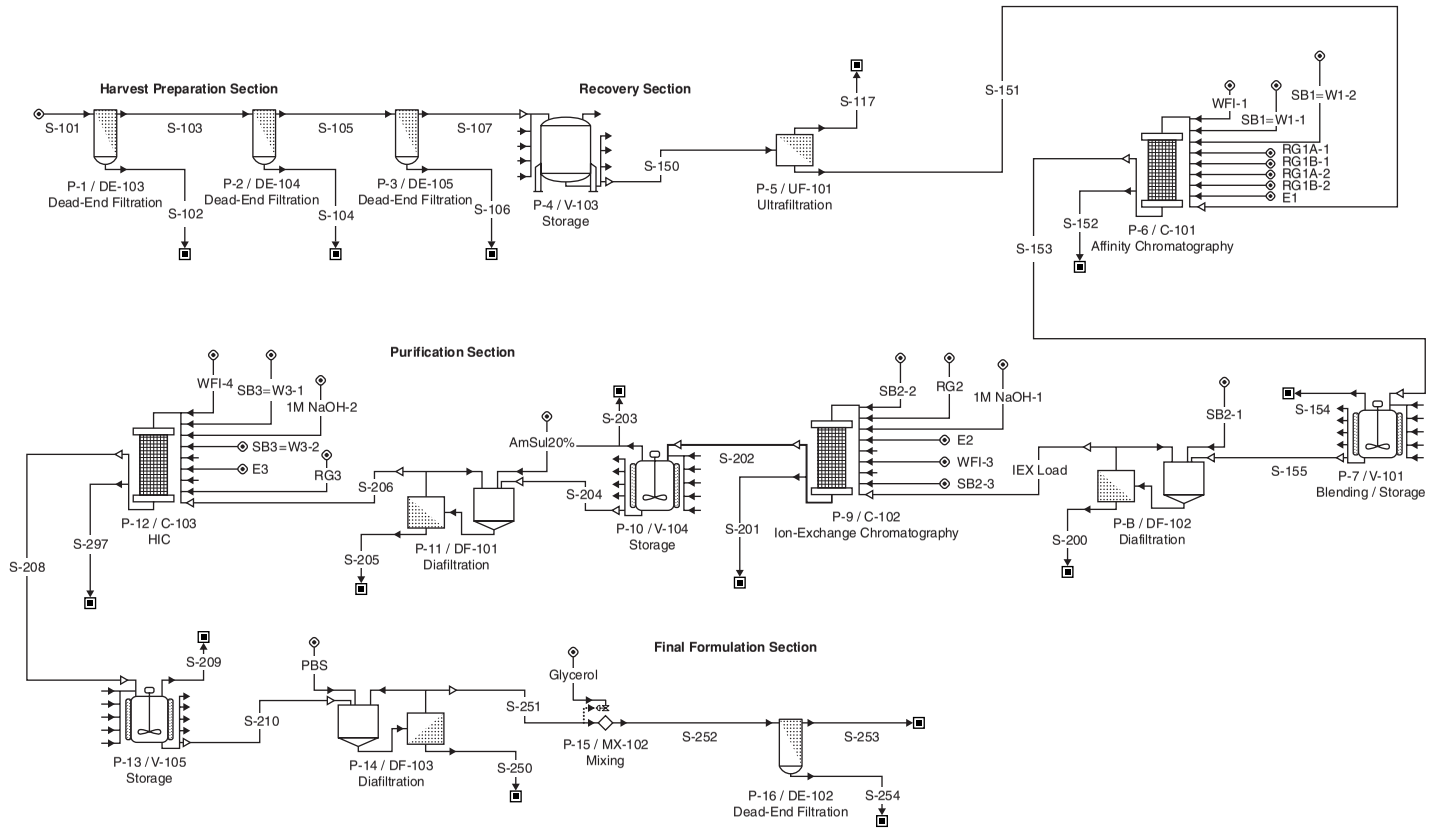


Figure 14.2 *Process flowsheet for rAAT case study*

tangential-flow filtration device that is cleaned in place between batches. Following filtration, an immobilized-lectin affinity chromatography using ConA resin is used to capture glycosylated proteins, applying five multi-step cycles in each of three columns simultaneously. The ConA Sepharose resin is scheduled to be replaced every 400 cycles. The combined pool from all 15 elution steps is concentrated twice, then exchanged into the equilibration buffer, using diafiltration, for the next purification step, which utilizes anion exchange (DEAE Sepharose) chromatography using a multi-step cycle in a single column. The DEAE chromatography step is used to remove molecules whose surface charge differs from that of the rAAT of interest. This resin is replaced every 100 cycles. The pool eluted from the DEAE column is also concentrated (2×) and is then exchanged into the equilibration buffer, via diafiltration, for the hydrophobic-interaction chromatography step. A single multi-step cycle is used in a single Octyl-Sepharose column to remove inactive forms of AAT; this resin is also expected to last 100 cycles before replacement. Lastly, the eluted pool containing the active rAAT is concentrated twice and diafiltered with phosphate-buffered saline. The batch is finally combined with glycerol to reduce water activity and stabilize the product during the final filling steps that are not included in the process described here.

Table 14.1 lists model assumptions used in the analysis. Critical parameters such as dynamic binding capacities, yields, and product purities as a function of the target protein concentration in the feed, flowrate during loading, and ionic strength were obtained experimentally, while other parameters were set at the default values in SuperPro Designer®. Cost estimations were based on the built-in model in SuperPro Designer®.

14.4 Discussion

Table 14.2 shows results from the economic analysis of the case-study design under the base-case conditions given in Table 14.1. A total of 53 batches are needed to produce 32.7 kg of active AAT per year. The corresponding unit production cost is $ 780/g, which would result in a net loss at a selling price of $ 280–390/g. It should be pointed out that this analysis neglects the complete operating costs associated with the upstream (bioreactor) steps as well as the final formulation (lyophilization, formulation, and filling). It is clear from the analysis that the affinity-chromatography step is the primary driver of the high cost. The costs associated with the ConA affinity purification in terms of capital equipment (∼30% of the total equipment purchase cost), ConA resin costs (∼76% of the consumables and ∼16% of the annual operating costs), ConA buffers (∼60% of the raw materials and ∼20% of the operating costs) contribute significantly to the economics and indicate a target area for process improvement.

The ConA chromatography step is also the longest unit procedure, taking up to 72% of the overall batch time and is the equipment-scheduling bottleneck for the process, while other major equipment units are idle for much of the time (Figure 14.3).

An environmental assessment of the case-study process was performed based on the SuperPro Designer® model. Figure 14.4 shows the Mass Index (*MI*) and Environmental Index (*EI*) of components (input and output) in the process. The Mass Index is derived from the mass balance and indicates how much of a certain component of the mass balance is consumed or formed per unit amount of final product produced (kg/kg rAAT). The *EI*

Table 14.1 *Operating parameters for case-study process flowsheet (base case)*

Process characteristic	Value	Unit
Preparation for ConA		
Concentration in ultrafiltration (UF)	10	times
Denaturation in UF	0	%
ConA separation		
Material removed	α-amylase	
Active rAAT binding capacity	0.25	mg/mL of resin
Load and elute velocity	50	cm/h
Active rAAT binding	100	%
Active rAAT Yield	95	%
Resin cost	4000	$/L
Resin replacement frequency	400	cycles
Wash and regeneration velocity	60	cm/h
Equilibration velocity	300	cm/h
Column volumes to elute the rAAT	3	Column volume (CV)
Column volumes remaining in product pool	1	CV
Methyl α-D-mannopyranoside	50	$/kg
Number of cycles	5	
Number of columns	3	
Preparation DEAE		
Concentration prior to diafiltration (DF)	2	times
Active rAAT denaturation in DF	0	%
DEAE separation		
Material removed	Low-MW protein	
Active rAAT binding capacity	20	mg/mL of resin
Load and elute velocity	150	cm/h
Active rAAT binding	100	%
Active rAAT yield	95	%
Resin cost	481	$/L
Resin replacement frequency	100	cycles
Wash, equilibration, and regeneration linear velocity	300	cm/h
Column volumes to elute the rAAT	10	CV
Column volumes remaining in product pool	2	CV
Number of cycles	1	
Preparation for octyl		
Concentration prior to DF	2	times
Denaturation in DF	5	%
Octyl separation		
Material removed	'inactive AAT'	
Active rAAT binding capacity	20	mg/mL of resin
Load and elute velocity	75	cm/h
Active rAAT binding	100	%
Active rAAT yield	95	%
Resin cost	1080	$/L
Resin replacement frequency	100	cycles
Wash, equilibration, and regeneration linear velocity	150	cm/h
Column volumes to elute the rAAT	8	CV
Column volumes remaining in product pool	4	CV
Number of cycles	1	
Final formulation		
Concentration prior to DF	2	times
Denaturation in DF	5	%

Table 14.2 *Summary economic evaluation for base-case process*

Batch Information	
Number of batches per year	53
Mass of active rAAT (kg/year)	32.7
Total capital investment	$ 21.1 million
Annual operating costs	$ 25.5 million
Raw materials	$ 8.37 million
ConA buffers	$ 5.03 million (60%)
Consumables	$ 5.45 million
Harvest dead-end filters	$ 1.12 million (20%)
ConA resin	$ 4.16 million (76%)
Waste treatment	$ 7.52 million
Unit-production cost	$ 780/g

connects the mass consumed or formed to the environmental *relevance* of a compound. The case-study process is characterized by a high material intensity as is typical for pharmaceutical/biotech processes; however, the substances involved in the process have only a low or medium environmental potential (expressed by relatively low *EI* values), which is also typical for bioprocesses.

For the case-study process the most environmentally relevant components are ammonium sulfate, Tris · HCl, NaOH, and sodium acetate. Tris · HCl, is used in buffers for the ConA

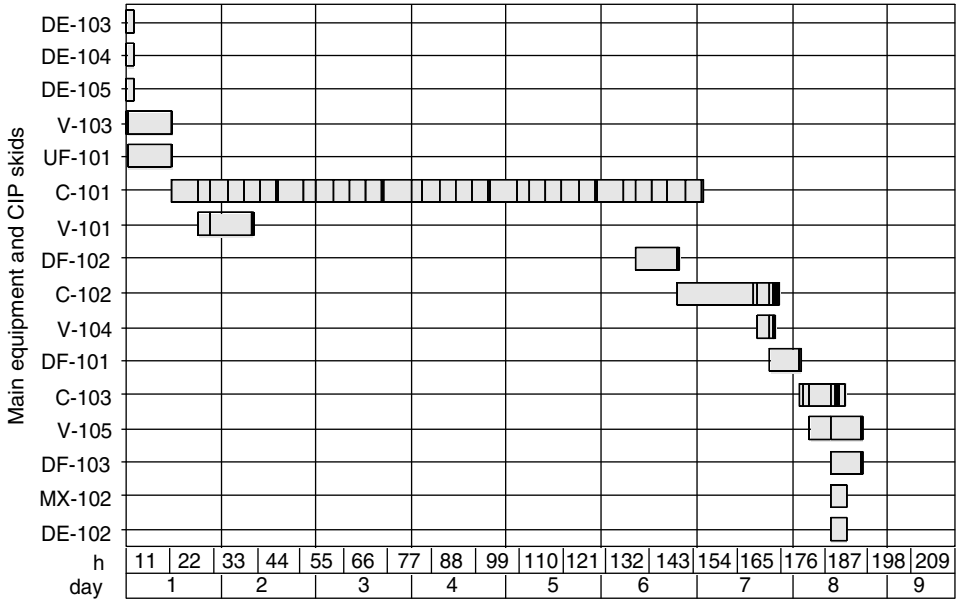


Figure 14.3 *Equipment utilization during each batch*

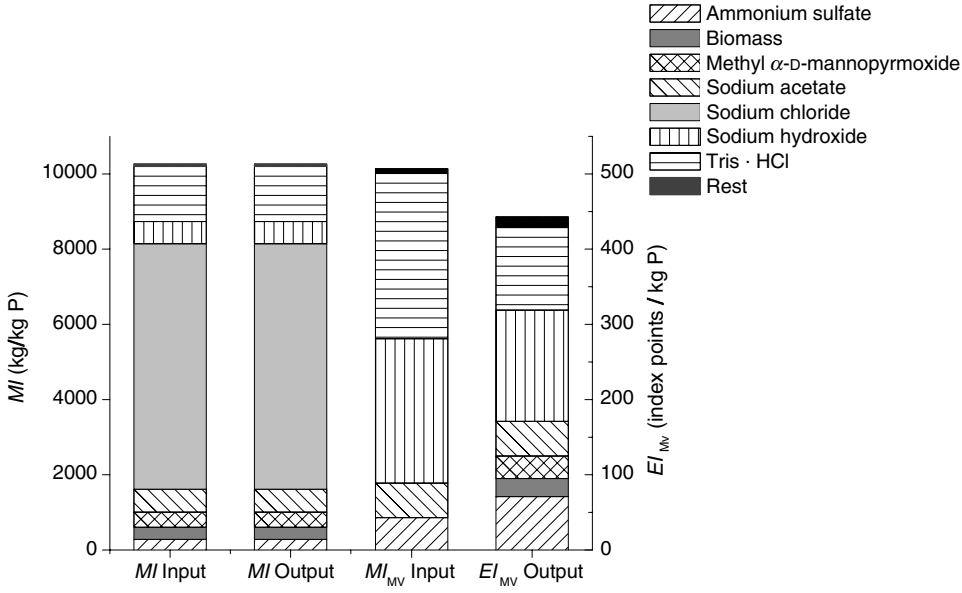


Figure 14.4 *Mass Indices and Environmental Indices (EI_{MV}) of the α-1-antitrypsin production excluding water*

separation (equilibration, wash, elution, and regeneration) and sodium acetate is used for the regeneration of the ConA column. Ammonium sulfate is added during the diafiltration step between the DEAE column and the Octyl column to reach a 1 M ammonium sulfate concentration. Ammonium sulfate is the salt most commonly used to control adsorption in hydrophobic-interaction chromatography due to the fact it has a high solubility and is inexpensive. Sodium hydroxide is used for cleaning/sterilization of the DEAE and Octyl chromatography resins and vessels used in the process.

14.5 Conclusions

A model was developed for the recovery and purification of recombinant AAT from transgenic rice-cell suspension cultures using a three-step chromatographic separation and intermediate diafiltration steps. Preliminary economic assessment of the base-case model indicates that the unit-production cost would be significantly higher than the wholesale price of human plasma-derived AAT on the market. Either improving the affinity purification step or using alternative, lower-cost chromatography or ultrafiltration steps would improve process economics. As with other biologics-manufacturing processes, the environmental impact is predicted to be low to moderate, although there is still the potential for improvement by investigating the process impact of alternatives to ammonium sulfate, Tris · HCl, NaOH, and sodium acetate.

Suggested Exercises

1. One of the most critical parameters in this model is the ConA resin's binding capacity (under operating conditions), which was initially set to 0.25 mg/mL in the base-case model. Keeping the product target throughput at 32.7 kg active rAAT/year and the ConA bed height at 0.5 m, determine the effect of ConA resin binding capacity [in Operating Data, LOAD-1 (PBA Column Loading) in Operation Cond's tab] on unit-production cost and number of batches per year. Be sure to keep the sizing of the ConA column in the Calculated (Design mode). Note that as ConA resin binding capacity is increased, the required number of ConA columns can be reduced and the number of cycles per batch can also be varied. For a given resin binding capacity and number of columns find the optimum Number of Cycles per Batch of ConA processing cycles (in Procedure Data under the Scheduling tab) which provide the lowest unit cost. Plot the unit cost as a function of ConA resin binding capacity.
2. Investigate the influence of ConA loading and elution linear velocity [also in Operating Data, LOAD-1 (PBA Column Loading) in Oper. Cond's tab], while keeping the overall throughput and column size/number/cycles constant, on unit cost and number of batches per year. Plot the unit cost as a function of ConA loading and elution linear velocity.
3. Investigate the influence of starting-material properties, particularly the concentration of active rAAT on the unit cost, keeping other critical parameters and product target throughput constant.

References

[14.1] Kim, S., Woo, J., Seo, E., Yu, M., Ryu, S. (2001): A 2.1 Å resolution structure of an uncleaved α-1-antitrypsin shows variability of the reactive center and other loops. *J. Mol. Biol.*, **306**, 109–119.

[14.2] Blank, C., Brantly, M. (1994): Clinical features and molecular characteristics of α-1-antitrypsin deficiency. *Ann. Allergy*, **72**, 105–121.

[14.3] Elliott, P., Pei, X., Dafforn, T., Lomas, D., (2000): Topography of a 2.0 A structure of α-1-antitrypsin reveals targets for rational drug design to prevent conformational disease. *Protein Sci.*, **9**, 1274–1281.

[14.4] Huggins, F. (2005): α-1-Proteinase inhibitor (human) (Aralast™). University of Utah. Health Science Center, Salt Lake City. Available at: http://uuhsc.utah.edu/pharmacy/bulletins/aralast.html (June 11, 2005).

[14.5] Rodriguez, R. (1998): Functional α-1-antitrypsin from rice cell culture: A new expression system for the biotech industry. IBC 4th Annual International Conference on Commercial Opportunities and Clinical Applications of Cloning and Transgenics, San Francisco.

[14.6] Terashima, M., Murai, Y., Kawamura, M., Nakanishi, S., Stoltz, T., Chen, L., Drohan, W., Rodriguez, R., Katoh, S. (1999): Production of functional human α-1-antitrypsin by plant cell culture. *Appl. Microbiol. Biotechnol.*, **52**, 516–523.

[14.7] Terashima, M., Ejiri, Y., Hashikawa, N., Yoshida, H. (1999): Effect of osmotic pressure on human α-1-antitrypsin production by plant cell culture. *Biochem. Eng. J.*, **4**, 31–36.

[14.8] Terashima, M., Ejiri, Y., Hashikawa, N., Yoshida, H. (2000): Effects of sugar concentration on recombinant human α-1-antitrypsin production by genetically engineered rice cell. *Biochem. Eng. J.*, **6**, 201–205.

[14.9] Terashima, M., Ejiri, Y., Hashikawa, N., Yoshida, H. (2001): Utilization of an alternative carbon source for efficient production of human α-1-antitrypsin by genetically engineered rice cell culture. *Biotechnol. Prog.*, **17**, 403–406.

[14.10] Huang, J., Sutliff, T., Wu, L., Nandi, S., Benge, K., Terashima, M., Ralston, A., Drohan, W., Huang, N., Rodriguez, R. (2001): Expression and purification of functional human α-1-antitrypsin from cultured plant cells. *Biotechnol. Prog.*, **17**, 126–133.

[14.11] Trexler, M., McDonald, K., Jackman, A. (2002): Bioreactor production of human α-1-antitrypsin using metabolically regulated plant cell cultures. *Biotechnol. Prog.*, **18**, 501–508.

[14.12] Trexler, M., McDonald, K., Jackman, A. (2005): A cyclical semi-continuous process for production of human α-1-antitrypsin using metabolically induced plant cell suspension cultures. *Biotechnol. Prog.*, **21**, 321–328.

[14.13] McDonald, K., Hong, L., Trombly, D., Xie, Q., Jackman, A. (2005): Production of Human α-1-antitrypsin from transgenic rice cell culture in a membrane bioreactor. *Biotechnol. Prog.*, **21**, 728–734.

15

Plasmid DNA

*Sindélia S. Freitas, José A. L. Santos, D. Miguel F. Prazeres**

15.1 Introduction

15.1.1 General

Gene therapy [15.1] and DNA vaccination [15.2, 15.3] belong to a new class of molecular therapies which use nucleic acids as therapeutic and prophylactic agents of cells. In many cases whole genes are delivered and expressed in the target human or nonhuman cells, yielding the corresponding therapeutic protein. Depending on the pathology, this protein may (i) replace a defective protein (e.g. cystic fibrosis [15.4]), or (ii) trigger the immune system in order to kill tumor cells [15.5] or immunize individuals against pathogens such as the malaria agent *Plasmodium falciparum* [15.6]. Both viral and nonviral vectors such as plasmid DNA (pDNA) have been used to efficiently deliver the therapeutic gene to target cells [15.7]. Plasmid DNA molecules are extra-chromosomal carriers of genetic information which have the ability to replicate autonomously. These vectors constitute an attractive gene-transfer system since they are safer and easier to produce when compared with viral vectors [15.1, 15.8]. However, since pDNA vectors are less effective in transfecting cells when compared with viral vectors [15.8], the full treatment or vaccination of one individual may require milligram amounts of pDNA. Clearly, large-scale pDNA-manufacturing processes are needed to meet the demand associated with the large number of gene therapy and DNA vaccine applications that are moving from the laboratory to clinical trials and eventually to the market. Plasmids are synthesized *in vivo* by the bacterium *Escherichia coli*. A typical pDNA-production process thus starts with a cell culture (fermentation) step and is followed by a sequence of downstream processing operations as schematized

* Corresponding author: prazeres@alfa.ist.utl.pt, ++351/218419133

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in Figure 15.1. A number of such processes have been described in the recent literature [15.9–15.15]. However, to the best of our knowledge, the economics of pDNA-production processes have not been addressed so far. Thus, the major objective of the study presented is to estimate the cost of the production of a pDNA therapeutic product.

15.1.2 Case Introduction

The case presented in this chapter is based, as is often the case, on a bench-scale process which has been developed specifically for the production and purification of pDNA vectors for gene therapy and DNA vaccination [15.16–15.18]. The majority of the production and purification data used here (productivity, step yields, stream purity, final quality) have been obtained at lab scale using an experimental 7067 bp (base pair) (approx. 4664 kDa) DNA vaccine against rabies. This vaccine encodes the rabies virus glycoprotein [15.17, 15.19]. This process has not been optimized for large-scale production and is used here as an illustrative exercise.

As a design basis we have assumed a plant capacity of around 141 g of purified pDNA produced per batch – this corresponds to a 1000-fold increase in the amount of pDNA obtained at lab scale in the cited study [15.17]. The plant is designed to operate 330 days a year, with a new batch initiated every 48 hours – this corresponds to 164 batches and 23.2 kg of pDNA per year. Guidelines and quality standards issued by regulatory agencies such as the Food and Drug Administration (FDA) and the European Medicines Evaluation Agency (EMEA) have been used to set up product specifications in terms of final purity [15.20–15.22]. Basically, the final bulk pDNA product should be free from proteins and RNA, while endotoxins (ETs) and genomic DNA (gDNA) should not exceed 0.05 μg per μg of pDNA and 0.1 endotoxin units per μg of pDNA, respectively. Finally, we have assumed that at the end of the process the bulk pDNA product will be distributed in vials, each containing a 2 mL pDNA dose in 2 ml of sterile PBS (phosphate-buffered saline) buffer. This roughly corresponds to the maximum single dose of a DNA vaccine which has been used in human trials [15.23, 15.24].

15.1.3 Process Description

The entire flowsheet for the production of pDNA is shown in Figure 15.2. The process is divided into five sections: Fermentation, primary recovery, intermediate recovery, final

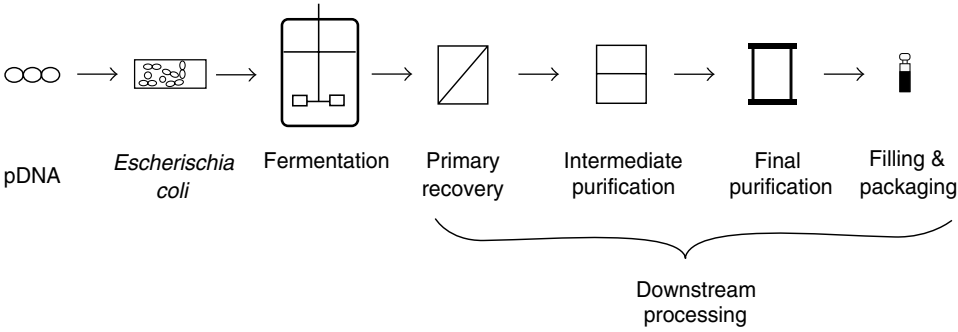


Figure 15.1 *Outline of a typical pDNA-production process*

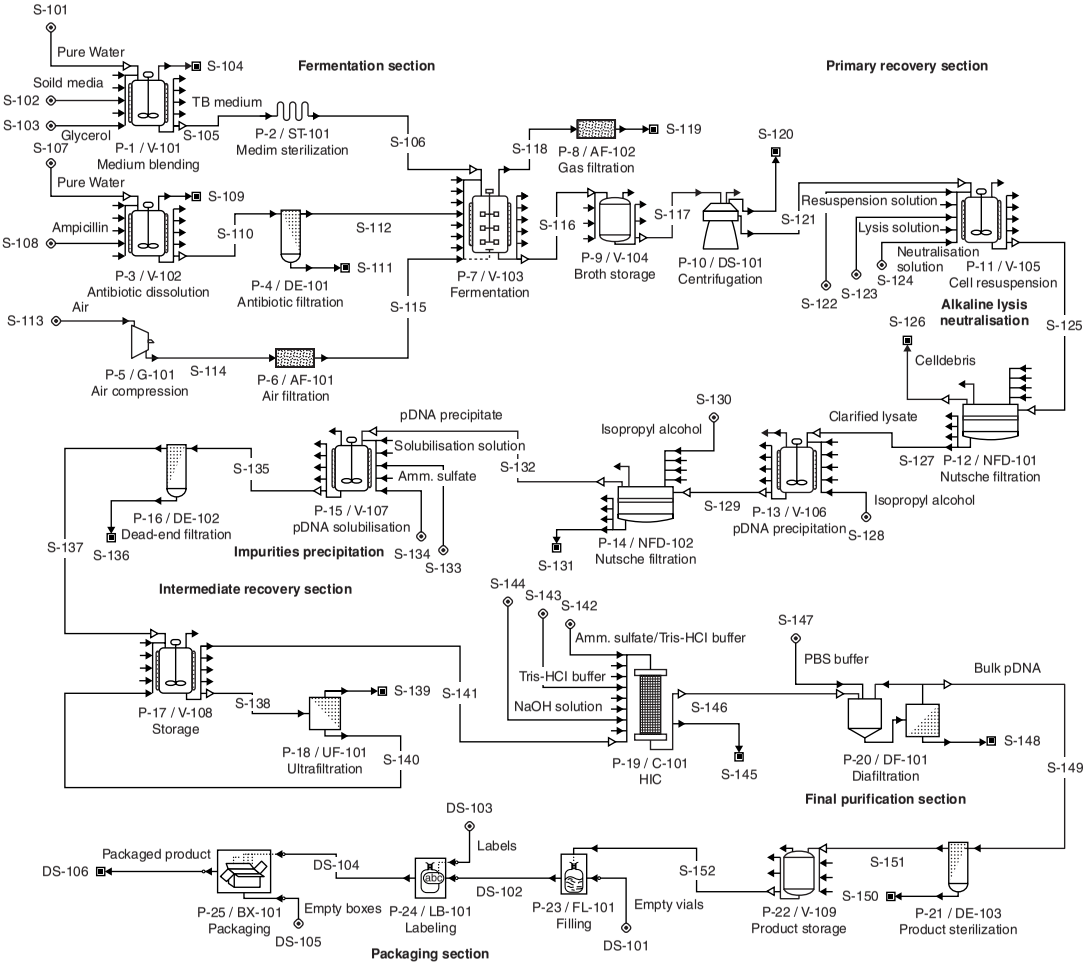


Figure 15.2 *Plasmid-DNA-production flowsheet*

purification, and filling and packaging section. The overall pDNA-recovery yield per batch is around 65% (141 g of pDNA are recovered out of the 216 g that are present in the cell lysate).

Bioreaction Section. Fermentation medium is prepared in a stainless steel tank (P-1) and sterilized in a continuous heat sterilizer (P-2). The axial compressor (P-5) and the absolute filter (P-6) provide sterile air to the fermenter at an average rate of 1.5 vvm. The *inoculum* is prepared in a fermenter train (not shown in this flowsheet) seeded with *E. coli* cells that host the vaccine DNA. The ampicillin solution that is used to prevent the growth of plasmid-free *E.coli* cells and other microorganisms is prepared in a stainless steel tank (P-3) and sterilized by filtration (P-4) because of its heat sensitivity. Fermentation is carried out in the fermenter V-103 for 24 hours at 37 °C. The final concentration of cells in the fermenter is 7 g/L (dry cell weight, DCW). At the end of fermentation the broth is transferred to the storage vessel P-9. After completing the fermentation, the equipment is washed and sterilized with steam in order to prepare it for the next batch.

Downstream Sections

(i) *Primary recovery section*

Cells are harvested in a disk stack centrifuge (P-10) at 14 300 *g* (98% yield assumed). During centrifugation the broth is concentrated approximately 20-fold from 4414 L to 204 L. The subsequent lysis of cells to release pDNA is probably the most critical and troublesome of all unit operations in the downstream processing. High amounts of intact, supercoiled pDNA must be released to the surrounding medium in order to guarantee high overall process yields. Other intracellular components such as RNA, gDNA (genomic DNA), endotoxins, and proteins also are released. Shear and chemical sensitivity of pDNA and gDNA molecules [15.25], as well as the high viscosity of the process streams due to the large concentration of nucleic acids [15.26], are of major concern during this stage [15.27].

After centrifugation, the cell paste is resuspended in 450 L of resuspension solution in a blending tank (P-11). Cell lysis is performed by adding the same volume of an alkaline solution [200 mM NaOH, 1% w/v sodium dodecyl sulfate (SDS)] with gentle stirring. Cell debris, gDNA, and proteins are precipitated by adding 187 L of pre-chilled 3 M potassium acetate (pH 5.5). The precipitate is removed by filtration (P-12). Operation temperature is maintained at 4 °C in order to avoid lysate degradation. The neutralized and clarified lysate is subjected to further purification.

(ii) *Intermediate recovery section*

This section has two objectives: to concentrate pDNA and to remove a large fraction of impurities before the final purification steps. The clarified lysate is transferred to a blending tank (P-13) and pDNA is precipitated by adding 0.7 vols of isopropyl alcohol. In a filtration unit (P-14) the supernatant is removed and the precipitated pDNA is washed with isopropyl alcohol in order to remove salt ions. The pDNA is then transferred to a tank (P-15) and re-dissolved in 300 L of 10 mM Tris•HCl buffer (pH 8.0). Solid ammonium sulfate is dissolved in this solution under gentle agitation up to a concentration of 2.5 M in order to precipitate protein, endotoxin, and RNA impurities. The final solution is filtred (P-16) in order to remove the precipitate. At

this stage the volume of the stream, which contains partially purified pDNA, is 368 L. An ultrafiltration step (P-18) is included to concentrate pDNA 10-fold, in preparation for the subsequent chromatographic operation.

(iii) *Final purification section*
The final purification is based on hydrophobic-interaction chromatography (HIC) which is carried out in column P-19. Adsorbent matrix is a commercial phenyl-Sepharose HIC gel (Amersham Pharmacia). HIC exploits the minimal interaction of pDNA (supercoiled and open-circle iso-forms) with the adsorbent matrix when compared with the remaining impurities (RNA, gDNA, and ETs) [15.17, 15.18]. Therefore, chromatography is run in the negative mode since impurities are adsorbed while the target molecule flows through the column. As a result, only 6% of the resin's dynamic capacity is used. Isocratic elution of bound material is carried out in a step mode with a low-ionic-strength buffer. Finally, column cleaning is performed with 1 M NaOH. The pDNA-containing fraction is then dialysed (P-20) against PBS buffer to remove ammonium sulfate. The retentate is sterilized by microfiltration (P-21) to assure the absence of contaminants before filling and packaging.

(iv) *Filling and packaging section*
The bulk pDNA product (about 2 mg in 2 mL) is filled in vials that are labeled and packed. Each individual pack of final product contains three vials. The impact of this section in the overall economic performance of the process was not taken into account in this case study. However, it should be noted that, for low-dose products, this can be a significant part of the final product cost.

Process Scheduling. The scheduling and equipment utilization for two consecutive batches is shown in Figure 15.3. The plant batch time is approximately 64 h, with a new batch initiated every 48 h. This batch start time roughly corresponds to the beginning of the purification section in the previous batch. The fermentation procedure (nutrient and ampicillin charge, growth, transfer of broth to storage, CIP – cleaning in place, and SIP – steaming in place) taking place in fermenter P-7, with a duration of approximately 32 h, is clearly identified in the chart as the time bottleneck.

15.2 Model Description

15.2.1 Bioreaction Section

The following overall reaction has been used to describe the conversion of nutrients into pDNA-containing biomass:

$$\begin{aligned} &CH_{2.67}O + 5.05CH_{1.91}O_{0.56}N_{0.23} + 2.68CH_{1.795}O_{0.3}N_{0.2} \\ &\quad +1.96O_2 \rightarrow 7.07CH_{1.77}O_{0.49}N_{0.24} + 1.66CO_2 + 2.31H_2O \end{aligned} \tag{15.1}$$

The empirical formulas for yeast extract ($CH_{1.91}O_{0.56}N_{0.23}$), tryptone ($CH_{1.795}O_{0.5}N_{0.2}$), and biomass ($CH_{1.88}O_{0.49}N_{0.24}$), in their reduced form, were taken from Doran [15.28]. The chemical formulation of glycerol was adopted in the reduced form as well. The coefficients were calculated by performing stoichiometric balances on C, H, O, and N elements [15.28]

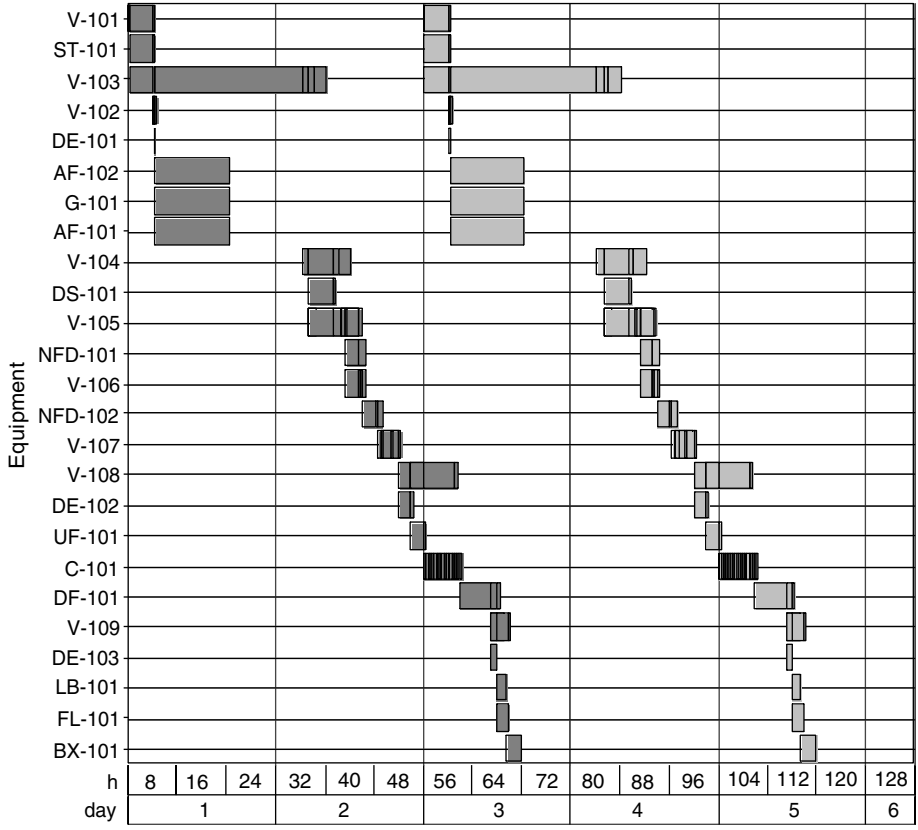


Figure 15.3 *Gantt chart for production scheduling of two consecutive batches of pDNA*

and by further assuming a respiratory coefficient equal to 0.845 (close to values reported by [15.29]) and an equal consumption of yeast extract and tryptone.

The *E. coli* cells obtained at the end of fermentation are assumed to have a typical composition in terms of the major components. On a dry cell weight basis and from calculations made with published data [15.30], this corresponds to 50% protein, 20% RNA, 16.7% endotoxins, 1.7% gDNA, and 10.9% of other components (small ions, lipids, carbohydrates, etc.). Plasmid DNA corresponds to 0.7% of the total dry cell weight [15.17].

15.2.2 Downstream Sections

Primary Recovery Section. The overall yield of each cell component in the primary recovery section (lysis and filtration) was estimated on the basis of literature data. Plasmid DNA and gDNA yields of 80 and 60%, respectively, were used on the basis of experimental data published by Ciccolini et al. [15.31]. The majority of endotoxins were removed in this section (94%) as indicated by the endotoxin analysis presented by Diogo et al. [15.17]. As

for protein and RNA, experimental data (Freitas, unpublished data) points to reductions of 99% and 66%, respectively.

Intermediate Recovery Section. The isopropyl alcohol precipitation step concentrates the pDNA 3-fold with an 86% yield ([15.17]; Freitas, unpublished data) and around 40% of the proteins are removed (Freitas, unpublished data). No relevant removal of RNA and endotoxins is achieved (both around 14%), because these molecules have hydrophobic groups that interact strongly with the aliphatic chains of isopropyl alcohol [15.32] and are precipitated with pDNA. No pDNA is lost in the subsequent ammonium sulfate precipitation step, which removes 99.8% of endotoxins [15.17], 97% of gDNA [15.33], 83% of protein (Freitas, unpublished data), and 97% of RNA. We have further assumed that this step has the ability to remove the remaining components of *E. coli*.

Final Purification Section. The core of the final purification section is the HIC operation, which is run in the negative mode to bind impurities while product passes through. On the basis of experimental data obtained in a scale-up study of this operation [15.18] the dimensions of this column were set as 40 cm diameter and 20 cm bed height. As an optimistic assumption, the maximum feed volume was assumed to correspond to 30% of the bed volume. Thus, five consecutive column cycles are needed to process the 37 L of the incoming stream (S-141). If a greater capacity of this step were desired there is room to add additional cycles without changing the column or loadings. Each cycle comprises five distinct operations: (i) equilibration with two bed volumes (BVs) of 1.5 M ammonium sulfate in 10 mM Tris solution (pH 8.0) at 150 cm/h, (ii) loading of 7.36 L of feed at 30 cm/h, (iii) washing with 0.63 BV of equilibration buffer at 30 cm/h (pDNA is recovered here with a 95.4% yield), (iv) elution of bound and weakly bound impurities with 1 BV of 10 mM Tris solution (pH 8.0) at 150 cm/h, and (v) column cleaning with 2 BV of 1 M NaOH. The total cycle time is 1.28 h.

15.3 Inventory Analysis

The overall material balance per batch is summarized in Table 15.1. Apart from pDNA and gases, all output materials end up in liquid-waste streams, which are disposed of after adequate treatment in order to minimize environmental impacts. Water is the major raw material used (94%), most of it for equipment cleaning. This is typical in the production of biopharmaceuticals (see, e.g., production of immunoglobulin G [15.34] and recombinant β-glucuronidase [15.35]). Yeast extract, tryptone, and glycerine are used as medium components in the fermentation step. Large amounts of isopropyl alcohol and ammonium sulfate are required in the downstream processing. Isopropyl alcohol is mainly used as a precipitating agent for the purpose of concentrating pDNA. Although it can be recycled after distillation for reuse in the process, this option has not been considered here. As will be seen later, isopropyl alcohol will have a negative environmental impact on the process. Ammonium sulfate is used mainly as a precipitating agent in an impurity-reduction step and as a buffer component in the chromatographic purification.

Table 15.1 *Overall material balances for pDNA production (kg/year). The overall pDNA recovery yield is 65.5%*

Component	Total inlet (kg/year)	Total outlet (kg/year)	Product (kg/year)
Ammonium sulfate	30 883	30 883	
Ampicillin	74	74	
Biomass	0	103	
Carbon dioxide	0	2131	
EDTA disodium	248	248	
Endotoxins	0	845	
gDNA	0	86	
Glucose	665	665	
Glycerine	3739	2841	
isopropyl alcohol	127 005	127 005	
Sodium dihydrogen phosphate	199	199	
pDNA	0	35.4	23.2
Potassium acetate	9037	9037	
Proteins	0	2531	
RNA	0	1012	
SDS	738	738	
Small molecules	0	552	
Sodium chloride	1448	1448	
Sodium hydroxide	6494	6494	
Tris•HCl	486	486	
Tryptone	8 856	6929	
Water	3 193 243	3 517 906	
Water for injection (WFI)	323 447		
Yeast extract	17 712	13 858	
Total (raw materials)	3 724 272	3 726 105	
Nitrogen	1 442 017	1 442 852	
Oxygen	437 768	436 188	
Total (raw materials + air)	5 604 057	5 605 145	

15.4 Economic Assessment

Table 15.2 shows the key economic evaluation results for this project. Economic evaluations were based on the following assumptions: (i) the entire direct fixed capital is depreciated linearly over a period of ten years, assuming a 10% salvage value for the entire plant, (ii) the project lifetime is 15 years, and (iii) 23.2 kg of final product will be produced per year. For a plant of this capacity, the total capital investment is around $ 24 million. The unit-production cost is $ 2.25/pack with 3 vials containing 2 mg of pDNA each ($ 0.38/mg of pDNA). Since pDNA products have not yet reached the market, we have tentatively assumed a selling price of $ 10.00/3-vial pack ($ 1.67/mg of pDNA). The project thus yields an after-tax internal rate of return (IRR) of 63% and a net present value (NPV) around of $ 117 million (assuming a discount interest of 7%). This economic picture could change by the inclusion of costs that were not accounted for in this case-study (filling and packaging section, R&D, process validation).

Table 15.2 *Key economic evaluation results for pDNA production*

Economic parameter	Value
Direct fixed capital (DFC)	$ 21.4 million
Total capital investment (TCI)	$ 23.9 million
Plant throughput	23.2 kg pDNA/year
Operating cost	$ 8.7 million/year
Unit-production cost (UPC)	$ 2.25/pack 6 mg of pDNA
Selling price	$ 10/pack 6 mg of pDNA
Revenues	$ 38.7 million/year
Gross profit	$ 30 million/year
Taxes (40%)	$ 12 million/year
Net profit	$ 20 million/year
Internal rate of return (IRR) (after taxes)	63%
Net present value (NPV) (at 7.0% interest)	$ 117 million

The total equipment purchase cost was estimated to be around $ 3.6 million. The cost of unlisted equipment (including the equipment in the *inoculum* preparation section) was assumed to represent 20% of the total equipment cost.

The breakdown of the annual operating cost (AOC) is shown in Figure 15.4. Note that the cost of utilities (electricity, steam, and cooling agents) is minimal, representing 0.5% of the AOC. Facility-dependent cost (45% of the AOC) is the principal operating cost in this process, as is typical for high-value products which are produced in small quantities [15.16]. Labor-related costs come next, accounting for 25% of the AOC. The annual cost of raw materials is around $ 1.0 million, 52% of which is associated with the fermentation medium (tryptone and yeast extract). The cost of consumables is 10% of the AOC, around $ 900 000. The chromatographic resin (Phenyl-Sepharose) used in the HIC step represents 60% of this value.

Waste treatment and disposal amount to $ 167 000 per year, approximately 2% of the AOC. Around 53% of this cost is designated to treatment of the aqueous waste and around 32% to the treatment of the organic waste generated in the pDNA precipitation with

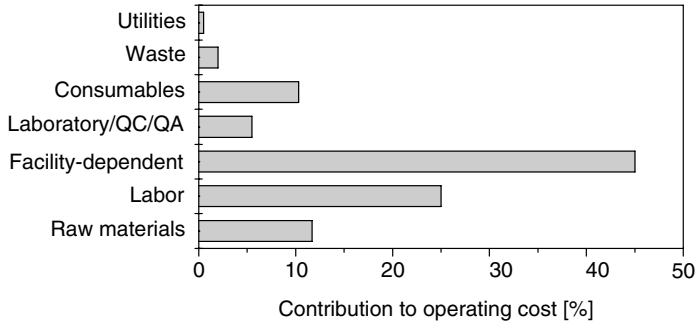


Figure 15.4 *Breakdown of the annual operating cost*

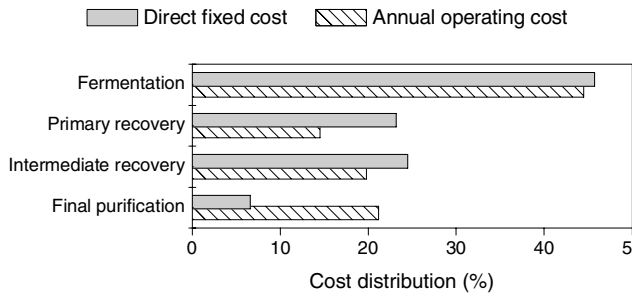


Figure 15.5 *Breakdown of annual operating cost per process section*

isopropyl alcohol. The disposal of the chromatographic resin cost represents 15% of this value.

Figure 15.5 shows a breakdown of the annual operating cost (AOC) and direct fixed capital (DFC) per process section. Note that the DFC is most elevated in the fermentation section as a result of the cost of equipment units, such as the fermenter vessel. The DFC costs in the primary and intermediate recovery sections are linked to the solid–liquid separation equipment (centrifuge, Nutsche filters) used, while in the final purification the chromatographic column accounts for most of the DFC costs.

The operating cost associated with the fermentation step is 44% of the AOC. Downstream processing (primary recovery, intermediate recovery, and final purification steps) accounts for 56% of the AOC. The operating costs of the fermentation section are related with the medium and labor-dependent cost, and the major cost in the final purification section is the chromatographic resin.

Table 15.3 shows economic parameters of this process. Assuming that pDNA is sold for $ 10.00/3-vial pack ($ 1.67 million/kg), the annual revenue will amount to $ 38.7 million for an annual production of 23 kg of pDNA. The return on investment (ROI) will be 89% with a payback time of 1.2 years. In order to determine the effect of the selling price on ROI and payback time, the selling price of the pack was allowed to vary from $ 5 to $ 20.

Table 15.3 *Profitability analysis for pDNA production*

A. Total capital investment (TCI)	$ 23.9 million
B. Revenue	23.2 kg/year
C. Selling price	$ 1 667/g
D. Revenue	$ 38.7 million/year
E. Annual operating cost	$ 8.7 million/year
F. Gross profit (D − E)	$ 30 million/year
G. Taxes (40%)	$ 12 million/year
H. Net profit (F − G + depreciation)	$ 20 million/year
Gross margin (F/D)	78%
Return on investment (H/A × 100)	89%
Payback time (A/H)	1.20 years

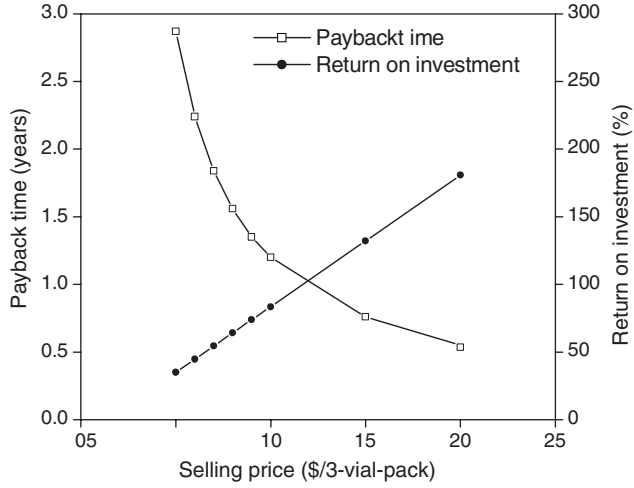


Figure 15.6 *Return on investment and payback time at different pDNA selling prices*

As shown in Figure 15.6, the ROI increases around 10% for every $ 1 increase in the selling price of each pack. In the other side, the payback time declines for every $ 1 increase. At the lowest selling price ($ 5) the payback time increased to around 2.8 years.

15.5 Environmental Assessment

On the basis of the input and output materials provided by SuperPro Designer® (Table 15.1), Environmental Indexes (*EI*) and Impact Group Indexes were calculated. The EIs connect the mass consumed or formed to the environmental relevance of a compound, and make it possible to identify the environmentally most crucial components of the mass balance. Figure 15.7 shows the results obtained for input and output components in the case study. Ammonium sulfate and isopropyl alcohol are clearly the materials that have a high impact on the environment. The *EI* of isopropyl alcohol could certainly be reduced by introducing an isopropyl alcohol recycling step in the process. For instances, if a 70% recycling of isopropyl alcohol is assumed, the environmental impact is reduced by approximately 50%, as shown in Figure 15.7. A further reduction in the environmental impact by 70% could be obtained if the isopropyl alcohol pDNA-concentrating step is replaced by an environmentally friendly alternative operation (e.g. ultra- or microfiltration).

The environmental impact of the output materials in different Impact Group Categories (component risk, organisms, air, water/soil) is shown in Figure 15.8 for the current process and for the alternative, isopropyl alcohol-recycle and isopropyl alcohol-free processes. Clearly, reduction or elimination of isopropyl alcohol significantly decreases the impact in all group categories. Elimination of isopropyl alcohol altogether would also eliminate the costs associated with the treatment of the corresponding organic waste (approximately 34% of the total $ 167 000/year costs associated with waste treatment and disposal, as seen in the previous section) and points to the benefits of aqueous processes.

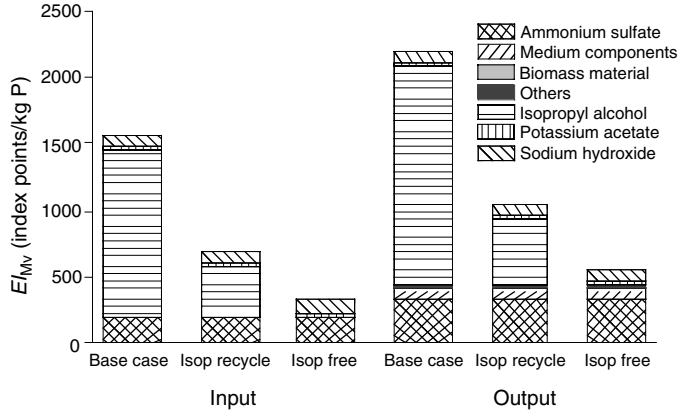


Figure 15.7 *Environmental Index (EI_{MV}) of the base-case process and of alternative processes which assume a 70% recycling of isopropyl alcohol (Isop recycle) or a replacement of the isopropyl alcohol precipitation for microfiltration step (Isop free)*

15.6 Discussion

In this chapter we have used the process simulator SuperPro Designer® to analyse a process for the production of a plasmid DNA product hosted in *E. coli* cells, and subsequent purification up to therapeutic grade. The inventory, economic, and environmental analyses performed have highlighted a number of possible improvements. This is not surprising

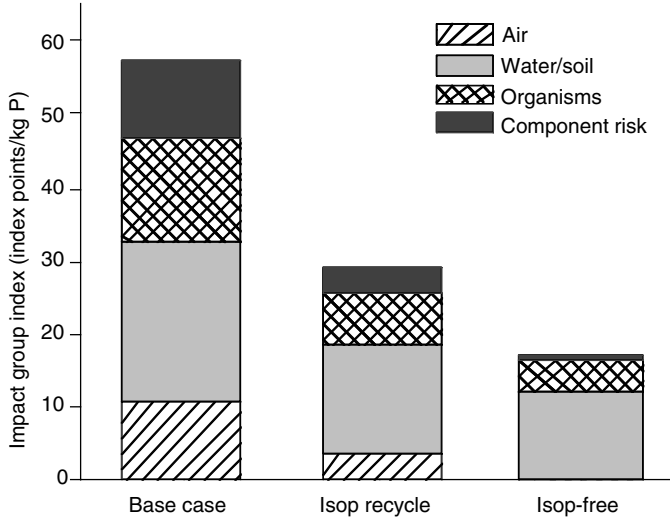


Figure 15.8 *Environmental impact of the output and contribution of the different impact groups for base-case process and for the alternative isopropyl alcohol-recycle (Isop recycle) and isopropyl alcohol-free processes (Isop free)*

given that the case-study process selected was established at lab scale and has not been optimized.

The identification of the fermentation procedures (32 h) as the time bottleneck in the process is common and expected, given the relative speediness of the individual steps in downstream processing (64 h). Reduction of fermentation time, however, is unlikely. However, there is definitely room for fermentation optimization in order to increase cell density (7 g dry cell weight/L) and associated pDNA productivity (49 mg/L). For instances, cell densities and pDNA productivity figures higher than 100 g/L and 100 mg/L, respectively, have been reported in the literature for fed-batch fermentations with a rich medium [15.36]. A higher fermentation productivity obtained in a fed-batch mode with a richer medium would nevertheless increase associated costs and eventually the duration. It should be kept in mind that the host-cell strain and pDNA construction will also impact the performance of the fermentation. The model assumes one fermenter. However, the use of the two smaller fermenters might be more efficient. Any changes in fermentation performance will have an impact downstream as product loading and product/impurity ratios will change. Thus, understanding the impact through simulation will help in robust process design.

Inventory analysis identified water, fermentation components (yeast extract and tryptone), isopropyl alcohol, and ammonium sulfate as the major raw materials, a picture which is not likely to change unless specific unit operations are replaced. Major raw-material costs (50%) are related to fermentation components.

A clear improvement in the downstream processing sections would be the replacement of the isopropyl alcohol pDNA precipitation, which is designed essentially as a concentration step for an equivalent membrane step. This would benefit the process by: (i) reducing the overall cost of raw materials (14%), (ii) reducing the environmental impact associated with the use and disposal of isopropyl alcohol (70%), and (iii) reducing costs associated with the treatment and disposal of liquid waste (32%).

Economic analysis for the case-study scenario considered (23 kg pDNA/year, 164 batches/year) indicate a unit-production cost of $ 375/g, a figure which falls within the production costs of recombinant biopharmaceuticals such as β-glucuronidase ($ 43/g, [15.35]), insulin ($ 42/g, [15.16]) or IgG ($ 910/g, [15.16]). Cost and profitability analysis indicates that pDNA can be economically produced even if selling prices lower that the one considered ($ 1667/g) are practiced.

15.7 Conclusions

The analysis presented in this chapter indicates that the production and purification of a therapeutic pDNA product is economically viable, even with a sub-optimized process. Process improvements will certainly reduce costs and environmental impact. As an educated guess, we may estimate selling prices to be in the range $ 500–$ 1,500/g of pDNA, with unit costs (UPC) lower than $ 500/g of pDNA.

Suggested Exercises

1. Replace the isopropyl alcohol precipitation step (tank P-13 and Nutsche filter P-14) with an equivalent, but environmentally friendly, microfiltration unit operation, and check

the impact of this change on process economics. Assume that the composition of input (S-127) and output (S-132) streams is the same.
2. Assume that an optimization of the fermentation step results in higher biomass (50 g/L). Check the impact on the overall process economics.
3. Model the use of two bioreactors that have together the same working volume as the reactor in the base model. Let them run in staggered mode and compare the results with the base case.

References

[15.1] Mountain, A. (2000): Gene therapy: The first decade. *Trends Biotechnol*., **18**, 119–128.

[15.2] Tighe, H., Corr, M., Roman, M., Raz, E. (1998): Gene vaccination: Plasmid DNA is more than just a blueprint. *Immunol. Today*, **19**, 89–97.

[15.3] Robinson, H., Ginsgerg, H., Davis, H., Johnston, S., Liu, M. (1997): The scientific future of DNA immunization. American Academy of Microbiology, Washington DC.

[15.4] Zabner, J., Cheng, S., Meeker, D., Launspach, J., Balfour, R., Perricone, M., Morris, J., Marshall, J., Fasbender, A., Smith, A. (1997): Comparison of DNA-lipid complexes and DNA alone for gene transfer to cystic fibrosis airway epithelia in vivo. *J. Clin. Invest.*, **100**, 1529–1537.

[15.5] Walther, W., Stein, U. (1999): Therapeutic genes for cancer gene therapy. *Mol. Biotechnol.*, **13**, 21–28.

[15.6] Tuteja, R. (2002): DNA vaccine against malaria: A long way to go. *Crit. Rev. Biochem. Mol. Biol.*, **37**, 29–54.

[15.7] Li, S., Huang, L. (2000): Nonviral gene therapy: Promises and challenges. *Gene Ther.*, **7**, 31–34.

[15.8] Luo, D., Saltzman, W. (2000): Synthetic DNA delivery systems. *Nature Biotechnol.*, **18**, 33–37.

[15.9] Horn, N., Meek, J., Budahazi, G., Marquet, M. (1995): Cancer gene therapy using plasmid DNA: Purification of DNA for human clinical trials. *Hum. Gene Ther.*, **6**, 565–573.

[15.10] Bhikhabhai, R. (2002): Plasmid DNA purification using divalent alkaline earth metal ions and two anion exchangers. US Patent 6 410 274.

[15.11] Varley, D., Hitchkock, A., Weiss, A., Horler, W., Cowell, R., Peddie, L., Sharpe, G., Thatcher, D., Hanak, J. (1998): Production of plasmid DNA for human gene therapy using modified alkaline cell lysis and expanded bed anion exchange chromatography. *Bioseparation*, **8**, 209–217.

[15.12] Lander, R., Winters, M., Meacle, J. (2002): Process for the scaleable purification of plasmid DNA. US Patent application number 0 012 990 A1.

[15.13] Lee, A., Sagar, S. (2002): Method for large scale plasmid purification. US Patent 6 197 553.

[15.14] Kepka, C., Lemmens, R., Vasi, J., Nyhammar, T., Gustavsson, P.E. (2004): Integrated process for purification of plasmid DNA using aqueous two-phase systems combined with membrane filtration and lid bead chromatography. *J. Chromatogr., A,* **1057**, 115–124.

[15.15] Teeters, M., Conrardy, S., Thomas, B., Root, T., Lightfoot, E. (2003): Adsorptive membrane chromatography for purification of plasmid DNA. *J. Chromatogr., A,* **989**, 165–173.

[15.16] Diogo, M., Quiroz, J., Monteiro, G., Martins, S., Ferreira, G., Prazeres, D. (2000): Purification of a cystic fibrosis plasmid vector for gene therapy using hydrophobic interaction chromatography. *Biotechnol. Bioeng.*, **68**, 576–583.

[15.17] Diogo, M., Ribeiro, S., Queiroz, J., Monteiro, G., Tordo, N., Perrin, P., Prazeres, D. (2001): Production, purification and analysis of an experimental DNA vaccine against rabies. *J. Gene Med.*, **3**, 577–584.

[15.18] Diogo, M., Ribeiro, S., Queiroz, J., Monteiro, G., Perrin, P., Tordo, N., Prazeres, D. (2000): Scale-up of hydrophobic interaction chromatography for the purification of a DNA vaccine against rabies. *Biotechnol. Lett.*, **22**, 1397–1400.

[15.19] Bahloul, C., Jacob, Y., Tordo, N., Perrin, P. (1998): DNA-based immunization for exploring the enlargement of immunological cross-reactivity against the lyssaviruses. *Vaccine*, **16**, 417–425.

[15.20] EMEA (1999): Note for guidance on the quality, preclinical and clinical aspects of gene transfer medicinal products. The European Agency for the Evaluation of Medicinal Products, London.

[15.21] USFDA (1996): Points to consider on plasmid DNA vaccines for preventive infectious disease indications. US FDA Center for Biologics Evaluation and Research, Rockville.

[15.22] USFDA (1996): Addendum to the points to consider in Human somatic cell and gene therapy (draft). US FDA Center for Biologics Evaluation and Research, Rockville.

[15.23] Timmerman, J., Singh, G., Hermanson, G., Hobart, P., Czerwinski, D., Taidi, B., Rajapaksa, R., Caspar, C., Van Beckhoven, A., Levy, R. (2002): Immunogenicity of a plasmid DNA vaccine encoding chimeric idiotype in patients with B-cell lymphoma. *Cancer. Res.*, **62**, 5845–5852.

[15.24] Conry, R., Curiel, D., Strong, T., Moore, S., Allen, K., Barlow, D., Shaw, D., LoBuglio, A. (2002): Safety and immunogenicity of a DNA vaccine encoding carcinoembryonic antigen and hepatitis B surface antigen in colorectal carcinoma patients. *Clin. Cancer Res.*, **8**, 2782–2787.

[15.25] Levy, M., Collins, I., Yim, S., Ward, J., Titchener-Hooker, N., Shamlou, P., Dunnill, P. (1999): Effect of shear on plasmid DNA solution. *Bioprocess Eng.*, **20**, 7–13.

[15.26] Ciccolini, L., Shamlou, P., Titchener-Hooker, N., Ward, J., Dunnill, P. (1998): Time course of SDS-alkaline lysis of recombinant bacterial cells for plasmid release. *Biotechnol. Bioeng.*, **60**, 768–770.

[15.27] Prazeres, D., Ferreira, G., Monteiro, G., Cooney, C., Cabral J. (1999): Large-scale production of pharmaceutical-grade plasmid DNA for gene therapy: Problems and bottlenecks. *Trends Biotechnol.*, **17**, 169–174.

[15.28] Doran, P. M. (1995): Bioprocess Engineering Principles. Academic Press, London.

[15.29] Kay, A., O'Kennedy, R., Ward, J., Keshavarz-Moore, E. (2003): Impact of plasmid size on cellular oxygen demand in *Escherichia coli*. *Biotechnol. Appl. Biochem.*, **38**, 1–7.

[15.30] Atkinson, B., Mavituna, F. (1991): Biochemical Engineering and Biotechnology Handbook. Macmillan Publishers Ltd, New York.

[15.31] Ciccolini, L., Shamlou, P., Titchener-Hooker, N. (2002): A mass balance study to assess the extent of contaminant removal achieved in the operations for the primary recovery of plasmid DNA from *Escherichia coli* cells. *Biotechnol. Bioeng.*, **77**, 796–805.

[15.32] Tseng, W., Ho, F. (2003): Enhanced purification of plasmid DNA using Q-Sepharose by modulation of alcohol concentrations. *J. Chromatogr. B: Biomed. Appl.*, **791**, 263–272.

[15.33] Martins, S., Prazeres, D., Cabral, J., Monteiro, G. (2003): Comparison of real-time polymerase chain reaction and hybridization assays for the detection of *Escherichia coli* genomic DNA in process samples and pharmaceutical-grade plasmid DNA products. *Anal. Biochem.*, **322**, 127–129.

[15.34] Petrides, D. (2003): Bioprocess design. In: Harrison, R.G., Todd, P.W., Rudge, S.R., Petrides, D.: Bioseparations Science and Engineering. Oxford University Press, Oxford, pp. 319–372.

[15.35] Evangelista, R., Kusnadi, A., Howard, J., Nikolov, Z. (1998): Process and economic evaluation of the extraction and purification of recombinant β-glucuronidase from transgenic corn. *Biotechnol. Prog.*, **14**, 607–614.

[15.36] Chen, W. (1999): Automated high-yield fermentation of plasmid DNA in *Escherichia coli*. US Patent 5 955 323.

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